

PACKING $k^2 - c$ UNIT SQUARES: A LOWER BOUND OF ORDER $k^{1/4}$ FOR THE DEFICIENCY

SUNGJOON RYU

ABSTRACT. Let $s(n)$ be the side of the smallest square into which n non-overlapping unit squares can be packed, and let $M(k)$ be the largest number of unit squares that can be placed in $[0, k]^2$ so that they are pairwise disjoint as closed sets; then $s(k^2 - c) = k$ exactly when $c < k^2 - M(k)$. Building on a previous paper [R], where $k^2 - M(k) \geq 0.0353 \log k$ is shown for every integer $k \geq 2$, we prove that $k^2 - M(k) \geq 0.1 k^{1/4}$ for every integer $k \geq 4.62 \cdot 10^{12}$ and that $\liminf_{k \rightarrow \infty} (k^2 - M(k))/k^{1/4} \geq 0.1696$. Consequently $s(k^2 - c) = k$ for every integer $k \geq \max(4.62 \cdot 10^{12}, 10^4 c^4 + 1)$, so the largest c with $s(k^2 - c) = k$ grows at least like $k^{1/4}$; by recent work on the wasted area it is $O(k^{3/5})$. The proof follows upward paths from the bottom side (and, by reflection, downward paths from the top side) that adapt to the orientation of the squares they cross and enter squares only through their bottom sides (top sides for the downward paths), organized over a continuum of scales, and it uses the lemmas of [R] on the cost of inclinations. The constants 0.1, $4.62 \cdot 10^{12}$ and 0.1696 rely on the computer-assisted Lemma 4.10 of [R], whose box computations have not yet been reproduced by an independent implementation. Without it, using the analytic Lemma 4.9 of [R] instead, the same argument gives $k^2 - M(k) \geq 0.1 k^{1/4}$ for every integer $k \geq 2.17 \cdot 10^{13}$ and the constant 0.1411. This paper has not been peer reviewed.

CONTENTS

1. Introduction and main results	3
2. Results used from [R]	6
2.1. Packings and the reduction	6
2.2. Elementary geometry	6
2.3. The sharpened fundamental lemma and the cost of inclinations	7
2.4. The form of the multiplicity bound used in this paper	10
2.5. Dependence on a computer-assisted result	11
3. Path flows	12
3.1. Parameters and constraints	12
3.2. Lines, windows and constants	13
3.3. Squares, frames and the floor	15
3.4. Rules of motion	16
3.5. Trajectories	18
3.6. Piecewise-affine structure	19
3.7. The master flow, the scale flows and the ceiling flows	22
3.8. Losses and line quantities	24
4. P1 and P2: the height identity and quantization at bottom sides	28
4.1. Notation and standing assumptions	28
4.2. Trajectories	30
4.3. P1: the height identity	31
4.4. Three geometric lemmas	32
4.5. P2: quantization at bottom sides	34
4.6. Squares of $\mathcal{Z}(s)$: the form used later	37

4.7. Remarks on the hypotheses and on degenerate cases	39
5. End collisions (P3)	40
5.1. Setting	40
5.2. Sources and delays	41
5.3. Pair waste and multiplicity	43
5.4. End collisions	44
5.5. The bound for end collisions	45
6. The shadow inequality (P4*)	46
6.1. Setting and notation	47
6.2. Trajectories and event points	48
6.3. The exceptional set	51
6.4. Statements	53
6.5. Crossings: proof of (a)	54
6.6. The bookkeeping identities: proofs of (b) and (c)	56
6.7. Square crossers: proof of (d)	58
6.8. Gap crossers: proof of (e)	58
6.9. The shadow inequality and the wall term	59
6.10. The order of M and T: proof of (g)	60
6.11. Ceiling flows and measurability in y	61
7. Deaths, merges and gap overlap (P5)	62
7.1. Points with two sources	63
7.2. Valley pairs	66
7.3. Deaths	67
7.4. Merges	68
7.5. Gap overlap	68
7.6. The bound on Λ_0 and the wall term	69
8. P6: the path version of [R, Lemma 4.14]	70
8.1. Setting, notation, and facts used from Section 3	70
8.2. Density	73
8.3. The frame of a pair and the hull slit	73
8.4. The vertical comparison	74
8.5. Disjointness of the vertical slits	76
8.6. Slit inequality, pair waste and multiplicity	77
8.7. Proof of Theorem 8.3	78
8.8. The scale form	79
8.9. Remarks	80
9. P7: the per-scale count	80
9.1. Setting, notation and statements	81
9.2. Parameter facts	83
9.3. Tilt terminations in the scale flows	84
9.4. Squares without alive bottom contacts	86
9.5. Measurability	88
9.6. The per-scale count	89
9.7. The lower bound for $ \mathcal{Z}(s) $	91
9.8. The ceiling versions	91
10. The integration inequality	92
10.1. Standing assumptions and notation	92
10.2. The constants	93
10.3. Results used	94
10.4. The inequality	95

10.5.	Proof of Theorem 10.3	95
10.6.	Remarks on the proof	98
11.	Proof of Theorems A and A_{13} and of Corollary A	99
11.1.	Upper bounds for the constants	99
11.2.	Monotonicity in k	99
11.3.	The explicit part of Theorem A	100
11.4.	The asymptotic constant	100
11.5.	Theorem A_{13} : the variant without the computer-assisted lemma	101
11.6.	Corollary A	102
12.	Constants and thresholds	102
12.1.	Conventions	103
12.2.	Fixed constants	103
12.3.	The parameters of Theorems A and A_{13}	104
12.4.	The asymptotic constant; the variant without [R, Lemma 4.10]	105
12.5.	Further thresholds and bounds at given k	105
12.6.	Sensitivity of the threshold	105
12.7.	The script	107
	Verification status	107
	Acknowledgements	108
	Use of AI	108
	Licence	108
	References	108

1. INTRODUCTION AND MAIN RESULTS

For $n \geq 1$ let $s(n)$ be the side of the smallest square that contains n unit squares with pairwise disjoint interiors (rotations allowed); trivially $s(k^2) = k$, and $s(n) \leq \lceil \sqrt{n} \rceil$. For $n = k^2 - c$ with c small compared with k it is natural to expect that one cannot do better than the $k \times k$ grid with c squares removed. Friedman's survey [Fri] contains the conjecture that $s(n^2 - c) = n$ implies $s((n+1)^2 - c) = n+1$, and Chung and Graham [CG09, concluding remarks] wrote that the same should hold for $n^2 - c$ unit squares when c is fixed and n is sufficiently large. Following the notes [Dan, file `s12/search/FRIEDMAN.md`], put

$$c^*(k) := \max\{c \in \mathbb{Z}_{\geq 0} : s(k^2 - c) = k\}.$$

Since s is non-decreasing, $s(k^2 - c) = k$ holds exactly for $0 \leq c \leq c^*(k)$.

The classical lower bound for the wasted area is that of Roth and Vaughan [RV]: their theorem [RV, p. 170, Theorem] states that the wasted area $w(\alpha)$ in a square of side α satisfies $w(\alpha) \gg (\alpha \|\alpha\|)^{1/2}$ whenever $\alpha(\alpha - \lfloor \alpha \rfloor) > 1/6$, where $\|\alpha\|$ is the distance from α to the nearest integer; this vanishes as $\alpha \rightarrow k^-$, and it gives no information on $c^*(k)$. In the previous paper [R] it is shown that $c^*(k) \rightarrow \infty$, with a logarithmic rate (see below). In the other direction, Erdős and Graham [EG] packed a square of side x with wasted area $O(x^{7/11})$, and this exponent was lowered several times (see the introductions of [McC, Bui]); the exponent $3/5$ was claimed by Chung and Graham [CG], an error in their argument was pointed out by Arslanov and Bui [AB], and the bound $O(x^{3/5})$ was re-established, independently and by different methods, in the preprints of Bui [Bui] and of McClenagan [McC]. Applied to x slightly below k , this gives $s(k^2 - Ck^{3/5}) < k$ for large k , that is, $c^*(k) = O(k^{3/5})$ (the refereed bound $O(x^{(3+\sqrt{2})/7} \log x)$ of [CG09] gives $c^*(k) = O(k^{(3+\sqrt{2})/7} \log k)$). In this paper we show that $c^*(k)$ grows at least like $k^{1/4}$.

Following [R], a *square* is a closed unit square, and a *closed packing* of $[0, k]^2$ is a finite family of squares contained in $[0, k]^2$ that are pairwise disjoint as closed sets. For a closed packing with N

squares, $W := k^2 - N$ is the area of $[0, k]^2$ not covered by the squares; it is an integer. Let $M(k)$ be the largest size of a closed packing of $[0, k]^2$ and $W_{\min}(k) := k^2 - M(k)$ the least value of W . By [R, Lemma 2.1] and its proof, for integers $k \geq 2$ and $0 \leq c < k^2$ one has $s(k^2 - c) = k$ if and only if $k^2 - c > M(k)$, that is, if and only if $c < W_{\min}(k)$; hence $c^*(k) = W_{\min}(k) - 1$. Lower bounds for W that hold for all closed packings of $[0, k]^2$ therefore give $s(k^2 - c) = k$ for all c below them.

In [R] it is shown that $W \geq \kappa \log k$ for every integer $k \geq 2$ and every closed packing, with $\kappa = 0.0353$ ($\kappa = 0.0319$ without the computer-assisted [R, Lemma 4.10]), and that $W \geq 1.99954 (\log k - 13.06675)/30.418$; in particular $c^*(k) \rightarrow \infty$. In this paper we prove a bound of order $k^{1/4}$ for large k . Throughout, $\log$ is the natural logarithm and $\delta := 10^{-5}$.

Theorem A. *Let $k \geq 4.62 \cdot 10^{12}$ be an integer. Then every closed packing of $[0, k]^2$ by N unit squares satisfies*

$$W = k^2 - N \geq 0.1 k^{1/4},$$

in particular $W \geq \lceil 0.1 k^{1/4} \rceil \geq 147$. Moreover

$$\liminf_{k \rightarrow \infty} \frac{W_{\min}(k)}{k^{1/4}} \geq c_{\infty} := \frac{16 \cdot 2^{-1/4} (1 - \lambda)^{5/4}}{5 \cdot 5.0001^{1/4} \sqrt{AA'}} = 0.1696524 \dots,$$

where $A := 4\sqrt{9/0.32}/(2 - 3 \cdot 10^{-6})$, $A' := A/\cos(1.5 \cdot 10^{-6}) + 10^{-8}$ and $\lambda := 2\delta + 2 \cdot 10^{-12}$.

The digits of c_{∞} shown are those of its decimal expansion, i.e. the value is rounded down; A is the constant of [R, Lemma 4.13] ($A = 4\sqrt{K_w^*/Q_*}/(2 - 3 \cdot 10^{-6})$ with $K_w^* = 9$, $Q_* = 0.32$).

The threshold $4.62 \cdot 10^{12}$ has little margin. With the parameters used in the proof, the last inequality of the argument (condition (i) of Proposition 11.2) holds exactly for $k \geq k_*$, where k_* is about $4.6167 \cdot 10^{12}$, and $4.62 \cdot 10^{12}$ is k_* rounded up to three significant digits ($4.62 \cdot 10^{12}/k_* - 1 < 7.3 \cdot 10^{-4}$; Section 12.6). If one of the constants entering the proof were larger than the tolerance stated in Section 12.6 (for instance $A' \leq 10.60905$ or $C_{\Lambda} \leq 1.0150$), this inequality would fail at $4.62 \cdot 10^{12}$ and the threshold would have to be raised.

Theorem A₁₃. *Let $k \geq 2.17 \cdot 10^{13}$ be an integer. Then every closed packing of $[0, k]^2$ satisfies $W \geq 0.1 k^{1/4}$, in particular $W \geq 216$. Moreover*

$$\liminf_{k \rightarrow \infty} \frac{W_{\min}(k)}{k^{1/4}} \geq c_{\infty}^{(13)} := \frac{16 \cdot 2^{-1/4} (1 - \lambda)^{5/4}}{5 \cdot 5.0001^{1/4} \sqrt{A_{13}A'_{13}}} = 0.1411593 \dots,$$

where $A_{13} := 4\sqrt{13/0.32}/(2 - 3 \cdot 10^{-6})$, $A'_{13} := A_{13}/\cos(1.5 \cdot 10^{-6}) + 10^{-8}$, and λ is as in Theorem A.

The proof of Theorem A₁₃ does not use [R, Lemma 4.10]; the value of $c_{\infty}^{(13)}$ is again rounded down. Its threshold has little margin as well: it is about $2.1696 \cdot 10^{13}$ rounded up (Section 12.6).

Corollary A. *Let $c \geq 0$ be an integer. Then $s(k^2 - c) = k$ for every integer $k \geq \max(4.62 \cdot 10^{12}, 10^4 c^4 + 1)$. In particular $c^*(k) \geq \lceil 0.1 k^{1/4} \rceil - 1$ for every integer $k \geq 4.62 \cdot 10^{12}$, and $s(k^2 - c) = k$ for every integer $k \geq 4.62 \cdot 10^{12}$ if $c \leq 146$. Without [R, Lemma 4.10] (by Theorem A₁₃), the same statements hold with $2.17 \cdot 10^{13}$ in place of $4.62 \cdot 10^{12}$ and, in the last one, with $c \leq 215$ in place of $c \leq 146$.*

Together with the upper bound recalled above, $c^*(k)$ thus lies between $\lceil 0.1 k^{1/4} \rceil - 1$ and $O(k^{3/5})$ for $k \geq 4.62 \cdot 10^{12}$ (the upper bound relying on the preprints [Bui, McC]).

The proofs of Theorems A, A₁₃ and Corollary A are given in Section 11; the numerical constants and thresholds are collected in Section 12, where the direction of rounding of every displayed value is stated. We do not prove bounds of the form $W \geq c_* k^{1/4}$ for all $k < k_0$, where $k_0 = 4.62 \cdot 10^{12}$ in Theorem A and $k_0 = 2.17 \cdot 10^{13}$ in Theorem A₁₃; Section 12 gives lower bounds for W at single values of k (Table 4).

Dependence on [R]. The proofs use the results of [R] listed in Section 2, where their statements are quoted with the numbering of [R]; their proofs are in [R]. A few facts about these proofs are also

used; they are listed in Remark 2.4. Among the results used, [R, Lemma 4.10] is computer-assisted: its proof is a branch-and-bound computation over boxes, whose verdicts were replayed with the same programs [LevR] but have not yet been reproduced by an independent implementation. It enters only through the multiplicity bound $K_w^* = 9$, and through it the constants A , A' and C_Λ (Definition 3.6). Thus Theorem A, the value of c_∞ , the threshold $4.62 \cdot 10^{12}$ and the form of Corollary A with this threshold depend on it. Theorem A₁₃ and the form of Corollary A with the threshold $2.17 \cdot 10^{13}$ do not: they use instead the analytic [R, Lemma 4.9] ($K_w = 13$) everywhere (Remark 2.2), whose proof contains only a short finite check of angle conditions, carried out in [R] by a small program.

Comparison with [R]. The bound of [R] holds for every $k \geq 2$ and grows like $\log k$, with asymptotic slope $1.99954/30.418 > 0.0657$; Theorems A and A₁₃ give the order $k^{1/4}$, but only for $k \geq k_0$. For a fixed integer $c \geq 0$, Corollary A gives $s(k^2 - c) = k$ as soon as $k \geq \max(4.62 \cdot 10^{12}, 10^4 c^4 + 1)$ (note $\log(4.62 \cdot 10^{12}) < 29.17$), whereas for $c \geq 1$ the bound of [R] gives it for $\log k > 15.22c + 13.07$ [R, Remark 7.4]. Compared with [R], the quantization in columns of [R, Section 5] is replaced by Sections 4–6, and [R, Lemma 4.14] by Section 8. The lemmas of [R, Section 4] quoted in Section 2 are used as stated there, with two exceptions explained in that section: the multiplicity bounds of [R, Lemmas 4.9 and 4.10] are applied to families of pairs that need not be edges of the visibility graphs of [R], as their proofs allow (Proposition 2.1), and for Theorem A₁₃ [R, Lemmas 4.13–4.15] are used with A₁₃ in place of A (Remark 2.2).

Idea of the proof. As in [R], we consider horizontal lines y with $y_0 \leq d(y) \leq (1 - \varepsilon)(\frac{k}{2} - 3)$, where $d(y) = \min(y, k - y)$, whose height is not close to an integer (the set H of Definition 3.5), and we integrate over them with the measure $d(y)^{-3/4} dy$. On such a line with little waste, either the line meets a square of inclination at least $\beta(y) = c_1 d(y)^{-3/4}$, and such lines are paid for by [R, Lemma 4.13]; or all squares on the line are nearly axis-parallel, and then the short chords cut from the ramps of squares near their bottom and top add up to almost a full unit ([R, Lemmas 3.4, 3.5 and 4.15]). The squares that carry these short chords form the sets $\mathcal{Z}(s)$ of Definition 3.5. In [R] such squares are controlled through vertical columns below them; here we use paths. A path starts at a point $(x, 0)$ of the bottom side of $[0, k]^2$ and moves upwards, following the orientation of the squares it meets: in a gap it keeps the direction n_X of the last square X it passed, it enters a square only through the relative interior of the bottom side, and it crosses the square parallel to its sides (Section 3). A path dies when its total length in gaps reaches δ ; of several paths that arrive at the same entry point only one continues (merge); a path that meets a square elsewhere than on its bottom side stops (end collision); and in the flow of scale s a path stops when it enters a square of inclination at least $\alpha(s) = c_0 s^{-1/2}$ (tilt). The height identity of Section 4 shows that a path that has crossed m squares of inclination less than $\alpha(s)$ with total gap less than δ is within $0.50001 \alpha(s)^2 q_y + \delta \approx 0.50001 c_0^2 + \delta$ of the integer m when it reaches a bottom side at height $q_y \approx s$; hence the ramps of a square whose bottom side is reached alive lie close to integers, and no path of the flow of scale s reaches the bottom side of a square of $\mathcal{Z}(s)$ alive. The shadow inequality of Section 6 turns this into a lower bound for the losses: for each scale s , the measure $N(s)$ of the set of tilt-terminated starting points is at least $|\mathcal{Z}(s)|/(\varepsilon s + b_W)$ minus the other losses (Section 9). Deaths, merges, end collisions and overlaps of paths in gaps cost $O(W/\delta)$ in total (Sections 5 and 7), and exits through the side walls cost $O(\sqrt{s})$ per scale. Tilt terminations are paid for by a version of [R, Lemma 4.14] for paths (Section 8): $\int \alpha(T^*(x)) dx + \int \alpha(\tilde{T}^*(x)) dx \leq A'W$, where $T^*(x)$ is the least scale at which the path from $(x, 0)$ is tilt-terminated, and $\tilde{T}^*$ is the same for paths from the top side. Integrating over the scales with the weight $|\alpha'(s)|$ and over the lines with the measure $d(y)^{-3/4} dy$ (Section 10) gives a left-hand side of order $k^{1/4}$ and a right-hand side $BW + O(\log k)$. The tilt threshold $c_0 s^{-1/2}$ is chosen so that the accumulated loss of height $\sum_i (1 - \cos \varphi_i) \leq \alpha(s)^2 q_y / (2 \cos \alpha(s))$ over about s crossed squares stays of order $c_0^2/2$. This is possible because paths enter squares only through bottom sides: a contact with another side of a

square ends the path once, as an end collision, and the end collisions cost $O(\delta W)$ in total (Section 5). (The columns of [R] use the threshold $1/(2s)$.)

Organisation. Section 2 collects the definitions and results of [R] that are used. Section 3 defines the parameters, the flows of paths and the quantities attached to them, and proves their basic properties. Section 4 proves the height identity and the quantization of the bottom-side entries. Section 5 bounds the end collisions. Section 6 proves the shadow inequality, with its exceptional heights, and the bound for the wall term. Section 7 bounds deaths, merges and the overlap of paths in gaps. Section 8 proves the path version of [R, Lemma 4.14]. Section 9 proves the per-scale count and the lower bound for $|\mathcal{Z}(s)|$. Section 10 proves the integration inequality (★) (Theorem 10.3). Section 11 deduces Theorems A, A₁₃ and Corollary A, and Section 12 collects the constants and thresholds.

Status. This paper has not been peer reviewed, and neither has [R]. How the arguments have been checked so far, by AI-based reviews and numerical tests only, is described at the end of the paper (Verification status).

2. RESULTS USED FROM [R]

In this section we collect the definitions and results of [R] (S. Ryu, *Packing $k^2 - c$ unit squares: $s(k^2 - c) = k$ for all large k* , Preprint, version 1.2, 2026) that are used in this paper. Each statement is quoted verbatim and carries the number it has in [R]; only cross-references are adapted (a reference such as 4.13 points to the quotation in this section). Letters that are bound inside a quoted statement (for instance $\alpha, \beta, \delta, \lambda, \varphi, \Lambda, X, Y, I, J$) have the meaning given there; they are unrelated to the objects with the same names in Sections 3–12 unless this is said explicitly. Proofs are in [R]. Throughout, $k \geq 4$ is an integer.

2.1. Packings and the reduction. The following definitions are those of [R, Sections 1 and 2]. For $n \geq 1$ let $s(n)$ denote the side of the smallest square that contains n unit squares with pairwise disjoint interiors (rotations allowed), and

$$c^*(k) := \max\{c \in \mathbb{Z}_{\geq 0} : s(k^2 - c) = k\}.$$

Throughout, a *square* is a closed unit square. A *closed packing* of $[0, k]^2$ is a finite family of squares contained in $[0, k]^2$ that are pairwise disjoint as closed sets. Put $M(k)$ for the maximum size of a closed packing of $[0, k]^2$ and, for a given closed packing, $W := k^2 - N$, the area of $[0, k]^2$ not covered by the squares.

Lemma 2.1 of [R]. *For integers $k \geq 2$ and $n \geq 1$: $s(n) < k$ if and only if $n \leq M(k)$. Moreover $M(k) \leq k^2 - 1$ (Corollary 3.3 below), so $c^*(k) = k^2 - M(k) - 1$.*

The last sentence of the proof of [R, Lemma 2.1] reads: “Finally, $s(n) \leq k$ always holds for $n \leq k^2$, so for integers $0 \leq c < k^2$ we have $s(k^2 - c) = k$ iff $k^2 - c > M(k)$.” We use the lemma in this form. Note that W is an integer.

2.2. Elementary geometry. The following definitions are those of [R, Section 3]. The *inclination* $a(S) \in [0, \pi/4]$ of a square S is the least angle between a side of S and a side of $[0, k]^2$; the inclination $\theta(S_1, S_2)$ between two squares is the least angle between their sides (this is the quotient metric on $\mathbb{R}/(\pi/2)\mathbb{Z}$, with values in $[0, \pi/4]$), and we write $S_0 := \partial[0, k]^2$ with $\theta(S, S_0) := a(S)$. Likewise, for a single side σ of $[0, k]^2$ we put $\theta(S, \sigma) = \theta(\sigma, S) := a(S)$. (These are the definitions of Roth and Vaughan.)

Lemma 3.1 of [R]. *Let S be a square with inclination a and lowest point at height v .*

- (a) *The vertical extent of S is $[v, v + \cos a + \sin a]$.*
- (b) *For $y \in \mathcal{M}(S) := [v + \sin a, v + \cos a]$ (the middle band, of length $\cos a - \sin a$) the line $y = \text{const}$ meets S in a segment of length $\sec a \geq 1$; for y in the ramp set $\mathcal{R}(S) := [v, v + \sin a) \cup (v + \cos a, v + \cos a + \sin a]$ the length is $< \sec a$. If $a = 0$, then $\mathcal{R}(S) = \emptyset$.*

(c) *Symmetrically, for x in the vertical middle band $\mathcal{M}'(S)$, an interval of length $\cos a - \sin a$ (defined in the proof; for $a = 0$ it is the horizontal extent of S), the vertical line through x meets S in a segment of length $\sec a$, entering S through a side of slope $\pm \tan a$; the lower endpoint of that segment has height in $[v, v + \sin a]$. The complement of $\mathcal{M}'(S)$ in the horizontal extent of S (the vertical ramp set) has total length $2 \sin a$.*

(In the proof of [R, Lemma 3.1], for $0 < a \leq \pi/4$ and up to a reflection, the vertices of S are $P_0 = (x_0, v)$, $P_1 = P_0 + (\cos a, \sin a)$, $P_3 = P_0 + (-\sin a, \cos a)$, $P_2 = P_1 + P_3 - P_0$, and $\mathcal{M}'(S) := [x_0, x_0 + \cos a - \sin a]$. Part (c) is not used in this paper.)

For $y \in [0, k]$ let $\omega(y)$ be the length of the part of the segment $[0, k] \times \{y\}$ not covered by squares (the *line waste*); by Fubini $\int_0^k \omega(y) dy = W$. Similarly, for a vertical segment σ we call the length of σ not covered by squares its *vertical waste*; integrating the vertical waste of $\{x\} \times [0, k]$ over x also gives W . Since a packing is finite, all functions of y (or of x) considered in [R] are piecewise continuous with finitely many pieces, and all sets of y (or of x) defined there are finite unions of intervals.

Lemma 3.2 of [R]. *Let $k \geq 2$. On every horizontal line, the squares of a closed packing of $[0, k]^2$ meet the segment $[0, k] \times \{y\}$ in pairwise disjoint closed intervals, and at most $k - 1$ of these have length ≥ 1 .*

Corollary 3.3 of [R]. *For $k \geq 2$ every closed packing of $[0, k]^2$ has $W \geq 1$; that is, $M(k) \leq k^2 - 1$.*

For a square S and $y \in \mathbb{R}$ let $c_S(y)$ be the length of the horizontal chord $S \cap (\mathbb{R} \times \{y\})$ (zero if the line misses S).

Lemma 3.4 of [R]. *Let $k \geq 2$, $y \in (0, k)$ and $0 < \alpha \leq \frac{\pi}{4}$, and suppose that every square meeting the line y has inclination $< \alpha$. Then*

$$\sum_{S: 0 < c_S(y) < 1} c_S(y) > 1 - \omega(y) - (k - 1)(\sec \alpha - 1),$$

the sum being over the squares S of the packing. Moreover, for a square S let $\Phi(S) := \{y : 0 < c_S(y) < 1\}$. If $a := a(S) > 0$, then $\Phi(S) \subset \mathcal{R}(S)$, $|\Phi(S)| = 2 \sin a \cos a$ and $\int_{\Phi(S)} c_S(y) dy = \sin a \cos a$; if $a(S) = 0$, then $\Phi(S) = \emptyset$.

Only the second part (“Moreover . . .”) of [R, Lemma 3.4] is used in this paper.

Lemma 3.5 of [R]. *Let $k \geq 2$ and $y \in (0, k)$, and put*

$$E(y) := \sum_{S: c_S(y) \geq 1} (c_S(y) - 1),$$

the sum being over the squares S of the packing. Then

$$\sum_{S: 0 < c_S(y) < 1} c_S(y) \geq 1 - \omega(y) - E(y).$$

The proof of [R, Lemma 3.5] uses [R, Lemma 3.2].

2.3. The sharpened fundamental lemma and the cost of inclinations. The following conventions are those of [R, Section 4]: throughout [R, Section 4], $k \geq 4$ (this is used, for instance, in [R, Lemma 4.8(b)] and in the proof of [R, Lemma 4.10]; in this paper $k \geq 4$ always holds), a closed packing of $[0, k]^2$ is fixed, $S_0 := \partial[0, k]^2$, and the *waste* is the set of points of $(0, k)^2$ that lie in no square; its area is W .

Lemma 4.1 of [R]. *Let I be an interval of length λ , let $0 \leq \varphi \leq 1$, and let $g \geq 0$ be a continuous, convex, piecewise linear function on I with finitely many break points, such that $|g'| \geq \varphi$ off the break points and the slope of g increases by at least 2 at every break point. Then $\int_I g \geq \varphi(2 - \varphi)\lambda^2/4$.*

Let (s, t) be Cartesian coordinates in the plane. For a square Z let $[\tau^-(Z), \tau^+(Z)]$ be its projection to the t -axis, and for $\tau^-(Z) \leq t \leq \tau^+(Z)$ let $l_Z(t)$ and $r_Z(t)$ be the least and the largest s with $(s, t) \in Z$. The vertices of Z at which $t = \tau^+(Z)$ or $t = \tau^-(Z)$ are its t -extreme vertices.

Lemma 4.2 of [R]. *The function l_Z is convex and r_Z is concave. Each of them is piecewise linear with at most one break point, at which the slope of l_Z increases, and that of r_Z decreases, by at least 2. Off the break points $l'_Z(t) = \tan \psi$, where $\psi \in (-\frac{\pi}{2}, \frac{\pi}{2})$ is the angle between the t -axis and the side of Z containing $(l_Z(t), t)$, and similarly for r_Z . Moreover $\tau^+(Z) - \tau^-(Z) \geq 1$.*

The rectangles $R(S_1, S_2)$. For a square S_1 and S_2 either another square or S_0 , let L be the segment from the centre c_1 of S_1 to the centre c_2 of S_2 , respectively to a point c_2 of S_0 nearest to c_1 (any choice will do), and let $R = R(S_1, S_2)$ be the closed rectangle of points at distance at most $\frac{1}{2}$ from the line through L whose orthogonal projection lies on L . This is the rectangle used by Roth and Vaughan. Let u be the unit vector along L and n a unit normal, and use the coordinates (s, t) of $c_1 + su + tn$; thus $R = \{0 \leq s \leq |L|, |t| \leq \frac{1}{2}\}$. (Here, and in the quoted statements below, s and t are these coordinates; they are unrelated to the scale s and the ray parameter t of Section 3.)

Proposition 4.3 of [R]. *Let $\theta := \theta(S_1, S_2)$, and let m' be the number of squares $Z \notin \{S_1, S_2\}$ of the packing that have a t -extreme vertex $v \in R$ with $|t(v)| < \frac{1}{2}$. Let $\mathcal{T} \subset (-\frac{1}{2}, \frac{1}{2})$ be the set of the numbers $t(v)$ for these vertices, and let $\lambda_1, \dots, \lambda_q$ be the lengths of the intervals into which the points of $\mathcal{T}$ cut $(-\frac{1}{2}, \frac{1}{2})$. Then the waste in R has area at least $\frac{\theta(2-\theta)}{4} \sum_r \lambda_r^2 \geq \theta(2-\theta)/(4(m'+1))$.*

Lemma 4.4 of [R]. *Let Z be a square, e a unit vector, and v a vertex of Z at which $p \mapsto p \cdot e$ is smallest on Z . For $0 \leq \nu \leq \frac{3}{4}$ the line $\{p : p \cdot e = v \cdot e + \nu\}$ meets Z in a segment of length at least $\min(2\nu, 1)$.*

[R, Lemma 4.4] is used only inside the proof of [R, Proposition 4.5].

Proposition 4.5 of [R]. *Let S_1 be a square and S_2 a square or S_0 , at distance less than d , with $\theta := \theta(S_1, S_2) \leq \theta_1$, where $d, \theta_1 \leq 10^{-5}$. Then $\sum_r \lambda_r^2 \geq Q_* := 0.32$ in Proposition 4.3, so the waste in $R(S_1, S_2)$ has area at least $Q_* \theta(2-\theta)/4$.*

Visibility graphs. Let $\mathcal{V}_h$ consist of the squares of the packing together with the left and the right side of $[0, k]^2$. Two distinct members P, Q of $\mathcal{V}_h$ are *consecutive on the line y* ($0 < y < k$) if both meet the segment $[0, k] \times \{y\}$ and the open segment between them on that line meets no square. Let G_h be the graph on $\mathcal{V}_h$ in which P and Q are adjacent if they are consecutive on a set of lines y of positive measure. Define $\mathcal{V}_v$ and G_v in the same way, with vertical lines and the bottom and the top side of $[0, k]^2$. For $0 < d \leq 10^{-4}$ put $\Lambda = \Lambda(d) := d + \sqrt{2}$ and $R_0 = R_0(d) := \sqrt{\Lambda^2 + \frac{1}{4}} < 1.5001$, and

$$K := 27, \quad q_0 := 10.$$

(The symbols $\Lambda(d)$, $R_0(d)$, $K = 27$ and q_0 are used only in this section, as in [R]; they are unrelated to the loss terms Λ_0 , $\Lambda^*(s)$ of Section 3 and to the multiplicity bound denoted K in Sections 5 and 7.)

Lemma 4.8 of [R]. *Let $0 < d \leq 10^{-4}$.*

- (a) *If S_1 is a square and S_2 is a square or S_0 , at distance $< d$, then $m' \leq q_0 - 1 = 9$ in Proposition 4.3; hence the waste in $R(S_1, S_2)$ has area at least $\theta(2-\theta)/(4q_0)$, $\theta = \theta(S_1, S_2)$.*
- (b) *Let $k \geq 4$ and let G be G_h or G_v . For an edge $e = \{P, Q\}$ of G whose members are at distance $< d$ put $R_e := R(P, Q)$ if both are squares, and $R_e := R(P, S_0)$ if P is a square and Q a side. Then every point lies in R_e for at most K such edges e (the two sides of $[0, k]^2$ that are vertices of G are at distance $k > d$).*

Part (a) is used in this paper; part (b) only supplies the definition of R_e . For a pair e we write f_e for the area of the waste in the rectangle R_e attached to e (as in the proof of [R, Lemma 4.13]).

Lemma 4.9 of [R]. *Let $0 < d \leq 10^{-4}$, let G be G_h or G_v , and define R_e for the edges e of G whose members are at distance $< d$ as in Lemma 4.8(b). Then every point of the waste lies in R_e for at most $K_w := 13$ such edges.*

The paragraph of [R] that follows the proof of [R, Lemma 4.9] reads: “The proof of Lemma 4.9 uses no property of G except that the only sides involved are two opposite ones; the bound holds for any family of such pairs at distance $< d$. A computer-assisted refinement of the same counting gives the bound 9, which is sharp for the number of all pairs at distance $< d$ (see the remark after the proof), though not known to be sharp for the edges of a single graph G .”

Lemma 4.10 of [R]. *Under the assumptions of Lemma 4.9, every point of the waste lies in R_e for at most $K_w^* := 9$ of the edges e considered there.*

Remark 4.11 of [R]. *The proof bounds the number of all pairs of squares (or of a square and one of two opposite sides) at distance less than d whose rectangles contain p , without using the graph G , and for this number the bound 9 is sharp [...]*

(We quote the first sentence of [R, Remark 4.11]; the rest of the remark describes a configuration showing the sharpness and is not used. In the remark, p is the point of the waste considered in the proof of [R, Lemma 4.10].)

Lemma 4.12 of [R]. *Let A, B be distinct squares, or let one of them be the left or the right side of $[0, k]^2$, and let E_{AB} be the set of $y \in [0, k]$ such that on the line y both A and B are met, A lies to the left of B , and no square is met strictly between them. For $y \in E_{AB}$ let $g(y)$ be the length of the gap between them, and let $\theta := \theta(A, B)$.*

(a) *All $y \in E_{AB}$ with $g(y) < 1$ lie in one component of E_{AB} .*

(b) *On every component J of E_{AB} the function g is convex, and $\int_I g \geq \theta(2 - \theta)|I|^2/4$ for every interval $I \subset J$. In particular, for $\eta > 0$ the set $\{y \in J : g(y) < \eta\}$ is an interval.*

The same holds with the roles of the axes exchanged (vertical lines, A below B , the bottom and the top side).

Lemma 4.13 of [R]. *Let $0 < \bar{\beta} \leq 1.5 \cdot 10^{-6}$, and let $Y \subset [0, k]$ and $\beta : Y \rightarrow [0, \bar{\beta}]$ be such that for every $y \in Y$: $\omega(y) < \frac{1}{2}$ and the line y meets a square of inclination $\geq \beta(y)$. Then*

$$\int_Y 2\beta(y) dy \leq AW, \quad A := \frac{4\sqrt{K_w^*/Q_*}}{2 - 3 \cdot 10^{-6}} < 10.6067.$$

Lemma 4.14 of [R]. *Let $0 < \delta \leq 10^{-5}$ and $0 < \bar{\beta} \leq 3 \cdot 10^{-6}$. Let $X, X' \subset [0, k]$. For $x \in X$ let $\bar{y}_x \in (0, \frac{k}{2} - 1]$ and $\beta(x) \in [0, \bar{\beta}]$ be such that the segment $\sigma_x = \{x\} \times [0, \bar{y}_x]$ has vertical waste $< \delta$ and meets a square of inclination $\geq \beta(x)$; for $x \in X'$ let $\bar{y}'_x, \beta'(x)$ satisfy the same with the segment $\sigma'_x = \{x\} \times [k - \bar{y}'_x, k]$. Then*

$$\int_X \beta(x) dx + \int_{X'} \beta'(x) dx \leq AW.$$

In this paper [R, Lemma 4.14] is used only inside the proof of [R, Lemma 4.15], where it is applied with $\bar{\beta} := \alpha$ and $\bar{y}_x := y = d(y)$ (or, for $y > \frac{k}{2}$, with $X := \emptyset$ and $\bar{y}'_x := k - y = d(y)$). Section 8 proves a version of it for the paths of Section 3.

For $y \in (0, k)$ let $d(y) := \min(y, k - y)$ be the distance to the nearer of the bottom and the top side.

Lemma 4.15 of [R]. *Let $0 < \delta \leq 10^{-5}$ and $0 < \alpha \leq 1.5 \cdot 10^{-6}$, let $y \in (0, k)$ with $d(y) \leq \frac{k}{2} - 1$, and suppose that every square meeting the line y has inclination $< \alpha$. Then, with $E(y)$ as in Lemma 3.5,*

$$E(y) \leq 0.50001 \alpha \left(A + \frac{\alpha}{\delta} \right) W.$$

The proof of [R, Lemma 4.15] begins with the inequality

$$(2.1) \quad \sec x - 1 \leq 0.50001 x^2 \quad (0 \leq x \leq 10^{-3}).$$

For completeness we include a proof: for $0 < x \leq 10^{-3}$ we have $\cos x \geq 1 - x^2/2 > 0$, hence $\sec x - 1 = (1 - \cos x)/\cos x \leq (x^2/2)/(1 - x^2/2) = x^2/(2 - x^2) \leq x^2/(2 - 10^{-6}) < 0.5000003 x^2$; for $x = 0$ there is nothing to prove. The proof of [R, Lemma 4.15] also uses [R, Lemma 4.14] and, through it, the pair estimates [R, Proposition 4.5], [R, Lemma 4.8(a)] and [R, Lemma 4.10] with $d = \delta$, exactly as the proof of [R, Lemma 4.14] does (the proof of [R, Lemma 4.13] uses them with $d = d' = 2\bar{\beta}$).

2.4. The form of the multiplicity bound used in this paper. The formal statements of [R, Lemmas 4.9 and 4.10] concern edges of the visibility graphs G_h and G_v . In Sections 5, 7 and 8 the pairs to which the bound is applied are pairs of squares that arise from the paths of Section 3 (for instance consecutive squares entered by a path, or the two squares from which two paths reach a common point), or a square and the bottom side (respectively the top side) of $[0, k]^2$; they need not be edges of G_h or G_v . We therefore use the bound in the following form, which is what the proofs of [R, Lemmas 4.9 and 4.10] establish according to the paragraph of [R] following the proof of [R, Lemma 4.9] and to [R, Remark 4.11], both quoted above.

Proposition 2.1 (multiplicity for families of pairs). *Let $0 < d \leq 10^{-4}$, and let σ, σ' be two opposite sides of $[0, k]^2$. Let $\mathfrak{F}$ be a family of distinct unordered pairs $\{P, Q\}$, where P is a square of the packing and Q is either another square of the packing or one of σ, σ' , such that the distance between P and Q is less than d . For $e = \{P, Q\} \in \mathfrak{F}$ put $R_e := R(P, Q)$ if Q is a square and $R_e := R(P, S_0)$ if Q is a side, and let f_e be the area of the waste in R_e . Then every point of the waste lies in R_e for at most $K_w^* = 9$ pairs $e \in \mathfrak{F}$; by the analytic argument of [R, Lemma 4.9] it lies in R_e for at most $K_w = 13$ pairs $e \in \mathfrak{F}$. Consequently*

$$\sum_{e \in \mathfrak{F}} f_e \leq K_w^* W = 9W, \quad \text{and also} \quad \sum_{e \in \mathfrak{F}} f_e \leq K_w W = 13W.$$

Proof. The pointwise bounds are the statements quoted above: the paragraph after the proof of [R, Lemma 4.9] states that the bound 13 “holds for any family of such pairs at distance $< d$ ”, where the sides involved are two opposite ones, and [R, Remark 4.11] states that the proof of [R, Lemma 4.10] bounds “the number of all pairs of squares (or of a square and one of two opposite sides) at distance less than d whose rectangles contain p ” by 9. (In that proof the count runs over the set I of all centres within distance $\rho_1 := \sqrt{(d + \frac{\sqrt{2}}{2})^2 + \frac{1}{4}}$ of p , and every pair counted has a member whose centre lies in I ; the proof bounds separately the pairs with both centres in I , the pairs of a square with centre in I and a side, and the pairs of a square with centre in I and a square with centre outside I , using only distances and angles, never the graph.) For the sum, the multiplicity function $p \mapsto \#\{e \in \mathfrak{F} : p \in R_e\}$ is a finite sum of indicator functions of closed rectangles, hence Borel, and by Tonelli’s theorem

$$\sum_{e \in \mathfrak{F}} f_e = \int_{\text{waste}} \#\{e \in \mathfrak{F} : p \in R_e\} dp \leq 9 |\text{waste}| = 9W,$$

and in the same way with 13; here $|\text{waste}| = W$ because the squares have total area N and the waste differs from $[0, k]^2 \setminus \bigcup \mathcal{P}$ by a null set. ($\mathfrak{F}$ is finite, since the packing is finite.) $\square$

When Proposition 2.1 is applied, the pairs are always unordered and each is counted once; in the applications, σ and σ' are the bottom and the top side of $[0, k]^2$. For a square P near a corner of $[0, k]^2$ the rectangle $R(P, S_0)$ may stand on a side other than σ, σ' (the side of $[0, k]^2$ nearest to the centre of P); this is the situation of [R, Lemmas 4.8(b) and 4.9], whose proofs allow it.

2.5. Dependence on a computer-assisted result.

Remark 2.2 (dependence on [R, Lemma 4.10]). The proof of [R, Lemma 4.10] is computer-assisted: it consists of a branch-and-bound computation over a cover of a parameter domain by boxes, with rigorous ball arithmetic (see [R, proof of Lemma 4.10]). In this paper [R, Lemma 4.10] enters only through the number $K_w^* = 9$, namely

- (i) through the constant $A = 4\sqrt{K_w^*/Q_*}/(2 - 3 \cdot 10^{-6})$ of [R, Lemmas 4.13–4.15];
- (ii) through Proposition 2.1 with the bound 9, in Sections 5, 7 and 8 (hence in the constants C_Λ and A' of Definition 3.6).

Theorem A and its constants c_∞ and $k_0 = 4.62 \cdot 10^{12}$ use [R, Lemma 4.10]. Theorem A_{13} does not: it uses the analytic [R, Lemma 4.9] ($K_w = 13$) in all places where $K_w^* = 9$ is used. In (ii) this is the second bound of Proposition 2.1. In (i) it rests on the following facts about the proofs in [R]. In the proof of [R, Lemma 4.13], [R, Lemma 4.10] is used only in the sentence “By Lemma 4.10 (with $d = d'$), every point of the waste lies in R_e for at most K_w^* of the pairs with $|Y_e^c| > 0$, so $\sum f_e \leq K_w^* W$, the sum over these pairs”, and these pairs are edges of G_h (“they are edges of G_h ”); in the proof of [R, Lemma 4.14] it is used only in “ $\sum_e f_e \leq K_w^* W$ (Lemma 4.10 with $d = \delta$)”, for pairs that are “edges of G_v whose members are at distance $< \delta$ ”; and the proof of [R, Lemma 4.15] uses [R, Lemma 4.10] only through [R, Lemma 4.14]. Replacing 9 by 13 in these sentences ([R, Lemma 4.9] applies to these edges), K_w^* by K_w in the formulas that follow them, and leaving the rest of the proofs unchanged (the last step of the proof of [R, Lemma 4.13] uses only $A \geq 2$), one obtains [R, Lemmas 4.13–4.15] with A replaced by

$$A_{13} := \frac{4\sqrt{K_w/Q_*}}{2 - 3 \cdot 10^{-6}} = \frac{4\sqrt{13/0.32}}{2 - 3 \cdot 10^{-6}} < 12.7476,$$

as recorded in [R, Remark 7.5]: “with the analytic bound $K_w = 13$ of Lemma 4.9 in place of K_w^* , the proof of Section 6 gives $A < 12.7476$ ”. Numerically $A = 10.60661762 \dots < 10.6067$ and $A_{13} = 12.74756790 \dots < 12.7476$ (digits truncated; the bounds are rounded up).

Remark 2.3 (where the results of [R] are used). [R, Lemma 2.1] (with [R, Corollary 3.3]) gives Corollary A from Theorems A and A_{13} (Section 11). [R, Lemma 3.1(a)(b)] gives the vertical extent of a square, the positions of the ramps and the middle chords used in Sections 3, 4, 5, 8 and 9. The second part of [R, Lemma 3.4] ($\Phi(S) \subset \mathcal{R}(S)$, $\int_{\Phi(S)} c_S = \sin a \cos a$, $\Phi(S) = \emptyset$ if $a(S) = 0$) is used in Sections 4, 9 and 10, and [R, Lemma 3.5] gives the short chords on lines of kind (iii) in Sections 9 and 10; [R, Lemma 3.2] enters through the proof of [R, Lemma 3.5]. [R, Proposition 4.5] and [R, Lemma 4.8(a)] give the lower bounds for the waste in the rectangles R_e (Sections 5, 7, 8). [R, Lemmas 4.1 and 4.12] are mentioned only for comparison, in Section 8, whose slit inequalities are proved directly; like [R, Lemma 4.2, Proposition 4.3 and Lemma 4.4], they are needed only for the statements and proofs in [R] of the results quoted here. [R, Lemmas 4.9, 4.10 and Remark 4.11] are used through Proposition 2.1. [R, Lemmas 4.13 and 4.15] bound the lines of kinds (ii) and (iii) in Section 10.

Remark 2.4 (facts taken from the proofs in [R]). Besides the statements quoted above, the following facts about the *proofs* in [R] (version 1.2) are used. They are not part of the quoted statements; each is stated where it is used, and can be checked in [R] at the place indicated.

- (a) The proofs of [R, Lemmas 4.9 and 4.10] use only distances and angles, the disjointness of the squares, the fact that p is a point of the waste, and the fact that the sides involved are two opposite ones; they do not use the visibility graphs ([R, the paragraph after the proof of Lemma 4.9, and Remark 4.11]). This is used in Proposition 2.1 and its proof.
- (b) In the proofs of [R, Lemmas 4.13–4.15], [R, Lemma 4.10] enters only through the two sentences quoted in Remark 2.2. Theorem A_{13} rests on this.

- (c) The proofs of [R, Lemmas 4.8(b), 4.9 and 4.10] allow the rectangle $R(P, S_0)$ of a square P near a corner to stand on any side of $[0, k]^2$, and the proof of [R, Lemma 4.10] shows that all such rectangles that contain a given point of the waste stand on one and the same side. This is used after Proposition 2.1 and in Remark 5.6.
- (d) The distance from a square of inclination a to a side of $[0, k]^2$ is the distance from its centre to that side minus $\frac{1}{2}(\cos a + \sin a)$ ([R, proof of Proposition 4.5(b)]; used in Remark 5.6).
- (e) The proof of [R, Lemma 4.15] begins with the inequality (2.1) and uses [R, Lemma 4.14] as described after [R, Lemma 4.15] above.

3. PATH FLOWS

Throughout this section $k \geq 4$ is an integer and $\mathcal{P}$ is a closed packing of $[0, k]^2$ with N squares; $W = k^2 - N$. As in [R], the *waste* is the set of points of $(0, k)^2$ that lie in no square of $\mathcal{P}$; its area is W . We write $|\cdot|$ for Lebesgue measure on $\mathbb{R}$ (for sets of starting abscissas $x \in (0, k)$ we also call it the *floor measure*), $e_1 = (1, 0)$, $e_2 = (0, 1)$, and $\ell_y := \mathbb{R} \times \{y\}$ for the horizontal line at height y . The letter π denotes the parameter vector of Definition 3.1, π_x a path, and π also the number; the meaning is always clear from the context.

3.1. Parameters and constraints.

Definition 3.1 (parameters). *Fixed constants.* $\delta := 10^{-5}$, $\bar{\beta} := 1.5 \cdot 10^{-6}$, $b_W := 2.0002$, and, from [R], $Q_* = 0.32$, $K_w^* = 9$, $K_w = 13$ (see Section 2). Some results of Sections 5–8 are stated for every $\delta \in (0, 10^{-5}]$; everywhere else $\delta = 10^{-5}$.

Free parameters. $\pi = (c_0, c_1, y_0, \varepsilon, \omega_0)$.

Derived quantities. For $s > 0$, $t > 0$ and $0 < y < k$ put

$$\begin{aligned} \alpha(s) &:= c_0 s^{-1/2}, & \alpha_{\max} &:= \alpha(y_0), & \hat{\beta}(t) &:= c_1 t^{-3/4}, \\ d(y) &:= \min(y, k - y), & \beta(y) &:= \hat{\beta}(d(y)), & b^* &:= \hat{\beta}((1 - \varepsilon)y_0), \\ y_1 &:= \frac{k}{2} - 3, & h_{\max} &:= \frac{k}{2} - 1, \\ w_0 &:= 0.50001 c_0^2 \left(1 + \frac{1.0001}{y_0}\right) + \delta + 1.0001 b^* + 10^{-12}, & h_0 &:= 1 - 2w_0. \end{aligned}$$

$\alpha(s)$ is the *tilt threshold* at scale s and $\beta(y)$ the *line threshold* at height y .

The parameter vectors used for the main theorems are

$$\pi^* := (0.3078, 0.2754, 42\,108\,000\,000, 5.8 \cdot 10^{-6}, 1.7 \cdot 10^{-5}) \quad (\text{Theorem A}),$$

$$\pi_{13}^* := (0.3674, 0.3012, 60\,000\,000\,000, 5 \cdot 10^{-6}, 1.4 \cdot 10^{-5}) \quad (\text{Theorem A}_{13}).$$

Definition 3.2 (constraints). The parameters are subject to the following conditions.

- (C0) $c_0, c_1 > 0$, $y_0 \in \mathbb{Z}_{>0}$, $0 < \varepsilon < 1$, $\omega_0 > 0$.
- (C1) $\alpha(y_0) \leq \bar{\beta} = 1.5 \cdot 10^{-6}$.
- (C2) $\hat{\beta}(y_0) \leq 1.5 \cdot 10^{-6}$.
- (C3) $c_1(1 - \varepsilon)^{-3/4} < c_0 y_0^{1/4}$ (not used in the proofs; see below).
- (C4) $\omega_0 < \frac{1}{2}$.
- (C5) $w_0 < \frac{1}{2}$.
- (C6) $\delta \leq \frac{1}{2} \cos 2\alpha_{\max}$.
- (C7) $y_0 + 1 \leq (1 - \varepsilon)y_1$.
- (Q1) $\alpha(y_0) \leq 10^{-3}$.
- (Q2) $b^* \leq 10^{-4}$.

The role of each condition is explained at the end of Section 3.2, after the sets of heights to which some of them refer have been defined.

Lemma 3.3 (elementary consequences). (a) (C1) holds if and only if $y_0 \geq (c_0/\bar{\beta})^2$. Under (C1), $\alpha_{\max} \leq 1.5 \cdot 10^{-6}$, (Q1) holds, and (C6) holds for every $\delta \in (0, 10^{-5}]$.
 (b) The functions α and $\hat{\beta}$ are decreasing. If (C2) holds and $\varepsilon \leq 0.9963$, then (Q2) holds.
 (c) (C3) holds if and only if $\hat{\beta}((1-\varepsilon)s) < \alpha(s)$ for every $s \geq y_0$.
 (d) Under (C0), (C1) and (C7), for every $s \in [y_0, y_1]$: $0 < \alpha(s) \leq \alpha_{\max} \leq \bar{\beta} < \pi/4$, $s+2 \leq h_{\max}$, and $s < k/2$; moreover $0 < h_{\max} < k$.
 (e) If (C7) holds for some k , it holds, with the same π , for every larger k .

Proof. (a) $c_0 y_0^{-1/2} \leq \bar{\beta}$ is equivalent to $y_0 \geq (c_0/\bar{\beta})^2$. If (C1) holds, then $\alpha_{\max} = \alpha(y_0) \leq 1.5 \cdot 10^{-6} \leq 10^{-3}$, and $2\alpha_{\max} \leq 3 \cdot 10^{-6}$ gives $\frac{1}{2} \cos 2\alpha_{\max} \geq \frac{1}{2}(1 - \frac{1}{2}(3 \cdot 10^{-6})^2) > 0.49 > 10^{-5} \geq \delta$.

(b) The first statement is clear. We have $b^* = (1-\varepsilon)^{-3/4} \hat{\beta}(y_0) \leq 1.5 \cdot 10^{-6} (1-\varepsilon)^{-3/4}$. If $\varepsilon \leq 0.9963$, then $(1-\varepsilon)^{3/4} \geq 0.0037^{3/4} > 0.015$, because $0.0037^3 = 5.0653 \cdot 10^{-8} > 5.0625 \cdot 10^{-8} = 0.015^4$; hence $b^* < 1.5 \cdot 10^{-6}/0.015 = 10^{-4}$.

(c) For $s > 0$, $\hat{\beta}((1-\varepsilon)s)/\alpha(s) = \frac{c_1}{c_0}(1-\varepsilon)^{-3/4}s^{-1/4}$ is strictly decreasing in s , so it is < 1 for all $s \geq y_0$ if and only if it is < 1 at $s = y_0$, which is (C3).

(d) As α is decreasing and $s \geq y_0$, $0 < \alpha(s) \leq \alpha(y_0) = \alpha_{\max} \leq \bar{\beta} < \pi/4$. From $s \leq y_1 = \frac{k}{2} - 3$ we get $s+2 \leq \frac{k}{2} - 1 = h_{\max} < k$ and $s < \frac{k}{2}$; and $h_{\max} \geq 1$ as $k \geq 4$.

(e) $(1-\varepsilon)y_1 = (1-\varepsilon)(\frac{k}{2} - 3)$ increases with k . $\square$

Lemma 3.4 (the parameter vectors). (a) π^* satisfies (C0)–(C7), (Q1) and (Q2) for every integer $k \geq 4.62 \cdot 10^{12}$. At π^* one has $(c_0/\bar{\beta})^2 = 205\,200^2 = 42\,107\,040\,000 \leq y_0$, $b^* < 10^{-8}$, $0.0473813 < w_0 < 0.0473814$ and $0.9052372 < h_0 < 0.9052374$.

(b) π_{13}^* satisfies (C0)–(C7), (Q1) and (Q2) for every integer $k \geq 2.17 \cdot 10^{13}$, and $w_0 < 0.0676$ there.

Proof. (a) (C0) is clear. (C1): $c_0^2 = 0.3078^2 = 0.09474084$ and $\bar{\beta}^2 y_0 = 2.25 \cdot 10^{-12} \cdot 42\,108\,000\,000 = 0.094743$, so $c_0^2 \leq \bar{\beta}^2 y_0$; also $c_0/\bar{\beta} = 205\,200$ exactly. (C2), (Q2): $y_0 \geq 10^{10}$ and $(1-\varepsilon)y_0 \geq 10^{10}$ give $\hat{\beta}(y_0), b^* \leq 0.2754 \cdot 10^{-7.5} < 0.2754/(3 \cdot 10^7) < 10^{-8}$. (C3): $c_1(1-\varepsilon)^{-3/4} \leq 0.2754/(1-5.8 \cdot 10^{-6}) < 0.2755$ and $c_0 y_0^{1/4} \geq 0.3078 \cdot 10^{2.5} > 97$. (C4) is clear. (C5): $0.50001 c_0^2 = 0.0473713674084$, $1.0001/y_0 < 10^{-10}$ and $1.0001 b^* < 1.0001 \cdot 10^{-8}$, so

$$w_0 > 0.0473713674 + 10^{-5} > 0.0473813,$$

$$w_0 < 0.0473713674085(1 + 10^{-10}) + 10^{-5} + 1.0001 \cdot 10^{-8} + 10^{-12} < 0.0473814,$$

which gives the stated bounds for w_0 and $h_0 = 1 - 2w_0$, and (C5). (C6) and (Q1) follow from (C1) by Lemma 3.3(a). (C7): for $k \geq 4.62 \cdot 10^{12}$, $(1-\varepsilon)y_1 \geq (1-10^{-5})(2.31 \cdot 10^{12} - 3) > 2 \cdot 10^{12} > y_0 + 1$.

(b) In the same way: $c_0^2 = 0.3674^2 = 0.13498276 \leq 0.135 = 2.25 \cdot 10^{-12} \cdot 6 \cdot 10^{10}$ gives (C1); $\hat{\beta}(y_0), b^* < 0.3012/(3 \cdot 10^7) < 1.1 \cdot 10^{-8}$ gives (C2) and (Q2); $c_1(1-\varepsilon)^{-3/4} < 0.302 < 97 < c_0 y_0^{1/4}$ gives (C3); $w_0 < 0.50001 \cdot 0.13498276(1 + 10^{-10}) + 10^{-5} + 1.0001 \cdot 1.1 \cdot 10^{-8} + 10^{-12} < 0.0676$ gives (C5); and for $k \geq 2.17 \cdot 10^{13}$, $(1-\varepsilon)y_1 \geq (1-10^{-5})(1.085 \cdot 10^{13} - 3) > 10^{13} > y_0 + 1$. $\square$

3.2. Lines, windows and constants.

Definition 3.5 (lines and windows). A square S meets the line y if $S \cap \ell_y \neq \emptyset$. For $y \in \mathbb{R}$, $\{y\}$ is the fractional part and $\text{dist}(y, \mathbb{Z}) = \min(\{y\}, 1 - \{y\})$. Put

$$H := \{y \in (0, k) : y_0 \leq d(y) \leq (1-\varepsilon)y_1, \{y\} \in [w_0, 1-w_0]\}, \quad H_b := H \cap (0, \frac{k}{2}), \quad H_t := H \cap (\frac{k}{2}, k).$$

Every $y \in H$ is of exactly one of the following kinds: (i) $\omega(y) \geq \omega_0$; (ii) $\omega(y) < \omega_0$ and the line y meets a square S with $a(S) \geq \beta(y)$; (iii) $\omega(y) < \omega_0$ and every square meeting the line y has $a(S) < \beta(y)$. Let $Y_i, Y_{ii}, Y_{iii} \subset H$ be the corresponding sets, $Y_b := Y_{iii} \cap (0, \frac{k}{2})$ and $Y_t := Y_{iii} \cap (\frac{k}{2}, k)$. For $s \in [y_0, y_1]$ let

$$\mathcal{W}_s := [(1-\varepsilon)s, s], \quad \mathcal{V}_s := [(1-\varepsilon)s - 1.0001, s + 1.0001] \quad (\text{so } |V_s| = \varepsilon s + b_W),$$

$\mathcal{Z}(s) := \{Z \in \mathcal{P} : \Phi(Z) \cap Y_b \cap \mathcal{W}_s \neq \emptyset\}$, $\tilde{\mathcal{Z}}(s) := \{Z \in \mathcal{P} : \Phi(Z) \cap Y_t \cap (k - \mathcal{W}_s) \neq \emptyset\}$, where $k - Y := \{k - y : y \in Y\}$ for a set $Y \subset \mathbb{R}$, and $\Phi(Z) = \{y : 0 < c_Z(y) < 1\}$ ([R, Lemma 3.4]). For a Borel set $Y \subset H$, $\ell(Y) := \int_Y d(y)^{-3/4} dy$ (the *line measure*).

Under (C7), $d(y) \leq (1 - \varepsilon)y_1 < \frac{k}{2} - 1$ on H ; in particular $\frac{k}{2} \notin H$, so $Y_{iii} = Y_b \sqcup Y_t$, and $d(y) = y$ on H_b .

Definition 3.6 (constants). With Q_*, K_w^*, K_w from [R] put

$$A := \frac{4\sqrt{K_w^*/Q_*}}{2 - 3 \cdot 10^{-6}}, \quad A' := \frac{A}{\cos \bar{\beta}} + 10^{-8}, \quad C_\Lambda := 1 + 1185.6 \delta^2;$$

$$A_{13} := \frac{4\sqrt{K_w/Q_*}}{2 - 3 \cdot 10^{-6}}, \quad A'_{13} := \frac{A_{13}}{\cos \bar{\beta}} + 10^{-8}, \quad C_\Lambda^{(13)} := 1 + 1712.6 \delta^2.$$

(A is the constant of [R, Lemma 4.13].) For the integration of Section 10 put

$$\text{inv } F := \frac{4(1 - \varepsilon)^{3/4}(1 - (1 - \varepsilon)^{3/4})}{3\varepsilon(1 + b_W/(\varepsilon y_0))}, \quad \Gamma := 1.00002 \left(2c_1 A y_0^{-1/2} + \frac{4}{5} c_1^2 \delta^{-1} y_0^{-5/4} \right),$$

$$\Omega_W := \int_{y_0}^{y_1} 2(s + 2) \tan \alpha(s) \frac{c_0}{2} s^{-3/2} ds, \quad K_W := \frac{4c_1}{c_0 \text{inv } F} \cdot 1.000001 c_0^2,$$

$$B := \frac{1}{\omega_0 y_0^{3/4}} + \frac{A}{2c_1} + \frac{2c_1}{c_0 \text{inv } F} \left(A' + \frac{\alpha(y_0) C_\Lambda}{\delta} \right) + \Gamma.$$

In the analytic variant (Theorem A₁₃), A, A', C_Λ are replaced by $A_{13}, A'_{13}, C_\Lambda^{(13)}$ in Γ and B .

Γ denotes the explicit expression above; Section 10 shows that it is an upper bound for the integral $\Gamma_H := \int_H 0.50001 \beta(y) (A + \beta(y)/\delta) d(y)^{-3/4} dy$. (The numbers Γ and Γ_H are not to be confused with the trajectories of Definition 3.16, which always carry a floor point as an index, as in Γ_x, Γ_x^0 .) Numerically (rounded up) $A < 10.6067$, $A' < 10.6068$, $C_\Lambda < 1.0000002$, $A_{13} < 12.7476$, $A'_{13} < 12.7476$ and $C_\Lambda^{(13)} < 1.0000002$; see Section 12. The constants C_Λ and A' are those of the bounds of Sections 7 and 8. The subscript W in b_W, Ω_W and K_W is only a label (window, wall), as in the wall time t_W of Definition 3.9; it does not refer to the waste area W , and K_W is unrelated to the multiplicity bounds $K_w = 13$ and $K_w^* = 9$ of [R].

The roles of the conditions of Definition 3.2 are as follows. (C0) is standing. (C1) gives $\alpha_{\max} \leq 1.5 \cdot 10^{-6}$ and $\alpha_{\max} \leq \bar{\beta}$, the hypotheses on $\alpha_{\max}$ in Sections 5, 7 and 8, and the bound $\tan x \leq 1.000001 x$ for $0 \leq x \leq \alpha_{\max}$ used for the wall term in Section 10; it implies (Q1) and (C6) (Lemma 3.3). (C2) makes all line thresholds $\beta(y)$, $y \in H$, at most $1.5 \cdot 10^{-6}$; this is the hypothesis of [R, Lemmas 4.13 and 4.15] in Section 10, and it bounds the inclinations of the squares of $\mathcal{Z}(s)$ (Sections 4 and 9). (C3) is equivalent to $\hat{\beta}((1 - \varepsilon)s) < \alpha(s)$ for all $s \geq y_0$ (Lemma 3.3); it is not used in any proof of this paper and is kept in the list only for reference (it holds at π^* , at π_{13}^* and at all other parameters used in Sections 11 and 12). The results below whose hypotheses include (C0)–(C6) or (C0)–(C7) hold without (C3). (C4) is the hypothesis $\omega(y) < \frac{1}{2}$ of [R, Lemma 4.13] for the lines of kind (ii). (C5) gives $h_0 > 0$, used in the lower bound for the measure of H in Section 10. (C6) is the hypothesis of the two-source lemma of Section 7, on which the bounds for deaths, merges and gap overlaps there rest; it is not used in Section 5. (C7) depends on k ; it gives $y_0 < y_1$ and is used in this section (Lemma 3.3(d) and the scale flows) and in Sections 10 and 11. (Q1) gives $\sec \alpha(s) - 1 \leq 0.50001 \alpha(s)^2$ for $s \geq y_0$, by the inequality (2.1) (Section 4). (Q2) gives $a(Z) < b^* \leq 10^{-4}$ and $\cos a(Z) + \sin a(Z) < 1.0001$ for the squares $Z \in \mathcal{Z}(s)$ (Section 4). Finally $0 < \delta \leq 10^{-5}$ is the hypothesis of [R, Proposition 4.5 and Lemmas 4.14 and 4.15] and of Sections 5–8.

3.3. Squares, frames and the floor.

Definition 3.7 (frames and sides). Let S be a square. Its *phase* $\varphi_S \in (-\pi/4, \pi/4]$ is the unique number in this interval that is congruent modulo $\pi/2$ to the angle between the positive x -axis and a side of S . Put

$$u_S := (\cos \varphi_S, \sin \varphi_S), \quad n_S := (-\sin \varphi_S, \cos \varphi_S),$$

let c_S be the centre, so that $S = \{c_S + \xi u_S + \zeta n_S : |\xi|, |\zeta| \leq \frac{1}{2}\}$, and $a(S) = |\varphi_S|$ is the inclination of $[R]$; for squares S_1, S_2 , $\theta(S_1, S_2) = \min_{m \in \mathbb{Z}} |\varphi_{S_1} - \varphi_{S_2} - m\pi/2|$. The *sides* of S are

$$\begin{aligned} \text{bot}(S) &:= \{c_S - \tfrac{1}{2}n_S + \xi u_S : |\xi| \leq \tfrac{1}{2}\}, & \text{top}(S) &:= \{c_S + \tfrac{1}{2}n_S + \xi u_S : |\xi| \leq \tfrac{1}{2}\}, \\ \{c_S + \tfrac{1}{2}u_S + \zeta n_S : |\zeta| \leq \tfrac{1}{2}\} & \text{(right side)}, & \{c_S - \tfrac{1}{2}u_S + \zeta n_S : |\zeta| \leq \tfrac{1}{2}\} & \text{(left side)}, \end{aligned}$$

with outer normals $-n_S, n_S, u_S, -u_S$; their *relative interiors* (relint) are given by $|\xi| < \frac{1}{2}$, resp. $|\zeta| < \frac{1}{2}$, and the vertices are $c_S \pm \frac{1}{2}u_S \pm \frac{1}{2}n_S$. The four relative interiors and the four vertices partition ∂S . The *line of* $\text{bot}(S)$ is $\{v : (v - c_S) \cdot n_S = -\frac{1}{2}\}$, and similarly for the other sides. (The centre c_S is a point, with coordinates $c_{S,x}, c_{S,y}$; the chord length $c_S(y)$ of Section 2 is a function of the height y and always carries its argument.)

The 45° convention. If the sides of S make the angle $\pi/4$ with the axes, then $\varphi_S = +\pi/4$; thus $n_S = (-1, 1)/\sqrt{2}$, $\text{bot}(S)$ is the lower right side (outer normal $(1, -1)/\sqrt{2}$), and the lower left side is the left side.

The floor. The floor is the pseudo-square X_0 with $\varphi_{X_0} := 0$, $u_{X_0} := e_1$, $n_{X_0} := e_2$, whose “top side” is $[0, k] \times \{0\}$ with relative interior $(0, k) \times \{0\}$. It is not a member of $\mathcal{P}$.

For X, Y squares or the floor we have

$$(3.1) \quad \begin{aligned} n_X \cdot e_2 &= \cos \varphi_X, & e_2 &= \cos \varphi_X n_X + \sin \varphi_X u_X, \\ n_X \cdot n_Y &= u_X \cdot u_Y = \cos(\varphi_Y - \varphi_X), & u_X \cdot n_Y &= -\sin(\varphi_Y - \varphi_X), \end{aligned}$$

and the point $c_S + \xi u_S + \zeta n_S$ has height $c_{S,y} + \xi \sin \varphi_S + \zeta \cos \varphi_S$. In particular the lowest point of S has height $v_S = c_{S,y} - \frac{1}{2}(\cos a(S) + \sin a(S))$, and the points of $\text{top}(S)$ have heights $\geq c_{S,y} + \frac{1}{2}(\cos a(S) - \sin a(S)) = v_S + \cos a(S) > v_S$. Since $S \subset [0, k]^2$ and the abscissas $\xi \mapsto c_{S,x} \mp \frac{1}{2} \sin \varphi_S + \xi \cos \varphi_S$ of the points of $\text{top}(S)$, resp. $\text{bot}(S)$, are strictly increasing in ξ , the points of relint $\text{top}(S)$ and of relint $\text{bot}(S)$ have abscissas in $(0, k)$.

Lemma 3.8 (reflection). *Let $\rho(x, y) := (x, k - y)$.*

- (a) $\rho\mathcal{P} := \{\rho S : S \in \mathcal{P}\}$ is a closed packing of $[0, k]^2$ with N squares and the same W , and ρ maps the waste of $\mathcal{P}$ onto the waste of $\rho\mathcal{P}$. For every square S : $a(\rho S) = a(S)$, $\varphi_{\rho S} \equiv -\varphi_S \pmod{\pi/2}$, $v_{\rho S} = k - (v_S + \cos a(S) + \sin a(S))$ and $c_{\rho S}(k - y) = c_S(y)$ for all y .
- (b) $\mathcal{R}(\rho S) = k - \mathcal{R}(S)$ and $\Phi(\rho S) = k - \Phi(S)$; the line waste and the excess E of $[R]$, Lemma 3.5] satisfy $\omega_{\rho\mathcal{P}}(y) = \omega_{\mathcal{P}}(k - y)$ and $E_{\rho\mathcal{P}}(y) = E_{\mathcal{P}}(k - y)$; $d(k - y) = d(y)$ and $\beta(k - y) = \beta(y)$; the line $k - y$ meets ρS if and only if the line y meets S .
- (c) Since $k \in \mathbb{Z}$: $\text{dist}(k - y, \mathbb{Z}) = \text{dist}(y, \mathbb{Z})$ for all y , hence $k - H = H$ and $H_t = k - H_b$. The kind of $y \in H$ for $\rho\mathcal{P}$ equals the kind of $k - y$ for $\mathcal{P}$; thus $Y_{iii}[\rho\mathcal{P}] = k - Y_{iii}[\mathcal{P}]$, $Y_b[\rho\mathcal{P}] = k - Y_t[\mathcal{P}]$, and $\rho Z \in \mathcal{Z}(s)[\rho\mathcal{P}]$ if and only if $Z \in \tilde{\mathcal{Z}}(s)[\mathcal{P}]$.

Proof. (a) ρ is an isometry of $\mathbb{R}^2$ mapping $[0, k]^2$ onto itself, so it maps squares to squares, preserves disjointness and areas, and maps $(0, k)^2$ onto itself. It maps the direction $(\cos \psi, \sin \psi)$ to $(\cos \psi, -\sin \psi)$, hence the side directions of ρS have angles congruent to $-\varphi_S$, and the least angle between a side and the axes is unchanged. The highest point of S has height $v_S + \cos a + \sin a$ ([R, Lemma 3.1(a)]) and is mapped to the lowest point of ρS . Finally $\rho S \cap \ell_{k-y} = \rho(S \cap \ell_y)$.

(b) With $v' := v_{\rho S} = k - v - \cos a - \sin a$ ($v := v_S$, $a := a(S)$), [R, Lemma 3.1(b)] gives $\mathcal{R}(\rho S) = [v', v' + \sin a] \cup (v' + \cos a, v' + \cos a + \sin a] = [k - v - \cos a - \sin a, k - v - \cos a] \cup (k - v - \sin a, k - v]$, which is $k - ([v, v + \sin a] \cup (v + \cos a, v + \cos a + \sin a]) = k - \mathcal{R}(S)$, including the open and closed

ends. The statements on Φ , ω , E and on meeting lines follow from $\rho S \cap \ell_{k-y} = \rho(S \cap \ell_y)$ and $\rho([0, k] \times \{y\}) \setminus \bigcup \mathcal{P} = ([0, k] \times \{k-y\}) \setminus \bigcup \rho \mathcal{P}$; $d(k-y) = d(y)$ is clear.

(c) $k - \mathbb{Z} = \mathbb{Z}$. If $y \in \mathbb{Z}$, both fractional parts vanish and $0 \notin [w_0, 1 - w_0]$ (as $w_0 > 0$); otherwise $\{k-y\} = 1 - \{y\}$ and $[w_0, 1 - w_0]$ is invariant under $t \mapsto 1 - t$. Together with $d(k-y) = d(y)$ this gives $k - H = H$. By (b), ω , the inclinations of the squares meeting a line and β are transported by $y \mapsto k - y$, so the kinds are; and $k - (\frac{k}{2}, k) = (0, \frac{k}{2})$. The last statement follows from $\Phi(\rho Z) = k - \Phi(Z)$: $(k - \Phi(Z)) \cap (k - Y_t) \cap \mathcal{W}_s \neq \emptyset$ if and only if $\Phi(Z) \cap Y_t \cap (k - \mathcal{W}_s) \neq \emptyset$. $\square$

3.4. Rules of motion.

Definition 3.9 (flows: ray steps and events). Let $0 < \alpha_F < \pi/4$, $\delta > 0$ and $0 < h_F < k$. The *flow* $F = F(\alpha_F, \delta, h_F)$ assigns to every floor point $x \in (0, k)$ a *path* π_x . A path moves through *states* (p, d, X, g) : $p \in [0, k]^2$ is the current point; X is the floor or a square, the *source* (the square last passed); $d = n_X$ is the direction; and $g \geq 0$ is the *cumulative gap*, the sum of the Euclidean lengths of the gap segments travelled so far. The initial state is $((x, 0), e_2, X_0, 0)$.

A *ray step* from the state (p, n_X, X, g) uses $r(t) := p + tn_X$, $t \geq 0$, and

$$t_S := \min\{t \geq 0 : r(t) \in Y \text{ for some square } Y \in \mathcal{P}, Y \neq X\} \quad (+\infty \text{ if there is none}),$$

$$t_W := \min\{t \geq 0 : r(t)_x \in \{0, k\}\} \quad (+\infty \text{ if there is none}), \quad t_D := \delta - g, \quad t_H := \frac{h_F - p_y}{\cos \varphi_X},$$

$$t^* := \min(t_S, t_W, t_D, t_H).$$

(The minimum defining t_S exists because the squares are closed and finite in number.) The first of the following rules that applies is applied; this is the *priority* $D > W > \text{contact} > H$.

- (D) If $t_D = t^*$, the path terminates at $r(t_D)$ (*death*, D).
- (W) Otherwise, if $t_W = t^*$, the path terminates at $r(t_W)$ (*wall*, W).
- (C) Otherwise, if $t_S = t^*$, put $q := r(t_S)$ and $g \leftarrow g + t_S$; let Y be the square containing q (it is unique, as the squares are disjoint). If $q \in \text{relint bot}(Y)$, then (Y, q) is an *entry candidate* and is processed by rule R1 (Definition 3.14) and then rule R2 (Definition 3.10). Otherwise the path terminates at q (*end collision*, E).
- (H) Otherwise ($t_H < \min(t_S, t_W, t_D)$) the path terminates at $r(t_H)$; it is *completed* (H).

A point $q = r(t_S)$ with $t^* = t_S$ is called a *first contact point*; it is a contact point in case (C), and the end point of a death or wall termination if $t_D = t_S$ or $t_W = t_S$.

Since contact has priority over H, a contact at height exactly h_F is processed; since D has priority over contact, a processed contact has $g + t_S < \delta$ (Lemma 3.17(f)). The rules R1 and R2 are defined below in the order in which they can be stated: first rule R2 and the passing rule, together with the *free paths*, which ignore R1 (Definition 3.10); then, by a recursion over the squares that uses the free paths, rule R1 and the paths of F (Definition 3.14). The terminations M (merge) and T (tilt) are introduced there. The terminations D, W, E, M, T are *losses*; H is not. (The letter d for the direction of a ray, with components d_x, d_y , is not to be confused with the function $d(y) = \min(y, k - y)$ of Definition 3.1.)

Definition 3.10 (rule R2 and passing; free paths). *Rule R2 (tilt)*. If a path is the winner at an entry candidate (Y, q) (Definition 3.14) and $a(Y) \geq \alpha_F$, it terminates at q (*tilt termination*, T). A contact processed by rule (C) at the relative interior of a left or right side, or at a vertex, is an end collision whatever $a(Y)$ is; tilt terminations occur only at entries through $\text{relint bot}(Y)$.

Passing. If the winner has $a(Y) < \alpha_F$, it *passes* Y : it traverses the *pass segment* $[q, q + n_Y]$, its state becomes $(q + n_Y, n_Y, Y, g)$ (the length travelled inside Y is not added to g), and if $(q + n_Y)_y \geq h_F$ it terminates at the *exit point* $q + n_Y$ with H; otherwise a new ray step begins.

Free paths. The *free path* π_x^f is defined by Definition 3.9 and the two rules above, every entry candidate being treated as won (rule R1 is ignored).

Lemma 3.11 (top sides are not hit first). *In every ray step of a free path or of a path of a flow, the first contact point is not in $\text{relint top}(Y)$ for any square Y . Consequently an end collision occurs at a point of the relative interior of the left or the right side of a square, or at a vertex.*

Proof. Let the step start at p with source X , so $d = n_X$ with $|\varphi_X| < \pi/4$ (X is the floor, or a square passed by the path, so $a(X) < \alpha_F < \pi/4$). Suppose $q = r(t_S) = c_Y + \frac{1}{2}n_Y + \xi u_Y$ with $|\xi| < \frac{1}{2}$. If $t_S = 0$, then $q = p$; if X is a square, $p \in X$ and $X \cap Y = \emptyset$, which is impossible; if X is the floor, $q = (x, 0)$ has height 0, so it is a lowest point of $Y \subset [0, k]^2$, whereas the points of $\text{top}(Y)$ have heights $> v_Y$. So $t_S > 0$. As $|\varphi_Y - \varphi_X| < \pi/4 + \pi/4 = \pi/2$, we have $n_Y \cdot d = \cos(\varphi_Y - \varphi_X) > 0$ by (3.1). For $0 < t_S - t$ small, $r(t) = q - (t_S - t)d$ satisfies $(r(t) - c_Y) \cdot n_Y = \frac{1}{2} - (t_S - t) \cos(\varphi_Y - \varphi_X) \in (-\frac{1}{2}, \frac{1}{2})$ and $|(r(t) - c_Y) \cdot u_Y| = |\xi - (t_S - t)d \cdot u_Y| < \frac{1}{2}$, so $r(t) \in Y$ with $t < t_S$, contradicting the minimality of t_S . The second statement follows since ∂Y is the disjoint union of the relative interiors of the four sides and the four vertices. $\square$

Lemma 3.12 (DAG lemma). *Let a path (free, or of a flow) pass a square X , entering it at $q_X \in \text{relint bot}(X)$, and let its next contact be a point $q_Y \in \text{relint bot}(Y)$ of a square Y . Then*

$$c_{Y,y} - c_{X,y} \geq \frac{1}{2}(\cos a(X) - \sin a(X)) + \frac{1}{2}(\cos a(Y) - \sin a(Y)) > 0.$$

Consequently the squares of the successive entry candidates of a path have strictly increasing centre heights. In particular they are distinct, a path has at most one entry candidate at each square, and two squares with the same centre height never both carry entry candidates of one path.

Proof. Write $q_X = c_X - \frac{1}{2}n_X + \tau_X u_X$ with $|\tau_X| < \frac{1}{2}$. Then $c_{X,y} = q_{X,y} + \frac{1}{2} \cos \varphi_X - \tau_X \sin \varphi_X \leq q_{X,y} + \frac{1}{2}(\cos a(X) + \sin a(X))$. In the same way $c_{Y,y} = q_{Y,y} + \frac{1}{2} \cos \varphi_Y - \tau_Y \sin \varphi_Y \geq q_{Y,y} + \frac{1}{2}(\cos a(Y) - \sin a(Y))$. The path goes from q_X to the exit point $q_X + n_X$ and then a distance $t \geq 0$ in the direction n_X , so $q_{Y,y} = q_{X,y} + (1+t) \cos \varphi_X \geq q_{X,y} + \cos a(X)$. Combining the three relations gives the inequality. Its first term is positive because $a(X) < \alpha_F < \pi/4$, and the second is ≥ 0 because $a(Y) \leq \pi/4$. The consequences follow by induction along the path (the first entry candidate comes from the floor, and every later one is the next contact after a pass). $\square$

Lemma 3.13 (branch maps). *Let $\sigma = (Z_1, \dots, Z_m)$ be a sequence of distinct squares with $a(Z_i) < \pi/4$ for $i < m$, and put $Z_0 := X_0$. Let $\tau_1^\sigma(x)$ be the u_{Z_1} -coordinate (relative to c_{Z_1}) of the intersection of the vertical line $\{x\} \times \mathbb{R}$ with the line of $\text{bot}(Z_1)$, and, for $2 \leq j \leq m$, let $\tau_j^\sigma(x)$ be the u_{Z_j} -coordinate of the intersection of the line of $\text{bot}(Z_j)$ with the line through $c_{Z_{j-1}} + \frac{1}{2}n_{Z_{j-1}} + \tau_{j-1}^\sigma(x)u_{Z_{j-1}}$ in the direction $n_{Z_{j-1}}$. Then each $\tau_j^\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is an affine bijection with slope $\prod_{i=1}^j \sec(\varphi_{Z_i} - \varphi_{Z_{i-1}}) \geq 1$. If the free path of x passes $Z_1, \dots, Z_{m-1}$ as its first $m-1$ passes and its next contact is a point $q \in \text{relint bot}(Z_m)$, then $q = c_{Z_m} - \frac{1}{2}n_{Z_m} + \tau_m^\sigma(x)u_{Z_m}$.*

Proof. For $i \leq m$ we have $|\varphi_{Z_{i-1}}| < \pi/4$ and $\varphi_{Z_i} \in (-\pi/4, \pi/4]$, so $|\varphi_{Z_i} - \varphi_{Z_{i-1}}| < \pi/2$. Let $L_1 = \{p_0 + su_1 : s \in \mathbb{R}\}$ and $L_2 = \{c_2 + ru_2 : r \in \mathbb{R}\}$ be lines with unit directions $u_i = (\cos \varphi_i, \sin \varphi_i)$, $|\varphi_2 - \varphi_1| < \pi/2$, and put $n_i := (-\sin \varphi_i, \cos \varphi_i)$. Since $n_1 \cdot n_2 = \cos(\varphi_2 - \varphi_1) \neq 0$, the line through $p_0 + su_1$ in the direction n_1 meets L_2 in exactly one point $c_2 + ru_2$; taking the inner product of $p_0 + su_1 + tn_1 = c_2 + ru_2$ with u_1 gives $(p_0 - c_2) \cdot u_1 + s = r \cos(\varphi_2 - \varphi_1)$, so r is an affine function of s with slope $\sec(\varphi_2 - \varphi_1) \geq 1$. The step from the floor is the case $L_1 = \mathbb{R} \times \{0\}$, $s = x$, $n_1 = e_2$; the passage from the line of $\text{bot}(Z)$ to the line of $\text{top}(Z)$ by $v \mapsto v + n_Z$ preserves the u_Z -coordinate relative to c_Z . Composing gives the slope. For the last statement: the first ray step of the free path is vertical, so its first entry point lies on $\{x\} \times \mathbb{R}$ and on the line of $\text{bot}(Z_1)$; the exit point is the entry point plus n_{Z_1} , with the same u_{Z_1} -coordinate; the next contact lies on the line through the exit point in the direction n_{Z_1} ; and so on by induction. $\square$

Definition 3.14 (rule R1 and the paths of a flow). Number the squares of $\mathcal{P}$ as $Y_1, \dots, Y_N$ so that $c_{Y_1,y} \leq c_{Y_2,y} \leq \dots \leq c_{Y_N,y}$ (squares with equal centre heights in a fixed arbitrary order). For $i = 1, \dots, N$ in turn and for every $q \in \text{relint bot}(Y_i)$ let $\mathcal{A}_{Y_i}(q)$ be the set of $x \in (0, k)$ such that

(Y_i, q) is an entry candidate of the free path π_x^f and x is the winner at every entry candidate of π_x^f preceding (Y_i, q) ; if $\mathcal{A}_{Y_i}(q) \neq \emptyset$, the winner $w(Y_i, q)$ is the element $x \in \mathcal{A}_{Y_i}(q)$ for which the pair $(g_x(q), x)$ is lexicographically smallest, where $g_x(q)$ is the cumulative gap of π_x^f at q (after the update in rule (C)). We say that x is the winner at (Y, q) if $x = w(Y, q)$.

The path π_x of F is π_x^f if x is the winner at every entry candidate of π_x^f ; otherwise it is the initial part of π_x^f up to the first entry candidate (Y, q) at which x is not the winner, and it terminates there at q (merge, M). In words, rule R1 (merge): among the paths that reach an entry candidate (Y, q) alive, the one with the lexicographically smallest (g, x) continues, and the others terminate at q with M; then rule R2 is applied to the winner.

Lemma 3.15 (R1 is well defined). (a) For every square Y and every $q \in \text{relint bot}(Y)$ the set $\mathcal{A}_Y(q)$ is finite.

(b) The recursion of Definition 3.14 is well defined: at stage i it uses only winners at squares Y_l with $l < i$. A path of F reaches an entry candidate (Y, q) without having terminated if and only if $x \in \mathcal{A}_Y(q)$, and up to its termination π_x coincides with π_x^f ; thus the paths of F obey Definition 3.9 with entry processing R1 followed by R2.

(c) At most one path of F passes, or is T-terminated at, a given entry point (Y, q) .

Proof. (a) Let $x \in \mathcal{A}_Y(q)$. Before q , the free path π_x^f passes squares $Z_1, \dots, Z_{m-1}$ with $a(Z_i) < \alpha_F < \pi/4$, which are distinct by Lemma 3.12, and q is its next contact. By Lemma 3.13, $q = c_Y - \frac{1}{2}n_Y + \tau_m^\sigma(x)u_Y$ with $\sigma = (Z_1, \dots, Z_{m-1}, Y)$, and as τ_m^σ is a bijection, x is determined by σ and q . There are only finitely many sequences of distinct squares, so $\mathcal{A}_Y(q)$ is finite and the lexicographic minimum exists (distinct x give distinct pairs).

(b) By Lemma 3.12, an entry candidate of π_x^f preceding (Y_i, q) lies on a square Y_l with $c_{Y_l, y} < c_{Y_i, y}$, hence $l < i$; so the winners used at stage i have been defined before. The rules D, W, E, H, R2 and the passing rule depend only on the state of the path itself; rule R1 stops losers and does not change the motion of winners. Hence, by induction along the path, π_x coincides with π_x^f until it terminates, and it reaches (Y, q) without having terminated exactly when x won all earlier candidates, i.e. when $x \in \mathcal{A}_Y(q)$.

(c) A path that passes, or is T-terminated at, (Y, q) is the winner $w(Y, q)$, which is unique. $\square$

The order in which R1 and R2 are applied at an entry candidate is discussed in Remark 3.34 at the end of this section.

3.5. Trajectories.

Definition 3.16 (trajectories). Let π_x be a path of a flow F (or a free path). Its *trajectory* Γ_x is the polygonal line formed, in order, by its *gap segments* $[p, p + t^*n_X]$, one for each ray step (of length $t^* \geq 0$), and its *pass segments* $[q, q + n_Y]$, one for each passed square. The last point of Γ_x is the *termination point* $e(x)$ (for a path completed after a pass, the exit point). The *termination height* is $\tau(x) := e(x)_y$, except that $\tau(x) := h_F$ if π_x is completed (H). The endpoints of the segments, including $(x, 0)$ and $e(x)$, are the *event points*. The path is *alive at height* y if $\tau(x) > y$. Each path has exactly one *termination type* $\star \in \{T, D, E, M, W, H\}$.

Lemma 3.17 (trajectories). Let π_x be a path of a flow $F = F(\alpha_F, \delta, h_F)$ or a free path, $x \in (0, k)$.

- (a) Every segment of Γ_x has a unit direction d with $d_y \geq \cos \alpha_F > 0$. Hence, if $z \neq z'$ are points of Γ_x and z comes before z' , then $z_y < z'_y$.
- (b) Every ray step starts at a point p with $p_x \in (0, k)$, $p_y < h_F$ and cumulative gap $g < \delta$. If its source X is a square, then $p \in \text{relint top}(X)$, $t_S > 0$ and $t^* > 0$.
- (c) The open part $\{p + tn_X : 0 < t < t^*\}$ of a gap segment lies in the waste.
- (d) If $t_S < \infty$, the point $r(t_S)$ lies in exactly one square Y , and $r(t_S) \in \partial Y$.
- (e) A pass segment $[q, q + n_Y]$ lies in Y and meets no other square; its open part lies in $\text{int } Y$; $q \in \text{relint bot}(Y)$ and $q + n_Y \in \text{relint top}(Y)$.

- (f) At a processed contact the cumulative gap is $g + t_S < \delta$. If a step ends with a death at a first contact point ($t_D = t_S$), the cumulative gap there equals δ .
- (g) No event point lies in the interior of a square. The points of Γ_x that lie in a square Z are first contact points on ∂Z , and the points of the pass segment of Z if π_x passes Z ; in the latter case the initial point of that pass segment is a first contact point.
- (h) π_x has at most $h_F / \cos \alpha_F + 1$ passes and at most $h_F / \cos \alpha_F + 2$ ray steps.

Proof. (a) The gap segments have directions n_X , where X is the floor ($\cos \varphi_X = 1$) or a passed square ($a(X) < \alpha_F$), and the pass segments have directions n_Y with Y passed; by (3.1), $n \cdot e_2 = \cos \varphi > \cos \alpha_F$. Along a segment of positive length the height is therefore strictly increasing.

(f) At the start of every ray step $g < \delta$: initially $g = 0$, and a new step begins only after a pass, which follows a processed contact. A processed contact has $t_S = t^*$ and $t_D \neq t^*$ (otherwise D applies), so $t_D > t_S$, i.e. $g + t_S < \delta$. If $t_D = t_S = t^*$, rule D applies at $r(t_D)$, where the cumulative gap is $g + t_D = \delta$.

(b) The first step starts at $(x, 0)$ with $x \in (0, k)$ and $g = 0$. Every later step starts at an exit point $p = q + n_Y$ of a pass that did not end with H, so $p_y < h_F$, $p \in \text{relint top}(Y)$ (by (e)), whose abscissas lie in $(0, k)$, and $g < \delta$ by (f). If X is a square, then $p \in X$ and $p \notin Y'$ for every square $Y' \neq X$; as the set of $t \geq 0$ with $r(t) \in \bigcup_{Y' \neq X} Y'$ is closed, $t_S > 0$. Moreover $t_D = \delta - g > 0$, $t_W > 0$ (as $p_x \in (0, k)$) and $t_H = (h_F - p_y) / \cos \varphi_X > 0$, so $t^* > 0$.

(c) Let $0 < t < t^*$. As $t < t^* \leq t_S$, $r(t)$ lies in no square $Y \neq X$; if X is a square, $(r(t) - c_X) \cdot n_X = \frac{1}{2} + t > \frac{1}{2}$, so $r(t) \notin X$. By (b), $p_x \in (0, k)$, and $t < t^* \leq t_W$ gives $r(t)_x \in (0, k)$. Finally $r(t)_y > p_y \geq 0$ and $r(t)_y < p_y + t^* \cos \varphi_X \leq p_y + t_H \cos \varphi_X = h_F < k$. Hence $r(t)$ lies in $(0, k)^2$ and in no square.

(d) $r(t_S)$ lies in a square by the definition of t_S , and in only one since the squares are disjoint. If $t_S > 0$, then $r(t) \notin Y$ for $0 \leq t < t_S$, so $r(t_S)$ is a limit of points outside Y and lies on ∂Y . If $t_S = 0$, the source is the floor by (b), $r(0) = (x, 0)$ has height 0, and no point of height 0 is interior to a square in $[0, k]^2$.

(e) $q = c_Y - \frac{1}{2}n_Y + \xi u_Y$ with $|\xi| < \frac{1}{2}$, so $q + tn_Y = c_Y + \xi u_Y + (t - \frac{1}{2})n_Y$, which lies in Y for $0 \leq t \leq 1$ and in $\text{int } Y$ for $0 < t < 1$; $q + n_Y \in \text{relint top}(Y)$. Other squares are disjoint from Y .

(g) The event points are: $(x, 0)$, of height 0, not interior to any square; entry points and exit points, on ∂Y by (e); end collision points, first contact points on ∂Y by (d); death points $r(t_D)$, which lie in no square if $t_D < t_S$ (as in (c)) and are first contact points if $t_D = t_S$; wall points, whose abscissa is 0 or k ; and the end points of completions, which are either gap points $r(t_H)$ with $t_H < t_S$ (in no square, as in (c)) or exit points. For the second statement: points of open gap parts lie in the waste by (c); points of the pass segment of a square $Y \neq Z$ lie in Y , hence not in Z ; the start point of a ray step is $(x, 0)$ (covered above, if in Z it is $r(0)$ with $t_S = 0$, a first contact point) or an exit point of a passed square X (if it lies in Z , then $Z = X$ and it is a point of the pass segment of Z); the end point $r(t^*)$ of a ray step lies in no square if $t^* < t_S$ and is a first contact point if $t^* = t_S$. Finally, a pass segment of Z starts at the entry point, which is the first contact point of the preceding ray step.

(h) An entry candidate is processed only if $t_S \leq t_H$, i.e. at height $\leq h_F$. Each pass raises the height by $\cos \varphi_Y \geq \cos \alpha_F$ and the gap segments do not decrease it, so the j -th entry point has height $\geq (j-1) \cos \alpha_F$. Hence $(j-1) \cos \alpha_F \leq h_F$ for every entry, which gives the bound on passes; every ray step except the first follows a pass. $\square$

3.6. Piecewise-affine structure.

Lemma 3.18 (piecewise-affine structure). *Let $F = F(\alpha_F, \delta, h_F)$ be a flow. There is a finite set $\Sigma_F \subset [0, k]$ containing 0 and k such that on every component I of $[0, k] \setminus \Sigma_F$ (a piece of F):*

- (a) *the combinatorial type of π_x is the same for all $x \in I$: the number of ray steps, the source of each step, the event that ends each step (D, W, contact, H), for each contact the square*

- met and the side (relative interior of the bottom, left or right side) that contains the contact point, for each entry candidate whether x is the winner and whether it then passes or is T -terminated, and the termination type;
- (b) the start point $p_j(x)$ and the start gap $g_j(x)$ of the j -th ray step, its length $t_j^*(x)$, its contact point $q_j(x)$ (if it ends with a contact) and the termination point $e(x)$ are affine functions of $x \in I$;
- (c) for every $x \in I$, the line that carries a ray step of π_x contains no vertex of a square. In particular no first contact point of π_x (Definition 3.9) is a vertex of a square: for $x \in I$ no entry candidate, no end collision and no D - or W -termination at a first contact point ($t_D = t_S$ or $t_W = t_S$) lies at a vertex. Consequently, first contact points at vertices of squares occur only for the finitely many $x \in \Sigma_F$.

The same holds for the free paths, with a finite set $\Sigma_F^f \subset \Sigma_F$ and without the $R1$ outcomes in (a).

Lemma 3.19 (expansion). *On a piece I of Lemma 3.18 (for F or for the free paths) let X_j be the source of the j -th ray step (X_0 the floor).*

- (a) $p'_j(x) = \lambda_j u_{X_j}$ with $\lambda_0 = 1$ and $\lambda_j = \prod_{i=1}^j \sec(\varphi_{X_i} - \varphi_{X_{i-1}}) \geq 1$.
- (b) If the j -th step ends with a contact at the relative interior of $\text{bot}(Y)$, then $q'_j(x) = \mu_j u_Y$ with $\mu_j = \lambda_j / \cos(\varphi_Y - \varphi_{X_j}) \geq \lambda_j \geq 1$; if the path passes Y , then $X_{j+1} = Y$ and $\lambda_{j+1} = \mu_j$.
- (c) If the j -th step ends with a contact at the relative interior of the left or the right side of Y , then $q'_j(x) = \nu n_Y$ with $\nu \neq 0$. (The derivative factors μ_j and ν are unrelated to the measures $\mu_Z(y)$ and ν_y of Definition 3.28 and Lemma 3.29.)
- (d) For $y \in \mathbb{R}$, on the set of $x \in I$ with $p_{j,y}(x) < y < p_{j,y}(x) + t_j^*(x) \cos \varphi_{X_j}$, the abscissa of the point of height y on the j -th gap segment is an affine function of x with derivative $\lambda_j / \cos \varphi_{X_j} \geq 1$. If the path passes Y after step j , then on the set of $x \in I$ with $q_{j,y}(x) < y < q_{j,y}(x) + \cos \varphi_Y$ the abscissa of the point of height y on the pass segment of Y is affine with derivative $\mu_j / \cos \varphi_Y \geq 1$.

Proof of Lemmas 3.18 and 3.19. Step 1: one ray step of the free paths. Let I be an open interval and $j \geq 0$, and suppose that for every $x \in I$ the free path π_x^f has a j -th ray step, with the same source X , with start point $p(x)$ and start gap $g(x)$ affine in x , and with $p'(x) = \lambda u_X$, $\lambda \geq 1$. Put $d := n_X$; since $p' \perp d$, the lines $\{p(x) + td : t \in \mathbb{R}\}$, $x \in I$, are parallel and are indexed by $\xi(x) := p(x) \cdot u_X$, an affine function with slope $\lambda > 0$.

(i) For each square $Y \neq X$ the lines that meet Y are those with ξ between the least and the largest of the four numbers $v \cdot u_X$, v a vertex of Y . Let C_1 be the finite set of $x \in I$ with $\xi(x) = v \cdot u_X$ for some vertex v of some square. On a component I_1 of $I \setminus C_1$ and for each $Y \neq X$, either no line meets Y , or every line meets the interior of Y in a non-degenerate segment that contains no vertex of Y . In the latter case the initial point of the segment (in the direction d) lies in the relative interior of a side σ_Y of Y ; it moves continuously with x and never meets a vertex, so σ_Y is the same for all $x \in I_1$. The line is not parallel to σ_Y (a segment along σ_Y would begin at a vertex), so the parameters $t_Y(x) \leq t'_Y(x)$ of the initial and final points are affine functions of x .

(ii) If X is a square, $p(x) \in X$ lies in no other square, so $0 \notin [t_Y(x), t'_Y(x)]$; the open sets $\{t_Y > 0\}$ and $\{t'_Y < 0\}$ cover I_1 , which is connected, so one of them is I_1 . If X is the floor, then $d = e_2$, $p(x) = (x, 0)$, and $t_Y(x) \geq 0$ is the height of the lowest point of Y on the vertical line through $(x, 0)$. In both cases the set $\mathcal{Y}$ of squares met at some $t \geq 0$ is the same for all $x \in I_1$, and $t_S(x) = \min_{Y \in \mathcal{Y}} t_Y(x)$ ($+\infty$ if $\mathcal{Y} = \emptyset$).

(iii) For $x \in I_1$ and distinct $Y, Y' \in \mathcal{Y}$, $t_Y(x) \neq t_{Y'}(x)$, since otherwise the point $p(x) + t_Y(x)d$ would lie in $Y \cap Y'$. Hence the sets $\{x \in I_1 : t_Y(x) < t_{Y'}(x) \ \forall Y' \in \mathcal{Y} \setminus \{Y\}\}$, $Y \in \mathcal{Y}$, are open, disjoint and cover I_1 ; one of them, for $Y = Y_S$, is I_1 . Thus the first square met and the side $\sigma := \sigma_{Y_S}$ are constant on I_1 , and $t_S = t_{Y_S}$ is affine.

(iv) $t_D = \delta - g$ and $t_H = (h_F - p_y)/\cos \varphi_X$ are affine. If $d_x = 0$ then $t_W \equiv +\infty$ (as $p_x \in (0, k)$ by Lemma 3.17(b)); if $d_x > 0$ then $t_W = (k - p_x)/d_x$, and if $d_x < 0$ then $t_W = -p_x/d_x$; in each case t_W is affine or $+\infty$.

(v) Let C_2 be the set of zeros in I_1 of those differences of two of the functions t_S, t_W, t_D, t_H that are finite and not identically zero; it is finite. On a component I_2 of $I_1 \setminus C_2$ all order relations ($<$, $=$) among the four functions are constant, so the rule applied (D, W, contact, H) is the same for all $x \in I_2$, and t^* is affine.

(vi) By (i), for $x \in I_2$ the line $\{p(x) + td : t \in \mathbb{R}\}$ contains no vertex of a square, and if $t_S(x) < \infty$, the first contact point $p(x) + t_S(x)d$ is the initial point of the segment in which this line meets Y_S , so it lies in relint σ and is not a vertex; this holds whichever rule (D, W or contact) is applied at it. If the rule is contact, $q(x) = p(x) + t_S(x)d$ is affine and lies in relint σ ; by Lemma 3.11, σ is the bottom, the left or the right side of Y_S , the same for all $x \in I_2$. Whether $a(Y_S) \geq \alpha_F$ is fixed. In the case of a pass, let C_3 be the zero (if any) of the affine function $q_y(x) + \cos \varphi_{Y_S} - h_F$ if it is not identically zero; on each component of $I_2 \setminus C_3$ it is fixed whether the pass ends with H, and otherwise the next step has source Y_S , start point $q(x) + n_{Y_S}$ and start gap $g(x) + t_S(x)$, all affine.

(vii) *Derivatives.* If $\sigma = \text{bot}(Y)$ ($Y = Y_S$), its line has direction u_Y , so $q' = \mu u_Y$ for some μ ; on the other hand $q' = p' + t'_S d = \lambda u_X + t'_S n_X$. Taking the inner product with u_X and using (3.1) gives $\mu \cos(\varphi_Y - \varphi_X) = \lambda$. As $|\varphi_X| < \pi/4$ and $\varphi_Y \in (-\pi/4, \pi/4]$, $\cos(\varphi_Y - \varphi_X) \in (0, 1]$, so $\mu = \lambda / \cos(\varphi_Y - \varphi_X) \geq \lambda$. After a pass the next start point is $q + n_Y$, with derivative μu_Y . If σ is a left or right side, its line has direction n_Y , so $q' = \nu n_Y$, and $\nu \neq 0$ because $q' \cdot u_X = \lambda \neq 0$. For the crossings: the point of height y on the gap segment is $p + tn_X$ with $t = (y - p_y)/\cos \varphi_X$, of abscissa $p_x - (y - p_y) \tan \varphi_X$, whose derivative is $\lambda \cos \varphi_X + \lambda \sin \varphi_X \tan \varphi_X = \lambda / \cos \varphi_X \geq 1$; the point of height y on the pass segment of Y has abscissa $q_x - (y - q_y) \tan \varphi_Y$, with derivative $\mu \cos \varphi_Y + \mu \sin \varphi_Y \tan \varphi_Y = \mu / \cos \varphi_Y \geq 1$.

Step 2: the free paths. Start with $I = (0, k)$, $j = 0$, $X = X_0$, $p(x) = (x, 0)$, $g = 0$, $\lambda = 1$, and apply Step 1 repeatedly. On each resulting interval the paths either terminate in the j -th step or pass and satisfy the hypotheses of Step 1 for $j + 1$ with $\lambda_{j+1} = \mu$. By Lemma 3.17(h) the number of steps is bounded, so after finitely many subdivisions we obtain a finite set $\Sigma_F^f \ni 0, k$ with properties (a)–(c) for the free paths, and Step 1(vii) gives Lemma 3.19 for them (with λ_j as stated, by induction).

Step 3: the outcomes of R1. For a free piece I (a component of $[0, k] \setminus \Sigma_F^f$) and a square Y carrying an entry candidate of the common type, write the contact point as $q^I(x) = c_Y - \frac{1}{2}n_Y + \tau^I(x)u_Y$, where τ^I is affine with slope ≥ 1 (Step 1(vii)), and let $g^I(x)$ be the cumulative gap at $q^I(x)$, also affine. We show by induction over the stages $i = 1, \dots, N$ of Definition 3.14 that for every free piece I the sets $A_{Y_i}(I) := \{x \in I : x \in \mathcal{A}_{Y_i}(q^I(x))\}$ and $L_{Y_i}(I) := \{x \in A_{Y_i}(I) : x \neq w(Y_i, q^I(x))\}$ are finite unions of intervals and points (empty if Y_i carries no candidate of the type of I); these two families of sets are used only in this proof. Assume this for all stages $l < i$. The candidates of the type of I that precede the one at Y_i lie on squares Y_l with $l < i$ (Lemma 3.12), so $A_{Y_i}(I) = I \setminus \bigcup_l L_{Y_l}(I)$, a finite union of intervals and points. A point $x \in A_{Y_i}(I)$ is in $L_{Y_i}(I)$ if and only if some $x' \neq x$ in $\mathcal{A}_{Y_i}(q^I(x))$ has $(g_{x'}, x') <_{\text{lex}} (g^I(x), x)$. Such an x' either lies in the finite set Σ_F^f , and since τ^I is injective and x' has at most one candidate at Y_i , it affects at most one $x \in I$; or it lies in a free piece I' (possibly $I' = I$) with $x' \in A_{Y_i}(I')$ and $q^{I'}(x') = q^I(x)$, i.e. $x' = \psi(x) := (\tau^{I'})^{-1}(\tau^I(x))$, an affine map defined on the interval $\{x \in I : \tau^I(x) \in \tau^{I'}(I')\}$. Therefore $L_{Y_i}(I)$ is the union of finitely many points and of the finitely many sets

$$\{x \in A_{Y_i}(I) \cap \text{dom } \psi : \psi(x) \in A_{Y_i}(I'), (g^{I'}(\psi(x)), \psi(x)) <_{\text{lex}} (g^I(x), x)\},$$

each defined by finitely many affine equalities and inequalities in x , hence a finite union of intervals and points. (For $I' = I$ we have $\psi = \text{id}$ and the strict inequality fails.)

Step 4: the flow. Let Σ_F consist of Σ_F^f and the endpoints of the intervals and the isolated points of all the sets $L_Y(I)$ and $A_Y(I)$. On each component of $[0, k] \setminus \Sigma_F$ all outcomes of R1 are constant, so by Definition 3.14 and Lemma 3.15(b) the path π_x is the free path truncated at the same candidate (or not truncated) for all x ; (a)–(c) and Lemma 3.19 follow from the free case (the termination point of an M-terminated path is the affine contact point). $\square$

3.7. The master flow, the scale flows and the ceiling flows.

Definition 3.20 (master flow). Assume (C0) and (C1). The *master flow* is $F^0 := F(\alpha_{\max}, \delta, h_{\max})$ with $\alpha_{\max} = \alpha(y_0)$ and $h_{\max} = \frac{k}{2} - 1$; by (C0) and Lemma 3.3(a), $0 < \alpha_{\max} \leq 1.5 \cdot 10^{-6} < \pi/4$, and $0 < h_{\max} < k$ as $k \geq 4$, so this is a flow. Its paths are π_x^0 , its trajectories Γ_x^0 , its termination heights $\tau_0(x)$. Rule R1 is decided only in F^0 . The *R1-passed entries* of π_x^0 are the entry candidates of π_x^0 at which x is the winner; we list them in path order as $(Z_1, q_1), \dots, (Z_J, q_J)$, $J = J(x) \geq 0$, and put $\eta_j := (q_j)_y$. For $j < J$, π_x^0 passes Z_j ; Z_J is passed or T-terminates π_x^0 .

Definition 3.21 (scale flows). Assume (C0), (C1) and (C7), and let $s \in [y_0, y_1]$. The path π_x^s of the *scale flow* F_s is π_x^0 truncated at the first, in path order, of the following events.

- (a) T_s : an R1-passed entry (Z_j, q_j) with $a(Z_j) \geq \alpha(s)$; π_x^s terminates at q_j with T.
- (b) H_s : either a ray step of π_x^0 , with start point p , direction d and parameters t_S, t_W, t_D (in F^0), such that $t_H^{(s)} := (s + 2 - p_y)/d_y < \min(t_S, t_W, t_D)$; then π_x^s terminates at $p + t_H^{(s)}d$ with H; or a pass of π_x^0 whose exit point has height $\geq s + 2$; then π_x^s terminates at the exit point with H.
- (c) The termination of π_x^0 (D, W, E, M or T), if no event (a) or (b) precedes it; π_x^s then terminates at the same point with the same type.

If π_x^0 is T-terminated at an R1-passed entry (Z_j, q_j) that is not preceded by an event (a) or (b), then (a) holds there as well, because $a(Z_j) \geq \alpha_{\max} \geq \alpha(s)$; such an *inherited* T termination is a T_s termination. Thus “T-terminated in F_s ” and “ T_s -terminated in F_s ” mean the same. We put $h_{F_s} := s + 2$, write $\tau_s(x)$ for the termination height of π_x^s ($\tau_s := s + 2$ for H), and

$$\mathcal{T}_s := \{x \in (0, k) : \pi_x^s \text{ is T-terminated}\}, \quad N(s) := |\mathcal{T}_s|.$$

If a ray step of π_x^0 ends with H in F^0 , then $t_H^{(s)} \leq t_H < \min(t_S, t_W, t_D)$ (as $s + 2 \leq h_{\max}$), so (b) applies; if π_x^0 is completed at an exit point, that exit point has height $\geq h_{\max} \geq s + 2$, so (b) applies there or earlier. Hence (c) never concerns a completion. A contact at height exactly $s + 2$ has $t_S = t_H^{(s)}$, so (b) does not apply in its step: as in F^0 , contact precedes H.

Definition 3.22 (the tilt time). Let (Z_j, q_j) , $j \leq J(x)$, be the R1-passed entries of π_x^0 and $\eta_j = (q_j)_y$. Put

$$s_j := \max(y_0, \eta_j - 2, (c_0/a(Z_j))^2) \text{ if } a(Z_j) > 0, \quad s_j := +\infty \text{ if } a(Z_j) = 0,$$

and $T^*(x) := \min_j s_j$ if this minimum is $\leq y_1$; otherwise $T^*(x)$ is undefined. Let $\text{dom } T^*$ be the set of x where $T^*(x)$ is defined. Only R1-passed entries enter: deaths, merges and end collisions do not.

Lemma 3.23 (scale flows). Assume (C0), (C1), (C7), let $x \in (0, k)$ and $s \in [y_0, y_1]$.

- (a) $\eta_1 < \eta_2 < \dots < \eta_J$. For $j \geq 2$, every point of Γ_x^0 before q_j has height $< \eta_j$; every point before q_1 has height $\leq \eta_1$.
- (b) π_x^s is T-terminated if and only if there is j with $a(Z_j) \geq \alpha(s)$ and $\eta_j \leq s + 2$; its termination point is then q_j for the least such j .
- (c) The trajectory of π_x^s is an initial part of Γ_x^0 . Every square passed by π_x^s is passed by π_x^0 and has inclination $< \alpha(s)$. Every D, W, E or M termination of F_s is a termination of F^0 of the same type at the same point. Every first contact point on the trajectory of π_x^s has height $\leq s + 2$.

- (d) If $y_0 \leq s \leq s' \leq y_1$, then $\mathcal{T}_s \subset \mathcal{T}_{s'}$.
- (e) $\{x \in \text{dom } T^* : T^*(x) \leq s\} = \mathcal{T}_s$.
- (f) There is a finite set $\Sigma_s \supset \Sigma_{F^0}$ such that on each component of $[0, k] \setminus \Sigma_s$ the truncation event of π_x^s (its type and the step at which it occurs) is constant and the termination point of π_x^s is affine in x ; Lemma 3.18(a)–(c) holds for F_s with these pieces. In particular $\mathcal{T}_s$ and, for every y and $\star$, the set $\{x : \pi_x^s \text{ is } \star\text{-terminated, } \tau_s(x) < y\}$ are finite unions of intervals and points.

Proof. (a) For $j \geq 2$, q_j is the first contact of the ray step that starts at the exit point p of Z_{j-1} ; this step has a square source, so $t_S > 0$ (Lemma 3.17(b)) and $p_y < \eta_j$; all earlier points have height $\leq p_y$ by Lemma 3.17(a), and $\eta_{j-1} < p_y$. Before q_1 there is no entry candidate (a lost candidate would have terminated the path), so the points before q_1 lie on the first, vertical, gap segment and have height $\leq \eta_1$.

(c) The first statement is the definition. A square passed by π_x^s is an R1-passed entry (Z_j, q_j) of π_x^0 at which (a) of Definition 3.21 does not hold, so $a(Z_j) < \alpha(s)$. D, W, E, M terminations of F_s come from (c) of Definition 3.21. Let q be a first contact point on the trajectory of π_x^s , ending a ray step that starts at p . If $p_y \geq s + 2$, then p (which has positive height) is an exit point of height $\geq s + 2$, and π_x^s would have terminated at p by (b); so $p_y < s + 2$. If $q_y > s + 2$, then $t_H^{(s)} < t_S$, and $t_S = t^* \leq \min(t_D, t_W)$, so $t_H^{(s)} < \min(t_S, t_W, t_D)$ and π_x^s would have terminated before q by (b). Hence $q_y \leq s + 2$.

(b) ($\Leftarrow$) Let j be least with $a(Z_j) \geq \alpha(s)$ and $\eta_j \leq s + 2$. Up to its first truncation, π_x^s coincides with π_x^0 , and π_x^0 reaches q_j and wins there. No event (a) precedes q_j : an earlier R1-passed entry Z_i ($i < j$) has $\eta_i < \eta_j \leq s + 2$ by (a), hence $a(Z_i) < \alpha(s)$ by the minimality of j . No event (b) precedes q_j : the exit points before q_j have height $< \eta_j \leq s + 2$ by (a); every ray step before q_j ends with a contact at an entry q_i of height $\eta_i \leq \eta_j \leq s + 2$, so $t_S \leq t_H^{(s)}$ in that step and (b) does not apply. Hence π_x^s reaches q_j with the state of π_x^0 , is the R1 winner there (R1 is decided in F^0), and is T-terminated at q_j by (a). ($\Rightarrow$) If π_x^s is T-terminated, its termination point is an R1-passed entry (Z_j, q_j) with $a(Z_j) \geq \alpha(s)$ (for an inherited termination, $a(Z_j) \geq \alpha_{\max} \geq \alpha(s)$), and $\eta_j \leq s + 2$ by (c). By the first part, the termination point is q_j for the least such j .

(d) As $\alpha(s') \leq \alpha(s)$ and $s + 2 \leq s' + 2$, the condition in (b) for s implies the condition for s' .

(e) For every j : if $a(Z_j) = 0$, then $s_j = +\infty > s$ and $a(Z_j) < \alpha(s)$. If $a(Z_j) > 0$, then, as $y_0 \leq s$, $s_j \leq s$ holds if and only if $\eta_j \leq s + 2$ and $(c_0/a(Z_j))^2 \leq s$, i.e. $a(Z_j) \geq c_0 s^{-1/2} = \alpha(s)$. Hence $\min_j s_j \leq s$ if and only if some j satisfies the condition of (b), i.e. if and only if $x \in \mathcal{T}_s$. Finally, $\min_j s_j \leq s \leq y_1$ means that $x \in \text{dom } T^*$ and $T^*(x) \leq s$.

(f) On a piece I of F^0 the R1-passed entries Z_j are fixed, and $\eta_j(x)$, the exit heights, $t_j^*(x)$ and $t_H^{(s)}(x)$ (for each ray step) are affine. Let Σ_s consist of Σ_{F^0} and the zeros in each piece of the finitely many affine functions $\eta_j - (s + 2)$, (exit height) $-(s + 2)$ and $t_H^{(s)} - \min(t_S, t_W, t_D)$ that are not identically zero. On each component of $[0, k] \setminus \Sigma_s$ the truth value of each of the conditions (a), (b), (c) of Definition 3.21 at each event of the common type is constant, hence so is the first truncation; the termination point is $q_j(x)$, $p_j(x) + t_H^{(s)}(x)d_j$, an exit point, or the termination point of π_x^0 , all affine. Thus Lemma 3.18(a),(b) for F_s follow from the same statements for F^0 ; and since the ray steps of π_x^s are initial parts of ray steps of π_x^0 (with the same lines) and each component of $[0, k] \setminus \Sigma_s$ lies in a piece of F^0 , Lemma 3.18(c) for F_s follows from Lemma 3.18(c) for F^0 . The last statement follows since on each such component the type is fixed and τ_s is affine or the constant $s + 2$. $\square$

By Lemma 3.23(c), Lemma 3.17(a) and (c)–(g) hold for the paths of F_s as well (with $h_F = s + 2$ and $\alpha_F := \alpha(s)$; the length t^* of a ray step of π_x^s is $t_H^{(s)}$ at an H_s truncation in a gap, and the length t^* of the ray step of π_x^0 otherwise): the trajectory of π_x^s is an initial part of Γ_x^0 , and the additional

end point $p + t_H^{(s)}d$ of an H termination in a gap has $t_H^{(s)} < \min(t_S, t_W, t_D)$, so, as in the proof of Lemma 3.17(c), it lies in the waste.

Definition 3.24 (ceiling flows). For any object or quantity Q defined from the floor flows of the packing $\mathcal{P}$ (paths, trajectories, termination heights, $N(s)$, T^* , and the losses and line quantities defined in Section 3.8 below, such as $\mathcal{D}$, $\mathcal{M}$, $\mathcal{E}$, Λ_0), its *ceiling version* $\tilde{Q}$ is the same quantity computed for the reflected packing $\rho\mathcal{P}$ and transported back by ρ : floor points x are unchanged and heights are mapped by $y \mapsto k - y$. Thus the ceiling master flow is $\tilde{F}^0 := \rho \circ F^0[\rho\mathcal{P}] \circ \rho$; its path $\tilde{\pi}_x^0 := \rho(\pi_x^0[\rho\mathcal{P}])$ starts at (x, k) and moves downwards; similarly $\tilde{F}_s$, $\tilde{N}(s) := N(s)[\rho\mathcal{P}]$, $\tilde{T}^* := T^*[\rho\mathcal{P}]$, $\tilde{\Lambda}_0 := \Lambda_0[\rho\mathcal{P}]$. A ceiling path is alive at height y if its termination height $\tilde{\tau}(x) := k - \tau[\rho\mathcal{P}](x)$ is $< y$.

By Lemma 3.8, $\rho\mathcal{P}$ is a closed packing of $[0, k]^2$ with the same N and W ; hence every statement of this paper about the floor flows of an arbitrary closed packing applies to the ceiling flows, with heights reflected.

Remark 3.25 (one R1 for all scales). All scale flows are truncations of the single flow F^0 , in which R1 is decided once. Therefore an entry point is passed or T-terminated by at most one path (Lemma 3.15(c)), simultaneously for F^0 and all F_s ; and $\mathcal{T}_s$ increases with s and equals $\{T^* \leq s\}$ (Lemma 3.23(d)(e)). These two facts are used in Sections 8 and 10. They are properties of this construction: if R1 were decided separately in each F_s , the winners could depend on s , and neither fact would follow.

3.8. Losses and line quantities. In this subsection $F \in \{F^0\} \cup \{F_s : s \in [y_0, y_1]\}$, with $h_F = h_{\max}$, resp. $h_F = s + 2$, paths π_x and termination heights $\tau = \tau_F$. For F_s , “T-terminated” means T_s -terminated (Definition 3.21).

Definition 3.26 (losses). For $\star \in \{T, D, E, M, W\}$ and $y \in \mathbb{R}$ put

$$L_\star^F(y) := |\{x \in (0, k) : \pi_x \text{ is } \star\text{-terminated in } F, \tau_F(x) < y\}|, \quad L^F(y) := \sum_\star L_\star^F(y).$$

For the master flow let $\mathcal{D}, \mathcal{M}, \mathcal{E}$ be the sets of x for which π_x^0 is D-, M-, E-terminated, and $L_E := |\mathcal{E}|$; their ceiling versions are $\tilde{\mathcal{D}}, \tilde{\mathcal{M}}, \tilde{\mathcal{E}}, \tilde{L}_E$. For a square Z let $m^F(Z)$ be the floor measure of the set of x such that π_x has an R1-passed entry at Z in F (i.e. passes Z or is T-terminated at $\text{bot}(Z)$ in F).

Corollary 3.27 (tilt losses of a scale flow). For $s \in [y_0, y_1]$ and every y , $L_T^{F_s}(y) \leq N(s)$.

Proof. By Definition 3.21 and Lemma 3.23(b), the set of x that are T-terminated in F_s is $\mathcal{T}_s$, so $L_T^{F_s}(y) = |\{x \in \mathcal{T}_s : \tau_s(x) < y\}| \leq |\mathcal{T}_s| = N(s)$. $\square$

Definition 3.28 (line quantities). Let $y \in (0, h_F)$. If $\tau(x) \geq y$, the trajectory Γ_x meets ℓ_y in exactly one point $P_y(x)$, with abscissa $X_y(x)$: indeed Γ_x is a connected compact polygonal line that starts at height 0 and ends at height $\geq \tau(x) \geq y$ (for a completed path the end has height $\geq h_F$), and heights increase strictly along it (Lemma 3.17(a)). Put $\text{Wa}_y := \ell_y \cap \text{waste}$, identified with the set of its abscissas; $|\text{Wa}_y| = \omega(y)$. For a square Z let

$$\begin{aligned} A_Z(y) &:= \{x : \tau(x) > y, P_y(x) \in \text{int } Z\}, & \mu_Z(y) &:= |A_Z(y)|, \\ A_{\text{gap}}(y) &:= \{x : \tau(x) > y, P_y(x) \in \text{Wa}_y\}, & F_{\text{gap}}(y) &:= |A_{\text{gap}}(y)|, \\ \text{Sh}(y) &:= \sum_{Z \in \mathcal{P}} (c_Z(y) - \mu_Z(y))_+, & \text{Ov}(y) &:= \sum_{Z \in \mathcal{P}} (\mu_Z(y) - c_Z(y))_+, \\ \text{Bd}(y) &:= k - L(y) - \sum_{Z \in \mathcal{P}} \mu_Z(y) - F_{\text{gap}}(y). \end{aligned}$$

Convention for μ_Z : a path is counted if it is alive at y in the strict sense $\tau(x) > y$ and crosses ℓ_y in the interior $\text{int } Z$ of Z in $\mathbb{R}^2$. (The set $\text{int } Z \cap \ell_y$ is contained in the relative interior of the

chord $Z \cap \ell_y$, and it is empty when ℓ_y contains a side of Z , although then $c_Z(y) = 1$.) Let ν_y be the image of the floor measure restricted to $A_{\text{gap}}(y)$ under $x \mapsto X_y(x)$. Finally, for a point q of the waste, $M_g(q)$ is the number of x such that q lies in the open part of a gap segment of Γ_x^0 , and $M_g := \sup_q M_g(q)$ (the *gap multiplicity*).

For a square Z , $\mu_Z(y) = 0$ for all y if no path of F passes Z (Lemma 3.29(a) below). More generally, by Lemma 3.29(a) every $x \in A_Z(y)$ passes Z and so has an R1-passed entry at Z ; hence $\mu_Z(y) \leq m^F(Z)$ for every y , and $\mu_Z \equiv 0$ if $m^F(Z) = 0$.

Lemma 3.29 (crossings and the gap density). *Let $y \in (0, h_F)$ and let x satisfy $\tau(x) > y$.*

- (a) $P_y(x) \in \text{int } Z$ if and only if π_x passes Z in F and $P_y(x)$ is a point of the open part of the pass segment of Z , i.e. $q_{Z,y}(x) < y < q_{Z,y}(x) + \cos \varphi_Z$ for its entry point $q_Z(x)$.
- (b) $P_y(x) \in \text{Wa}_y$ if and only if $P_y(x)$ lies on the open part of a gap segment of Γ_x .
- (c) $A_{\text{gap}}(y)$ is the union of finitely many points and finitely many disjoint open intervals J_c ; on each J_c , $P_y(x)$ lies on the open part of the gap segment of a fixed ray step with a fixed source X_c , and X_y is affine on J_c with slope $\kappa_c \geq 1$. Consequently ν_y is absolutely continuous, with density

$$\rho_y = \sum_c \kappa_c^{-1} \mathbf{1}_{X_y(J_c)} \quad (\text{almost everywhere}),$$

which vanishes almost everywhere outside Wa_y , and $F_{\text{gap}}(y) = \nu_y(\mathbb{R}) = \int_{\text{Wa}_y} \rho_y$. (The density always carries the height as a subscript, ρ_y ; the letter ρ alone denotes the reflection of Lemma 3.8.)

- (d) $A_Z(y)$ is the union of finitely many points and finitely many open intervals on each of which X_y is affine with slope ≥ 1 .

Proof. Since $\tau(x) > y$, the point $P_y(x)$ is not the termination point (whose height is $\geq \tau(x)$, also for completed paths). The other event points are $(x, 0)$, of height $0 < y$, and entry and exit points, which lie on the boundaries of squares (Lemma 3.17(e)). Hence $P_y(x)$ is either an entry or exit point, or a point of the open part of a gap segment (which lies in the waste, Lemma 3.17(c)), or a point of the open part of a pass segment (which lies in int of the passed square, Lemma 3.17(e)). As the interiors of the squares, their boundaries and the waste are pairwise disjoint, (a) and (b) follow.

(c) Let I be a piece of F (Lemma 3.18, Lemma 3.23(f)). For each ray step j of the common type (for F_s , t_j^* is the length of the step in F_s , as stated after Lemma 3.23), the set of $x \in I$ with $p_{j,y}(x) < y < p_{j,y}(x) + t_j^*(x) \cos \varphi_{X_j}$ is an open interval, given by two affine strict inequalities; for such x , the path continues above height y , so $\tau(x) > y$, and by (b) $x \in A_{\text{gap}}(y)$. By (b), $A_{\text{gap}}(y) \cap I$ is the union of these intervals, which are disjoint because the height is strictly increasing along Γ_x . There are finitely many pieces, and Σ_F (resp. Σ_s) is finite. On each interval the slope of X_y is $\kappa_c = \lambda_j / \cos \varphi_{X_j} \geq 1$ by Lemma 3.19(d). The image of Lebesgue measure on an interval J under an injective affine map with slope κ is $|\kappa|^{-1}$ times Lebesgue measure on the image; summing over c gives the density. The images lie in Wa_y by (b).

(d) In the same way, by (a), on a piece I the set $A_Z(y) \cap I$ is empty unless Z is passed in the common type, and then it is the open interval given by $q_{Z,y}(x) < y < q_{Z,y}(x) + \cos \varphi_Z$; the slope is $\mu / \cos \varphi_Z \geq 1$ by Lemma 3.19(d). $\square$

Definition 3.30 (gap overlap and loss terms). With ρ_y as in Lemma 3.29 put

$$\text{Ov}_{\text{gap}}(y) := \int_{\text{Wa}_y} (\rho_y - 1)_+ dq, \quad U(y) := \int_{\text{Wa}_y} (1 - \rho_y)_+ dq$$

(these do not depend on the version of ρ_y),

$$\Lambda^F(y) := [L_D^F + L_E^F + L_M^F + L_W^F](y) + \text{Ov}_{\text{gap}}^F(y),$$

$$\Lambda_0 := |\mathcal{D}| + |\mathcal{M}| + L_E + \sup_{0 < y < h_{\max}} \text{Ov}_{\text{gap}}^{F^0}(y), \quad \Lambda^*(s) := \Lambda_0 + 2(s+2) \tan \alpha(s),$$

and, for the ceiling flows, $\tilde{\Lambda}_0 := \Lambda_0[\rho\mathcal{P}]$ and $\tilde{\Lambda}^*(s) := \tilde{\Lambda}_0 + 2(s+2) \tan \alpha(s)$.

Since $\text{Ov}_{\text{gap}}(y) \leq \int_{W_{a_y}} \rho_y = F_{\text{gap}}(y) \leq k$ and $\mathcal{D}, \mathcal{M}, \mathcal{E}$ are disjoint subsets of $(0, k)$, we have $\Lambda_0 \leq 2k < \infty$. The term $2(s+2) \tan \alpha(s)$ in $\Lambda^*(s)$ accounts for wall terminations of F_s ; it is not of the form $O(W/\delta)$ and is integrated separately in Section 10.

Lemma 3.31 (measurability). *(a) The sets $\mathcal{D}, \mathcal{M}, \mathcal{E}, \mathcal{T}_s, \text{dom } T^*$, and, for every y and every square Z , the sets $\{x : \pi_x \text{ is } \star\text{-terminated in } F, \tau_F(x) < y\}$, $A_Z(y)$ and $A_{\text{gap}}(y)$ are finite unions of intervals and points.*

(b) T^ is a Borel function on the Borel set $\text{dom } T^*$, and $\{(x, s) : x \in \text{dom } T^*, T^*(x) \leq s\}$ is a Borel subset of $\mathbb{R}^2$.*

(c) $s \mapsto N(s)$ is non-decreasing on $[y_0, y_1]$, hence Borel.

(d) For fixed F , the functions $y \mapsto L_\star^F(y), \mu_Z(y), F_{\text{gap}}(y), c_Z(y), \omega(y), \text{Sh}(y), \text{Ov}(y)$ and $\text{Bd}(y)$ are Borel on $(0, h_F)$.

Proof. (a) On each piece of F^0 the termination type is constant, and on each piece of F_s as well (Lemma 3.23(f)); τ is affine or constant on pieces; there are finitely many pieces and finitely many exceptional points. This gives the statements for $\mathcal{D}, \mathcal{M}, \mathcal{E}$ and the loss sets; for $\mathcal{T}_s$ see Lemma 3.23(f); for $A_Z(y)$ and $A_{\text{gap}}(y)$ see Lemma 3.29. On a piece I of F^0 the R1-passed entries Z_j are fixed and η_j is affine, so each s_j is a continuous piecewise affine function on I with at most three pieces, and $\min_j s_j$ is a continuous piecewise affine function with finitely many pieces; hence $\text{dom } T^* \cap I = \{x \in I : \min_j s_j(x) \leq y_1\}$ is a finite union of intervals.

(b) By the above, T^* is continuous on $\text{dom } T^* \cap I$ for each piece I , and the finitely many remaining points form a Borel set; so T^* is Borel. The set in question is the preimage of the closed set $\{(t, s) : t \leq s\}$ under the Borel map $(x, s) \mapsto (T^*(x), s)$ on $\text{dom } T^* \times \mathbb{R}$.

(c) This is Lemma 3.23(d).

(d) Fix a piece I of F (Lemma 3.18, resp. Lemma 3.23(f)) and consider the planar sets

$$\{(x, y) : x \in I, 0 < y < h_F, \pi_x \star\text{-terminated}, \tau(x) < y\},$$

$$\{(x, y) : x \in I, 0 < y < h_F, x \in A_Z(y)\}, \quad \{(x, y) : x \in I, 0 < y < h_F, x \in A_{\text{gap}}(y)\}.$$

On I the termination type is constant and τ is affine or constant. By Lemma 3.29(a),(b), for $x \in I$ the condition $x \in A_Z(y)$ holds if and only if Z is passed in the common type and $q_{Z,y}(x) < y < q_{Z,y}(x) + \cos \varphi_Z$, and the condition $x \in A_{\text{gap}}(y)$ holds if and only if $p_{j,y}(x) < y < p_{j,y}(x) + t_j^*(x) \cos \varphi_{X_j}$ for one of the finitely many ray steps j of the common type; the functions of x occurring here are affine on I (Lemma 3.18(b)). Hence each of the three sets is a finite union of sets of the form $\{(x, y) : x \in I, l_1(x, y) \diamond_1 0, \dots, l_m(x, y) \diamond_m 0\}$ with affine functions l_i and relations $\diamond_i \in \{<, \leq, =\}$. Each such set is the intersection of finitely many open or closed half-planes (the strip $I \times \mathbb{R}$ is the intersection of two open half-planes, and an equation is the intersection of two closed half-planes); it is a convex polyhedral set, and it is Borel because half-planes are open or closed. For each of the finitely many floor points $x \in \Sigma_F \cap (0, k)$ (resp. $\Sigma_s \cap (0, k)$) the set of $y \in (0, h_F)$ that satisfy the defining condition is an open half-line, an open interval or a finite union of open intervals (the height ranges of the open parts of the segments of Γ_x), intersected with $(0, h_F)$; adding these finitely many Borel subsets of vertical lines keeps the three planar sets, taken over all of $(0, k)$, Borel. By Tonelli's theorem the measures of their sections at height y , which are $L_\star^F(y), \mu_Z(y)$ and $F_{\text{gap}}(y)$, are Borel functions of y . Next, if $a(Z) > 0$, then $c_Z(y)$ is the length of the horizontal section of a convex polygon, a continuous piecewise linear function of y with finitely many pieces; if $a(Z) = 0$, then c_Z is the indicator function of the closed interval $[v_Z, v_Z + 1]$ (the chord has length 1 also at the heights of the two horizontal sides). In both cases c_Z is Borel. Further $\omega(y) = k - \sum_Z c_Z(y)$ for $y \in [0, k]$, since the chords are pairwise disjoint subsets of $[0, k] \times \{y\}$; and Sh, Ov and Bd are finite sums and positive parts of the functions above. $\square$

Definition 3.32 (exceptional heights). For $F \in \{F^0\} \cup \{F_s\}$ let Exc_F be the set of $y \in (0, h_F)$ for which there are a piece I of F (a component of $[0, k] \setminus \Sigma_{F^0}$, resp. of $[0, k] \setminus \Sigma_s$) and an event point of the common type of the paths π_x , $x \in I$, whose height is the constant function y on I . Put $\text{Exc} := \text{Exc}_{F^0}$.

Lemma 3.33 (exceptional heights). (a) Exc_F is finite.

(b) If $y \in (0, h_F) \setminus \text{Exc}_F$, then only finitely many x outside the finite exceptional set of the pieces $(\Sigma_{F^0}$, resp. Σ_s) have an event point at height y .

(c) $\text{Exc}_{F_s} \subset \text{Exc} \cap (0, s+2)$ for every $s \in [y_0, y_1]$. Thus Exc is one finite set of heights that serves for all scale flows at once.

Proof. (a) There are finitely many pieces, and the paths of a piece have finitely many event points (Lemma 3.17(h)). (b) On each piece the height of each event point is affine in x and not the constant y , so it equals y for at most one x . (c) The pieces of F_s are subintervals of the pieces of F^0 . An event point of π_x^s is an event point of π_x^0 , or the end point $p + t_H^{(s)}d$ of an H termination in a gap, whose height is $s+2 \notin (0, s+2)$. An affine function that is constant on a subinterval of a piece of F^0 is constant on the whole piece. Finally $\text{Exc}_{F_s} \subset (0, s+2)$ by definition. $\square$

Section 6 describes the heights in Exc and shows that they cannot be omitted from the shadow inequality.

Remark 3.34 (the order “M before T”). At an entry candidate (Y, q) with $a(Y) \geq \alpha_F$, rule R1 is applied before rule R2: the winner is T-terminated and the others are M-terminated. Consider the opposite order, in which every path that reaches such a candidate alive is T-terminated and R1 is applied only at candidates with $a(Y) < \alpha_F$. At a candidate with $a(Y) \geq \alpha_F$ all arriving paths terminate at q in both orders, and at a candidate with $a(Y) < \alpha_F$ the two orders agree; the rules D, W, E, H do not depend on labels. By induction over the centre heights, the two orders produce the same trajectories, termination points and termination heights; only the labels T and M differ. Hence every quantity that is defined from trajectories and termination heights alone (such as those of Definition 3.28) is the same for both orders; for the shadow inequality of Section 6 see Theorem 6.9(g). The injectivity used in Section 8 comes from rule R1 at the *lower* member of each consecutive pair of entries (a passed square, or the floor), not from this order. The order is used in Section 10, for the inclusion $\{x : \pi_x^s \text{ is T-terminated in } F_s\} \subset \{T^* \leq s\}$ (Lemma 3.23(e)): with the order “M before T”, a path that is T-terminated in F^0 is the R1 winner at its terminating entry, so that entry is one of the R1-passed entries from which T^* is defined (Definition 3.22).

Remark 3.35 (degenerate cases). *Vertices and grazing.* Let a ray first meet a square at a vertex; this happens in particular for a ray running along the line of a side. By the priority $D > W > \text{contact} > H$ the path terminates there with D if $t_D = t_S$, otherwise with W if $t_W = t_S$ (the vertex then lies on a side wall of the container), and otherwise with an end collision E, since a vertex does not lie in $\text{relint bot}(Y)$. By Lemma 3.18(c), which holds for the scale flows by Lemma 3.23(f), all this happens only for the finitely many $x \in \Sigma_F$ of each flow F (resp. $x \in \Sigma_s$ for F_s). *Ties.* Coincidences such as $t_D = t_S$ can occur for sets of x of positive measure (for example for paths crossing a horizontal gap of constant width); they are resolved by the priority $D > W > \text{contact} > H$, and a tie in g at a merge is resolved by x . *Squares on the floor.* If an axis-parallel square Y rests on the floor, then for $(x, 0) \in \text{relint bot}(Y)$ the first step has $t_S = 0$ and an entry candidate $(Y, (x, 0))$ with $g = 0$; no other path reaches a point of height 0, so x wins and passes Y (as $a(Y) = 0 < \alpha_F$). The two abscissas of the end points of $\text{bot}(Y)$, and the abscissa of a vertex of a tilted square on the floor, give end collisions with $\tau = 0$; these are finitely many x . *Walls.* The floor points 0 and k are excluded; a path reaches a wall only through rule W, and open gap parts and pass segments stay in $(0, k) \times \mathbb{R}$. *Heights h_F and $s+2$.* A contact at height exactly h_F (in F^0) or $s+2$ (in F_s) is processed. *Inclination $\pi/4$.* By the 45° convention such a square has a bottom side; as $\alpha_F < \pi/4$ it is never passed: at an entry candidate on its bottom side the R1 winner is T-terminated and the

other arriving paths are M-terminated, and a first contact elsewhere gives E (or D or W at a tie). *Exceptional heights* are treated by Lemma 3.33.

4. P1 AND P2: THE HEIGHT IDENTITY AND QUANTIZATION AT BOTTOM SIDES

This section proves the height identity P1 (Theorem 4.2 and Corollary 4.4) and the quantization P2 at bottom sides (Theorem 4.11), and derives from them the form of P2 that is used in Section 9 (Theorem 4.13): if $Z \in \mathcal{Z}(s)$, then $Z \subset \mathbb{R} \times V_s$, no trajectory of the scale flow F_s meets the bottom side of Z , every trajectory of F_s meets Z at most in its own termination point, and $\mu_Z^{F_s} \equiv 0$ (under the conditions stated there). The ceiling versions follow by reflection. Besides the definitions of Section 3 and the well-definedness of R1 (Lemmas 3.12 and 3.15), the proofs use only [R, Lemma 3.1], the second part of [R, Lemma 3.4], and the inequality $\sec x - 1 \leq 0.50001 x^2$ for $0 \leq x \leq 10^{-3}$ from the proof of [R, Lemma 4.15], quoted as (2.1) in Section 2.

4.1. Notation and standing assumptions. Throughout this section $k \geq 4$ is an integer, $\mathcal{P}$ is a closed packing of $[0, k]^2$ with N squares, and the parameters of Section 3 satisfy $\delta > 0$, $c_0, c_1 > 0$, $y_0 > 0$, $0 < \varepsilon < 1$ and $\omega_0 > 0$. The letter s always denotes a number in $[y_0, y_1]$, where $y_1 = k/2 - 3$. Further conditions on the parameters are stated in each result that uses them. We recall the notions of Section 3 on which the proofs rely.

Squares. For a square S let $\varphi_S \in (-\frac{\pi}{4}, \frac{\pi}{4}]$ be its phase angle (a square whose sides make the angle $\frac{\pi}{4}$ with the axes has $\varphi_S = \frac{\pi}{4}$), $u_S = (\cos \varphi_S, \sin \varphi_S)$, $n_S = (-\sin \varphi_S, \cos \varphi_S)$, and c_S its centre, so that

$$S = \{c_S + \xi u_S + \zeta n_S : |\xi|, |\zeta| \leq \tfrac{1}{2}\};$$

$a(S) = |\varphi_S| \in [0, \frac{\pi}{4}]$ is the inclination of S . The bottom and top sides of S are the closed segments

$$\text{bot}(S) = \{c_S - \tfrac{1}{2}n_S + \xi u_S : |\xi| \leq \tfrac{1}{2}\}, \quad \text{top}(S) = \{c_S + \tfrac{1}{2}n_S + \xi u_S : |\xi| \leq \tfrac{1}{2}\},$$

and their relative interiors $\text{relint bot}(S)$, $\text{relint top}(S)$ are given by $|\xi| < \frac{1}{2}$. The floor is the pseudo-square X_0 with $\varphi_{X_0} = 0$, $n_{X_0} = e_2 := (0, 1)$ and “top side” $[0, k] \times \{0\}$; it is not a square. The height of a point z is its second coordinate z_y . Since $|\varphi_S| \leq \frac{\pi}{4}$,

$$(4.1) \quad n_S \cdot e_2 = \cos \varphi_S \geq \cos \frac{\pi}{4} > 0$$

for every square S and for $S = X_0$. We write $\ell_y = \mathbb{R} \times \{y\}$; $c_S(y)$ is the length of $S \cap \ell_y$; v_S is the height of the lowest point of S ; $\Phi(S) = \{y : 0 < c_S(y) < 1\}$; and $\mathcal{R}(S)$ is the ramp set of [R, Lemma 3.1]: with $a = a(S)$ and $v = v_S$,

$$\mathcal{R}(S) = [v, v + \sin a] \cup (v + \cos a, v + \cos a + \sin a], \quad \mathcal{R}(S) = \emptyset \text{ if } a = 0.$$

For $y \in \mathbb{R}$, $\{y\} \in [0, 1)$ is the fractional part of y and $\text{dist}(y, \mathbb{Z}) = \min(\{y\}, 1 - \{y\})$.

Flows. (Definitions 3.9, 3.14 and 3.10.) A flow $F = F(\alpha_F, \delta, h_F)$ moves a path π_x from each floor point $(x, 0)$, $x \in (0, k)$. The state (p, d, X, g) consists of the current point p , the direction $d = n_X$, the last traversed square X (initially the floor X_0) and the cumulative gap g ; the initial state is $((x, 0), e_2, X_0, 0)$. In this section a pass of Section 3 is called a *traversal* and a pass segment a *traversal segment*. We number the ray steps of π_x by $j = 0, 1, 2, \dots$. Step j starts at the point p_j , with last traversed square X_j and with cumulative gap G_j (the start gap g_j of Lemma 3.18; here the letters g_i are reserved for the lengths of the gap segments, see Theorem 4.2); its ray is $r_j(t) = p_j + t n_{X_j}$ ($t \geq 0$), and

$$t_{S,j} = \min\{t \geq 0 : r_j(t) \in Y \text{ for a square } Y \neq X_j\} \quad (= +\infty \text{ if there is no such } t),$$

$$t_{W,j} = \min\{t \geq 0 : (r_j(t))_x \in \{0, k\}\}, \quad t_{D,j} = \delta - G_j, \quad t_{H,j} = \frac{h_F - p_{j,y}}{\cos \varphi_{X_j}},$$

$$t_j^* = \min(t_{S,j}, t_{W,j}, t_{D,j}, t_{H,j}).$$

At t_j^* the first applicable rule in the following list is applied: D, if $t_{D,j} = t_j^*$ (termination at $r_j(t_j^*)$); W, if $t_{W,j} = t_j^*$ (termination at $r_j(t_j^*)$); contact, if $t_{S,j} = t_j^*$; H otherwise (the path is completed at $r_j(t_j^*)$). At a contact, the point $q = r_j(t_{S,j})$ lies in a unique square Y , the cumulative gap becomes $G_j + t_{S,j}$, and (Y, q) is an *entry candidate* if $q \in \text{relint bot}(Y)$; otherwise the path terminates at q with E. An entry candidate is processed by R1 (among all paths that reach (Y, q) alive, the one with the lexicographically least pair (g, x) wins and the others terminate at q with M; R1 is well defined, see Definition 3.14 and Lemmas 3.12 and 3.15), then by R2 (the winner terminates at q with T if $a(Y) \geq \alpha_F$); otherwise the winner *traverses* Y : it moves along the segment $[q, q + n_Y]$, its state becomes $(q + n_Y, n_Y, Y, G_j + t_{S,j})$, and it is completed (H) if $(q + n_Y)_y \geq h_F$ and starts its next ray step otherwise. If step j ends with the traversal of a square, we call that square X_{j+1} and put

$$q_j := r_j(t_{S,j}) \text{ (entry point),} \quad p_{j+1} := q_j + n_{X_{j+1}} \text{ (exit point),} \quad G_{j+1} := G_j + t_{S,j}.$$

The *trajectory* Γ_x of π_x is the polygonal path formed, in this order, by the gap segments $[p_j, r_j(t_j^*)]$ and the traversal segments $[q_j, p_{j+1}]$; its last point is the termination point $e(x)$. For $z, z' \in \Gamma_x$ we say that z *precedes* z' if z comes before z' along Γ_x ; by Lemma 4.1 (F1) below this holds if and only if $z_y < z'_y$.

First-contact points. If $t_j^* = t_{S,j} < \infty$, the point $r_j(t_{S,j})$ is called the *first-contact point of step j* , whatever rule (D, W or contact) is applied there. It is the last point of the gap segment of step j and therefore lies on Γ_x . The number $G_j + t_{S,j}$ is called the *cumulative gap at $r_j(t_{S,j})$* ; at a contact it is the new value of the state variable g .

Master flow and scale flows. (Definitions 3.20 and 3.21.) Put $\alpha(s) = c_0 s^{-1/2}$. The master flow is $F^0 = F(\alpha_{\max}, \delta, h_{\max})$ with $\alpha_{\max} = \alpha(y_0)$ and $h_{\max} = k/2 - 1$; R1 is decided in F^0 only. For F^0 to be a flow we need $0 < \alpha_{\max} < \pi/4$ and $0 < h_{\max} < k$; the latter holds as $k \geq 4$, and the former is assumed whenever F^0 or F_s is used in this section (it holds under (Q1) below, since then $\alpha_{\max} \leq 10^{-3}$, and hence under the constraint (C1) of Section 3). We write π_x^0 and Γ_x^0 for the path and the trajectory of x in F^0 , and use the notation $p_j, X_j, G_j, q_j, \dots$ for π_x^0 . For $s \in [y_0, y_1]$ the path of x in the scale flow F_s is π_x^0 truncated at the first point of Γ_x^0 that is a point of type (T_s) or (H_s) below, or, if there is no such point, at the termination point of π_x^0 :

(T_s) the point q of an entry candidate (Y, q) at which π_x^0 wins R1 and $a(Y) \geq \alpha(s)$;

(H_s) a point $r_j(t)$ of the gap segment of a step j with $t = t_{H,j}^{(s)} := (s + 2 - p_{j,y})/\cos \varphi_{X_j} < \min(t_{S,j}, t_{W,j}, t_{D,j})$; or an exit point p_{j+1} with $p_{j+1,y} \geq s + 2$.

We write Γ_x^s for the trajectory of x in F_s ; it is the part of Γ_x^0 from $(x, 0)$ up to the truncation point. The F_s path of x *traverses* a square X if the traversal segment of X is contained in Γ_x^s , and it *enters* X if it reaches an entry candidate (X, q) alive and wins R1 there.

Ceiling flows. Let $\rho(x, y) = (x, k - y)$. Then $\rho\mathcal{P} = \{\rho S : S \in \mathcal{P}\}$ is a closed packing of $[0, k]^2$ (Lemma 4.7). The ceiling flows are $\tilde{F}^0 = \rho \circ F^0[\rho\mathcal{P}] \circ \rho$ and $\tilde{F}_s = \rho \circ F_s[\rho\mathcal{P}] \circ \rho$: the trajectory of x in $\tilde{F}_s$ is the image under ρ of the trajectory of x in $F_s[\rho\mathcal{P}]$, and the first-contact points, entry candidates, entries, termination points and termination types of $\tilde{F}_s$ are the images under ρ of those of $F_s[\rho\mathcal{P}]$. The ceiling quantities are defined through ρ ; for instance $\mu_Z^{\tilde{F}_s}(y) = \mu_Z^{F_s[\rho\mathcal{P}]}(k - y)$ and $\text{Sh}^{\tilde{F}_s}(y) = \text{Sh}^{F_s[\rho\mathcal{P}]}(k - y)$.

Quantities. (Section 3.) For $F \in \{F^0, F_s\}$, $\tau_F(x)$ is the termination height of the path of x (with the convention of Section 3 for completed paths); for an E, D or W termination it is the height of the termination point. For $\tau_F(x) \geq y$, $P_y(x)$ is the point of $\Gamma_x \cap \ell_y$, which is unique by Lemma 4.1 (F1). With $|\cdot|$ the Lebesgue measure,

$$\mu_Z^F(y) = |\{x \in (0, k) : \tau_F(x) > y, P_y(x) \in \text{int } Z\}|, \quad \text{Sh}^F(y) = \sum_{S \in \mathcal{P}} (c_S(y) - \mu_S^F(y))_+.$$

Parameters and classes of lines. (Section 3.) Put $\hat{\beta}(t) = c_1 t^{-3/4}$ for $t > 0$ (a decreasing function), $d(y) = \min(y, k - y)$, $\beta(y) = \hat{\beta}(d(y))$, and

$$b^* := \hat{\beta}((1 - \varepsilon)y_0) = c_1((1 - \varepsilon)y_0)^{-3/4}, \quad w_0 := 0.50001 c_0^2 \left(1 + \frac{1.0001}{y_0}\right) + \delta + 1.0001 b^* + 10^{-12}.$$

Let $\omega(y)$ be the line waste, and

$$H := \{y \in (0, k) : y_0 \leq d(y) \leq (1 - \varepsilon)y_1, \{y\} \in [w_0, 1 - w_0]\},$$

$$Y_{iii} := \{y \in H : \omega(y) < \omega_0 \text{ and } a(S) < \beta(y) \text{ for every } S \in \mathcal{P} \text{ meeting } \ell_y\},$$

$$Y_b := Y_{iii} \cap (0, k/2), Y_t := Y_{iii} \cap (k/2, k), \mathcal{W}_s := [(1 - \varepsilon)s, s], V_s := [(1 - \varepsilon)s - 1.0001, s + 1.0001],$$

$$\mathcal{Z}(s) := \{Z \in \mathcal{P} : \Phi(Z) \cap Y_b \cap \mathcal{W}_s \neq \emptyset\}, \quad \tilde{\mathcal{Z}}(s) := \{Z \in \mathcal{P} : \Phi(Z) \cap Y_t \cap (k - \mathcal{W}_s) \neq \emptyset\},$$

where $k - Y := \{k - y : y \in Y\}$ for $Y \subset \mathbb{R}$. Since $w_0 \geq \delta > 0$,

$$(4.2) \quad \{y\} \in [w_0, 1 - w_0] \iff \text{dist}(y, \mathbb{Z}) \geq w_0 \quad (y \in \mathbb{R});$$

indeed both conditions fail if $y \in \mathbb{Z}$, and if $y \notin \mathbb{Z}$ both say $\min(\{y\}, 1 - \{y\}) \geq w_0$.

Conditions. The following conditions on the parameters are used; each result states which of them it needs.

$$(Q1) \quad \alpha(y_0) = c_0 y_0^{-1/2} \leq 10^{-3}.$$

$$(Q2) \quad b^* = c_1((1 - \varepsilon)y_0)^{-3/4} \leq 10^{-4}.$$

$$(C2) \quad \hat{\beta}(y_0) = c_1 y_0^{-3/4} \leq 1.5 \cdot 10^{-6}.$$

(C2) is one of the constraints of Section 3, and (Q1) follows from the constraint (C1), $\alpha(y_0) \leq 1.5 \cdot 10^{-6}$, of Section 3.

4.2. Trajectories. The next lemma restates, in the notation of this section, parts of Lemma 3.17 and of its proof: (F1) is part (a) there, (F2) follows from the proofs of parts (b) and (c), (F3) is part (d), (F4) is part (e), (F5) is part (f) together with part (b), and (F6) is part (g). We repeat the short proofs in the form used below.

Lemma 4.1. *Let π_x be a path of a flow F , with the notation of Subsection 4.1.*

- (F1) *Every gap or traversal segment of Γ_x has the form $\{z + tn : 0 \leq t \leq t_0\}$ with $t_0 \geq 0$ and $n = n_X$ for a square X or for the floor, and n has positive second component. Hence the height is strictly increasing along Γ_x , distinct points of Γ_x have distinct heights, and Γ_x meets every horizontal line in at most one point.*
- (F2) *If $0 < t < t_{S,j}$, then $r_j(t)$ lies in no square. If $j \geq 1$, then $p_j \in \text{top}(X_j)$ and $t_{S,j} > 0$.*
- (F3) *If $t_{S,j} < \infty$, then $r_j(t_{S,j})$ lies in exactly one square, and on the boundary of that square.*
- (F4) *If step j ends with the traversal of X_{j+1} , then $q_j \in \text{relint bot}(X_{j+1})$, $p_{j+1} \in \text{relint top}(X_{j+1})$, and the segment $[q_j, p_{j+1}]$ is contained in X_{j+1} and meets no other square.*
- (F5) *$G_j < \delta$ for every step j of π_x . If $t_j^* = t_{S,j}$, then the cumulative gap at the first-contact point $r_j(t_{S,j})$ satisfies $G_j + t_{S,j} \leq \delta$, with equality only if rule D is applied there.*
- (F6) *Let Z be a square and $z \in \Gamma_x \cap Z$. Then*
 - (i) *z is the first-contact point of some step j (so $t_j^* = t_{S,j}$) and Z is the square containing it, or*
 - (ii) *$Z = X_{j+1}$ for some j and $z \in [q_j, p_{j+1}]$.*

In case (ii), the entry point q_j is a point of type (i) and lies in $\text{relint bot}(Z)$.

Proof. (F4) Let $Y = X_{j+1}$, with centre c and frame u, n . Since (Y, q_j) is an entry candidate, $q_j = c - \frac{1}{2}n + \xi u$ with $|\xi| < \frac{1}{2}$. For $0 \leq \lambda \leq 1$ we get $q_j + \lambda n = c + (\lambda - \frac{1}{2})n + \xi u \in Y$, and for $\lambda = 1$ this is $p_{j+1} = c + \frac{1}{2}n + \xi u \in \text{relint top}(Y)$. Since the squares of $\mathcal{P}$ are pairwise disjoint, the segment $[q_j, p_{j+1}] \subset Y$ meets no other square.

(F1) The gap segment of step j is $\{p_j + t n_{X_j} : 0 \leq t \leq t_j^*\}$ and the traversal segment of X_{j+1} is $\{q_j + t n_{X_{j+1}} : 0 \leq t \leq 1\}$; the vectors $n_{X_j}, n_{X_{j+1}}$ are unit vectors with positive second component by (4.1). Hence the height increases strictly along every segment of positive length, and since consecutive segments share their endpoints, it increases strictly along Γ_x . Two distinct points of Γ_x are separated along Γ_x by a part of positive length, so they have distinct heights.

(F2) Let $0 < t < t_{S,j}$. For a square $Y \neq X_j$ we have $r_j(t) \notin Y$ by the minimality of $t_{S,j}$. If $j \geq 1$, then $p_j \in \text{top}(X_j)$ by (F4) (applied to step $j-1$), that is $(p_j - c_{X_j}) \cdot n_{X_j} = \frac{1}{2}$; hence $(r_j(t) - c_{X_j}) \cdot n_{X_j} = \frac{1}{2} + t > \frac{1}{2}$ and $r_j(t) \notin X_j$. Moreover, for $j \geq 1$ the point p_j lies in X_j , and the finitely many squares $Y \neq X_j$ are compact and disjoint from X_j ; so the distance from p_j to their union is positive, and $t_{S,j} > 0$.

(F3) The minimum defining $t_{S,j}$ is attained, since there are finitely many squares and they are closed; so $r_j(t_{S,j})$ lies in a square $Y \neq X_j$, and in no other square because the squares are disjoint. If $t_{S,j} > 0$, then $r_j(t) \notin Y$ for $0 < t < t_{S,j}$ by (F2), so $r_j(t_{S,j})$ is a limit of points outside Y and lies on ∂Y . If $t_{S,j} = 0$, then $j = 0$ by (F2) and $r_0(0) = (x, 0)$; the points $(x, -\eta)$, $\eta > 0$, are not in $Y \subset [0, k]^2$, so again $(x, 0) \in \partial Y$.

(F5) $G_0 = 0 < \delta$. If step j ends with a traversal, the rule applied at $t_j^* = t_{S,j}$ is the contact rule, so rule D was not applicable: $t_{D,j} \neq t_j^*$, that is $t_{S,j} < t_{D,j} = \delta - G_j$, and $G_{j+1} = G_j + t_{S,j} < \delta$. By induction $G_j < \delta$ for every step. If $t_j^* = t_{S,j}$, then $t_{S,j} \leq t_{D,j} = \delta - G_j$; equality means $t_{D,j} = t_j^*$, and then rule D is applied.

(F6) The points of Γ_x are the points $r_j(t)$, $0 \leq t \leq t_j^*$, of the gap segments and the points of the traversal segments. Let $z \in \Gamma_x \cap Z$.

- $z = r_j(t)$ with $t > 0$. Since $t \leq t_j^* \leq t_{S,j}$, (F2) shows that z lies in no square unless $t = t_j^* = t_{S,j}$; then z is the first-contact point of step j , and Z is the square containing it by (F3). This is case (i).
- $z = r_j(0) = p_j$ with $j \geq 1$. Then $p_j \in X_j$ by (F4), so $Z = X_j$ and $z \in [q_{j-1}, p_j]$. This is case (ii).
- $z = r_0(0) = (x, 0)$. Then $t_{S,0} = 0$, hence $t_0^* = 0 = t_{S,0}$, and z is the first-contact point of step 0. This is case (i).
- z lies on a traversal segment $[q_j, p_{j+1}]$. By (F4) it lies in X_{j+1} and in no other square, so $Z = X_{j+1}$. This is case (ii).

In case (ii), step j ended with the contact rule at $q_j = r_j(t_{S,j})$, so $t_j^* = t_{S,j}$ and q_j is the first-contact point of step j ; it lies in Z , and in $\text{relint bot}(Z)$ by (F4). $\square$

4.3. P1: the height identity.

Theorem 4.2 (P1, height identity). *Let π_x be a path of a flow F that traverses the squares $X_1, \dots, X_m$ ($m \geq 0$), in this order, and let $q := r_m(t_{S,m})$ be the first-contact point of step m (so $t_m^* = t_{S,m}$). Put $\varphi_i := \varphi_{X_i}$ for $1 \leq i \leq m$, $\varphi_0 := 0$, and $g_i := t_{S,i}$ for $0 \leq i \leq m$. Then g_i is the length of the gap segment of step i ,*

$$(4.3) \quad q = (x, 0) + \sum_{i=0}^m g_i n_{X_i} + \sum_{i=1}^m n_{X_i},$$

that is,

$$(4.4) \quad q_y = \sum_{i=1}^m \cos \varphi_i + \sum_{i=0}^m g_i \cos \varphi_i, \quad q_x = x - \sum_{i=1}^m \sin \varphi_i - \sum_{i=0}^m g_i \sin \varphi_i,$$

and the cumulative gap at q is $G_m + t_{S,m} = \sum_{i=0}^m g_i$.

Proof. For $i < m$, step i ends with the traversal of X_{i+1} . Hence $t_i^* = t_{S,i} = g_i$, the gap segment of step i is $[p_i, q_i]$ with $q_i = p_i + g_i n_{X_i}$, and

$$p_{i+1} = q_i + n_{X_{i+1}}, \quad G_{i+1} = G_i + g_i.$$

Moreover $q = p_m + g_m n_{X_m}$. Starting from $p_0 = (x, 0)$ and $G_0 = 0$ and adding these relations for $i = 0, \dots, m-1$ gives (4.3) and $G_m + t_{S,m} = \sum_{i=0}^m g_i$. Since $|n_{X_i}| = 1$, g_i is the length of the segment from p_i to $p_i + g_i n_{X_i}$. Inserting $n_{X_i} = (-\sin \varphi_i, \cos \varphi_i)$, which also holds for $i = 0$ because $n_{X_0} = (0, 1)$ and $\varphi_0 = 0$, gives (4.4). $\square$

Remark 4.3. The identity is a restatement of the definition of the trajectory: neither R1, nor R2, nor the height bound h_F is used. It describes every point of Γ_x , but in two different forms. Suppose that the path has traversed $X_1, \dots, X_m$ before the point $z \in \Gamma_x$ considered.

- (a) If $z = r_m(t)$ with $0 \leq t \leq t_m^*$ is a point of the gap segment of step m , then z satisfies (4.3) with g_m replaced by t .
- (b) If $z = q_m + \lambda n_{X_{m+1}}$ with $0 \leq \lambda \leq 1$ is a point of the traversal segment of X_{m+1} , then

$$z = (x, 0) + \sum_{i=0}^m g_i n_{X_i} + \sum_{i=1}^m n_{X_i} + \lambda n_{X_{m+1}}.$$

Thus (4.3) holds verbatim, with the partial gap length, at the points of gap segments; at the points of a traversal segment the additional term $\lambda n_{X_{m+1}}$ appears. Both formulas follow from the relations in the proof of Theorem 4.2.

Corollary 4.4 (P1, corollary). *In the situation of Theorem 4.2, let $0 < \alpha < \pi/2$ and assume $a(X_i) < \alpha$ for $1 \leq i \leq m$. Put $g := \sum_{i=0}^m g_i$. Then*

$$(4.5) \quad -(\sec \alpha - 1) q_y \leq -\sum_{i=1}^m (1 - \cos \varphi_i) \leq q_y - m \leq \sum_{i=0}^m g_i \cos \varphi_i \leq g \leq \delta.$$

If moreover $\alpha \leq 10^{-3}$, then $(\sec \alpha - 1) q_y \leq 0.50001 \alpha^2 q_y$. Consequently $q_y - m \in [-0.50001 \alpha^2 q_y, \delta]$, and $q_y - m \in [-0.50001 \alpha^2 q_y, \delta]$ if rule D is not applied at q (in particular, if q is the point of an entry candidate).

Proof. By (4.4),

$$q_y - m = -\sum_{i=1}^m (1 - \cos \varphi_i) + \sum_{i=0}^m g_i \cos \varphi_i.$$

Since $g_i \geq 0$ and $0 < \cos \varphi_i \leq 1$, the second sum lies in $[0, g]$, and the first sum is nonnegative. This gives the second, third and fourth inequalities in (4.5); the fifth is (F5) of Lemma 4.1, because g is the cumulative gap at q by Theorem 4.2. For the first inequality let $1 \leq i \leq m$. Since $|\varphi_i| = a(X_i) < \alpha < \pi/2$ and $\sec$ is even and increasing on $[0, \pi/2)$,

$$1 - \cos \varphi_i = \cos \varphi_i (\sec \varphi_i - 1) \leq \cos \varphi_i (\sec \alpha - 1).$$

Summing over i and using $\sum_{i=1}^m \cos \varphi_i \leq q_y$, which holds by (4.4) because the remaining terms are nonnegative, we get $\sum_{i=1}^m (1 - \cos \varphi_i) \leq (\sec \alpha - 1) q_y$. (For $m = 0$ the sums over $i \geq 1$ are empty and $q_y = g_0 \in [0, g]$.) If $\alpha \leq 10^{-3}$, then $\sec \alpha - 1 \leq 0.50001 \alpha^2$ by the inequality (2.1), and $q_y \geq 0$, so $(\sec \alpha - 1) q_y \leq 0.50001 \alpha^2 q_y$. Finally, by (F5) the cumulative gap g at q equals δ only if rule D is applied at q ; at the point of an entry candidate the contact rule is applied. $\square$

4.4. Three geometric lemmas.

Lemma 4.5 (bottom side). *Let Z be a square, $\varphi = \varphi_Z$ and $a = a(Z)$.*

- (a) $v_Z = c_{Z,y} - \frac{1}{2}(\cos a + \sin a)$, and the heights of the points of $\text{bot}(Z)$ fill exactly the interval $[v_Z, v_Z + \sin a]$. In particular the lowest point of Z lies on $\text{bot}(Z)$, and

$$q_y - \sin a \leq v_Z \leq q_y \quad \text{for every } q \in \text{bot}(Z) \text{ (endpoints included).}$$

- (b) If $a > 0$, the closure of $\mathcal{R}(Z)$ is $[v_Z, v_Z + \sin a] \cup [v_Z + \cos a, v_Z + \cos a + \sin a]$. If $a = 0$, then $\mathcal{R}(Z) = \emptyset$.

Proof. (a) The point $c_Z + \xi u_Z + \zeta n_Z$ has height $c_{Z,y} + \xi \sin \varphi + \zeta \cos \varphi$. Since $\cos \varphi > 0$, the minimum over $|\xi|, |\zeta| \leq \frac{1}{2}$ is attained at $\zeta = -\frac{1}{2}$ and $\xi = -\frac{1}{2} \operatorname{sgn} \varphi$ (any ξ if $\varphi = 0$), and equals $c_{Z,y} - \frac{1}{2} \cos \varphi - \frac{1}{2} |\sin \varphi| = c_{Z,y} - \frac{1}{2}(\cos a + \sin a)$. On $\operatorname{bot}(Z)$, where $\zeta = -\frac{1}{2}$, the height $c_{Z,y} - \frac{1}{2} \cos a + \xi \sin \varphi$ is an affine function of $\xi \in [-\frac{1}{2}, \frac{1}{2}]$ whose range is $[c_{Z,y} - \frac{1}{2} \cos a - \frac{1}{2} \sin a, c_{Z,y} - \frac{1}{2} \cos a + \frac{1}{2} \sin a] = [v_Z, v_Z + \sin a]$. The remaining assertions follow. Only $\cos \varphi > 0$ and $|\sin \varphi| = \sin a$ were used, so the convention $\varphi = \frac{\pi}{4}$ for squares at 45° plays no role.

(b) This follows from the formula for $\mathcal{R}(Z)$ in [R, Lemma 3.1(b)], with $v = v_Z$: for $a > 0$ both intervals have the positive length $\sin a$, so their closures are the closed intervals displayed. $\square$

Lemma 4.6 (the auxiliary inequality). *Let $f(a) := \sin a + 1 - \cos a$ for $a \in [0, \frac{\pi}{4}]$.*

(a) *f is strictly increasing on $[0, \frac{\pi}{4}]$, $f(0) = 0$ and $f(\frac{\pi}{4}) = 1$.*

(b) *There is a unique $a^* \in (0, \frac{\pi}{4}]$ with $f(a^*) = 1.0001 a^*$, and for $a \in [0, \frac{\pi}{4}]$*

$$f(a) \leq 1.0001 a \iff a \leq a^*.$$

(c) $2.000133 \cdot 10^{-4} < a^* < 2.000134 \cdot 10^{-4}$, that is, $a^* = 2.000133 \dots \cdot 10^{-4}$. In particular $f(a) \leq 1.0001 a$ for $0 \leq a \leq 2 \cdot 10^{-4}$; for these a this also follows directly from $f(a) \leq a + a^2/2$.

Proof. (a) $f'(a) = \cos a + \sin a > 0$ on $[0, \frac{\pi}{4}]$, and $f(\frac{\pi}{4}) = \frac{\sqrt{2}}{2} + 1 - \frac{\sqrt{2}}{2} = 1$.

(b) Let $h(a) := f(a) - 1.0001 a$. Then $h(0) = 0$, $h'(a) = (\cos a + \sin a) - 1.0001$, $h'(0) = -10^{-4} < 0$, and $h''(a) = \cos a - \sin a > 0$ for $0 \leq a < \frac{\pi}{4}$, so h is strictly convex on $[0, \frac{\pi}{4}]$. Also $h(\frac{\pi}{4}) = 1 - 1.0001 \cdot \frac{\pi}{4} > 1 - 1.0001 \cdot 0.7854 > 0.2$. Since $h'(0) < 0$, we have $h(a) < 0$ for all sufficiently small $a > 0$, so by the intermediate value theorem h has a zero $a^* \in (0, \frac{\pi}{4})$. If $0 < a < a^*$, strict convexity gives $h(a) < (1 - \frac{a}{a^*})h(0) + \frac{a}{a^*}h(a^*) = 0$. If $a^* < a \leq \frac{\pi}{4}$, put $\lambda = a^*/a \in (0, 1)$; then $0 = h(a^*) = h((1 - \lambda) \cdot 0 + \lambda a) < (1 - \lambda)h(0) + \lambda h(a) = \lambda h(a)$, so $h(a) > 0$. Hence $\{a \in [0, \frac{\pi}{4}] : h(a) \leq 0\} = [0, a^*]$, and a^* is the only zero of h in $(0, \frac{\pi}{4}]$.

(c) For $t > 0$, integrating the inequality $\cos t \leq 1$ repeatedly gives

$$t - \frac{t^3}{6} \leq \sin t \leq t - \frac{t^3}{6} + \frac{t^5}{120}, \quad \frac{t^2}{2} - \frac{t^4}{24} \leq 1 - \cos t \leq \frac{t^2}{2},$$

and therefore

$$\frac{t}{2} - \frac{t^2}{6} - \frac{t^3}{24} - 10^{-4} \leq \frac{h(t)}{t} \leq \frac{t}{2} - \frac{t^2}{6} + \frac{t^4}{120} - 10^{-4}.$$

For $t_1 = 2.000133 \cdot 10^{-4}$ the right-hand side is less than $-1.7 \cdot 10^{-11}$, and for $t_2 = 2.000134 \cdot 10^{-4}$ the left-hand side is greater than $3.2 \cdot 10^{-11}$ (direct evaluation; both bounds are rounded towards zero). Hence $h(t_1) < 0 < h(t_2)$, and by (b), $t_1 < a^* < t_2$. Finally, $\sin a \leq a$ and $1 - \cos a \leq a^2/2$ give $f(a) \leq a + a^2/2 \leq 1.0001 a$ for $0 \leq a \leq 2 \cdot 10^{-4}$. $\square$

Parts (b)–(d) of the next lemma, and the first sentence of part (a), are also contained in Lemma 3.8; the second sentence of part (a), which describes ρ on the bottom and the top side, is new. We include the proofs.

Lemma 4.7 (reflection). *Let $\rho(x, y) = (x, k - y)$.*

(a) *For every square $Z \subset [0, k]^2$, ρZ is a square in $[0, k]^2$ with $a(\rho Z) = a(Z) =: a$, $v_{\rho Z} = k - (v_Z + \cos a + \sin a)$ and $c_{\rho Z}(k - y) = c_Z(y)$ for all y . If $a < \frac{\pi}{4}$, then $\varphi_{\rho Z} = -\varphi_Z$ and $\rho(\operatorname{bot}(\rho Z)) = \operatorname{top}(Z)$, $\rho(\operatorname{relint} \operatorname{bot}(\rho Z)) = \operatorname{relint} \operatorname{top}(Z)$.*

(b) $\mathcal{R}(\rho Z) = k - \mathcal{R}(Z)$ and $\Phi(\rho Z) = k - \Phi(Z)$.

(c) $\operatorname{dist}(k - y, \mathbb{Z}) = \operatorname{dist}(y, \mathbb{Z})$ for all $y \in \mathbb{R}$; hence $\{k - y\} \in [w_0, 1 - w_0] \iff \{y\} \in [w_0, 1 - w_0]$.

(d) $\rho \mathcal{P}$ is a closed packing of $[0, k]^2$ with N squares; $\omega_{\rho \mathcal{P}}(k - y) = \omega_{\mathcal{P}}(y)$; a square ρS of $\rho \mathcal{P}$ meets ℓ_{k-y} if and only if S meets ℓ_y , and $a(\rho S) = a(S)$; and $\beta(k - y) = \beta(y)$. Consequently

$$k - H = H, \quad Y_{iii}[\rho \mathcal{P}] = k - Y_{iii}[\mathcal{P}], \quad Y_b[\rho \mathcal{P}] = k - Y_t[\mathcal{P}],$$

and, for every square $Z \in \mathcal{P}$ and every s ,

$$Z \in \tilde{\mathcal{Z}}(s)[\mathcal{P}] \iff \rho Z \in \mathcal{Z}(s)[\rho \mathcal{P}].$$

Here $[\mathcal{P}]$ and $[\rho\mathcal{P}]$ indicate the packing with respect to which a set is formed. The integrality of k is used only in (c) and, through (c), in (d).

Proof. (a) ρ is an isometry of $\mathbb{R}^2$ that maps $[0, k]^2$ onto itself and maps horizontal and vertical lines to horizontal and vertical lines. Hence ρZ is a square in $[0, k]^2$, and the least angle between a side of ρZ and a side of $[0, k]^2$ equals that of Z , so $a(\rho Z) = a(Z)$. By [R, Lemma 3.1(a)] the highest point of Z has height $v_Z + \cos a + \sin a$; ρ maps it to the lowest point of ρZ , so $v_{\rho Z} = k - (v_Z + \cos a + \sin a)$. Since $\rho Z \cap \ell_{k-y} = \rho(Z \cap \ell_y)$, the chords have equal lengths. For the last assertion write $c = c_Z$, $\varphi = \varphi_Z$, $u = u_Z$, $n = n_Z$; then

$$\rho(c + \xi u + \zeta n) = \rho(c) + \xi(\cos \varphi, -\sin \varphi) + \zeta(-\sin \varphi, -\cos \varphi) = \rho(c) + \xi u' - \zeta n',$$

with $u' = (\cos(-\varphi), \sin(-\varphi))$ and $n' = (-\sin(-\varphi), \cos(-\varphi))$. If $|\varphi| < \frac{\pi}{4}$, then $-\varphi \in (-\frac{\pi}{4}, \frac{\pi}{4})$ is the phase angle of ρZ and (u', n') its frame; the point of Z with frame coordinates (ξ, ζ) is mapped to the point of ρZ with frame coordinates $(\xi, -\zeta)$. Hence ρ maps $\text{top}(Z)$ ($\zeta = \frac{1}{2}$) onto $\text{bot}(\rho Z)$ ($\zeta = -\frac{1}{2}$), preserving relative interiors; as ρ is an involution, $\rho(\text{bot}(\rho Z)) = \text{top}(Z)$.

(b) If $a = 0$, both ramp sets are empty. Otherwise, by (a) and [R, Lemma 3.1(b)], with $v = v_Z$ and $v' = v_{\rho Z} = k - v - \cos a - \sin a$,

$$\begin{aligned} \mathcal{R}(\rho Z) &= [v', v' + \sin a) \cup (v' + \cos a, v' + \cos a + \sin a] \\ &= [k - v - \cos a - \sin a, k - v - \cos a) \cup (k - v - \sin a, k - v] \\ &= k - ((v + \cos a, v + \cos a + \sin a] \cup [v, v + \sin a)) = k - \mathcal{R}(Z), \end{aligned}$$

including the open and closed ends. The identity for Φ follows from $c_{\rho Z}(k - y) = c_Z(y)$.

(c) Since $k \in \mathbb{Z}$, $k - \mathbb{Z} = \mathbb{Z}$. If $y \in \mathbb{Z}$, both distances are 0. Otherwise $\{k - y\} = 1 - \{y\}$, and $\min(\{y\}, 1 - \{y\})$ is invariant under $\{y\} \mapsto 1 - \{y\}$. The second assertion follows from the first and (4.2).

(d) ρ maps $[0, k]^2$ bijectively onto itself and preserves disjointness, so $\rho\mathcal{P}$ is a closed packing with N squares. Since $([0, k] \times \{k - y\}) \setminus \bigcup \rho\mathcal{P} = \rho(([0, k] \times \{y\}) \setminus \bigcup \mathcal{P})$, the line wastes agree. Also $\rho S \cap \ell_{k-y} = \rho(S \cap \ell_y)$ and $a(\rho S) = a(S)$ by (a), and $d(k - y) = \min(k - y, y) = d(y)$, so $\beta(k - y) = \beta(y)$. Each of the three conditions defining H is invariant under $y \mapsto k - y$ (the last one by (c)), so $k - H = H$. Comparing the definitions, $k - y \in Y_{iii}[\rho\mathcal{P}]$ if and only if $y \in Y_{iii}[\mathcal{P}]$. Hence

$$Y_b[\rho\mathcal{P}] = Y_{iii}[\rho\mathcal{P}] \cap (0, k/2) = (k - Y_{iii}[\mathcal{P}]) \cap (0, k/2) = k - (Y_{iii}[\mathcal{P}] \cap (k/2, k)) = k - Y_t[\mathcal{P}].$$

Finally, by (b),

$$\Phi(\rho Z) \cap Y_b[\rho\mathcal{P}] \cap \mathcal{W}_s = (k - \Phi(Z)) \cap (k - Y_t[\mathcal{P}]) \cap \mathcal{W}_s = k - (\Phi(Z) \cap Y_t[\mathcal{P}] \cap (k - \mathcal{W}_s)),$$

so one of these sets is empty if and only if the other one is. $\square$

4.5. P2: quantization at bottom sides.

Lemma 4.8 (quantization, general form). *Let $0 < \alpha < \pi/2$. Let π_x be a path of a flow F that traverses the squares $X_1, \dots, X_m$ ($m \geq 0$), all with $a(X_i) < \alpha$, and suppose that the first-contact point q of step m lies on the closed side $\text{bot}(Z)$ of a square Z (whatever rule is applied at q). Let g be the cumulative gap at q , so that $g \leq \delta$ by (F5); put $a := a(Z)$ and $K := (\sec \alpha - 1)q_y$ (a local abbreviation, used only in this lemma and its proof). Then*

$$(4.6) \quad [v_Z, v_Z + \sin a] \subset [m - K - \sin a, m + g + \sin a],$$

$$(4.7) \quad [v_Z + \cos a, v_Z + \cos a + \sin a] \subset [m + \cos a - K - \sin a, m + \cos a + g + \sin a],$$

and every y in the closure of $\mathcal{R}(Z)$ satisfies

$$\text{dist}(y, \mathbb{Z}) < K + \delta + \sin a + (1 - \cos a).$$

Proof. By Theorem 4.2 and Corollary 4.4, $m - K \leq q_y \leq m + g$, and by Lemma 4.5(a), $q_y - \sin a \leq v_Z \leq q_y$. Hence

$$\begin{aligned} v_Z &\geq q_y - \sin a \geq m - K - \sin a, & v_Z + \sin a &\leq q_y + \sin a \leq m + g + \sin a, \\ v_Z + \cos a &\geq q_y + \cos a - \sin a \geq m + \cos a - K - \sin a, \\ v_Z + \cos a + \sin a &\leq q_y + \cos a + \sin a \leq m + \cos a + g + \sin a, \end{aligned}$$

which proves (4.6) and (4.7). If $a = 0$, then $\mathcal{R}(Z) = \emptyset$ and there is nothing more to prove. Let $a > 0$ and put $w := K + \delta + \sin a + (1 - \cos a)$. As $K \geq 0$, $\delta > 0$, $g \leq \delta$ and $1 - \cos a > 0$,

$$K + \sin a < w, \quad g + \sin a < w, \quad K + \sin a + (1 - \cos a) < w, \quad g + \sin a - (1 - \cos a) < w.$$

By Lemma 4.5(b) the closure of $\mathcal{R}(Z)$ is the union of the two intervals on the left of (4.6) and (4.7). For y in the first one, (4.6) gives $-(K + \sin a) \leq y - m \leq g + \sin a$, so $|y - m| < w$. For y in the second one, (4.7) and $m + \cos a = (m + 1) - (1 - \cos a)$ give

$$-(K + \sin a + (1 - \cos a)) \leq y - (m + 1) \leq g + \sin a - (1 - \cos a),$$

so $|y - (m + 1)| < w$. Since $m, m + 1 \in \mathbb{Z}$, $\text{dist}(y, \mathbb{Z}) < w$ in both cases. $\square$

Remark 4.9. The integer near which the ramps of Z lie is the number m of squares traversed before q (or $m + 1$ for the upper ramp). The quantization comes from the integer height 0 of the floor and from the fact that each traversal adds $\cos \varphi_i$, which is close to 1, to the height.

Lemma 4.10 (scale flows). *Let $s \in [y_0, y_1]$ and $x \in (0, k)$.*

- (a) Γ_x^s is an initial part of Γ_x^0 ; up to its termination point the F_s path coincides with π_x^0 (same steps, first-contact points and cumulative gaps). A point of Γ_x^0 that precedes a point of Γ_x^s lies on Γ_x^s .
- (b) Every square X traversed by the F_s path of x satisfies $a(X) < \alpha(s)$.
- (c) If $p_j \in \Gamma_x^s$ and the F_s path does not terminate at p_j , then $p_{j,y} < s + 2$. Every first-contact point of π_x^0 that lies on Γ_x^s has height $\leq s + 2$.
- (d) Let Z be a square and $z \in \Gamma_x^s \cap Z$. Then z is of type (i) or (ii) in Lemma 4.1 (F6) for π_x^0 , and in case (ii) the entry point $q_j \in \text{relint bot}(Z)$ lies on Γ_x^s . At every first-contact point on Γ_x^s the cumulative gap is $\leq \delta$, with equality only if rule D is applied there.
- (e) A truncation point of type (T_s) lies in $\text{relint bot}(Y)$ for the square Y of its entry candidate, and in no other square. A truncation point of type (H_s) lies in no square, or it is the exit point of a square traversed by the F_s path and lies in no other square.

Proof. (a) This is the definition of F_s (Subsection 4.1).

(b) The path π_x^0 traverses X , so it won R1 at the entry candidate (X, q) , where $q \in \text{relint bot}(X)$ is the entry point. If $a(X) \geq \alpha(s)$, then q is a point of type (T_s) , so the truncation point precedes or equals q , and Γ_x^s contains no point of the traversal segment of X other than q . This contradicts the assumption that X is traversed by the F_s path.

(c) For $j = 0$, $p_{0,y} = 0 < s + 2$. For $j \geq 1$, p_j is an exit point; if $p_{j,y} \geq s + 2$, it is a point of type (H_s) , and the F_s path would terminate at p_j or before. This proves the first assertion. Now let $q = r_j(t_{S,j})$, with $t_j^* = t_{S,j}$, lie on Γ_x^s . If $q = p_j$, then $t_{S,j} = 0$, so $j = 0$ by (F2) and $q_y = 0$. Otherwise p_j precedes q , so the F_s path does not terminate at p_j and $p_{j,y} < s + 2$; thus $t_{H,j}^{(s)} > 0$. If we had $t_{H,j}^{(s)} < \min(t_{S,j}, t_{W,j}, t_{D,j})$, the point $r_j(t_{H,j}^{(s)})$ of the gap segment of step j would be a point of type (H_s) preceding q , and q would not lie on Γ_x^s . Since $t_{S,j} = t_j^* \leq \min(t_{W,j}, t_{D,j})$, we get $t_{H,j}^{(s)} \geq t_{S,j}$, and hence

$$q_y = p_{j,y} + t_{S,j} \cos \varphi_{X_j} \leq p_{j,y} + t_{H,j}^{(s)} \cos \varphi_{X_j} = s + 2.$$

(d) By (a), $\Gamma_x^s \subset \Gamma_x^0$, so Lemma 4.1 (F6) applies to z and π_x^0 . In case (ii), q_j precedes or equals z , so $q_j \in \Gamma_x^s$ by (a). The statement on the cumulative gap is (F5) for π_x^0 .

(e) A point of type (T_s) is the point $q \in \text{relint bot}(Y)$ of an entry candidate (Y, q) ; it lies in no other square by disjointness. Let the truncation point be of type (H_s) . If it is an exit point p_{j+1} , then the traversal segment $[q_j, p_{j+1}]$ precedes it, so X_{j+1} is traversed by the F_s path, and p_{j+1} lies in X_{j+1} only (F4). Otherwise it is $r_j(t)$ with $t = t_{H,j}^{(s)} < \min(t_{S,j}, t_{W,j}, t_{D,j}) \leq t_{S,j}$. If $t > 0$, this point lies in no square by (F2). If $t = 0$ (so $j \geq 1$, because $p_{0,y} = 0 < s + 2$), then $p_{j,y} = s + 2$ and the point is the exit point p_j , which is covered by the previous case. ($t < 0$ is impossible, because $r_j(t)$ lies on the gap segment.) $\square$

For a square Z , $s \in [y_0, y_1]$ and a point q we put

$$(4.8) \quad w(s, Z; q) := 0.50001 \alpha(s)^2 q_y + \delta + \sin a(Z) + (1 - \cos a(Z)) = 0.50001 \alpha(s)^2 q_y + \delta + f(a(Z)),$$

with f as in Lemma 4.6. When q is clear from the context we also write $w(s, Z)$.

Theorem 4.11 (P2, quantization at bottom sides). *Assume (Q1). Let $s \in [y_0, y_1]$ and let Z be a square.*

- (a) *Let $q \in \text{bot}(Z)$ (the closed side, endpoints included) lie on the trajectory Γ_x^s of an F_s path. Then q is a first-contact point of π_x^0 , $q_y \leq s + 2$, and every y in the closure of $\mathcal{R}(Z)$ satisfies*

$$\text{dist}(y, \mathbb{Z}) < w(s, Z; q) = 0.50001 \alpha(s)^2 q_y + \delta + \sin a(Z) + (1 - \cos a(Z)).$$

Here $0.50001 \alpha(s)^2 q_y = 0.50001 c_0^2 (1 + \frac{q_y - s}{s}) \leq 0.50001 c_0^2 (1 + \frac{2}{s})$.

- (b) *If moreover $a(Z) \leq a^*$ (Lemma 4.6), then*

$$w(s, Z; q) \leq 0.50001 c_0^2 \left(1 + \frac{q_y - s}{s}\right) + \delta + 1.0001 a(Z).$$

For $a(Z) \in (a^, \frac{\pi}{4}]$ this inequality is false.*

- (c) (Ceiling.) *Let q lie on $\rho(\text{bot}(\rho Z))$ and on the trajectory of an $\tilde{F}_s$ path. Then $k - q_y \leq s + 2$, and every y in the closure of $\mathcal{R}(Z)$ satisfies*

$$\text{dist}(y, \mathbb{Z}) < 0.50001 \alpha(s)^2 (k - q_y) + \delta + \sin a(Z) + (1 - \cos a(Z)).$$

If $a(Z) < \frac{\pi}{4}$, then $\rho(\text{bot}(\rho Z)) = \text{top}(Z)$.

Part (a) applies in particular to every entry candidate (Z, q) reached by an F_s path that has not terminated before q , before R1 is applied at (Z, q) ; to E-contacts at the two endpoints of $\text{bot}(Z)$; and to D- and W-terminations at points of $\text{bot}(Z)$.

Proof. (a) By Lemma 4.10(d), q is of type (i) or (ii) for π_x^0 . In case (ii), q lies on the traversal segment $[q_j, q_j + n_Z]$ of Z , whose points $q_j + \lambda n_Z$ have frame coordinate $\zeta = -\frac{1}{2} + \lambda$ with respect to Z ; since $q \in \text{bot}(Z)$ means $\zeta = -\frac{1}{2}$, we get $\lambda = 0$ and $q = q_j$, which is of type (i). So q is the first-contact point of some step m of π_x^0 . The squares $X_1, \dots, X_m$ traversed by π_x^0 before q have their traversal segments before q on Γ_x^0 , hence on Γ_x^s by Lemma 4.10(a); so they are traversed by the F_s path and $a(X_i) < \alpha(s)$ by Lemma 4.10(b). Since α is decreasing and $s \geq y_0$, (Q1) gives $0 < \alpha(s) \leq \alpha(y_0) \leq 10^{-3} < \pi/2$. Lemma 4.8, applied with $\alpha := \alpha(s)$, gives $\text{dist}(y, \mathbb{Z}) < (\sec \alpha(s) - 1) q_y + \delta + f(a(Z))$ for y in the closure of $\mathcal{R}(Z)$, and $(\sec \alpha(s) - 1) q_y \leq 0.50001 \alpha(s)^2 q_y$ by Corollary 4.4. The bound $q_y \leq s + 2$ is Lemma 4.10(c). Finally $\alpha(s)^2 q_y = c_0^2 q_y / s = c_0^2 (1 + \frac{q_y - s}{s})$, and $q_y - s \leq 2$.

(b) By (4.8) and (a), $w(s, Z; q) - 0.50001 c_0^2 (1 + \frac{q_y - s}{s}) - \delta = f(a(Z))$. By Lemma 4.6(b), $f(a(Z)) \leq 1.0001 a(Z)$ holds if and only if $a(Z) \leq a^*$.

(c) By the definition of the ceiling flow, ρq lies on $\text{bot}(\rho Z)$ and on the trajectory of an $F_s[\rho \mathcal{P}]$ path, and ρZ is a square of the closed packing $\rho \mathcal{P}$ (Lemma 4.7(d)). Condition (Q1) does not involve the packing. By (a), applied to $\rho \mathcal{P}$, ρZ and ρq , whose height is $k - q_y$: $k - q_y \leq s + 2$, and $\text{dist}(y', \mathbb{Z}) < 0.50001 \alpha(s)^2 (k - q_y) + \delta + f(a(\rho Z))$ for every y' in the closure of $\mathcal{R}(\rho Z)$. By Lemma 4.7(a)–(c), $a(\rho Z) = a(Z)$, the closure of $\mathcal{R}(\rho Z)$ is k minus the closure of $\mathcal{R}(Z)$, and $\text{dist}(k - y, \mathbb{Z}) = \text{dist}(y, \mathbb{Z})$; taking $y' = k - y$ gives the assertion. The last sentence is Lemma 4.7(a).

The final assertion of the theorem holds because each of the listed points is a point of $\text{bot}(Z)$ on Γ_x^s : the point of an entry candidate reached by the F_s path is a first-contact point on Γ_x^s , and so are the E-, D- and W-termination points mentioned. $\square$

4.6. Squares of $\mathcal{Z}(s)$: the form used later.

Lemma 4.12 (window). *Assume (Q2) or (C2). Let $s \in [y_0, y_1]$ and $Z \in \mathcal{Z}(s)$, and let $y^* \in \Phi(Z) \cap Y_b \cap \mathcal{W}_s$. Put $b_s := \hat{\beta}(\max\{(1-\varepsilon)s, y_0\})$. Then:*

- (a) $0 < a(Z) < \beta(y^*) \leq b_s \leq \min\{b^*, \hat{\beta}(y_0)\}$, and $b_s \leq 10^{-4}$.
- (b) $Z \subset \mathbb{R} \times ((1-\varepsilon)s - 1.0001, s + 1.0001) \subset \mathbb{R} \times V_s$. In particular every point of Z has height $< s + 1.0001 < s + 2$.
- (c) $y^* \in \mathcal{R}(Z)$ and $\text{dist}(y^*, \mathbb{Z}) \geq w_0$.
- (d) $w(s, Z; q) < w_0 - 10^{-12}$ for every point $q \in Z$.

Proof. (a) Since $y^* \in \Phi(Z)$, the set $\Phi(Z)$ is nonempty, so $a(Z) > 0$ by the second part of [R, Lemma 3.4]. Since $c_Z(y^*) > 0$, the square Z meets ℓ_{y^*} , and $y^* \in Y_{iii}$ gives $a(Z) < \beta(y^*)$. As $y^* \in Y_b \subset H$ and $y^* < k/2$, we have $d(y^*) = y^* \geq y_0$; and $y^* \in \mathcal{W}_s$ gives $y^* \geq (1-\varepsilon)s > 0$. Since $\hat{\beta}$ is decreasing, $\beta(y^*) = \hat{\beta}(y^*) \leq \hat{\beta}(\max\{(1-\varepsilon)s, y_0\}) = b_s$. Moreover $b_s \leq \hat{\beta}(y_0)$, and $b_s \leq \hat{\beta}((1-\varepsilon)s) \leq \hat{\beta}((1-\varepsilon)y_0) = b^*$ because $s \geq y_0$. Under (Q2), $b_s \leq b^* \leq 10^{-4}$; under (C2), $b_s \leq \hat{\beta}(y_0) \leq 1.5 \cdot 10^{-6} < 10^{-4}$.

(b) Let $a = a(Z)$. By [R, Lemma 3.1(a)] the vertical extent of Z is $[v_Z, v_Z + \cos a + \sin a]$, and it contains y^* because $c_Z(y^*) > 0$. Since $\cos a \leq 1$ and $\sin a \leq a$, (a) gives $\cos a + \sin a \leq 1 + a < 1.0001$. Hence $v_Z \geq y^* - (\cos a + \sin a) > y^* - 1.0001 \geq (1-\varepsilon)s - 1.0001$ and $v_Z + \cos a + \sin a \leq y^* + (\cos a + \sin a) < y^* + 1.0001 \leq s + 1.0001$.

(c) As $a(Z) > 0$, the second part of [R, Lemma 3.4] gives $\Phi(Z) \subset \mathcal{R}(Z)$, so $y^* \in \mathcal{R}(Z)$. Since $y^* \in H$, $\{y^*\} \in [w_0, 1 - w_0]$, and (4.2) gives $\text{dist}(y^*, \mathbb{Z}) \geq w_0$.

(d) Let $q \in Z$. By (b), $0 \leq q_y < s + 1.0001$, so

$$0.50001 \alpha(s)^2 q_y = 0.50001 c_0^2 \frac{q_y}{s} \leq 0.50001 c_0^2 \left(1 + \frac{1.0001}{s}\right) \leq 0.50001 c_0^2 \left(1 + \frac{1.0001}{y_0}\right).$$

By (a), $a(Z) < b_s \leq 10^{-4} < a^*$, so Lemma 4.6 gives $f(a(Z)) \leq 1.0001 a(Z) < 1.0001 b_s \leq 1.0001 b^*$. Adding, $w(s, Z; q) < 0.50001 c_0^2 (1 + 1.0001/y_0) + \delta + 1.0001 b^* = w_0 - 10^{-12}$. $\square$

Theorem 4.13 (P2 for the squares of $\mathcal{Z}(s)$). *Assume (Q1), and (Q2) or (C2). Let $s \in [y_0, y_1]$ and $Z \in \mathcal{Z}(s)$.*

- (a) (Inclination and window.) $0 < a(Z) < \hat{\beta}(\max\{(1-\varepsilon)s, y_0\}) \leq \min\{b^*, \hat{\beta}(y_0)\}$, $a(Z) < 10^{-4}$, and $Z \subset \mathbb{R} \times V_s$.
- (b) (No contact with the bottom side.) *No point of the closed side $\text{bot}(Z)$ lies on the trajectory of any F_s path. In particular, no F_s path reaches an entry candidate (Z, q) ; hence no F_s path enters or traverses Z , the set of x whose F_s path enters Z is empty, and no M- or T-termination of F_s occurs at a point of Z . No F_s path terminates with D or W at a point of $\text{bot}(Z)$, and no F_s path makes an E-contact at one of the two endpoints of $\text{bot}(Z)$.*
- (c) (Trajectories meet Z only at their ends.) *Every F_s trajectory Γ_x^s meets Z in at most one point. Such a point is the termination point $e(x)$ of the F_s path; this termination is of type E, D or W, it is also the termination of π_x^0 , and $e(x) \in \partial Z \setminus \text{bot}(Z)$.*
- (d) (No through-paths.) $\mu_Z^{F_s}(y) = 0$ for every $y \in \mathbb{R}$. The same holds if $\text{int } Z$ is replaced by Z in the definition of $\mu_Z^{F_s}$, and also if the condition $\tau_{F_s}(x) > y$ is replaced by $\tau_{F_s}(x) \geq y$; in the last variant with Z in place of $\text{int } Z$, the set $\{x : \tau_{F_s}(x) \geq y, P_y(x) \in Z\}$ has at most $2(N+1)$ elements.
- (e) (Shadow.) $\text{Sh}^{F_s}(y) \geq \sum_{Z' \in \mathcal{Z}(s)} c_{Z'}(y)$ for every $y \in \mathbb{R}$; no height is excluded.

- (f) (Ceiling.) If $Z \in \tilde{\mathcal{Z}}(s)$, then $\rho Z \in \mathcal{Z}(s)[\rho\mathcal{P}]$ and (a)–(e) hold for the packing $\rho\mathcal{P}$, the square ρZ and the flow $F_s[\rho\mathcal{P}]$. Equivalently: $0 < a(Z) < 10^{-4}$ and $Z \subset \mathbb{R} \times (k - V_s)$; no point of $\text{top}(Z) = \rho(\text{bot}(\rho Z))$ lies on the trajectory of any $\tilde{F}_s$ path, and no $\tilde{F}_s$ path enters Z ; every $\tilde{F}_s$ trajectory meets Z in at most one point, which is its termination point, of type E, D or W, and lies in $\partial Z \setminus \text{top}(Z)$; $\mu_Z^{\tilde{F}_s} \equiv 0$ in all the readings of (d); and $\text{Sh}^{\tilde{F}_s}(y) \geq \sum_{Z' \in \tilde{\mathcal{Z}}(s)} c_{Z'}(y)$ for every y .

Proof. Fix $y^* \in \Phi(Z) \cap Y_b \cap \mathcal{W}_s$.

(a) This is Lemma 4.12(a),(b).

(b) Suppose that a point $z \in \text{bot}(Z)$ lies on Γ_x^s . By Theorem 4.11(a) (which uses (Q1)), z is a first-contact point of π_x^0 , and every y in the closure of $\mathcal{R}(Z)$ satisfies $\text{dist}(y, \mathbb{Z}) < w(s, Z; z)$. By Lemma 4.12(d), $w(s, Z; z) < w_0$. But $y^* \in \mathcal{R}(Z)$ and $\text{dist}(y^*, \mathbb{Z}) \geq w_0$ by Lemma 4.12(c), a contradiction. The remaining assertions of (b) are special cases. Indeed, the point q of an entry candidate (Z, q) reached by an F_s path is a point of $\text{bot}(Z)$ on its trajectory; entering Z , traversing Z , and M- or T-terminations at a point of Z all require such an entry candidate (M and T occur only at points of entry candidates, and a point of Z lies in no other square); and the termination points and E-contact points mentioned are points of $\text{bot}(Z)$ on the trajectory.

(c) Let $z \in \Gamma_x^s \cap Z$. By Lemma 4.10(d), z is of type (i) or (ii) for π_x^0 . In case (ii) the entry point $q_j \in \text{relint bot}(Z)$ would lie on Γ_x^s , which (b) excludes. So $z = r_j(t_{S,j})$ is the first-contact point of a step j of π_x^0 , Z is the square containing it, and $z \notin \text{bot}(Z)$ by (b). Consider the rule applied by π_x^0 at z : it is D if $t_{D,j} = t_j^*$, else W if $t_{W,j} = t_j^*$, else the contact rule; in the last case (Z, z) is not an entry candidate because $z \notin \text{relint bot}(Z)$, so the rule is E. In all three cases π_x^0 terminates at z , so z is the last point of Γ_x^0 ; as $z \in \Gamma_x^s$ and Γ_x^s is an initial part of Γ_x^0 , z is also the last point of Γ_x^s , that is $z = e(x)$, and $\Gamma_x^s \cap Z = \{z\}$. The F_s path is not truncated at z by (T_s) or (H_s) : by Lemma 4.10(e), a (T_s) point lies in $\text{relint bot}(Y)$ for the only square Y containing it, which would be Z , while $z \notin \text{bot}(Z)$; and an (H_s) point lies in no square or in a square traversed by the F_s path only, which would again be Z , while Z is not traversed by (b). Hence the termination of the F_s path at z is the E, D or W termination of π_x^0 . Finally $z \in \partial Z$ by (F3), and $z \notin \text{bot}(Z)$.

(d) For every x , by (c), $\Gamma_x^s \cap Z$ is empty or consists of the point $e(x) \in \partial Z$; in particular $\Gamma_x^s \cap \text{int } Z = \emptyset$. Hence no x satisfies $P_y(x) \in \text{int } Z$, whatever y and whichever aliveness condition is used; so $\mu_Z^{F_s}(y) = 0$, also in the variant with $\tau_{F_s}(x) \geq y$. Now replace $\text{int } Z$ by Z . If $\tau_{F_s}(x) > y$ and $P_y(x) \in Z$, then $P_y(x) = e(x)$ by (c), and the termination is of type E, D or W, so $\tau_{F_s}(x)$ is the height of $e(x)$, which is y ; a contradiction. So the set is empty in the variant with $\tau_{F_s}(x) > y$. In the variant with $\tau_{F_s}(x) \geq y$, every x in the set satisfies $e(x) = P_y(x) \in \partial Z \cap \ell_y$. Since $a(Z) > 0$ by (a), no side of Z is horizontal, and so ℓ_y meets ∂Z in at most two points: indeed $Z \cap \ell_y$ is empty or a segment $[A, B]$ (possibly a single point), and if a point strictly between A and B lay on a side σ of Z , then A and B , which lie in the closed half-plane bounded by the line of σ that contains Z , would both lie on that line, so ℓ_y would be the line of σ and σ would be horizontal. It remains to show that at most $N + 1$ paths of F_s terminate at a given point $z \in Z$. By the proof of (c), such a path terminates at the first-contact point z of its last step j , so $z = p_j + t n_{X_j}$ with $t = t_{S,j} \geq 0$, where X_j is the floor or the last square traversed. Let $x \neq x'$ be two such paths with the same $X := X_j$ (with their own p, t and p', t'). If X is the floor, then $p = (x, 0)$ and $p' = (x', 0)$, and $z = (x, t) = (x', t')$ gives $x = x'$. If X is a square, then $p, p' \in \text{top}(X)$ by (F2), so $p - p'$ is parallel to u_X , while $p - p' = (t' - t) n_X$ is parallel to n_X ; hence $p = p'$. Both paths then left X at p , so both traversed X after the entry candidate $(X, p - n_X)$, and both won R1 there (R1 is decided in F^0 , and the F_s paths are initial parts of the F^0 paths). This is impossible, because R1 has exactly one winner at each entry candidate (Definition 3.14, Lemma 3.15). So distinct paths terminating at z have distinct X , and there are at most $N + 1$ choices of X . Altogether the set has at most $2(N + 1)$ elements, and it has measure 0.

(e) By (d), $\mu_{Z'}^{F_s}(y) = 0$ for every $Z' \in \mathcal{Z}(s)$ and every y . As all terms of the sum defining $\text{Sh}^{F_s}(y)$ are nonnegative,

$$\text{Sh}^{F_s}(y) = \sum_{S \in \mathcal{P}} (c_S(y) - \mu_S^{F_s}(y))_+ \geq \sum_{Z' \in \mathcal{Z}(s)} (c_{Z'}(y) - 0)_+ = \sum_{Z' \in \mathcal{Z}(s)} c_{Z'}(y).$$

(f) By Lemma 4.7(d), $\rho Z \in \mathcal{Z}(s)[\rho\mathcal{P}]$, and $\rho\mathcal{P}$ is a closed packing of $[0, k]^2$ with N squares. The conditions (Q1), (Q2), (C2) do not involve the packing, so (a)–(e) hold for $\rho\mathcal{P}$, ρZ and $F_s[\rho\mathcal{P}]$. By definition, the trajectories, entries and termination points of $\tilde{F}_s$ are the images under ρ of those of $F_s[\rho\mathcal{P}]$. By Lemma 4.7(a), $a(Z) = a(\rho Z) < 10^{-4} < \frac{\pi}{4}$, so $\rho(\text{bot}(\rho Z)) = \text{top}(Z)$ and $\rho(\partial(\rho Z) \setminus \text{bot}(\rho Z)) = \partial Z \setminus \text{top}(Z)$; and $\rho Z \subset \mathbb{R} \times V_s$ gives $Z \subset \mathbb{R} \times (k - V_s)$. The measure statements follow from $\mu_Z^{\tilde{F}_s}(y) = \mu_{\rho Z}^{F_s[\rho\mathcal{P}]}(k - y)$, and similarly for the variants of (d). For the shadow, using $c_{\rho S}(k - y) = c_S(y)$ and $\mathcal{Z}(s)[\rho\mathcal{P}] = \{\rho Z' : Z' \in \tilde{\mathcal{Z}}(s)\}$ (Lemma 4.7(d)),

$$\text{Sh}^{\tilde{F}_s}(y) = \text{Sh}^{F_s[\rho\mathcal{P}]}(k - y) \geq \sum_{Z'' \in \mathcal{Z}(s)[\rho\mathcal{P}]} c_{Z''}(k - y) = \sum_{Z' \in \tilde{\mathcal{Z}}(s)} c_{Z'}(y). \quad \square$$

4.7. Remarks on the hypotheses and on degenerate cases.

Remark 4.14 (what is used). (1) Of the definition of $\mathcal{Z}(s)$, Lemma 4.12 and Theorem 4.13 use only the existence of $y^* \in \Phi(Z)$ with $(1 - \varepsilon)s \leq y^* \leq s$, $y_0 \leq y^* < k/2$, $\{y^*\} \in [w_0, 1 - w_0]$, and $a(S) < \beta(y^*)$ for every $S \in \mathcal{P}$ meeting ℓ_{y^*} . The conditions $\omega(y^*) < \omega_0$ and $d(y^*) \leq (1 - \varepsilon)y_1$ are not used. Under (Q2), the bound $y^* \geq y_0$ is used only for the inequality $\beta(y^*) \leq b_s$ in Lemma 4.12(a); the other conclusions hold without it, since $y^* \geq (1 - \varepsilon)s$ gives $a(Z) < \hat{\beta}((1 - \varepsilon)s) \leq b^* \leq 10^{-4}$.

- (2) No relation between the inclination of $Z \in \mathcal{Z}(s)$ and $\alpha(s)$ is needed. A square of $\mathcal{Z}(s)$ may or may not satisfy $a(Z) \geq \alpha(s)$, that is, it may or may not be a square at which an entering F_s path would be terminated with T; by Theorem 4.13(b), no F_s path reaches an entry candidate of Z at all. In particular, no inequality between $\hat{\beta}((1 - \varepsilon)s)$ and $\alpha(s)$ is assumed anywhere in this section.
- (3) Condition (Q1) is used only in Theorem 4.11(a),(c), through the inequality (2.1), and hence in Theorem 4.13(b)–(f). If (Q1) is dropped but $\alpha(s) < \pi/2$, Lemma 4.8 still gives the conclusion of Theorem 4.11(a) with the term $(\sec \alpha(s) - 1)q_y$ in place of $0.50001 \alpha(s)^2 q_y$.
- (4) Conditions (Q2) and (C2) are used only in Lemma 4.12, to obtain $a(Z) < 10^{-4}$. This gives the window (b) and, through Lemma 4.6, the bound $f(a(Z)) \leq 1.0001 a(Z)$ used in (d). Either of the two conditions suffices. Since (C2) is a constraint of Section 3 and (Q1) follows from (C1), Theorem 4.13 holds under the constraints (C1) and (C2) of that section, without (Q2).
- (5) The value of $\delta > 0$ is not used; it enters only through w_0 . The integrality of k is used only in the ceiling statements, through Lemma 4.7(c),(d).
- (6) The constant 1.0001 in V_s and in the term $0.50001 c_0^2(1 + 1.0001/y_0)$ of w_0 has a single source: the vertical extent $\cos a(Z) + \sin a(Z)$ of a square $Z \in \mathcal{Z}(s)$ is less than 1.0001.

Remark 4.15 (degenerate cases). (1) *Vertices and grazing contacts.* A contact at a vertex of a square, in particular the contact of a ray running along the line of a side, whose first point in the square is a vertex, is an E-contact and not an entry candidate (if instead rule D or W applies at that first-contact point, it is a D- or W-termination, see (2) and (3)); by Lemma 3.18(c), first-contact points at vertices occur only for finitely many x . Lemma 4.8 and Theorem 4.11(a) hold at all points of the closed side $\text{bot}(Z)$, including its endpoints; so for $Z \in \mathcal{Z}(s)$ the endpoints of $\text{bot}(Z)$ lie on no F_s trajectory (Theorem 4.13(b)), and E-contacts at the other two vertices of Z are termination points (Theorem 4.13(c),(d)).

- (2) *Death ties* ($t_{D,j} = t_{S,j} = t_j^*$). The cumulative gap at the first-contact point is then exactly δ (F5). Lemma 4.8 allows $g \leq \delta$ and its conclusion is strict, so death ties on $\text{bot}(Z)$ are excluded for $Z \in \mathcal{Z}(s)$; death ties at other points of ∂Z are termination points.
- (3) *Wall ties* ($t_{W,j} = t_{S,j} = t_j^*$, rule W). The point is a first-contact point and is treated exactly like a death tie.
- (4) *Contacts at height exactly $s+2$* . Contacts have priority over H, so such a contact is processed; Theorem 4.11(a) allows $q_y = s+2$. For $Z \in \mathcal{Z}(s)$ every point of Z has height $< s+1.0001$ (Lemma 4.12(b)).
- (5) *Ties in R1* (equal cumulative gaps). Theorem 4.11 concerns every path that reaches a point of $\text{bot}(Z)$, in particular every path that reaches an entry candidate alive, before R1 is applied; so the outcome of R1 plays no role.
- (6) *Axis-parallel squares* ($a(Z) = 0$). Then $\mathcal{R}(Z) = \emptyset$, so Theorem 4.11(a) is vacuous, and $\Phi(Z) = \emptyset$, so $Z \notin \mathcal{Z}(s)$. For such squares a set of paths of positive measure can terminate on $\text{bot}(Z)$: if the bottom side of an axis-parallel square Z has height δ and no other square meets the strip below it, every floor path starting below Z reaches $\text{bot}(Z)$ with cumulative gap δ and terminates there with D. This is why the count in Theorem 4.13(d) uses $a(Z) > 0$.
- (7) *Contacts reached directly from the floor* ($m = 0$). Then $q_y = g_0 \leq \delta$, and Lemma 4.8 applies with $m = 0$.
- (8) *Squares touching the floor*. If $(x, 0) \in Z$, the point $(x, 0)$ is the first-contact point of step 0, with $t_{S,0} = 0$ (Lemma 4.1 (F6)). For $Z \in \mathcal{Z}(s)$ this point would be the lowest point of Z , which lies on $\text{bot}(Z)$ (Lemma 4.5(a)), and it is excluded by Theorem 4.13(b). No lower bound on $(1 - \varepsilon)s - 1.0001$ is needed for this.
- (9) *Squares at 45°* . Lemma 4.5 does not depend on the convention $\varphi = \frac{\pi}{4}$. If $a(Z) = \frac{\pi}{4}$, then $f(a(Z)) = 1$, so $w(s, Z; q) > \frac{1}{2} \geq \text{dist}(y, \mathbb{Z})$ and Theorem 4.11(a) holds trivially. Squares of $\mathcal{Z}(s)$ have $a(Z) < 10^{-4}$.
- (10) *Exceptional heights*. Theorem 4.13(d),(e) hold at every height y ; no exceptional set is removed.
- (11) *Ceiling*. The ceiling statements follow by reflection (Lemma 4.7), using that k is an integer.

5. END COLLISIONS (P3)

This section bounds the measure of the set of floor points whose path ends with an end collision (event E). It also proves facts that are used again in Section 7: the basic properties of sources and delays (Lemma 5.2), the fact that a source carries at most one path through a given point (Lemma 5.3), the separation of the floor flow from the ceiling flow (Lemma 5.4), and the form of the multiplicity bound of [R] that is used in both sections (Lemma 5.5). Results of [R] are quoted in Section 2; a reference such as [R, Lemma 4.10] points to that quotation.

5.1. Setting.

Standing assumptions. Throughout this section and Section 7:

- $k \geq 4$ is an integer, $\mathcal{P}$ is a closed packing of $[0, k]^2$, and $W = k^2 - |\mathcal{P}|$. The *waste* is the set of points of $(0, k)^2$ that lie in no square of $\mathcal{P}$; its area is W .
- $0 < \delta \leq 10^{-5}$, $0 < \alpha_{\max} \leq 1.5 \cdot 10^{-6}$ and $h_{\max} = k/2 - 1$. (For the parameters of Section 3, $\alpha_{\max} = \alpha(y_0)$, and $\alpha_{\max} \leq 1.5 \cdot 10^{-6}$ is constraint (C1).)
- $F^0 = F(\alpha_{\max}, \delta, h_{\max})$ is the master flow (Definition 3.20) and $\tilde{F}^0$ the ceiling master flow (Definition 3.24). For $x \in (0, k)$ we write π_x for the path of F^0 that starts at $(x, 0)$ (denoted π_x^0 in Section 3).

Squares and sides. We use the frames of Definition 3.7: a square S has the phase $\varphi_S \in (-\pi/4, \pi/4]$, the frame $u_S = (\cos \varphi_S, \sin \varphi_S)$, $n_S = (-\sin \varphi_S, \cos \varphi_S)$ and the centre c_S , and a point z lies in S if and only if $|(z - c_S) \cdot u_S| \leq \frac{1}{2}$ and $|(z - c_S) \cdot n_S| \leq \frac{1}{2}$. The sides $\text{bot}(S)$, $\text{top}(S)$, the right side and the left side are given by $(z - c_S) \cdot n_S = -\frac{1}{2}$, $(z - c_S) \cdot n_S = \frac{1}{2}$, $(z - c_S) \cdot u_S = \frac{1}{2}$ and $(z - c_S) \cdot u_S = -\frac{1}{2}$,

with outer normals $-n_S, n_S, u_S, -u_S$; the right and the left side are the *lateral sides*. The relative interior relint σ of a side σ is σ without its end points; the four relative interiors and the four vertices partition ∂S . The inclination is $a(S) = |\varphi_S|$, and $\theta(S_1, S_2) \in [0, \pi/4]$ is the distance of φ_{S_1} and φ_{S_2} in $\mathbb{R}/(\pi/2)\mathbb{Z}$; if $|\varphi_{S_1} - \varphi_{S_2}| \leq \pi/2$, then $\theta(S_1, S_2) = \min(|\varphi_{S_1} - \varphi_{S_2}|, \pi/2 - |\varphi_{S_1} - \varphi_{S_2}|)$. The floor is the pseudo-square X_0 with $\varphi_{X_0} = 0$, $u_{X_0} = e_1$, $n_{X_0} = e_2$. We write $\sigma_{\text{bot}} := [0, k] \times \{0\}$ and $\sigma_{\text{top}} := [0, k] \times \{k\}$ for the bottom and the top side of the container, $S_0 := \partial[0, k]^2$, and $\theta(S, \sigma) := a(S)$ for a square S and a side σ of $[0, k]^2$, as in [R]. For $X, Y \in \mathcal{P} \cup \{X_0\}$ and $\Delta := \varphi_Y - \varphi_X$, by (3.1),

$$(5.1) \quad u_X \cdot u_Y = n_X \cdot n_Y = \cos \Delta, \quad u_Y \cdot n_X = \sin \Delta, \quad n_Y \cdot u_X = -\sin \Delta.$$

Paths. We recall the rules of Definitions 3.9, 3.14 and 3.10 in the form used below. A ray step of a path starts in a state (p, n_X, X, g) , where either $X = X_0$, $p = (x, 0)$ and $g = 0$, or X is the last square passed by the path (the *source*) and $p \in \text{relint top}(X)$; $g < \delta$ is the cumulative gap. With $r(t) = p + t n_X$ the step has the parameters $t_S, t_W, t_D = \delta - g, t_H = (h_{\max} - p_y)/\cos \varphi_X$ and the length $t^* = \min(t_S, t_W, t_D, t_H)$, and it ends with the first applicable of the rules D, W, contact and H. A contact at $q = r(t_S) \in Y$ is an *entry candidate* (Y, q) if $q \in \text{relint bot}(Y)$, and an end collision E otherwise. At an entry candidate, R1 selects one winner among the paths that reach it alive, and the other ones stop there with M. The winner stops with T if $a(Y) \geq \alpha_{\max}$; otherwise it passes Y along the pass segment $[q, q + n_Y]$ and, unless it is completed there (if $q + n_Y$ has height $\geq h_{\max}$), starts its next ray step at $q + n_Y \in \text{relint top}(Y)$.

Facts from Section 3. We use the following results of Section 3.

- (S1) (Lemmas 3.12 and 3.17(a)) Heights increase strictly along every trajectory, and the squares of the successive entry candidates of a path are distinct; in particular a path passes each square at most once.
- (S2) (Lemma 3.18) There is a finite set $\Sigma := \Sigma_{F^0} \subset [0, k]$, containing 0 and k , such that on every component I of $[0, k] \setminus \Sigma$ (a *piece*) the combinatorial type of π_x does not depend on $x \in I$ (the sources and the events of all steps, the squares and the sides met by the contacts, the outcomes of R1, the termination type), the start point $p_j(x)$, the start gap $g_j(x)$, the length $t_j^*(x)$ and the contact point $q_j(x)$ of the j -th ray step are affine functions of $x \in I$, and for no $x \in I$ is a first contact point of π_x (a contact, or the end point of a D or W termination with $t_D = t_S$ or $t_W = t_S$) a vertex of a square.
- (S3) (Lemma 3.19(a),(b)) On a piece, if X is the source of the j -th step, then $p_j'(x) = \lambda_j u_X$ with $\lambda_j \geq 1$; if the step ends with a contact in $\text{relint bot}(Y)$, then $q_j'(x) = \mu_j u_Y$ with $\mu_j = \lambda_j / \cos(\varphi_Y - \varphi_X) \geq \lambda_j \geq 1$.
- (S4) (Lemma 3.17(b),(c),(f)) Every ray step starts at a point p with $p_x \in (0, k)$ and cumulative gap $g < \delta$; if its source X is a square, then $p \in \text{relint top}(X)$ and $t_S > 0$. The open part $\{p + t n_X : 0 < t < t^*\}$ of a gap segment lies in the waste. At a processed contact, $g + t_S < \delta$.
- (S5) (Lemma 3.15(c)) At most one path passes a given entry point (Y, q) .

By (S2), every set of floor points that is defined below through the type of the path is, up to a finite set, a finite union of pieces; in particular it is Lebesgue measurable. We write $|\cdot|$ for Lebesgue measure (on $(0, k)$, on a line or in the plane, according to the context) and $\mathcal{H}^1$ for length.

The ceiling flow. Let $\rho(x, y) := (x, k - y)$. By Lemma 3.8, $\rho\mathcal{P}$ is a closed packing of $[0, k]^2$ with the same W ; $\tilde{F}^0$ is the master flow of $\rho\mathcal{P}$ transported by ρ , and the ceiling versions $\tilde{\mathcal{E}}, \tilde{\mathcal{D}}, \dots$ of all quantities are computed in this way (Definition 3.24). Every statement below about F^0 holds for an arbitrary closed packing; applied to $\rho\mathcal{P}$ and transported by ρ , it gives the corresponding statement for $\tilde{F}^0$. The reflection ρ is an isometry, maps the waste of $\rho\mathcal{P}$ onto the waste of $\mathcal{P}$ and S_0 onto itself, and preserves inclinations, the quotient distance θ , distances and areas.

5.2. Sources and delays.

Definition 5.1 (reaching a point). Let $x \in (0, k)$ and consider a ray step of π_x with start state (p, n_X, X, g) and length t^* . For $t \geq 0$ we say that π_x *reaches* the point $q := p + t n_X$ *from the source* X *with delay* t

- *in the gap case*, if $0 < t < t^*$, that is, if q lies in the open part of the gap segment of the step;
- *in the contact case*, if $t = t^* = t_S$ and the step ends with a contact, that is, $t_D > t^*$ and $t_W > t^*$.

A square X is a *source* of q if some path reaches q from the source X .

Lemma 5.2 (sources and delays). *Let π_x reach q from the source X with delay t , in a ray step with start state (p, n_X, X, g) .*

- $g + t < \delta$; in particular $0 \leq t < \delta$, and $t > 0$ if X is a square.
- If X is a square, then $a(X) < \alpha_{\max}$; in particular $|\varphi_X| < \alpha_{\max}$.
- $p \in X$ (for $X = X_0$: $p \in \sigma_{\text{bot}}$), $p_y \leq q_y \leq h_{\max}$, and $q_y < h_{\max}$ in the gap case.
- In the gap case, q is a point of the waste in $(0, k) \times (0, h_{\max})$.
- If X is a square, then $(q - c_X) \cdot n_X = \frac{1}{2} + t$ and $(q - c_X) \cdot u_X = (p - c_X) \cdot u_X \in (-\frac{1}{2}, \frac{1}{2})$. If $X = X_0$, then $q = (x, t)$.
- The path π_x reaches q in at most one of its steps and with at most one delay; and π_x has at most one step with source X .

Proof. (a) In the gap case $t < t^* \leq t_D = \delta - g$. In the contact case $g + t = g + t_S < \delta$ by (S4). As $g \geq 0$, $t < \delta$. If X is a square, then $t_S > 0$ by (S4), so $t = t_S > 0$ in the contact case, and $t > 0$ in the gap case by definition.

(b) A square is passed only by the winner of R1 at an entry candidate on it, and only if R2 does not stop that winner, that is, only if its inclination is $< \alpha_{\max}$.

(c) If X is a square, $p \in \text{relint top}(X) \subset X$; if $X = X_0$, $p = (x, 0) \in \sigma_{\text{bot}}$. The second coordinate of n_X is $\cos \varphi_X > 0$, so $p_y \leq q_y = p_y + t \cos \varphi_X$. Since $t \leq t^* \leq t_H$, we get $q_y \leq p_y + t_H \cos \varphi_X = h_{\max}$, and $q_y < h_{\max}$ if $t < t^*$.

(d) By (S4) and (c).

(e) For a square X , $p \in \text{relint top}(X)$ means $(p - c_X) \cdot n_X = \frac{1}{2}$ and $(p - c_X) \cdot u_X \in (-\frac{1}{2}, \frac{1}{2})$, and $q - p = t n_X$. For $X = X_0$, $p = (x, 0)$ and $n_{X_0} = e_2$.

(f) By (S1) two different points of the trajectory have different heights, and within one step $t \mapsto p + t n_X$ is injective; so q is reached at most once. The source of a step is the floor (for the first step only) or the square passed just before the step, and each square is passed at most once by (S1); so at most one step has source X . $\square$

By Definition 5.1, the gap multiplicity $M_g(q)$ of Definition 3.28, the number of x such that q lies in the open part of a gap segment of the trajectory of π_x , is the number of $x \in (0, k)$ such that π_x reaches q in the gap case.

Lemma 5.3 (one source, one path). (a) *If two paths π_x and $\pi_{x'}$ start ray steps at the same point $p \in \text{relint top}(X)$ of a square X after passing X , then $x = x'$.*

- For every point q and every $X \in \mathcal{P} \cup \{X_0\}$ there is at most one $x \in (0, k)$ such that π_x reaches q from the source X .

Proof. (a) A pass of X moves a path from its entry point $z \in \text{relint bot}(X)$ to $z + n_X$. So both paths passed X from the entry point $(X, p - n_X)$, and $x = x'$ by (S5).

(b) Let π_x and $\pi_{x'}$ reach q from the source X with delays t, t' in steps with start points p, p' . If $X = X_0$, then $p = (x, 0)$, $p' = (x', 0)$ and $q = p + t e_2 = p' + t' e_2$, so $x = x'$. If X is a square, then $p, p' \in \text{top}(X)$, so $p - p'$ is a multiple of u_X ; and $p - p' = (t' - t) n_X$ is a multiple of n_X . Hence $p = p'$, and (a) gives $x = x'$. $\square$

Lemma 5.4 (floor squares and ceiling squares). *Call a square a floor square if it is the source of a point reached by a path of F^0 , or if it contains the contact point of a ray step of F^0 that ends with a contact. Define ceiling squares in the same way with $\tilde{F}^0$. Then every floor square meets the half-plane $\{y \leq h_{\max}\}$, every ceiling square meets $\{y \geq k - h_{\max}\}$, and no square is both a floor square and a ceiling square.*

Proof. If X is the source of a point q reached by π_x in a step with start point p , then $p \in X$ and $p_y \leq h_{\max}$ by Lemma 5.2(c). A contact point q of a step ending with a contact is reached in the contact case, so $q_y \leq h_{\max}$ by the same lemma. Applying this to $\rho\mathcal{P}$ and reflecting, every ceiling square meets $\{y \geq k - h_{\max}\}$. By [R, Lemma 3.1(a)] a square of inclination a has vertical extent $\cos a + \sin a \leq \sqrt{2}$, which is less than $2 = (k - h_{\max}) - h_{\max}$. So no square meets both half-planes. $\square$

5.3. Pair waste and multiplicity. For a square S_1 and for S_2 a square or S_0 , $R(S_1, S_2)$ is the rectangle of Roth and Vaughan defined in Section 2 (before [R, Proposition 4.3]). As in [R, Lemma 4.8(b)], for a pair e of two squares S_1, S_2 we put $R_e := R(S_1, S_2)$, and for a pair $e = \{S, \sigma\}$ of a square S and a side σ of $[0, k]^2$ we put $R_e := R(S, S_0)$; f_e is the area of the waste in R_e . We use [R, Lemma 4.8(a)]: if $0 < d \leq 10^{-4}$ and S_1 is a square and S_2 a square or S_0 , at distance $< d$, then $f \geq \theta(2 - \theta)/40$ for the waste f in $R(S_1, S_2)$, $\theta = \theta(S_1, S_2)$; and the multiplicity bound in the following form.

Lemma 5.5 (multiplicity at waste points, for families of pairs). *Let $0 < d \leq 10^{-4}$, and let $\mathcal{F}$ be a set of unordered pairs, each of which consists either of two squares of $\mathcal{P}$ at distance $< d$, or of a square of $\mathcal{P}$ and one of the two sides $\sigma_{\text{bot}}, \sigma_{\text{top}}$ at distance $< d$. Then every point of the waste lies in R_e for at most K pairs $e \in \mathcal{F}$, where $K = 9$ by [R, Lemma 4.10 and Remark 4.11], and $K = 13$ by [R, Lemma 4.9] and the paragraph following its proof. (In this section and in Section 7, K denotes this multiplicity bound, $K = K_w^* = 9$ or $K = K_w = 13$; it is not the constant $K = 27$ of [R, Lemma 4.8(b)].) Consequently*

$$(5.2) \quad \sum_{e \in \mathcal{F}} f_e \leq K W.$$

Proof. This is Proposition 2.1 with $\sigma = \sigma_{\text{bot}}$, $\sigma' = \sigma_{\text{top}}$ and $\mathfrak{F} = \mathcal{F}$. The formal statements of [R, Lemmas 4.9 and 4.10] concern the edges of a visibility graph G_h or G_v , whereas the pairs of $\mathcal{F}$ need not be such edges. The form used here is the one that [R] states for the proofs of these lemmas: after the proof of [R, Lemma 4.9], that the bound 13 “holds for any family of such pairs at distance $< d$ ”, the only sides involved being two opposite ones; and in [R, Remark 4.11], that the proof of [R, Lemma 4.10] bounds “the number of *all* pairs of squares (or of a square and one of two opposite sides) at distance less than d whose rectangles contain p ” by 9, p being a point of the waste. In $\mathcal{F}$ the only sides are σ_{bot} and σ_{top} , which are opposite. Since $f_e = \int_{\text{waste}} \mathbf{1}_{R_e}$, summation gives $\sum_{e \in \mathcal{F}} f_e = \int_{\text{waste}} \#\{e \in \mathcal{F} : p \in R_e\} dp \leq K W$. $\square$

Remark 5.6 (rectangles of a square and a side near a corner). Let $e = \{S, \sigma_{\text{bot}}\}$ with S at distance $< d$ from σ_{bot} . Then S is at distance $< d$ from $S_0 \supset \sigma_{\text{bot}}$, so [R, Lemma 4.8(a)] applies to (S, S_0) and gives $f_e \geq a(S)(2 - a(S))/40$, because $\theta(S, S_0) = a(S)$. The rectangle $R_e = R(S, S_0)$ stands on a side of $[0, k]^2$ nearest to the centre of S . If S is close to a corner, this may be the left or the right side rather than σ_{bot} . Nothing changes in this case. The distance from S to a side of $[0, k]^2$ is the distance from its centre to that side minus $\frac{1}{2}(\cos a(S) + \sin a(S))$ (proof of [R, Proposition 4.5], part (b)); so S is at distance $< d$ from the side on which R_e stands as well. [R, Lemma 4.8(a)] makes no assumption on the side on which $R(S, S_0)$ stands, and the proofs of [R, Lemmas 4.9 and 4.10] treat the rectangle $R(P, S_0)$ of a pair of a square P and a side whatever side of $[0, k]^2$ it stands on (the proof of [R, Lemma 4.10] shows that all such rectangles containing a given waste point stand on one and the same side); see also the paragraph after Proposition 2.1. The same applies to pairs $\{S, \sigma_{\text{top}}\}$.

Remark 5.7 (dependence on [R, Lemma 4.10]). [R, Lemma 4.10] is computer-assisted in [R] (Remark 2.2). Every constant below, in this section and in Section 7, that is computed with $K = 9$ depends on it; the constants computed with $K = 13$ use only [R, Lemma 4.9]. Both values are given.

5.4. End collisions. By Lemma 3.11, a first contact point never lies in the relative interior of a top side, and an end collision occurs at a vertex or at a point of the relative interior of a lateral side. The next lemma determines the lateral side.

Lemma 5.8 (the lateral side that is hit). *Consider a ray step with direction $d = n_X$ that ends with a contact at a point $q = r(t_S)$ of the relative interior of a lateral side σ of a square Y , with outer normal o . Then $o \cdot d < 0$. Consequently $\varphi_Y \neq \varphi_X$, and σ is the left side of Y if $\varphi_Y > \varphi_X$ and the right side of Y if $\varphi_Y < \varphi_X$; we call this side $\sigma_{X,Y}$. An end collision of a step with source X at a square Y occurs at a vertex of Y or in relint $\sigma_{X,Y}$.*

Proof. Write $Y = \{z : |(z - c_Y) \cdot u_Y| \leq \frac{1}{2}, |(z - c_Y) \cdot n_Y| \leq \frac{1}{2}\}$. At $q \in \text{relint } \sigma$ the defining inequality $(z - c_Y) \cdot o \leq \frac{1}{2}$ is an equality and the other three are strict; they remain strict in a small disc $B(q, r_0)$, so that

$$Y \cap B(q, r_0) = \{z \in B(q, r_0) : (z - q) \cdot o \leq 0\}.$$

If $t_S = 0$, the source is the floor by (S4) and $q = (x, 0)$; the points $q - \varepsilon n_Y$, $0 < \varepsilon$ small, lie in $\sigma \subset Y$ (as σ has the direction n_Y and $q \in \text{relint } \sigma$) and have negative height, which contradicts $Y \subset [0, k]^2$. So $t_S > 0$. For $0 < \varepsilon < \min(t_S, r_0)$ the point $q - \varepsilon d = r(t_S - \varepsilon)$ lies in $B(q, r_0)$ and not in Y , because t_S is the least parameter at which the ray meets a square other than X (and $Y \neq X$). So $(q - \varepsilon d - q) \cdot o > 0$, that is, $o \cdot d < 0$.

The outer normals of the lateral sides are $\pm u_Y$, and $u_Y \cdot d = \sin(\varphi_Y - \varphi_X)$ by (5.1). If $\varphi_Y = \varphi_X$ both products vanish, which is impossible. Otherwise exactly one of $\pm u_Y$ has negative product with d : $-u_Y$ (left side) if $\varphi_Y > \varphi_X$, and u_Y (right side) if $\varphi_Y < \varphi_X$. The last statement follows with Lemma 3.11. $\square$

Let $\mathcal{E} \subset (0, k)$ be the set of x such that π_x terminates with E, and $L_E := |\mathcal{E}|$; $\tilde{\mathcal{E}}$ and $\tilde{L}_E$ are their ceiling versions (Definitions 3.26 and 3.24). Let $V \subset (0, k)$ be the set of x such that π_x has a contact at a vertex of a square; by (S2), $V \subset \Sigma$ is finite. For $X \in \mathcal{P} \cup \{X_0\}$ and a square Y with $\varphi_Y \neq \varphi_X$, let

$$E_{X,Y} := \{x \in (0, k) : \pi_x \text{ terminates with E at a point of relint } \sigma_{X,Y}, \text{ in a step with source } X\}.$$

By Lemma 5.8,

$$(5.3) \quad \mathcal{E} \subset V \cup \bigcup_{(X,Y)} E_{X,Y},$$

the union being over the finitely many ordered pairs (X, Y) with $E_{X,Y} \neq \emptyset$. For such a pair, X is the floor or a square with $|\varphi_X| < \alpha_{\max}$ (Lemma 5.2(b)).

Lemma 5.9 (one source and one target). *Let $X \in \mathcal{P} \cup \{X_0\}$ and $Y \in \mathcal{P}$ with $E_{X,Y} \neq \emptyset$, and put $\Delta := \varphi_Y - \varphi_X$. Then:*

- (a) *if X is a square, $\text{dist}(X, Y) < \delta$; if $X = X_0$, $\text{dist}(Y, \sigma_{\text{bot}}) < \delta$;*
- (b) *$|E_{X,Y}| \leq \delta \tan |\Delta|$.*

Proof. For $x \in E_{X,Y}$ let $p(x)$, $q(x)$ and $t(x)$ be the start point, the contact point and the delay of the step of π_x with source X ; the step is unique by Lemma 5.2(f), and π_x reaches $q(x)$ from X with delay $t(x)$ in the contact case.

(a) $q(x) \in Y$, $p(x) \in X$ (resp. $p(x) \in \sigma_{\text{bot}}$), and $|q(x) - p(x)| = t(x) < \delta$ by Lemma 5.2(a),(c).

(b) Put $\xi(x) := (p(x) - c_X) \cdot u_X$ if X is a square and $\xi(x) := x$ if $X = X_0$. Since $q(x) - p(x)$ is parallel to n_X , $\xi(x) = (q(x) - c_X) \cdot u_X$, respectively $\xi(x) = q(x)_x$.

ξ is injective on $E_{X,Y}$. For $X = X_0$ this is clear. If X is a square and $\xi(x) = \xi(x')$, then $p(x)$ and $p(x')$ are points of $\text{top}(X)$ with the same u_X -coordinate, so $p(x) = p(x')$, and Lemma 5.3(a) gives $x = x'$.

ξ expands. By (S2), $E_{X,Y} \setminus \Sigma$ is the union of the pieces I that it contains. On such an I the step with source X has a fixed index j , and by (S3) ξ is affine with $\xi' = \lambda_j \geq 1$ (for $X = X_0$, $\xi' = 1$). The intervals $\xi(I)$ are pairwise disjoint by injectivity, so

$$|E_{X,Y}| = \sum_I |I| \leq \sum_I |\xi(I)| = \mathcal{H}^1\left(\bigcup_I \xi(I)\right) \leq \mathcal{H}^1(\xi(E_{X,Y})).$$

Location of the contact points. By Lemma 5.2(a),(e), $q(x)$ lies on the line ℓ of $\sigma_{X,Y}$ and satisfies $\frac{1}{2} < (q(x) - c_X) \cdot n_X < \frac{1}{2} + \delta$ if X is a square, respectively $0 \leq q(x)_y < \delta$ if $X = X_0$. The line ℓ has the direction n_Y . Moving along ℓ at unit speed, $(z - c_X) \cdot n_X$ changes at the rate $n_Y \cdot n_X = \cos \Delta > 0$ and $(z - c_X) \cdot u_X$ at the rate $n_Y \cdot u_X = -\sin \Delta$, by (5.1) (for $X = X_0$: z_y and z_x , at the rates $\cos \varphi_Y$ and $-\sin \varphi_Y$; here $\Delta = \varphi_Y$). Here $\cos \Delta > 0$ because $|\varphi_X| < \alpha_{\max}$ and $|\varphi_Y| \leq \pi/4$. Hence the points of ℓ satisfying the above condition form a segment of length $\delta / \cos \Delta$, and the values of $(z - c_X) \cdot u_X$ (resp. z_x) on this segment form an interval of length

$$\frac{\delta}{\cos \Delta} |\sin \Delta| = \delta \tan |\Delta|.$$

As $\xi(E_{X,Y})$ is contained in this interval, $|E_{X,Y}| \leq \mathcal{H}^1(\xi(E_{X,Y})) \leq \delta \tan |\Delta|$. $\square$

The bound of Lemma 5.9(b) is written with $|\varphi_Y - \varphi_X|$ and not with $\theta(X, Y)$: the phase difference can exceed $\pi/4$ when Y is inclined by nearly $\pi/4$ (then $\theta(X, Y) = \pi/2 - |\Delta|$). This includes squares with $\varphi_Y = \pi/4$, whose bottom side is, by the 45° convention of Definition 3.7, the lower right side; such squares are never sources, since $\alpha_{\max} < \pi/4$.

5.5. The bound for end collisions.

Proposition 5.10 (P3). *Under the standing assumptions of Section 5.1, let K be the constant of Lemma 5.5 and $c_E := 41.933$. Then*

$$L_E + \tilde{L}_E \leq 2 c_E K \delta W.$$

With $K = 9$ (using [R, Lemma 4.10]) this gives $L_E + \tilde{L}_E \leq 754.8 \delta W$; with $K = 13$ (using only [R, Lemma 4.9]) it gives $L_E + \tilde{L}_E \leq 1090.3 \delta W$. Both numbers are rounded up.

Proof. Step 1: the ratio c_E . Let $X \in \mathcal{P} \cup \{X_0\}$ and $Y \in \mathcal{P}$ with $E_{X,Y} \neq \emptyset$, $\Delta = \varphi_Y - \varphi_X$ and $\theta := \theta(X, Y)$ (for $X = X_0$, $\theta = \theta(Y, S_0) = a(Y)$). We claim that $\theta > 0$ and

$$(5.4) \quad 40 \tan |\Delta| \leq c_E \theta(2 - \theta).$$

Indeed $\varphi_X \neq \varphi_Y$ (Lemma 5.8), $|\varphi_X| < \alpha_{\max}$ and $\varphi_Y \in (-\pi/4, \pi/4]$, so $0 < |\Delta| < \pi/4 + \alpha_{\max} < \pi/2$ and $\theta = \min(|\Delta|, \pi/2 - |\Delta|) > 0$.

- If $|\Delta| \leq \pi/4$, then $\theta = |\Delta|$. On $(0, \pi/4]$ the function $\tan \theta / (\theta(2 - \theta))$ is the product of the positive increasing functions $\tan \theta / \theta$ and $1/(2 - \theta)$, so

$$\frac{40 \tan |\Delta|}{\theta(2 - \theta)} \leq \frac{40}{(\pi/4)(2 - \pi/4)} = 41.93109 \dots$$

- If $\pi/4 < |\Delta| < \pi/4 + \alpha_{\max}$, then $\theta = \pi/2 - |\Delta| \in (\pi/4 - \alpha_{\max}, \pi/4)$. As $x(2 - x)$ is increasing on $[0, 1]$ and $\tan$ is increasing,

$$\frac{40 \tan |\Delta|}{\theta(2 - \theta)} < \frac{40 \tan(\pi/4 + \alpha_{\max})}{(\pi/4 - \alpha_{\max})(2 - \pi/4 + \alpha_{\max})} \leq 41.9313$$

for $\alpha_{\max} \leq 1.5 \cdot 10^{-6}$; the middle term increases with $\alpha_{\max}$ (its numerator increases, and the derivative of its denominator with respect to $\alpha_{\max}$ is $\pi/2 - 2 - 2\alpha_{\max} < 0$), and its value at $\alpha_{\max} = 1.5 \cdot 10^{-6}$ is 41.93124...

In both cases the ratio is at most $41.933 = c_E$.

Step 2: one pair. Let (X, Y) be as in Step 1, and let $e := \{X, Y\}$ if X is a square and $e := \{Y, \sigma_{\text{bot}}\}$ if $X = X_0$. By Lemma 5.9(a) the members of e are at distance $< \delta \leq 10^{-4}$. By [R, Lemma 4.8(a)] with $d = 10^{-4}$, applied to (X, Y) , respectively to (Y, S_0) (Remark 5.6), $f_e \geq \theta(2 - \theta)/40$. With Lemma 5.9(b) and (5.4),

$$(5.5) \quad |E_{X,Y}| \leq \delta \tan |\Delta| \leq c_E \delta \frac{\theta(2 - \theta)}{40} \leq c_E \delta f_e.$$

Step 3: the floor flow. Let $\mathcal{Q}_b^E$ be the set of the pairs e of Step 2, over all ordered pairs (X, Y) with $E_{X,Y} \neq \emptyset$. A pair of two squares arises from at most two ordered pairs, (X, Y) and (Y, X) , and a pair $\{Y, \sigma_{\text{bot}}\}$ from the single ordered pair (X_0, Y) . Since V is finite, (5.3) and (5.5) give

$$L_E \leq \sum_{(X,Y)} |E_{X,Y}| \leq 2 c_E \delta \sum_{e \in \mathcal{Q}_b^E} f_e.$$

Step 4: the ceiling flow. Apply Step 3 to $\rho\mathcal{P}$ and transport the pairs to $\mathcal{P}$ by ρ . The pair $\{\rho Y, \sigma_{\text{bot}}\}$ of $\rho\mathcal{P}$ becomes $\{Y, \sigma_{\text{top}}\}$. Since ρ is an isometry that maps S_0 onto itself, it maps $R(\rho S_1, \rho S_2)$ onto $R(S_1, S_2)$ (for a pair with a side, choosing as nearest point of S_0 the image of the chosen one), and it maps the waste of $\rho\mathcal{P}$ onto the waste of $\mathcal{P}$; so the values f_e and the distances are unchanged. Thus $\tilde{L}_E \leq 2 c_E \delta \sum_{e \in \mathcal{Q}_c^E} f_e$, where $\mathcal{Q}_c^E$ is a set of pairs of squares of $\mathcal{P}$ at distance $< \delta$ and of pairs $\{Y, \sigma_{\text{top}}\}$ with $\text{dist}(Y, \sigma_{\text{top}}) < \delta$.

Step 5: summation. The two sets $\mathcal{Q}_b^E$ and $\mathcal{Q}_c^E$ are disjoint. A pair of two squares in $\mathcal{Q}_b^E$ consists of floor squares (each is a source or contains a contact point of F^0), and a pair of two squares in $\mathcal{Q}_c^E$ consists of ceiling squares, so these pairs differ by Lemma 5.4. The pairs with a side differ by the side. Hence $\mathcal{F} := \mathcal{Q}_b^E \cup \mathcal{Q}_c^E$ is a set of pairs as in Lemma 5.5 (with $d = 10^{-4}$), and by (5.2)

$$L_E + \tilde{L}_E \leq 2 c_E \delta \sum_{e \in \mathcal{F}} f_e \leq 2 c_E K \delta W.$$

Step 6: numbers. For $K = 9$, $18 \cdot 41.933 = 754.794 \leq 754.8$. For $K = 13$, $26 \cdot 41.933 = 1090.258 \leq 1090.3$. $\square$

Remark 5.11 (what the proof uses). The proof of Proposition 5.10 uses R1 only through Lemma 5.3, the expansion $\lambda_j \geq 1$ of (S3), the classification of end collisions (Lemmas 3.11 and 5.8), [R, Lemma 4.8(a)] and Lemma 5.5. It does not use the hypothesis $\delta \leq \frac{1}{2} \cos 2\alpha_{\text{max}}$ (constraint (C6)) and none of the results of Section 7. The degenerate cases are covered as follows (see also Remark 3.35). A ray that runs along a lateral side of Y ($\varphi_Y = \varphi_X$) meets Y first at a vertex, so the corresponding x lie in the finite set V . A square resting on the floor gives contacts with delay 0 from the floor; these are entry candidates or vertex contacts (a contact with delay 0 is never in the relative interior of a lateral side, by the proof of Lemma 5.8). Contacts at height exactly h_{max} are processed (contact precedes H) and are included. A ray step never starts on a side wall of the container ((S4)). Ties between t_S, t_W, t_D, t_H (which may occur for sets of x of positive measure) are decided by the priority rule and do not affect Lemma 5.2.

6. THE SHADOW INEQUALITY (P4*)

This section works with one horizontal line ℓ_y and one flow F , which is either the master flow F^0 or a scale flow F_s . On ℓ_y we compare the chord of each square with the floor measure of the paths that cross the square there. The main result, Theorem 6.9, says that the total shadow $\text{Sh}^F(y)$ is at most the tilt loss below y plus the other losses $\Lambda^F(y)$, for every height y outside a finite set Exc that does not depend on s . Proposition 6.10 bounds the wall part of the losses of F_s . The remaining part Λ_0 is bounded in Section 7. Section 9 uses the result in the form of Corollary 6.11.

The arguments use the definitions and the structure lemmas of Section 3. They use no result of [R].

6.1. Setting and notation. Throughout this section $\mathcal{P}$ is a closed packing in $[0, k]^2$, where $k \geq 4$ is an integer, and the parameters are those of Section 3. The letter F denotes the master flow F^0 (Definition 3.20) or a scale flow F_s with $s \in [y_0, y_1]$ (Definition 3.21). We put

$$\alpha_{F^0} := \alpha_{\max}, \quad h_{F^0} := h_{\max} = \frac{k}{2} - 1, \quad \alpha_{F_s} := \alpha(s), \quad h_{F_s} := s + 2.$$

Since $y_0 \leq s \leq y_1 = \frac{k}{2} - 3$ and $\alpha_{\max} = \alpha(y_0)$, we have

$$(6.1) \quad 0 < \alpha(s) \leq \alpha_{\max} < \frac{\pi}{4}, \quad \delta > 0, \quad 2 < h_{F_s} \leq h_{F^0} = h_{\max} < k.$$

We recall the notions of Section 3 that are used below, in the form in which we use them.

- (N1) *Frames.* A square S has centre c_S and phase angle $\varphi_S \in (-\frac{\pi}{4}, \frac{\pi}{4}]$. Put $u_S = (\cos \varphi_S, \sin \varphi_S)$ and $n_S = (-\sin \varphi_S, \cos \varphi_S)$. Then S is the set of points $c_S + \xi u_S + \zeta n_S$ with $|\xi|, |\zeta| \leq \frac{1}{2}$. Its bottom side, top side and lateral sides are
- $$\text{bot}(S) = \{c_S - \frac{1}{2}n_S + \xi u_S\}, \quad \text{top}(S) = \{c_S + \frac{1}{2}n_S + \xi u_S\}, \quad \{c_S \pm \frac{1}{2}u_S + \zeta n_S\},$$
- with $|\xi| \leq \frac{1}{2}, |\zeta| \leq \frac{1}{2}$. The relative interiors of $\text{bot}(S)$ and $\text{top}(S)$ are given by $|\xi| < \frac{1}{2}$. The inclination is $a(S) = |\varphi_S| \in [0, \frac{\pi}{4}]$. The floor is treated as a pseudo-square X_0 with $\varphi_{X_0} = 0$, $u_{X_0} = e_1$, $n_{X_0} = e_2$.
- (N2) *Ray steps and events* (Definition 3.9). The path π_x of a floor point $x \in (0, k)$ in F^0 starts in the state $((x, 0), e_2, X_0, 0)$. In a state (p, n_X, X, g) it follows the ray $r(t) = p + tn_X$, $t \geq 0$. Here t_S is the least $t \geq 0$ with $r(t)$ in a square $Y \neq X$ ($+\infty$ if there is none), t_W the least $t \geq 0$ with $r(t)_x \in \{0, k\}$ ($+\infty$ if there is none), $t_D = \delta - g$, $t_H = (h_{\max} - p_y)/(n_X)_y$, and $t^* = \min(t_S, t_W, t_D, t_H)$. At t^* the first applicable rule in the order D ($t_D = t^*$), W ($t_W = t^*$), contact ($t_S = t^*$), H is applied. A contact at $q = r(t_S)$ belongs to exactly one square Y . It is an *entry candidate* (Y, q) if $q \in \text{relint bot}(Y)$; otherwise the path terminates there with label E.
- (N3) *Entry processing* (Definitions 3.14 and 3.10). At an entry candidate (Y, q) of F^0 , let $\mathcal{A}_Y(q)$ be the set of floor points whose paths reach (Y, q) alive. Rule R1 keeps the element of $\mathcal{A}_Y(q)$ for which $(g_x(q), x)$ is lexicographically smallest, the *R1 winner*. Here $g_x(q)$ is the cumulative gap at q . Every other element of $\mathcal{A}_Y(q)$ terminates at q with label M. Rule R2: if $a(Y) \geq \alpha_{\max}$, the winner terminates at q with label T (order “M before T”). Otherwise the winner *traverses* Y (it passes Y , in the terminology of Section 3) along the pass segment $[q, q + n_Y]$ and continues from the exit point $q + n_Y \in \text{relint top}(Y)$ in the state $(q + n_Y, n_Y, Y, g)$; it is completed (label H) if the exit point has height $\geq h_{\max}$. R1 is decided in F^0 only. Since pairs (g, x) with different x are different and the lexicographic order is total, *an entry candidate has at most one R1 winner*.
- (N4) *Scale flows* (Definition 3.21). For $s \in [y_0, y_1]$ the path of x in F_s is its path in F^0 , cut at the first of the following points:
- (T_s) an entry candidate at which the path is the R1 winner and $a(Y) \geq \alpha(s)$ (label T);
 - (H_s) a point of a ray step at height $s + 2$ with $t_H^{(s)} := (s + 2 - p_y)/(n_X)_y < \min(t_S, t_W, t_D)$, or an exit point of height $\geq s + 2$ (label H);
- if the F^0 path terminates before reaching such a point, the F_s path terminates there as well, with the same label. An F^0 termination with label T that is reached before any cut is an entry candidate with an R1 winner and $a(Y) \geq \alpha_{\max} \geq \alpha(s)$, so it is a T_s point. We write $\mathcal{T}_s$ for the set of x that terminate with label T in F_s , and $N(s) := |\mathcal{T}_s|$.
- (N5) *Trajectories* (described in detail in §6.2). $\Gamma_x = \Gamma_x^F$ is the trajectory of π_x in F , $e(x)$ its termination point, and $\tau(x) = \tau_F(x)$ the height of $e(x)$, except that $\tau(x) := h_F$ for label H. The path is *alive at* y if $\tau(x) > y$. Every x has exactly one termination type $\star \in \{T, D, E, M, W, H\}$, which we call its *label*.

- (N6) *Losses.* $L_\star(y) := |\{x : x \text{ has label } \star, \tau(x) < y\}|$ for $\star \in \{T, D, E, M, W\}$, and $L(y) := \sum_{\star \in \{T, D, E, M, W\}} L_\star(y)$. For F^0 , $\mathcal{D}, \mathcal{M}, \mathcal{E}$ are the sets of x with label D, M, E, and $L_E := |\mathcal{E}|$. Here $|\cdot|$ is Lebesgue measure.
- (N7) *Lines.* $\ell_y := \mathbb{R} \times \{y\}$, identified with $\mathbb{R}$ by $(q, y) \mapsto q$. The waste is $(0, k)^2 \setminus \bigcup_{Z \in \mathcal{P}} Z$, and $\text{Wa}_y := \ell_y \cap \text{waste}$. The line waste is $\omega(y) := |([0, k] \times \{y\}) \setminus \bigcup_{Z \in \mathcal{P}} Z|$, and $c_Z(y) := |Z \cap \ell_y|$ is the length of the closed chord.
- (N8) *Line quantities.* Let $y \in (0, h_F)$. If $\tau(x) \geq y$, then $P_y(x)$ denotes the unique point of Γ_x on ℓ_y (Theorem 6.9(a)) and $X_y(x)$ its abscissa. Put

$$\begin{aligned} A_Z(y) &:= \{x : \tau(x) > y, P_y(x) \in \text{int } Z\}, & \mu_Z(y) &:= |A_Z(y)|, \\ A_{\text{gap}}(y) &:= \{x : \tau(x) > y, P_y(x) \in \text{Wa}_y\}, & F_{\text{gap}}(y) &:= |A_{\text{gap}}(y)|. \end{aligned}$$

ν_y is the image of Lebesgue measure on $A_{\text{gap}}(y)$ under X_y , and ρ_y is its density (it exists by Theorem 6.9(e)). Then

$$\begin{aligned} \text{Sh}(y) &:= \sum_{Z \in \mathcal{P}} (c_Z(y) - \mu_Z(y))_+, & \text{Ov}(y) &:= \sum_{Z \in \mathcal{P}} (\mu_Z(y) - c_Z(y))_+, \\ \text{Ov}_{\text{gap}}(y) &:= \int_{\text{Wa}_y} (\rho_y(q) - 1)_+ dq, & \Lambda^F(y) &:= [L_D + L_E + L_M + L_W](y) + \text{Ov}_{\text{gap}}(y), \end{aligned}$$

$$\text{and } \Lambda_0 := |\mathcal{D}| + |\mathcal{M}| + L_E + \sup_{0 < y < h_{\max}} \text{Ov}_{\text{gap}}^{F^0}(y), \quad \Lambda^*(s) := \Lambda_0 + 2(s+2) \tan \alpha(s).$$

We also use two quantities that appear only in this section:

$$(6.2) \quad \text{Bd}(y) := k - L(y) - \sum_{Z \in \mathcal{P}} \mu_Z(y) - F_{\text{gap}}(y), \quad U(y) := \int_{\text{Wa}_y} (1 - \rho_y(q))_+ dq.$$

$U(y)$ is the part of the line waste that the gap crossers do not use. A superscript F (as in Sh^F , L_T^F , τ_F) indicates the flow when necessary. All sums over Z are finite. The sets in (N6)–(N8) are finite unions of intervals and points (Lemma 6.15), so all these measures are defined. By Theorem 6.9(e), $\text{Ov}_{\text{gap}}(y) \leq \int_{\text{Wa}_y} \rho_y = F_{\text{gap}}(y) \leq k$, hence $\Lambda_0 < \infty$.

6.2. Trajectories and event points. Section 3 also records most of the facts of this subsection and of Sections 6.3–6.8. We restate them in the form used here, with proofs, so that the proof of Theorem 6.9 can be read on its own; the statements agree with those of Section 3, under the same hypotheses and conventions. The correspondence is as follows. Lemma 6.2 restates Lemma 3.23(f). Definition 6.3 is Definition 3.16. Lemma 6.4(i),(iii) are contained in Lemmas 3.17(a), 3.23(c) and 3.18(b) (with Lemma 3.23(f) for F_s), and Lemma 6.4(ii) is Lemma 3.17(h), proved here also for F_s with $h_{F_s} = s + 2$. The first assertion of Lemma 6.5(0) and Lemma 6.5(i)–(iii) are contained in Lemma 3.17(b),(c),(e),(g) and the remark after Lemma 3.23. Definition 6.6 defines the sets Exc_F and Exc of Definition 3.32, and Lemma 6.7(i)–(iii) is Lemma 3.33. Lemma 6.14(i),(ii),(iv) are contained in Definition 3.28 and in Lemma 3.29(a),(b) and its proof. Lemma 6.15 extends Lemma 3.31(a). Theorem 6.9(e) contains Lemma 3.29(c); the bound $L_T^{F_s} \leq N(s)$ in Theorem 6.9(f) is Corollary 3.27; Theorem 6.9(g) proves in detail the statement of Remark 3.34; and Remark 6.22 is Lemma 3.31(d), extended to Ov_{gap} and U . We first collect the structural facts from Section 3 that are used.

Lemma 6.1 (structure). *The following facts are contained in Lemmas 3.12, 3.15, 3.18 and 3.19.*

- (i) *Along every path of F^0 , the centre heights of the squares at which the path has entry candidates increase strictly (Lemma 3.12). Hence a path has at most one entry candidate on each square, and it traverses each square at most once. Each set $\mathcal{A}_Y(q)$ is finite, and $R1$ is well defined (Section 3).*
- (ii) *There is a finite set $\Sigma_0 \subset [0, k]$ containing 0 and k with the following property on every component I of $[0, k] \setminus \Sigma_0$. The combinatorial type of the F^0 path of x is the same for all*

$x \in I$. The type consists of the number of ray steps, the source of each step (the floor or the last traversed square), the rule ending each step (D , W , contact or H), for a contact the square and the side or vertex contacted, for an entry candidate the outcomes of $R1$ and $R2$, and the label. Moreover the start $p_j(x)$, the cumulative gap $g_j(x)$ at the start, the length $t_j^*(x)$ and the end point $q_j(x) = p_j(x) + t_j^*(x)n_{X_j}$ of step j are affine functions of $x \in I$. Here X_j is the source of step j .

- (iii) On such an I , if step j has source X , then $p_j' = \lambda_j u_X$ with $|\lambda_j| \geq 1$ (for the floor $\lambda_0 = 1$). If step j ends at an entry candidate on $\text{bot}(Y)$, then $q_j' = \mu_j u_Y$ with $\mu_j = \lambda_j / \cos(\varphi_Y - \varphi_X)$, so $|\mu_j| \geq |\lambda_j| \geq 1$. If Y is then traversed, the next step has $\lambda_{j+1} = \mu_j$.

The components of $[0, k] \setminus \Sigma_0$ are the *pieces* of F^0 . For the scale flows we need the following refinement.

Lemma 6.2 (pieces of F_s). *Let $s \in [y_0, y_1]$. There is a finite set $\Sigma_s \supseteq \Sigma_0$ with the following property on every component I' of $[0, k] \setminus \Sigma_s$ (a piece of F_s). For all $x \in I'$, the F_s path of x is the F^0 path cut at the same step and by the same rule: T_s , H_s inside a ray step, H_s at an exit point, or the termination of the F^0 path. Consequently the combinatorial type of the F_s path is constant on I' . The start, the length and the end point of every ray step of the F_s path, all entry and exit points, and the termination point are affine functions of $x \in I'$.*

Proof. Fix a piece I of F^0 and use the notation of Lemma 6.1 on I . We list the points of the F^0 path at which F_s may cut, in the order of the path.

- H_s inside step j . The function $t_{H,j}^{(s)}(x) := (s + 2 - p_{j,y}(x)) / (n_{X_j})_y$ is affine on I . Suppose the rule ending step j in F^0 is D , W or contact. Then one of t_S, t_W, t_D equals t_j^* and none is smaller, so $\min(t_S, t_W, t_D) = t_j^*$. The condition for H_s therefore reads $t_{H,j}^{(s)}(x) < t_j^*(x)$, an affine strict inequality. If instead the rule ending step j is H , then $t_H < \min(t_S, t_W, t_D)$ and $t_{H,j}^{(s)} \leq t_H$ because $s + 2 \leq h_{\max}$. So the condition for H_s holds for all $x \in I$.
- T_s at the end of step j . The condition is that step j ends at an entry candidate on a square Y_j , the path is the $R1$ winner there, and $a(Y_j) \geq \alpha(s)$. By Lemma 6.1(ii) it holds for all $x \in I$ or for none.
- H_s at the exit point of Y_j . The exit point $q_j + n_{Y_j}$ has height $q_{j,y}(x) + \cos \varphi_{Y_j}$, so the condition is the affine inequality $q_{j,y}(x) + \cos \varphi_{Y_j} \geq s + 2$.

Let Σ_s consist of Σ_0 together with, for every piece I , the zeros in I of those of the finitely many affine functions $t_{H,j}^{(s)} - t_j^*$ and $q_{j,y} + \cos \varphi_{Y_j} - (s + 2)$ that do not vanish identically on I . On a component I' of $[0, k] \setminus \Sigma_s$ each of these functions vanishes identically or has constant sign, so every condition has the same truth value for all $x \in I'$. The F_s path is cut at the first point, in the order of the path, at which a condition holds. If no condition holds, it follows the F^0 path to its termination. Hence the step and the rule of the cut are the same for all $x \in I'$. The termination point of the F_s path is $q_j(x)$ for T_s , $p_j(x) + t_{H,j}^{(s)}(x)n_{X_j}$ for H_s inside step j , the exit point $q_j(x) + n_{Y_j}$ for H_s at an exit point, and the termination point of the F^0 path otherwise. All of these are affine on I' . The remaining data are those of F^0 . $\square$

We put $\Sigma_F := \Sigma_0$ if $F = F^0$ and $\Sigma_F := \Sigma_s$ if $F = F_s$. (The set Σ_s constructed here may differ by finitely many points from the set of Lemma 3.23(f); only its finiteness and the stated properties are used, and the exceptional sets below do not depend on this choice.)

Definition 6.3 (trajectory, event points). Let $x \in (0, k)$. The trajectory Γ_x of π_x in F is the polygonal line consisting, in order, of the following segments.

- For each ray step of the path in F , the *gap segment* $[p, p + t^*d]$. Here p is the start of the step, $d = n_X$ with X the source ($d = e_2$ for the floor), and $t^* \geq 0$ is the parameter at which the step ends in F . For a step of F_s cut by H_s , $t^* = t_H^{(s)}$.

- For each square Y traversed in F , the *pass segment* $[q, q + n_Y]$, where $q \in \text{relint bot}(Y)$ is the entry point.

The last point of Γ_x is $e(x)$. The *event points* of Γ_x are the end points of its segments, including $(x, 0)$ and $e(x)$. The *open part* of a segment $[a, b]$ is $[a, b] \setminus \{a, b\}$; it is empty if $a = b$.

The segments alternate: gap segment, pass segment, gap segment, and so on. The last segment is a gap segment, or a pass segment if the path is completed at an exit point. This is the trajectory of Section 3, described in the form used below.

Lemma 6.4. *Let $x \in (0, k)$.*

- (i) *Every square traversed by π_x in F has $a(Y) < \alpha_F$. Every segment of Γ_x has a direction d equal to e_2 or to n_Y for such a square, so $d_y > \cos \alpha_F > 0$. The height is strictly increasing along Γ_x .*
- (ii) *Γ_x has at most $2h_F / \cos \alpha_F + 3$ segments. It is a compact connected polygonal line from $(x, 0)$ to $e(x)$.*
- (iii) *On each piece of F , the number and the kinds of the segments of Γ_x are the same for all x , and every event point is an affine function of x .*

Proof. (i) In F^0 a square is traversed only by an R1 winner with $a(Y) < \alpha_{\max}$ (rule R2). If a square is traversed in F_s , then it is traversed in F^0 by the R1 winner, and the entry is not a T_s point; hence $a(Y) < \alpha(s)$. Since $|\varphi_Y| < \alpha_F$, we get $(n_Y)_y = \cos \varphi_Y > \cos \alpha_F$, and $(e_2)_y = 1 > \cos \alpha_F$. A segment of positive length therefore has strictly increasing height, and a segment of length zero is a single point.

(ii) Let $Y_1, \dots, Y_m$ be the squares traversed in F , with entry heights $\eta_1, \dots, \eta_m$. The exit height of Y_i is $\eta_i + \cos \varphi_{Y_i} > \eta_i + \cos \alpha_F$, and by (i) η_{i+1} is at least this exit height. Since $\eta_1 \geq 0$, this gives $\eta_m \geq (m-1) \cos \alpha_F$. On the other hand $\eta_m \leq h_F$. In F^0 a contact is processed only if $t_S = t^* \leq t_H$, so its height is at most $h_{\max}$. In F_s , suppose an entry point q_j of the F^0 path has height $> s + 2$. In the step ending at q_j we have $\min(t_S, t_W, t_D) = t_S$ and $t_H^{(s)} < t_S$, so the F_s path is cut by H_s inside this step or earlier. Thus $m \leq h_F / \cos \alpha_F + 1$, and there are at most $2m + 1$ segments.

(iii) This follows from Lemmas 6.1(ii) and 6.2. Note that an exit point is the corresponding entry point plus the constant vector n_Y . $\square$

Lemma 6.5 (position of trajectories). *Let $x \in (0, k)$, and let Γ_x be its trajectory in F .*

- (0) *The start p of every ray step satisfies $0 < p_x < k$. If the source X is a square, then $p_y \geq \cos \varphi_X \geq \cos \frac{\pi}{4} > 0$.*
- (i) *The open part of every gap segment lies in the waste.*
- (ii) *The open part of the pass segment of a square Y lies in $\text{int } Y$.*
- (iii) *No event point lies in the interior of a square. Every event point other than $(x, 0)$ and $e(x)$ is an entry point or an exit point of a traversed square Y , and it lies on ∂Y .*

Proof. (0) For the floor, $p = (x, 0)$ with $0 < x < k$. Otherwise $p \in \text{relint top}(X)$ and $X \subset [0, k]^2$. Suppose $p_x = 0$. The linear function $z \mapsto z_x$ is ≥ 0 on X and vanishes at p , so it attains its minimum over X at p . Since p is an interior point of the segment $\text{top}(X)$, the function is constant on $\text{top}(X)$. Then $(u_X)_x = \cos \varphi_X = 0$, which is impossible because $|\varphi_X| \leq \frac{\pi}{4}$. In the same way $p_x \neq k$. Moreover $p - n_X \in \text{bot}(X) \subset X \subset [0, k]^2$, so $p_y \geq (n_X)_y = \cos \varphi_X \geq \cos \frac{\pi}{4}$.

(i) Let $[p, p + t^*d]$ be a gap segment with $t^* > 0$ and source X , and let $0 < t < t^*$. Every ray step ends in F at a parameter $t^* \leq \min(t_S, t_W, t_D)$. In F^0 this holds because $t^* = \min(t_S, t_W, t_D, t_H)$. In F_s , a step cut by H_s has $t^* = t_H^{(s)} < \min(t_S, t_W, t_D)$, and any other step is a step of F^0 . Consequently:

- $r(t)$ lies in no square $Y \neq X$, because $t < t_S$;
- if X is a square, then $r(t) \notin X$, because $(r(t) - c_X) \cdot n_X = \frac{1}{2} + t > \frac{1}{2}$;

- $0 < r(t)_x < k$: by (0), $0 < p_x < k$; $r(t')_x \notin \{0, k\}$ for $0 \leq t' < t_W$; and $t < t_W$; so the claim follows by continuity;
- $0 < r(t)_y < k$: $r(t)_y > p_y \geq 0$ because $d_y > 0$. Also $r(t)_y$ is less than the end height of the step, which is at most $h_F < k$. In F^0 this is because $t^* \leq t_H$. In F_s the end height is $s + 2$ if the step is cut by H_s . Otherwise it is at most $s + 2$, because a step of F^0 whose end height exceeds $s + 2$ satisfies $t_H^{(s)} < t^* \leq \min(t_S, t_W, t_D)$ and is cut by H_s .

Hence $r(t) \in (0, k)^2 \setminus \bigcup_Z Z$, i.e. $r(t)$ lies in the waste.

(ii) Write $q = c_Y - \frac{1}{2}n_Y + \xi u_Y$ with $|\xi| < \frac{1}{2}$. For $0 < t < 1$ we have $q + tn_Y = c_Y + \xi u_Y + (t - \frac{1}{2})n_Y$ with $|\xi| < \frac{1}{2}$ and $|t - \frac{1}{2}| < \frac{1}{2}$, so $q + tn_Y \in \text{int } Y$.

(iii) First, two observations about a ray step with start p and source X . (1) For $0 < t < t_S$, the point $r(t)$ lies in no square: not in X by the computation in (i), and not in any other square because $t < t_S$. (2) The contact point $r(t_S)$, if it is reached, lies in exactly one square Y , and it lies on ∂Y . Indeed, if $t_S > 0$, it is a limit of the points $r(t)$, $t < t_S$, which are outside Y . If $t_S = 0$, the source is the floor and the point has height 0, so it is not interior to Y .

Since the segments alternate, every event point other than $(x, 0)$ and $e(x)$ is the common end point of a gap segment and the next pass segment (an entry point) or of a pass segment and the next gap segment (an exit point). Entry points lie in $\text{relint bot}(Y)$ and exit points in $\text{relint top}(Y)$, so both lie on ∂Y . The squares are disjoint, so such points lie in no other square. The point $(x, 0)$ has height 0, while $\text{int } Z \subset (0, k)^2$. It remains to consider $e(x)$, according to its label:

- T or M: $e(x)$ is an entry candidate point in $\text{relint bot}(Y) \subset \partial Y$.
- E: $e(x) = r(t_S)$ lies on ∂Y by (2).
- D: $e(x) = r(t_D)$ with $t_D \leq t_S$. If $t_D < t_S$, it lies in no square by (1) (note $t_D = \delta - g > 0$). If $t_D = t_S$, it is the contact point, on ∂Y by (2).
- W: $e(x)$ has abscissa 0 or k , so it is not in $(0, k)^2$.
- H inside a ray step: $e(x) = r(t)$ with $t < t_S$. If $t > 0$, it lies in no square by (1). If $t = 0$, it is the start of the step, i.e. $(x, 0)$ or an exit point.
- H at an exit point: $e(x)$ is an exit point. □

6.3. The exceptional set. By Lemma 6.4(iii), on a piece I of F the event points of Γ_x are given by finitely many affine functions $x \mapsto e(x)$ on I . We call them the *event-point functions* on I .

Definition 6.6 (exceptional set). For a flow F let Exc_F be the set of $y \in (0, h_F)$ such that, on some piece I of F , some event-point function has the constant height y on I . Put

$$\text{Exc} := \text{Exc}_{F^0}.$$

These are the sets Exc_F and Exc of Section 3. For $y \in (0, h_F)$ let Ξ_y^F be the set of $x \in (0, k) \setminus \Sigma_F$ such that Γ_x has an event point at height y .

The set Exc_F does not depend on the choice of Σ_F . Replace Σ_F by a finite superset: the new pieces are subintervals of the old ones, and the event-point functions are restricted. An affine function is constant on a non-empty open subinterval if and only if it is constant on the whole interval, so Exc_F does not change. Two admissible choices of Σ_F have their union as a common finite superset.

Lemma 6.7. (i) Exc_F is finite.

(ii) If $y \in (0, h_F) \setminus \text{Exc}_F$, then Ξ_y^F is finite.

(iii) For every $s \in [y_0, y_1]$, $\text{Exc}_{F_s} \subseteq \text{Exc} \cap (0, s + 2)$. Thus the single finite set Exc serves as exceptional set for F^0 and for all F_s .

(iv) Every element of Exc is either the height of the bottom side or of the top side of a square Z with $\varphi_Z = 0$, or the constant height of the end points of a ray step that ends with rule D, on some piece of F^0 .

Proof. (i) There are finitely many pieces and finitely many event-point functions on each. Each function with constant height contributes one value.

(ii) Let I be a piece and e an event-point function on I . Its height e_y is affine and, since $y \notin \text{Exc}_F$, not identically equal to y . Hence $e_y(x) = y$ for at most one $x \in I$. There are finitely many pairs (I, e) .

(iii) By definition $\text{Exc}_{F_s} \subseteq (0, s+2)$. Let I' be a piece of F_s ; it lies in a piece I of F^0 . By Lemma 6.2, every event point of the F_s trajectory on I' is of one of two kinds. Either it is an event point of the F^0 trajectory, i.e. the restriction to I' of an event-point function e of F^0 on I ; this covers the start, the entry and exit points, the end points of steps that are not cut, a T_s point (the end point of a step of F^0), an H_s point at an exit point, and an inherited termination point. Or it is the end point of a step cut by H_s inside the step, whose height is $s+2 \notin (0, s+2)$. So let $y \in \text{Exc}_{F_s}$ come from an F^0 event-point function e with $e_y \equiv y$ on I' . Since e_y is affine on I , $e_y \equiv y$ on I , and $y \in (0, s+2) \subseteq (0, h_{\max})$. Hence $y \in \text{Exc}$.

(iv) Let I be a piece of F^0 , and let e be an event-point function on I whose height is a constant $y \in (0, h_{\max})$. By Lemma 6.4(iii), the kind of the event point is the same for all $x \in I$. Let $p(x)$ be the start of the relevant ray step and X its source, so that $p' = \lambda u_X$ with $|\lambda| \geq 1$ (Lemma 6.1(iii)). We go through the kinds.

- *The start $(x, 0)$.* Its height is $0 \notin (0, h_{\max})$.
- *End point of a step ending with rule H .* Its height is $h_{\max} \notin (0, h_{\max})$.
- *End point of a step ending with rule D .* This is the second alternative of (iv).
- *End point of a step ending at a contact.* By Lemma 6.1(ii) the contacted square Y and the contacted side or vertex are the same for all $x \in I$. (This also follows from the affinity of e on I . The contact points lie on the boundaries of the disjoint compact squares, so by continuity a single square Y occurs. Moreover $e(I)$ is a non-degenerate segment, because $e' \cdot u_X = \lambda \neq 0$ as shown below, and a non-degenerate segment contained in ∂Y lies in one side of Y .) It is not a vertex. Indeed, the transversal coordinate $p(x) \cdot u_X$ has derivative $\lambda \neq 0$, so the ray lines $p(x) + \mathbb{R}n_X$, $x \in I$, are pairwise distinct parallel lines, and at most one of them contains a given vertex. So the end point $e(x) = p(x) + t(x)n_X$ lies on the line of a fixed side σ of Y for all $x \in I$. Let m be the outer unit normal of σ and v_σ a unit vector along σ . Differentiating $(e(x) - c_Y) \cdot m = \frac{1}{2}$ gives $e' \cdot m = 0$, so $e' = \kappa v_\sigma$ for some κ . Also $e' = \lambda u_X + t'n_X$, so $e' \cdot u_X = \lambda \neq 0$ and therefore $\kappa \neq 0$.
 - If σ is a lateral side, then $v_\sigma = \pm n_Y$ and the height has derivative $\pm \kappa \cos \varphi_Y \neq 0$, because $\cos \varphi_Y \geq \cos \frac{\pi}{4}$. This case does not occur.
 - If σ is the bottom side, then $v_\sigma = \pm u_Y$ and the height has derivative $\pm \kappa \sin \varphi_Y$. It vanishes only if $\varphi_Y = 0$, and then the height is the height $c_{Y,y} - \frac{1}{2}$ of the bottom side. This covers the entry points, including the termination points with labels T and M. If σ is the top side, the same argument shows that the height is constant only if $\varphi_Y = 0$, and then it is the height $c_{Y,y} + \frac{1}{2}$ of the top side.
- *Exit points.* The exit point is $q(x) + n_Y$, where q is the entry point; it has the same derivative as q . By the previous item, its height is constant only if $\varphi_Y = 0$, and then it equals the height of $\text{top}(Y)$.
- *End point of a step ending with rule W .* The end point has abscissa 0 or k ; being affine on I , it is the same wall $\{z_x = w\}$, $w \in \{0, k\}$, for all $x \in I$. If $\varphi_X = 0$, the ray is vertical with abscissa $p_x \in (0, k)$ (Lemma 6.5(0)) and never meets a wall. So $\sin \varphi_X \neq 0$. The end point is $p + t n_X$ with $p_x - t \sin \varphi_X = w$. Differentiating gives $\lambda \cos \varphi_X = t' \sin \varphi_X$. The height $p_y + t \cos \varphi_X$ therefore has derivative

$$\lambda \sin \varphi_X + \lambda \frac{\cos^2 \varphi_X}{\sin \varphi_X} = \frac{\lambda}{\sin \varphi_X} \neq 0,$$

and this case does not occur.

Every event point is of one of these kinds, which proves (iv). $\square$

Lemma 6.8 (vertices). *Let v be a vertex of a square Y and $x \in (0, k)$. If $v \in \Gamma_x$, then $v = e(x)$, and the label of x is one of the following:*

- *E: the last ray step reaches v at $t_S < \min(t_W, t_D)$;*
- *D: $t_D = t_S$ at v ;*
- *W: $t_W = t_S < t_D$ at v ; then v lies on a wall.*

The set Υ_F of $x \in (0, k)$ such that Γ_x contains a vertex of some square is finite.

Proof. A point of a pass segment of a square Z has the form $c_Z + \xi u_Z + \zeta n_Z$ with $|\xi| < \frac{1}{2}$. So it is not a vertex of Z , and it lies in no other square. In particular entry points and exit points are not vertices. The open part of a gap segment lies in the waste (Lemma 6.5(i)). Hence $v = (x, 0)$ or $v = e(x)$, and in both cases v is the end point of a gap segment. Indeed, if $v = (x, 0)$, then $t_S = 0$ for the first ray step because $v \in Y$, so the first gap segment is $\{v\}$. If $v = e(x)$, the last segment is a gap segment, since otherwise $e(x)$ would be an exit point.

Let $r(t) = p + tn_X$ be the ray step whose gap segment ends at v , and let t^* be its end parameter in F . If the source X is a square, then $Y \neq X$: the point $r(0) = p \in \text{relint top}(X)$ is not a vertex of X , and $r(t) \notin X$ for $t > 0$. By the definition of t_S , $r(t) \notin Y$ for $t < t_S$, and $v \in Y$; hence $t^* \geq t_S$. As $t^* \leq \min(t_S, t_W, t_D)$ (proof of Lemma 6.5(i)), we get $t^* = t_S$. Rule H requires $t^* < \min(t_S, t_W, t_D)$ and cannot apply. So the step ends with D (if $t_D = t_S$), with W (if $t_W = t_S < t_D$; then v has abscissa 0 or k), or at a contact at v . A contact at v is not an entry candidate because $v \notin \text{relint bot}(Y)$; it is an E-termination. In each case the path terminates at v , so $v = e(x)$. If $v = (x, 0)$, then $t_D = \delta > 0 = t_S$ and $t_W = +\infty$ for the vertical ray, so the label is E.

For finiteness, fix a piece I of F and a step j . For $x \in I$, the gap segment of step j lies on the line $p_j(x) + \mathbb{R}n_{X_j}$, whose transversal coordinate $p_j(x) \cdot u_{X_j}$ is affine with derivative $\lambda_j \neq 0$ (Lemma 6.1(iii); for F_s the same functions are used on the sub-pieces). A vertex v lies on this line if and only if $p_j(x) \cdot u_{X_j} = v \cdot u_{X_j}$, which holds for at most one $x \in I$. There are finitely many pieces, steps and vertices, and Σ_F is finite. $\square$

6.4. Statements.

Theorem 6.9 (P4*). *Let $k \geq 4$ be an integer, let $\mathcal{P}$ be a closed packing in $[0, k]^2$, and let the parameters be those of Section 3. Let $F \in \{F^0\} \cup \{F_s : s \in [y_0, y_1]\}$, and let Exc be the finite set of Definition 6.6; it is the same for all these flows.*

- (a) *Crossings. Let $y \in (0, h_F)$. For every x with $\tau(x) \geq y$, the trajectory Γ_x meets ℓ_y in exactly one point $P_y(x)$. If $y \notin \text{Exc}$, there is a finite set of floor points outside of which $\tau(x) \neq y$, and $\tau(x) > y$ implies that $P_y(x) \in \text{Wa}_y$ or $P_y(x) \in \text{int } Z \cap \ell_y$ for a square Z . The set $\text{int } Z \cap \ell_y$ is empty or the relative interior of the chord $Z \cap \ell_y$.*
- (b) *Bookkeeping. For every $y \in [0, k]$,*

$$k = \sum_{Z \in \mathcal{P}} c_Z(y) + \omega(y),$$

and $|\text{Wa}_y| = \omega(y)$ if $0 < y < k$. For every $y \in (0, h_F)$,

$$k = L(y) + \sum_{Z \in \mathcal{P}} \mu_Z(y) + F_{\text{gap}}(y) + \text{Bd}(y), \quad \text{Bd}(y) \geq 0,$$

and $\text{Bd}(y) = 0$ if $y \notin \text{Exc}$. Here $\text{Bd}(y)$ is the measure of the set of x that either terminate at height exactly y or are alive at y and cross ℓ_y at an entry point or an exit point (Remark 6.16).

- (c) *Shadow identity. For every $y \in (0, h_F)$,*

$$\text{Sh}(y) - \text{Ov}(y) = L(y) + F_{\text{gap}}(y) - \omega(y) + \text{Bd}(y).$$

In particular $\text{Sh}(y) = L(y) + F_{\text{gap}}(y) - \omega(y) + \text{Ov}(y)$ if $y \notin \text{Exc}$.

- (d) Square crossers. For every $y \in (0, h_F)$ and every square Z , $\mu_Z(y) \leq |\text{int } Z \cap \ell_y| \leq cz(y)$. Hence $\text{Ov} \equiv 0$ on $(0, h_F)$.
- (e) Gap crossers. For every $y \in (0, h_F)$ the measure ν_y is absolutely continuous, its density ρ_y vanishes almost everywhere outside Wa_y , and

$$F_{\text{gap}}(y) - \omega(y) = \text{Ov}_{\text{gap}}(y) - U(y) \leq \text{Ov}_{\text{gap}}(y).$$

- (f) Shadow inequality. For every $y \in (0, h_F) \setminus \text{Exc}$,

$$\text{Sh}^F(y) = L_T^F(y) + \Lambda^F(y) - U^F(y) \leq L_T^F(y) + \Lambda^F(y).$$

If $F = F_s$, then $L_T^{F_s}(y) \leq N(s)$ for every y (this is also Corollary 3.27), and for $0 < y < s+2$ the term $\Lambda^{F_s}(y)$ is bounded as in Proposition 6.10.

- (g) Order of M and T. The order “M before T” of rule R2 is not used in (a)–(f). Under the order “T before M” (Definition 6.19), the trajectories, termination points and termination heights of all paths of F^0 and of every F_s are unchanged; only the labels T and M change. Statements (a)–(f) hold verbatim with the labels of that order.

Proposition 6.10 (wall term). In the setting of Theorem 6.9, let $s \in [y_0, y_1]$ and $0 < y < s+2$. Then

$$L_D^{F_s}(y) \leq |\mathcal{D}|, \quad L_M^{F_s}(y) \leq |\mathcal{M}|, \quad L_E^{F_s}(y) \leq L_E, \quad L_W^{F_s}(y) \leq 2y \tan \alpha(s), \quad \text{Ov}_{\text{gap}}^{F_s}(y) \leq \text{Ov}_{\text{gap}}^{F^0}(y),$$

and consequently

$$\Lambda^{F_s}(y) \leq |\mathcal{D}| + |\mathcal{M}| + L_E + \text{Ov}_{\text{gap}}^{F^0}(y) + 2y \tan \alpha(s) \leq \Lambda_0 + 2(s+2) \tan \alpha(s) = \Lambda^*(s).$$

Corollary 6.11 (the form used in Section 9). In the setting of Theorem 6.9, for every $s \in [y_0, y_1]$ and every $y \in (0, s+2) \setminus \text{Exc}$,

$$\text{Sh}^{F_s}(y) \leq N(s) + \Lambda^*(s).$$

Remark 6.12 (what is used later). What Section 9 needs from this section is the content of Corollary 6.11 in the following weak form: for each s there is a finite set, here $\text{Exc} \cap (0, s+2)$, outside of which the inequality holds for $y \in (0, s+2)$. Two further facts established in this section are not needed there: that the exceptional set can be chosen independent of s , and that the functions of y involved are Borel (Remark 6.22). The size of Λ_0 is estimated in Section 7 (Corollary 7.2); this section does not bound it.

Remark 6.13 (hypotheses used). Besides the definitions of the flows and Lemma 6.1, the proofs in this section use only (6.1). They do not use the smallness of δ and $\alpha_{\max}$, nor any result of [R]. The uniqueness of the R1 winner (N3) is used in part (d) and nowhere else in (a)–(f).

From rule R2, the proofs of (a)–(e) and of the identity in (f) use only that every square traversed in F has $a(Y) < \alpha_F$ (Lemma 6.4(i)). These statements depend on the labels only through $L = \sum_{\star} L_{\star}$ and through $L_T + \Lambda = L + \text{Ov}_{\text{gap}}$. So the clause of R2 by which tilt terminations occur only at bottom-side entries (side and vertex contacts give E, whatever the inclination) is not used in them; it enters only through the definitions of $N(s)$ and of the losses in (f) and in Proposition 6.10.

Squares with $a(Y) = \pi/4$ need no separate treatment: the bottom side is fixed by the convention of Section 3, and the arguments use only $|\varphi_Y| \leq \frac{\pi}{4}$.

6.5. Crossings: proof of (a).

Lemma 6.14 (crossings). Let $y \in (0, h_F)$ and $x \in (0, k)$.

- (i) If $\tau(x) \geq y$, then $\Gamma_x \cap \ell_y$ consists of one point $P_y(x)$. This holds for every x , including $x \in \Sigma_F$.
- (ii) Suppose $\tau(x) > y$ and $P_y(x)$ is not an event point. Then exactly one of the following holds: $P_y(x)$ lies in the open part of a gap segment and $P_y(x) \in \text{Wa}_y$; or $P_y(x)$ lies in the open part of the pass segment of a square Y traversed in F , and $P_y(x) \in \text{int } Y$.

- (iii) If $\tau(x) = y$, then $P_y(x) = e(x)$; in particular $P_y(x)$ is an event point.
- (iv) If $\tau(x) > y$ and $P_y(x)$ is an event point, then $P_y(x)$ is an entry point or an exit point of a traversed square Y , and $P_y(x) \in \partial Y$. In particular $P_y(x) \notin \text{Wa}_y \cup \bigcup_Z \text{int } Z$.
- (v) If $y \notin \text{Exc}_F$ and $x \notin \Sigma_F \cup \Xi_y^F$, then $\tau(x) \neq y$; and if $\tau(x) > y$, then $P_y(x) \in \text{Wa}_y \cup \bigcup_Z \text{int } Z$.

Proof. (i) By Lemma 6.4(ii), Γ_x is a compact connected polygonal line from $(x, 0)$ to $e(x)$, and by Lemma 6.4(i) the height is strictly increasing along it. The height of $e(x)$ is $\tau(x) \geq y$ if the label is not H. If the label is H, the height of $e(x)$ is h_F (completion inside a ray step) or at least h_F (completion at an exit point), and $h_F > y$. Since $0 < y \leq$ the height of $e(x)$, the intermediate value theorem gives a point of Γ_x at height y . Strict monotonicity makes it unique. The argument does not use the pieces.

(ii) Since $P_y(x)$ is not an event point, it lies in the open part of a segment. By Lemma 6.5(i), the open part of a gap segment lies in the waste, so then $P_y(x) \in \ell_y \cap \text{waste} = \text{Wa}_y$. By Lemma 6.5(ii), the open part of the pass segment of Y lies in $\text{int } Y$. The waste and the interiors of the squares are disjoint, so exactly one case occurs.

(iii) The label is not H, since $\tau = h_F > y$ for label H. So $e(x)$ has height $\tau(x) = y$, and $P_y(x) = e(x)$ by uniqueness.

(iv) The point $P_y(x)$ has height y . So it is not $(x, 0)$, whose height is 0. It is not $e(x)$ either: the height of $e(x)$ is $\tau(x) > y$ if the label is not H, and at least $h_F > y$ if it is. By Lemma 6.5(iii), $P_y(x)$ is an entry point or an exit point of a traversed square Y , and it lies on ∂Y . A point of ∂Y is not in the waste, not in $\text{int } Y$, and not in any other square.

(v) If $\tau(x) = y$, or if $\tau(x) > y$ and $P_y(x)$ is an event point, then Γ_x has an event point at height y (by (iii)), so $x \in \Xi_y^F$. Otherwise (ii) applies. $\square$

Proof of Theorem 6.9(a). The first sentence is Lemma 6.14(i). Let $y \in (0, h_F) \setminus \text{Exc}$. Then $y \notin \text{Exc}_F$ by Lemma 6.7(iii), and the set $\Sigma_F \cup \Xi_y^F$ is finite by Lemma 6.7(ii). Now Lemma 6.14(v) gives the second sentence.

For the last sentence, let Z be a square. The chord $Z \cap \ell_y$ is a compact segment, and $\text{int } Z \cap \ell_y$ is a relatively open convex subset of it. Suppose $\text{int } Z \cap \ell_y$ contains a point z_0 , and let z be a point of the relative interior of the chord. If $z \neq z_0$, choose a point z_1 of the chord such that z lies strictly between z_0 and z_1 ; this is possible because z is a relative interior point. The open segment between the interior point z_0 and the point z_1 of the convex set Z lies in $\text{int } Z$. So $z \in \text{int } Z$. Hence $\text{int } Z \cap \ell_y$ is either empty or the relative interior of the chord. It is empty, for example, when ℓ_y contains a horizontal side of Z . $\square$

Lemma 6.15 (measurability for fixed y). *Let $y \in (0, h_F)$. Each of the sets*

$$\mathcal{L}(y) := \{x : \tau(x) < y\}, \quad \{x : \text{label } \star, \tau(x) < y\} \ (\star \in \{T, D, E, M, W\}), \quad A_Z(y), \quad A_{\text{gap}}(y),$$

$$\mathcal{B}(y) := \{x : \tau(x) \geq y\} \setminus \left(\bigcup_{Z \in \mathcal{P}} A_Z(y) \cup A_{\text{gap}}(y) \right)$$

is a finite union of intervals and points.

Proof. Let I be a piece of F . On I the label is constant, and τ is affine: it is the height of the termination point, an affine function by Lemma 6.4(iii), or the constant h_F for label H. Hence $\{x \in I : \text{label } \star, \tau(x) < y\}$ is empty or an interval, and so is $\mathcal{L}(y) \cap I$.

Let $x \in I$ with $\tau(x) > y$. By Lemma 6.14(ii) and (iv), $P_y(x) \in \text{int } Z$ holds if and only if $P_y(x)$ lies in the open part of the pass segment $[q_Z(x), q_Z(x) + n_Z]$ of Z . Since the height is strictly increasing along this segment, this means that the paths of I traverse Z (a property of the type, the same for all $x \in I$) and

$$q_{Z,y}(x) < y < q_{Z,y}(x) + \cos \varphi_Z.$$

Similarly, $P_y(x) \in \text{Wa}_y$ holds if and only if, for some gap segment of Γ_x with start $p_j(x)$, length $t_j^*(x)$ and direction d_j ,

$$p_{j,y}(x) < y < p_{j,y}(x) + t_j^*(x) d_{j,y}.$$

These are finitely many strict affine inequalities, together with $\tau(x) > y$. So $A_Z(y) \cap I$ and $A_{\text{gap}}(y) \cap I$ are finite unions of open intervals, and $\mathcal{B}(y) \cap I$ is a finite union of intervals and points. Adding the finitely many points of Σ_F proves the lemma. $\square$

6.6. The bookkeeping identities: proofs of (b) and (c).

Proof of Theorem 6.9(b). First identity. Fix $y \in (0, h_F)$. The sets $\mathcal{L}(y)$, $A_Z(y)$ ($Z \in \mathcal{P}$), $A_{\text{gap}}(y)$ and $\mathcal{B}(y)$ of Lemma 6.15 partition $(0, k)$. Indeed, every x has $\tau(x) < y$ or $\tau(x) \geq y$, and in the second case $P_y(x)$ is defined. The sets $A_Z(y)$ and $A_{\text{gap}}(y)$ are subsets of $\{\tau > y\}$. They are pairwise disjoint, because the interiors of distinct squares are disjoint and the waste meets no square. Finally $\mathcal{B}(y)$ is the rest of $\{\tau \geq y\}$. The set $\mathcal{L}(y)$ is the disjoint union of the sets $\{\text{label } \star, \tau < y\}$, $\star \in \{T, D, E, M, W\}$; label H does not occur in it, since then $\tau = h_F > y$. Hence $|\mathcal{L}(y)| = L(y)$, and

$$k = L(y) + \sum_Z \mu_Z(y) + F_{\text{gap}}(y) + |\mathcal{B}(y)|.$$

Comparing with (6.2) gives $\text{Bd}(y) = |\mathcal{B}(y)| \geq 0$. If $y \notin \text{Exc}$, then $y \notin \text{Exc}_F$ by Lemma 6.7(iii). By Lemma 6.14(v), $\mathcal{B}(y) \subseteq \Sigma_F \cup \Xi_y^F$, which is finite by Lemma 6.7(ii). Hence $\text{Bd}(y) = 0$.

Second identity. Let $y \in [0, k]$. Since $Z \subset [0, k]^2$, the closed chords $Z \cap \ell_y$ lie in $[0, k] \times \{y\}$. They are pairwise disjoint compact intervals, possibly empty or single points, because the squares are disjoint. By additivity of Lebesgue measure on $[0, k] \times \{y\}$,

$$k = \sum_Z c_Z(y) + \left| ([0, k] \times \{y\}) \setminus \bigcup_Z Z \right| = \sum_Z c_Z(y) + \omega(y).$$

If $0 < y < k$, then $\text{Wa}_y = ((0, k) \times \{y\}) \setminus \bigcup_Z Z$. It differs from $([0, k] \times \{y\}) \setminus \bigcup_Z Z$ at most in the points $(0, y)$ and (k, y) , so $|\text{Wa}_y| = \omega(y)$. $\square$

Proof of Theorem 6.9(c). For each Z , $(c_Z - \mu_Z)_+ - (\mu_Z - c_Z)_+ = c_Z - \mu_Z$. Summing over the finitely many squares and using (b),

$$\text{Sh}(y) - \text{Ov}(y) = \sum_Z c_Z(y) - \sum_Z \mu_Z(y) = (k - \omega(y)) - (k - L(y) - F_{\text{gap}}(y) - \text{Bd}(y)),$$

which is the identity. If $y \notin \text{Exc}$, then $\text{Bd}(y) = 0$ by (b). $\square$

Remark 6.16 (the term Bd and degenerate configurations).

- (1) *The set $\mathcal{B}(y)$.* By Lemma 6.14(iii) and (iv),

$$\mathcal{B}(y) = \{x : \tau(x) = y\} \cup \{x : \tau(x) > y, P_y(x) \text{ is an entry point or an exit point}\}.$$

A termination point at height $y \in (0, h_F)$ has one of the following labels. With T or M it lies on ∂Y . With E it lies on ∂Y . With D it lies in the waste if $t_D < \min(t_S, t_W)$, on ∂Y if $t_D = t_S$, and on a wall if $t_D = t_W$. With W it lies on a wall. Label H is impossible, since then $\tau = h_F > y$. So $\text{Bd}(y)$ counts the paths that terminate at height y (in the waste, on a wall, or on the boundary of a square), and the alive paths that cross ℓ_y at an entry or exit point. By (b), $\text{Bd}(y) = 0$ for $y \notin \text{Exc}$. At heights of Exc it can be positive (Proposition 6.17).

- (2) *Boundaries of squares.* A trajectory meets the boundary of a square only at event points (Lemma 6.5). An alive path crosses ℓ_y at a boundary point only at an entry or exit point (Lemma 6.14(iv)). The chord $c_Z(y)$ is the length of the closed chord. For example, at the height of the bottom side of a square with $\varphi_Z = 0$ we have $c_Z = 1$, while $\text{int } Z \cap \ell_y = \emptyset$ and hence $\mu_Z = 0$.

- (3) *Vertices.* A vertex of a square lies on a trajectory only as its termination point, with label E, D or W, and only for the finitely many $x \in \Upsilon_F$ (Lemma 6.8). The case W occurs when the ray reaches a wall exactly at a vertex of a square touching the wall ($t_W = t_S < t_D$); the priority of W over contact decides the label.
- (4) *Squares on the floor.* Let Y be a square with $\varphi_Y = 0$ whose bottom side lies on the floor, and let $x \in (0, k)$ with $(x, 0) \in \text{relint bot}(Y)$. The first ray step of x has $t_S = 0 < t_D = \delta$, $t_W = +\infty$ and $t_H = h_F > 0$, so $(Y, (x, 0))$ is an entry candidate with cumulative gap 0. Every point reached by a ray step with a square source has positive height (Lemma 6.5(0)). A ray step with the floor as source and start $(x', 0)$ runs along the vertical half-line above x' . Hence $\mathcal{A}_Y((x, 0)) = \{x\}$, and x is the R1 winner. Since $a(Y) = 0 < \alpha_F$, the path traverses Y (in F^0 and in every F_s). Its first gap segment is the single point $(x, 0)$. For $0 < y < 1$ (note $h_F \geq 1$) the path is alive at y , and $P_y(x) = (x, y) \in \text{int } Y$, so $x \in A_Y(y)$. If the end points of $\text{bot}(Y)$ have abscissae in $(0, k)$, these two floor points are vertex contacts, with label E and $\tau = 0$. A square with $a(Y) > 0$ meets the floor at most in its lowest vertex $(x_v, 0)$; if $x_v \in (0, k)$, then x_v has label E and $\tau = 0$. These finitely many floor points are counted in $L(y)$ for all $y > 0$ and have measure zero.
- (5) *Walls.* Floor points lie in $(0, k)$, so no path starts on a wall. The open parts of gap segments have abscissae in $(0, k)$ (Lemma 6.5(i)), so no trajectory runs along a wall. Points with label W lie on a wall and therefore not in the waste.

Proposition 6.17 (exceptional heights are needed). *Let $k \geq 4$ be an integer, let the parameters be as in Section 3, and let $0 < b < \min(\delta, 1)$. Let $\mathcal{P}$ consist of the three squares*

$$Z_i := \left[i + \frac{i+1}{1000}, i + 1 + \frac{i+1}{1000} \right] \times [b, b + 1], \quad i = 0, 1, 2.$$

Then $b \in \text{Exc}$, and for $F = F^0$ and for every $F = F_s$, $s \in [y_0, y_1]$,

$$\text{Sh}^F(b) = 3, \quad L_T^F(b) + \Lambda^F(b) = 0, \quad \text{Bd}^F(b) = 3.$$

So the inequality in Theorem 6.9(f) fails at $y = b$. On the other hand $\text{Sh}^F(y) = 0$ for $b < y < \min(b+1, h_F)$. The example concerns μ_Z as defined in (N8), with crossing points in $\text{int } Z$. If crossing points in the closed square Z were counted instead, one would get $\mu_{Z_i}(b) = 1$ and $\text{Sh}^F(b) = 0$.

Proof. Consecutive squares are at distance $\frac{1}{1000}$, and all three lie in $[0, 4]^2 \subseteq [0, k]^2$, so $\mathcal{P}$ is a closed packing. Note $h_F \geq 1 > b$, since $h_{F^0} = \frac{k}{2} - 1 \geq 1$ and $h_{F_s} = s + 2 > 2$. Let $J \subset (0, k)$ be the union of the three open intervals $(i + \frac{i+1}{1000}, i + 1 + \frac{i+1}{1000})$, so $|J| = 3$. No square meets the strip $\mathbb{R} \times [0, b)$, so every path starts with a vertical ray step.

- Let $x \in J$, with x in the i -th interval. The first ray step meets Z_i at $t_S = b$, with $t_D = \delta > b$, $t_W = +\infty$ and $t_H = h_F > b$. So $(Z_i, (x, b))$ is an entry candidate. Only the vertical ray of x reaches (x, b) , so x is the R1 winner. Since $a(Z_i) = 0 < \alpha_F$, the path traverses Z_i , and $\tau(x) > b$. Its crossing with ℓ_b is the entry point $(x, b) \in \partial Z_i$, so $x \in \mathcal{B}(b)$.
- The six end points of the three intervals are vertex contacts at height b , with label E and $\tau = b$.
- Let $x \in (0, k)$ lie outside the closure of J . The vertical line through x meets no square, so $t_S = t_W = +\infty$. The path terminates with D at height δ or is completed at height h_F ; both heights exceed b . So $P_b(x) = (x, b) \in \text{Wa}_b$. These x form a set of measure $k - 3$.

Hence at $y = b$: $L(b) = 0$; $\mu_{Z_i}(b) = 0$ because $\text{int } Z_i \cap \ell_b = \emptyset$; $F_{\text{gap}}(b) = k - 3$ and $\text{Bd}(b) = 3$. Also $c_{Z_i}(b) = 1$ and $\omega(b) = k - 3$. The gap crossers have $X_b(x) = x$, so $\rho_b = 1$ almost everywhere on Wa_b , and $\text{Ov}_{\text{gap}}(b) = U(b) = 0$. Therefore $\text{Sh}(b) = 3$, while $L_T(b) = 0$ and $\Lambda(b) = 0$.

On every piece of F contained in J (such pieces exist because Σ_F is finite), the entry point (x, b) has the constant height $b \in (0, h_F)$. So $b \in \text{Exc}_F$; for $F = F^0$ this gives $b \in \text{Exc}$.

For $b < y < \min(b + 1, h_F)$, every $x \in J$ is alive at y and $P_y(x) = (x, y) \in \text{int } Z_i$. So $\mu_{Z_i}(y) \geq 1 = c_{Z_i}(y)$, and $\text{Sh}(y) = 0$.

If crossing points in the closed square were counted, then at $y = b$ the floor points of J would be counted, while the six vertex points are not alive at b . This would give $\mu_{Z_i}(b) = 1 = c_{Z_i}(b)$ and $\text{Sh}(b) = 0$. $\square$

6.7. Square crossers: proof of (d).

Proof of Theorem 6.9(d). Fix $y \in (0, h_F)$ and a square Z .

Step 1: the paths of $A_Z(y)$ traverse Z . Let $x \in A_Z(y)$. The point $P_y(x) \in \text{int } Z$ is not an event point (Lemma 6.5(iii)), and it is not in the waste. By Lemma 6.14(ii), it therefore lies in the open part of the pass segment of a square Y traversed by π_x in F , and $P_y(x) \in \text{int } Y$. Since $\text{int } Y \cap \text{int } Z = \emptyset$ for $Y \neq Z$, we get $Y = Z$. So π_x traverses Z in F . A traversal in F_s is a traversal in F^0 , and in F^0 only the R1 winner of an entry candidate traverses. Hence x is the R1 winner at an entry candidate $(Z, q_Z(x))$ of F^0 , where $q_Z(x) \in \text{relint bot}(Z)$. This entry point is unique, because a path has at most one entry candidate on Z (Lemma 6.1(i)). Finally $P_y(x) = q_Z(x) + t n_Z$ with $0 < t < 1$.

Step 2: injectivity. Let $x, x' \in A_Z(y)$ with $P_y(x) = P_y(x')$. Write $q_Z(x) = c_Z - \frac{1}{2}n_Z + \xi u_Z$, $q_Z(x') = c_Z - \frac{1}{2}n_Z + \xi' u_Z$, and $P_y(x) = q_Z(x) + t n_Z = q_Z(x') + t' n_Z$. Taking scalar products with u_Z and with n_Z gives $\xi = \xi'$ and $t = t'$. So $q_Z(x) = q_Z(x') =: q$, and both x and x' are R1 winners at (Z, q) . An entry candidate has at most one R1 winner (N3), so $x = x'$. Thus X_y is injective on $A_Z(y)$.

Step 3: expansion. By the proof of Lemma 6.15, $A_Z(y) \setminus \Sigma_F$ is a finite union of pairwise disjoint open intervals J . Each J is contained in a piece of F on which the paths traverse Z at the same step j . On J the entry point $q_Z(x)$ is the end point $q_j(x)$ of step j , and Lemma 6.1(iii) gives $q'_j = \mu u_Z$ with $|\mu| \geq 1$. For F_s , J lies in a piece of F^0 and the same functions are used. Since $P_y(x) = q_Z(x) + t(x)n_Z$ has height y , we have $t(x) = (y - q_{Z,y}(x))/\cos \varphi_Z$ and

$$X_y(x) = q_{Z,x}(x) - (y - q_{Z,y}(x)) \tan \varphi_Z.$$

Hence

$$X'_y = \mu \cos \varphi_Z + \mu \sin \varphi_Z \tan \varphi_Z = \frac{\mu}{\cos \varphi_Z}, \quad |X'_y| \geq |\mu| \geq 1.$$

So X_y is affine on J with slope of absolute value at least 1, and $|X_y(J)| = |X'_y| |J| \geq |J|$.

Step 4: summation. By Step 2 the intervals $X_y(J)$ are pairwise disjoint, and by Step 1 they are contained in $\text{int } Z \cap \ell_y$, viewed as a subset of $\mathbb{R}$. The set Σ_F is finite. Therefore

$$\mu_Z(y) = \sum_J |J| \leq \sum_J |X_y(J)| \leq |\text{int } Z \cap \ell_y| \leq |Z \cap \ell_y| = c_Z(y).$$

Hence $(\mu_Z(y) - c_Z(y))_+ = 0$ for every Z , and $\text{Ov}(y) = 0$. $\square$

Step 2 is the only place in (a)–(f) where the uniqueness of the R1 winner is used.

6.8. Gap crossers: proof of (e).

Proof of Theorem 6.9(e). Fix $y \in (0, h_F)$.

Step 1: decomposition. For a piece I of F and a ray step j of the paths of I , let

$$J_{I,j} := \{x \in I : \tau(x) > y, p_{j,y}(x) < y < p_{j,y}(x) + t_j^*(x) d_{j,y}\}.$$

This is an open interval, possibly empty. By the proof of Lemma 6.15, $A_{\text{gap}}(y) \setminus \Sigma_F$ is the union of the sets $J_{I,j}$. These sets are pairwise disjoint. Different pieces are disjoint. For a fixed x , the point $P_y(x)$ lies in the open part of at most one segment, because the height is strictly increasing along Γ_x and the open parts of different segments therefore have disjoint ranges of heights. Let c run over the pairs (I, j) with $J_c := J_{I,j} \neq \emptyset$.

Step 2: slopes. Let $c = (I, j)$, let X be the source of step j , and recall $p'_j = \lambda u_X$ with $|\lambda| \geq 1$. For $x \in J_c$ we have $P_y(x) = p_j(x) + t n_X$ with $t = (y - p_{j,y}(x)) / \cos \varphi_X$. So on J_c

$$X_c(x) := X_y(x) = p_{j,x}(x) - (y - p_{j,y}(x)) \tan \varphi_X, \quad X'_c = \lambda \cos \varphi_X + \lambda \sin \varphi_X \tan \varphi_X = \frac{\lambda}{\cos \varphi_X},$$

and $|X'_c| \geq 1$.

Step 3: density. Under an affine map of slope $a \neq 0$, Lebesgue measure on an interval J is mapped to $|a|^{-1}$ times Lebesgue measure on the image interval. Since Σ_F is finite,

$$\nu_y = \sum_c (X_c)_\#(|\cdot|_{J_c}) = \left(\sum_c |X'_c|^{-1} \mathbf{1}_{X_c(J_c)} \right) dq.$$

So ν_y is absolutely continuous, with density $\rho_y = \sum_c |X'_c|^{-1} \mathbf{1}_{X_c(J_c)}$ almost everywhere. By the definition of $A_{\text{gap}}(y)$, $X_c(J_c) \subseteq \text{Wa}_y$. Hence $\rho_y = 0$ almost everywhere outside Wa_y . Further, ρ_y is bounded, and

$$F_{\text{gap}}(y) = |A_{\text{gap}}(y)| = \nu_y(\mathbb{R}) = \int_{\text{Wa}_y} \rho_y dq.$$

Step 4: conclusion. By (b), $\omega(y) = |\text{Wa}_y| = \int_{\text{Wa}_y} 1 dq$. All integrals are finite, because ρ_y is bounded and $|\text{Wa}_y| \leq k$. Writing $\rho_y - 1 = (\rho_y - 1)_+ - (1 - \rho_y)_+$,

$$F_{\text{gap}}(y) - \omega(y) = \int_{\text{Wa}_y} (\rho_y - 1) dq = \text{Ov}_{\text{gap}}(y) - U(y) \leq \text{Ov}_{\text{gap}}(y).$$

Another version of ρ_y , equal to this one almost everywhere, gives the same $\text{Ov}_{\text{gap}}(y)$ and $U(y)$. $\square$

Part (e) uses no bound on the density ρ_y ; the size of Ov_{gap} is estimated in Section 7.

6.9. The shadow inequality and the wall term.

Proof of Theorem 6.9(f). Let $y \in (0, h_F) \setminus \text{Exc}$. By (b), $\text{Bd}(y) = 0$, and by (d), $\text{Ov}(y) = 0$. So (c) gives $\text{Sh}(y) = L(y) + F_{\text{gap}}(y) - \omega(y)$, and (e) turns this into $\text{Sh}(y) = L(y) + \text{Ov}_{\text{gap}}(y) - U(y)$. Since $L = L_T + L_D + L_E + L_M + L_W$, this is $\text{Sh}(y) = L_T(y) + \Lambda(y) - U(y)$, and $U(y) \geq 0$.

Let $F = F_s$. By (N4), a path of F_s terminates with label T only at a T_s point. This is either a cut of type T_s , or an inherited F^0 termination with label T, which is itself a T_s point. In both cases $x \in \mathcal{T}_s$. Hence $\{x : \text{label T in } F_s, \tau_{F_s}(x) < y\} \subseteq \mathcal{T}_s$ and $L_T^{F_s}(y) \leq |\mathcal{T}_s| = N(s)$ for every y . The bound for Λ^{F_s} is Proposition 6.10, whose proof follows. $\square$

Lemma 6.18 (F_s and F^0). *Let $s \in [y_0, y_1]$ and $x \in (0, k)$.*

- (i) $\Gamma_x^{F_s}$ is an initial part of $\Gamma_x^{F^0}$. Its pass segments are pass segments of $\Gamma_x^{F^0}$, and its gap segments are those of $\Gamma_x^{F^0}$, except that the last one may be an initial part of the corresponding gap segment of $\Gamma_x^{F^0}$.
- (ii) $\tau_{F_s}(x) \leq s + 2$ and $\tau_{F_s}(x) \leq \tau_{F^0}(x)$.
- (iii) If x has label D, M, E or W in F_s , then it has the same label and the same termination point in F^0 .
- (iv) Every square traversed in F_s has $a(Y) < \alpha(s)$.

Proof. (i) and (iii) follow from (N4). The F_s path is the F^0 path cut at an entry point (T_s), at a point inside a ray step or at an exit point (H_s), or at the termination of the F^0 path. The cuts carry the labels T and H, so the labels D, M, E and W of F_s come from terminations of F^0 , at the same point. Part (iv) is Lemma 6.4(i).

(ii) Every ray step of F_s ends at height at most $s + 2$ (proof of Lemma 6.5(i)), and every entry point of F_s has height at most $s + 2$ (proof of Lemma 6.4(ii)). A termination with a label other than H is the end point of a ray step or an entry point, so it has height $\leq s + 2$. With label H, $\tau_{F_s} = s + 2$ by definition. This proves $\tau_{F_s}(x) \leq s + 2$.

If x has label H in F^0 , then $\tau_{F^0}(x) = h_{\max} \geq s + 2 \geq \tau_{F_s}(x)$. Otherwise $\tau_{F^0}(x)$ is the height of the last point of $\Gamma_x^{F^0}$. By (i) and Lemma 6.4(i), this is at least the height of the last point $e^{F_s}(x)$ of $\Gamma_x^{F_s}$. That height equals $\tau_{F_s}(x)$ if the label in F_s is not H, and it is at least $s + 2 = \tau_{F_s}(x)$ if the label is H, the cut being at height $s + 2$ inside a ray step or at an exit point of height $\geq s + 2$. $\square$

Proof of Proposition 6.10. Let $s \in [y_0, y_1]$ and $0 < y < s + 2$. Note that $y < h_{\max}$, so the quantities of F^0 at y are defined.

Step 1: deaths, merges, end collisions. By Lemma 6.18(iii), $\{x : \text{label D in } F_s, \tau_{F_s}(x) < y\} \subseteq \mathcal{D}$, so $L_D^{F_s}(y) \leq |\mathcal{D}|$. In the same way $L_M^{F_s}(y) \leq |\mathcal{M}|$ and $L_E^{F_s}(y) \leq L_E$.

Step 2: gap overlap. Let $x \in A_{\text{gap}}^{F_s}(y)$, so $\tau_{F_s}(x) > y$ and $P_y^{F_s}(x) \in \text{Wa}_y$. By Lemma 6.18(ii), $\tau_{F^0}(x) \geq \tau_{F_s}(x) > y$. By Lemma 6.18(i), $\Gamma_x^{F_s} \subseteq \Gamma_x^{F^0}$, so the unique point of $\Gamma_x^{F^0}$ on ℓ_y is $P_y^{F^0}(x) = P_y^{F_s}(x)$. Hence $x \in A_{\text{gap}}^{F^0}(y)$, with the same crossing abscissa. So $\nu_y^{F_s}(B) \leq \nu_y^{F^0}(B)$ for every Borel set $B \subseteq \mathbb{R}$. Both densities are bounded (proof of (e)). Taking $B = \{\rho_y^{F_s} > \rho_y^{F^0}\}$ gives $\int_B (\rho_y^{F_s} - \rho_y^{F^0}) \leq 0$ with a positive integrand, so $|B| = 0$, i.e. $\rho_y^{F_s} \leq \rho_y^{F^0}$ almost everywhere. Therefore

$$\text{Ov}_{\text{gap}}^{F_s}(y) = \int_{\text{Wa}_y} (\rho_y^{F_s} - 1)_+ \leq \int_{\text{Wa}_y} (\rho_y^{F^0} - 1)_+ = \text{Ov}_{\text{gap}}^{F^0}(y) \leq \sup_{0 < y' < h_{\max}} \text{Ov}_{\text{gap}}^{F^0}(y').$$

Step 3: walls. Let x have label W in F_s with $\tau(x) := \tau_{F_s}(x) < y$. By Lemma 6.18(iv), every segment of $\Gamma_x^{F_s}$ has direction e_2 or n_Y with $|\varphi_Y| < \alpha(s)$, so its direction d satisfies $|d_x| \leq d_y \tan \alpha(s)$. Summing over the segments of $\Gamma_x^{F_s}$, which goes from $(x, 0)$ to $e(x)$ and gains the height $\tau(x)$, we get $|e(x)_x - x| \leq \tau(x) \tan \alpha(s) < y \tan \alpha(s)$. Since $e(x)_x \in \{0, k\}$, this gives $x < y \tan \alpha(s)$ or $k - x < y \tan \alpha(s)$. Hence $L_W^{F_s}(y) \leq 2y \tan \alpha(s)$.

Step 4: summation. Adding Steps 1–3,

$$\Lambda^{F_s}(y) = [L_D + L_E + L_M + L_W]^{F_s}(y) + \text{Ov}_{\text{gap}}^{F_s}(y) \leq |\mathcal{D}| + |\mathcal{M}| + L_E + \text{Ov}_{\text{gap}}^{F^0}(y) + 2y \tan \alpha(s),$$

and the right-hand side is at most $\Lambda_0 + 2y \tan \alpha(s) \leq \Lambda_0 + 2(s + 2) \tan \alpha(s)$, because $0 < y < s + 2$ and $\tan \alpha(s) > 0$. $\square$

Proof of Corollary 6.11. Let $s \in [y_0, y_1]$ and $y \in (0, s + 2) \setminus \text{Exc}$. Since $h_{F_s} = s + 2$, Theorem 6.9(f) applies to $F = F_s$ and gives $\text{Sh}^{F_s}(y) \leq L_T^{F_s}(y) + \Lambda^{F_s}(y)$. By (f), $L_T^{F_s}(y) \leq N(s)$, and by Proposition 6.10, $\Lambda^{F_s}(y) \leq \Lambda^*(s)$. $\square$

6.10. The order of M and T: proof of (g).

Definition 6.19 (order “T before M”). The master flow $F^{0, \text{TM}}$ is defined like F^0 in (N2)–(N3), with one change at entry candidates (Y, q) with $a(Y) \geq \alpha_{\max}$: there every path that reaches (Y, q) alive terminates at q with label T, and R1 is not applied. At entry candidates with $a(Y) < \alpha_{\max}$, R1 and the traversal rule are applied as before, by the same induction over the squares in increasing order of centre height. For $s \in [y_0, y_1]$, the scale flow F_s^{TM} is the path of $F^{0, \text{TM}}$ cut at the first of the following points: an entry candidate reached alive with $a(Y) \geq \alpha(s)$ (label T, whether or not the path would win R1 there); an H_s point as in (N4); or the termination of the path of $F^{0, \text{TM}}$, with its label.

The induction is well defined for $F^{0, \text{TM}}$ because Lemma 6.1(i) is a statement about the geometry of a single path that traverses only squares with $a(Y) < \alpha_{\max}$, and such paths are the same in both orders.

Proof of Theorem 6.9(g). Step 1: the master flows. We show by induction over the squares, in increasing order of centre height (ties in the fixed order used for R1), the following claim. For every square Y and every $q \in \text{relint bot}(Y)$, the set $\mathcal{A}_Y(q)$ of floor points that reach (Y, q) alive is the same in F^0 and in $F^{0, \text{TM}}$. Moreover, each of these paths either terminates at q in both flows or traverses Y in the same state in both flows.

Assume the claim for all squares before Y . Apart from R1 and R2, every rule (D, W, the classification of contacts, E, H, traversal) depends only on the current state (p, d, X, g) of the path. Let x reach (Y, q) alive in one of the two flows. Its path up to q consists of ray steps, which are determined by the state, and of traversals of squares Y' at entry candidates (Y', q') . By Lemma 6.1(i) these squares Y' have smaller centre height than Y , so by the induction hypothesis x traverses each of them in the same state in the other flow as well. Hence x reaches (Y, q) alive in the other flow, and $\mathcal{A}_Y(q)$ is the same in both flows. If $a(Y) \geq \alpha_{\max}$, every path of $\mathcal{A}_Y(q)$ terminates at q in both flows: in F^0 the R1 winner with label T and the others with label M, in $F^{0, \text{TM}}$ all with label T. If $a(Y) < \alpha_{\max}$, both flows apply R1 to the same set with the same cumulative gaps $g_x(q)$ (the paths agree up to q), so they have the same winner. The winner traverses Y in the same state, and the other paths terminate at q with label M. This proves the claim.

Consequently, every floor point has the same trajectory and the same termination point in F^0 and in $F^{0, \text{TM}}$. The labels agree, except that at entry candidates with $a(Y) \geq \alpha_{\max}$ some paths with label M in F^0 have label T in $F^{0, \text{TM}}$.

Step 2: the scale flows. Fix s and x . By Step 1 the master path of x is the same in both orders: the same trajectory, the same entry candidates, reached alive or not in the same way. Let C be the first entry candidate on it that is reached alive and has $a(Y_C) \geq \alpha(s)$, if there is one. The H_s points and the master termination point are the same in both orders, since they depend only on the trajectory. Before C there is no T_s point in F_s , because a T_s point is such an entry candidate. Hence, in both orders, the path is cut or terminates at the first of C , the first H_s point and the master termination. At C , F_s^{TM} gives label T. In F_s , the path gives label T if it is the R1 winner at C (T_s). Otherwise C is an entry candidate where the path loses R1, so the F^0 path terminates at C with label M, and F_s inherits this. Elsewhere the labels agree. So every x has the same trajectory, termination point and termination height in F_s and in F_s^{TM} , and only labels M of F_s can become labels T.

Step 3: consequences. The pieces Σ_0, Σ_s serve for both orders. Trajectories coincide, and on a piece the labels of the order T before M are constant, because they are determined by the type. The following quantities are defined from trajectories, termination points, termination heights and aliveness only, so they coincide for the two orders: $P_y, A_Z, \mu_Z, A_{\text{gap}}, F_{\text{gap}}, c_Z, \omega, \nu_y, \rho_y, \text{Sh}, \text{Ov}, \text{Ov}_{\text{gap}}, U, \text{Bd}$ and Exc . So does $L = \sum_{\star} L_{\star}$, because every x with $\tau(x) < y$ is counted once whatever its label among T, D, E, M, W. The labels D, E and W are the same in both orders. Only L_T and L_M change, and $L_T^{\text{TM}} - L_T = L_M - L_M^{\text{TM}} \geq 0$.

Hence (a)–(e) hold for the order T before M: every quantity occurring in them is the same for the two orders. So does the identity in (f), because $L_T + \Lambda = L + \text{Ov}_{\text{gap}}$ is the same for the two orders. The bound $L_T^{\text{TM}}(y) \leq N^{\text{TM}}(s) := |\{x : \text{label T in } F_s^{\text{TM}}\}|$ holds trivially. In Proposition 6.10, the cuts of F_s^{TM} carry the labels T and H, so the labels M of F_s^{TM} are inherited from $F^{0, \text{TM}}$. Thus $L_M^{F_s^{\text{TM}}}(y) \leq |\mathcal{M}^{\text{TM}}|$, where $\mathcal{M}^{\text{TM}} \subseteq \mathcal{M}$ is the set of points with label M in $F^{0, \text{TM}}$. Steps 1–3 of that proof are otherwise unchanged, and $\Lambda_0^{\text{TM}} \leq \Lambda_0$. $\square$

Remark 6.20 (where the order is used). The order M before T is used later, in Section 10, through the inclusion $\mathcal{T}_s \subseteq \{T^* \leq s\}$. The function T^* is defined from the entry candidates at which the path is the R1 winner (Section 3). So the inclusion needs that the terminal entry of every $x \in \mathcal{T}_s$ be an entry at which x is the R1 winner. Under the order M before T this holds by (N4). Under the order T before M, the set of points with label T in F_s^{TM} may also contain R1 losers. The order M before T is sufficient for the inclusion. We make no claim about whether the later arguments could be carried out with the order T before M.

6.11. Ceiling flows and measurability in y .

Remark 6.21 (ceiling flows). Theorem 6.9, Proposition 6.10 and Corollary 6.11 hold for every closed packing. Let $\varrho(x, y) := (x, k - y)$ be the reflection of Lemma 3.8 (denoted ρ there and in the other

sections); in this remark we write ϱ for it, to keep it apart from the densities ρ_y . The reflected packing $\varrho(\mathcal{P})$ is a closed packing in $[0, k]^2$, and the ceiling flows of Section 3 are $\tilde{F} = \varrho \circ F[\varrho(\mathcal{P})] \circ \varrho$. Translated back to the original coordinates, a ceiling path moves downwards. Its termination height $\tilde{\tau}(x)$ is the original height of its termination point, with $\tilde{\tau}(x) := k - h_F$ for completed paths. It is alive at y if $\tilde{\tau}(x) < y$. The ceiling losses are $\tilde{L}_\star(y) := |\{x : \text{label } \star, \tilde{\tau}(x) > y\}|$, and $c_Z(y) = c_{\varrho Z}(k - y)$, $\omega(y) = \omega_{\varrho(\mathcal{P})}(k - y)$. With these conventions every quantity of a ceiling flow at height y equals the corresponding quantity of the floor flow of $\varrho(\mathcal{P})$ at height $k - y$. Hence, with $\widetilde{\text{Exc}} := k - \text{Exc}[\varrho(\mathcal{P})]$:

- $\widetilde{\text{Sh}}(y) = \tilde{L}_T(y) + \tilde{\Lambda}(y) - \tilde{U}(y) \leq \tilde{L}_T(y) + \tilde{\Lambda}(y)$ for $y \in (k - h_F, k) \setminus \widetilde{\text{Exc}}$;
- $\tilde{\Lambda}^{\tilde{F}_s}(y) \leq \tilde{\Lambda}_0 + 2(k - y) \tan \alpha(s) \leq \tilde{\Lambda}_0 + 2(s + 2) \tan \alpha(s)$ for $k - s - 2 < y < k$.

Here $\tilde{\Lambda}_0$ is Λ_0 computed for $\varrho(\mathcal{P})$.

Remark 6.22 (Borel measurability in y). For a fixed flow F , the functions

$$y \mapsto L_\star(y), \mu_Z(y), F_{\text{gap}}(y), c_Z(y), \omega(y), \text{Sh}(y), \text{Ov}(y), \text{Ov}_{\text{gap}}(y), U(y), \text{Bd}(y)$$

are Borel on $(0, h_F)$. This is not used in the later sections (Remark 6.12).

Proof. (1) Let $A_\bullet(y)$ be one of the sets $\{\text{label } \star, \tau < y\}$, $A_Z(y)$, $A_{\text{gap}}(y)$, and put $\mathbf{A}_\bullet := \{(x, y) : 0 < y < h_F, x \in A_\bullet(y)\}$. For x in a piece I , the conditions that define $x \in A_\bullet(y)$ in the proof of Lemma 6.15 are finitely many strict inequalities between y and affine functions of x . So $\mathbf{A}_\bullet \cap (I \times (0, h_F))$ is a finite union of open sets. For each of the finitely many $x \in \Sigma_F$, the set $\{y : x \in A_\bullet(y)\}$ is a finite union of intervals, determined by the height ranges of the finitely many segments of Γ_x . Hence $\mathbf{A}_\bullet$ is a Borel set, and by Tonelli's theorem $y \mapsto |A_\bullet(y)| = \int \mathbf{1}_{\mathbf{A}_\bullet}(x, y) dx$ is Borel. This covers $L_\star$, μ_Z and F_{gap} ; then $\text{Bd} = k - L - \sum_Z \mu_Z - F_{\text{gap}}$.

(2) $c_Z(y) = \int \mathbf{1}_Z(q, y) dq$ is Borel by Tonelli's theorem, and $\omega = k - \sum_Z c_Z$ by (b). Sh and Ov are finite sums of positive parts of Borel functions.

(3) For a piece I and a ray step j let $\mathbf{J}_{I,j} := \{(x, y) : x \in I, 0 < y < h_F, \tau(x) > y, p_{j,y}(x) < y < p_{j,y}(x) + t_j^*(x)d_{j,y}\}$, an open set. The map $\Psi_{I,j}(x, y) := (p_{j,x}(x) - (y - p_{j,y}(x)) \tan \varphi_{X_j}, y)$ is affine in (x, y) . The derivative of its first component in x is $\lambda_j / \cos \varphi_{X_j} \neq 0$, so $\Psi_{I,j}$ is an affine bijection of $\mathbb{R}^2$, and $\mathbf{Q}_{I,j} := \Psi_{I,j}(\mathbf{J}_{I,j})$ is open. Put

$$\bar{\rho}(q, y) := \sum_{I,j} \left| \frac{\cos \varphi_{X_j}}{\lambda_j} \right| \mathbf{1}_{\mathbf{Q}_{I,j}}(q, y).$$

For fixed y , the slice of $\mathbf{Q}_{I,j}$ at height y is $X_c(J_c)$ with $c = (I, j)$ as in the proof of (e). So $\bar{\rho}(\cdot, y)$ is a version of ρ_y . The functions $(q, y) \mapsto (\bar{\rho}(q, y) - 1)_+ \mathbf{1}_{\text{waste}}(q, y)$ and $(q, y) \mapsto (1 - \bar{\rho}(q, y))_+ \mathbf{1}_{\text{waste}}(q, y)$ are Borel, and Tonelli's theorem shows that Ov_{gap} and U are Borel. $\square$

7. DEATHS, MERGES AND GAP OVERLAP (P5)

We keep the standing assumptions and the notation of Section 5.1, and we use sources and delays (Definition 5.1) and Lemmas 5.2–5.5.

Let $\mathcal{D}$ and $\mathcal{M} \subset (0, k)$ be the sets of x such that π_x terminates with D, respectively with M, and let $\tilde{\mathcal{D}}, \tilde{\mathcal{M}}$ be their ceiling versions (Definitions 3.26 and 3.24). For $y \in (0, h_{\max})$, $\text{Wa}_y = \ell_y \cap \text{waste}$, $A_{\text{gap}}(y)$, X_y , the density ρ_y of the gap crossers and the gap overlap $\text{Ov}_{\text{gap}}^{F_0}(y) = \int_{\text{Wa}_y} (\rho_y - 1)_+$ are those of Definitions 3.28 and 3.30 and Lemma 3.29; a path is *alive at y* if its termination height exceeds y (Definition 3.16). Finally (Definition 3.30)

$$\Lambda_0 := |\mathcal{D}| + |\mathcal{M}| + L_E + \sup_{0 < y < h_{\max}} \text{Ov}_{\text{gap}}^{F_0}(y),$$

and $\tilde{\Lambda}_0 := \Lambda_0[\rho\mathcal{P}]$. We put $Q_* := 0.32$, the constant of [R, Proposition 4.5].

Theorem 7.1 (P5). *Assume the standing assumptions of Section 5.1. Let K be the constant of Lemma 5.5 and*

$$S_K := (1 + 10^{-11}) \frac{4K}{Q_*(2 - 3 \cdot 10^{-6})}, \quad \text{so that} \quad S_9 \leq 56.2501, \quad S_{13} \leq 81.2502$$

(rounded up). Then:

$$(D) \quad |\mathcal{D}| + |\tilde{\mathcal{D}}| \leq (1 + S_K \delta^2) \frac{W}{\delta};$$

$$(M) \quad |\mathcal{M}| + |\tilde{\mathcal{M}}| \leq \frac{4}{\cos(\pi/4 + \alpha_{\max})} S_K \delta W;$$

$$(G) \quad \sup_y \text{Ov}_{\text{gap}}^{F^0}(y) + \sup_y \text{Ov}_{\text{gap}}^{\tilde{F}^0}(y) \leq \frac{S_K}{\cos \alpha_{\max}} \delta W, \text{ the suprema being over all heights at which the gap overlaps are defined.}$$

Numerically (all values rounded up): with $K = 9$ (using [R, Lemma 4.10]),

$$|\mathcal{D}| + |\tilde{\mathcal{D}}| \leq C_D \frac{W}{\delta}, \quad |\mathcal{M}| + |\tilde{\mathcal{M}}| \leq C_M \delta W, \quad \sup_y \text{Ov}_{\text{gap}}^{F^0} + \sup_y \text{Ov}_{\text{gap}}^{\tilde{F}^0} \leq C_G \delta W,$$

with $C_D = 1 + 56.2501 \delta^2$, $C_M = 318.20$, $C_G = 56.2501$; with $K = 13$ (using only [R, Lemma 4.9]) the same holds with $C_D = 1 + 81.26 \delta^2$, $C_M = 459.7$, $C_G = 81.26$.

Corollary 7.2. *Under the standing assumptions,*

$$\Lambda_0 + \tilde{\Lambda}_0 \leq C_\Lambda \frac{W}{\delta},$$

where $C_\Lambda = 1 + 1185.6 \delta^2 < 1.0000002$ (Definition 3.6) if $K = 9$ is used; if $K = 13$ is used, the same holds with $C_\Lambda^{(13)} = 1 + 1712.6 \delta^2 < 1.0000002$ in place of C_Λ .

Remark 7.3 (the hypothesis $\delta \leq \frac{1}{2} \cos 2\alpha_{\max}$). Under the standing assumptions, $\delta \leq 10^{-5} < 0.4999999 < \frac{1}{2} \cos(3 \cdot 10^{-6}) \leq \frac{1}{2} \cos 2\alpha_{\max}$; this is constraint (C6) of Definition 3.2, which follows from (C1) (Lemma 3.3(a)). It is a hypothesis of the geometric Lemmas 7.5 and 7.8 below; Lemma 7.7, Lemma 7.11 and Theorem 7.1 depend on it through these two lemmas. Section 5 does not use it.

The proof of Theorem 7.1 rests on one geometric fact (Lemmas 7.7 and 7.8): a point is reached by at most two paths; if it is reached by two, their sources are two squares X_1, X_2 with $\varphi_{X_1} > \varphi_{X_2}$, the point lies in a rectangle of width $\delta \tan \theta / \cos \theta$ and height δ above the upper left vertex of X_1 ($\theta = \varphi_{X_1} - \varphi_{X_2}$), and the two squares are at distance less than $\delta \tan \theta$. The rest consists of summing the areas and lengths of these rectangles with the pair waste of [R].

7.1. Points with two sources.

Lemma 7.4 (floor and square sources). *No point is reached both from the floor X_0 and from a square.*

Proof. Let q be reached from X_0 with delay t_0 and from a square X with delay t . By parts (a) and (e) of Lemma 5.2, $q = (x, t_0)$ with $t_0 < \delta$, and $t > 0$. The start point of the second step is $p := q - t n_X$, a point of $\text{relint top}(X)$. The point $p - n_X$ lies in $\text{bot}(X) \subset X$, and its height is

$$q_y - (t + 1) \cos \varphi_X < \delta - \cos \alpha_{\max} < 0,$$

since $|\varphi_X| < \alpha_{\max}$ (Lemma 5.2(b)) and $\delta \leq 10^{-5} < \cos \alpha_{\max}$. This contradicts $X \subset [0, k]^2$. $\square$

The next lemma is purely geometric; it is stated with its own parameters $\alpha_{\max}$ and δ .

Lemma 7.5 (two sources). *Let $0 < \alpha_{\max} < \pi/4$ and $\delta > 0$ satisfy*

$$\delta \leq \frac{1}{2} \cos 2\alpha_{\max}.$$

Let X_1, X_2 be two disjoint closed unit squares with $|\varphi_{X_1}|, |\varphi_{X_2}| < \alpha_{\max}$, and let q be a point such that $q - t_i n_{X_i} \in \text{relint top}(X_i)$ with $0 < t_i < \delta$ for $i = 1, 2$. Then $\varphi_{X_1} \neq \varphi_{X_2}$. Label the squares so that $\varphi_{X_1} > \varphi_{X_2}$, and put $\theta := \varphi_{X_1} - \varphi_{X_2} \in (0, 2\alpha_{\max})$ and $\tau_i := (q - c_{X_i}) \cdot u_{X_i}$. Then

$$\xi_1 := \tau_1 + \frac{1}{2} < \frac{1}{2} \sin \theta, \quad \xi_2 := \frac{1}{2} - \tau_2 < \frac{1}{2} \sin \theta.$$

In particular, under the standing assumptions this applies to any two distinct square sources X_1, X_2 of a point q , with their delays t_1, t_2 (Lemma 5.2(a),(b),(e)).

So q lies just above the left end of $\text{top}(X_1)$ and just above the right end of $\text{top}(X_2)$.

Proof. Write u_i, n_i, c_i, φ_i for $u_{X_i}, n_{X_i}, c_{X_i}, \varphi_{X_i}$. Since $q - t_i n_i \in \text{relint top}(X_i)$, we have $(q - c_i) \cdot n_i = \frac{1}{2} + t_i$ and $\tau_i \in (-\frac{1}{2}, \frac{1}{2})$. Note $|\varphi_1 - \varphi_2| < 2\alpha_{\max} < \pi/2$, so $\cos(\varphi_1 - \varphi_2) \geq \cos 2\alpha_{\max}$.

Put $z := q - \frac{1}{2}n_1$. In the frame of X_1 , $(z - c_1) \cdot u_1 = \tau_1 \in (-\frac{1}{2}, \frac{1}{2})$ and $(z - c_1) \cdot n_1 = t_1 \in (0, \frac{1}{2}]$; hence $z \in X_1$. In the frame of X_2 , by (5.1),

$$(z - c_2) \cdot n_2 = \frac{1}{2} + t_2 - \frac{1}{2} \cos(\varphi_1 - \varphi_2) \in [t_2, t_2 + \frac{1}{2}(1 - \cos 2\alpha_{\max})], \quad (z - c_2) \cdot u_2 = \tau_2 - \frac{1}{2} \sin(\varphi_2 - \varphi_1).$$

Because $0 < t_2 < \delta \leq \frac{1}{2} \cos 2\alpha_{\max}$, the n_2 -coordinate $(z - c_2) \cdot n_2$ lies in $(0, \frac{1}{2}]$. *This is where the hypothesis $\delta \leq \frac{1}{2} \cos 2\alpha_{\max}$ enters the proof* (and in the same step with the roles of X_1 and X_2 exchanged). Since $X_1 \cap X_2 = \emptyset$, $z \notin X_2$, and therefore

$$(7.1) \quad |\tau_2 - \frac{1}{2} \sin(\varphi_2 - \varphi_1)| > \frac{1}{2}.$$

Exchanging the roles of X_1 and X_2 (with $z' := q - \frac{1}{2}n_2 \in X_2$) gives

$$(7.2) \quad |\tau_1 - \frac{1}{2} \sin(\varphi_1 - \varphi_2)| > \frac{1}{2}.$$

If $\varphi_1 = \varphi_2$, (7.1) says $|\tau_2| > \frac{1}{2}$, which is false. So $\varphi_1 \neq \varphi_2$; let $\varphi_1 > \varphi_2$ and $s := \sin \theta \in (0, 1)$. Since $\tau_2 > -\frac{1}{2}$, we have $\tau_2 + \frac{s}{2} > -\frac{1}{2}$, so (7.1) gives $\tau_2 + \frac{s}{2} > \frac{1}{2}$, that is, $\xi_2 < \frac{s}{2}$. Since $\tau_1 < \frac{1}{2}$, we have $\tau_1 - \frac{s}{2} < \frac{1}{2}$, so (7.2) gives $\tau_1 - \frac{s}{2} < -\frac{1}{2}$, that is, $\xi_1 < \frac{s}{2}$. $\square$

Remark 7.6 (the hypothesis cannot be weakened). The hypothesis $\delta \leq \frac{1}{2} \cos 2\alpha_{\max}$ cannot be replaced by a weaker bound on δ in Lemma 7.5. Let $0 < \alpha_{\max} < \pi/4$ and $\delta > \frac{1}{2} \cos 2\alpha_{\max}$. Choose $\theta \in (0, 2\alpha_{\max})$ with $\frac{1}{2} \cos \theta < \delta$ (possible by continuity), then $t \in (\frac{1}{2} \cos \theta, \delta)$, put $v := \sin \theta (t - \frac{1}{2} \cos \theta) > 0$, and choose $\eta \in (0, \min(v, 1))$. In coordinates with $q = (0, 0)$ let

$$X_2 := [\eta - 1, \eta] \times [-1 - t, -t], \quad X_1 := \{v_1 + a u - b n : a, b \in [0, 1]\},$$

where $u = (\cos \theta, \sin \theta)$, $n = (-\sin \theta, \cos \theta)$ and $v_1 := -\frac{1}{2} \sin \theta u - t n$. Then $q - t e_2 = (0, -t) \in \text{relint top}(X_2)$, since $0 \in (\eta - 1, \eta)$, and $q - t n = v_1 + \frac{1}{2} \sin \theta u \in \text{relint top}(X_1)$, since $\frac{1}{2} \sin \theta \in (0, 1)$; both delays equal $t \in (0, \delta)$. The first coordinate of v_1 is $-\frac{1}{2} \sin \theta \cos \theta + t \sin \theta = v$, and every point of X_1 has first coordinate $v + a \cos \theta + b \sin \theta \geq v > \eta$, while $X_2 \subset \{x \leq \eta\}$; so $X_1 \cap X_2 = \emptyset$. (Moreover the segment from $(0, -t)$ to q has first coordinate $0 < v$ and misses X_1 , and the points $q - s n = (s \sin \theta, -s \cos \theta)$, $0 \leq s \leq t$, have second coordinate $> -t$ and miss X_2 .) Rotating the configuration about q by $-\theta/2$ gives phases $\theta/2$ and $-\theta/2$, of absolute value $< \alpha_{\max}$, and the frames used above become the frames of these phases. All hypotheses of Lemma 7.5 hold except $\delta \leq \frac{1}{2} \cos 2\alpha_{\max}$, but $\xi_1 = \frac{1}{2} \sin \theta$, so the conclusion $\xi_1 < \frac{1}{2} \sin \theta$ fails. Under the standing assumptions, δ is far below the threshold (Remark 7.3).

Lemma 7.7 (at most two paths through a point). *For every point q , at most two $x \in (0, k)$ have the property that π_x reaches q (in the gap case or in the contact case). If there are two, their sources are two distinct squares X_1, X_2 , and q, X_1, X_2 and the two delays satisfy the hypotheses of Lemma 7.5. In particular $M_g(q) \leq 2$ for every q .*

Proof. Let $x_1, \dots, x_m$ be distinct and let π_{x_i} reach q from the source X_i with delay t_i (unique by Lemma 5.2(f)). By Lemma 5.3(b) the sources X_i are pairwise distinct. So at most one of them is the floor, and if one of them is the floor, then none is a square by Lemma 7.4, and $m = 1$. Otherwise all X_i are squares with $|\varphi_{X_i}| < \alpha_{\max}$ and $0 < t_i < \delta$ (Lemma 5.2(a),(b)), and Lemma 7.5 applies to each pair (Remark 7.3); in particular the phases are pairwise distinct. Suppose $m \geq 3$ and let A, B, C be three of the sources with $\varphi_A > \varphi_B > \varphi_C$. Applied to (A, B) , where B has the smaller phase, Lemma 7.5 gives $\tau_B > \frac{1}{2} - \frac{1}{2} \sin(\varphi_A - \varphi_B) > 0$, because $0 < \varphi_A - \varphi_B < 2\alpha_{\max} < \pi/2$. Applied to (B, C) , where B has the larger phase, it gives $\tau_B < -\frac{1}{2} + \frac{1}{2} \sin(\varphi_B - \varphi_C) < 0$. This is a contradiction, so $m \leq 2$, and if $m = 2$ both sources are squares. $\square$

Lemma 7.8 (two sources: refined form). *In the situation of Lemma 7.5, with $\varphi_{X_1} > \varphi_{X_2}$ and the notation there, let v_1 be the upper left vertex of X_1 and v_2 the upper right vertex of X_2 :*

$$v_1 := c_{X_1} - \frac{1}{2}u_{X_1} + \frac{1}{2}n_{X_1}, \quad v_2 := c_{X_2} + \frac{1}{2}u_{X_2} + \frac{1}{2}n_{X_2}.$$

Then $\xi_1 = (q - v_1) \cdot u_{X_1}$, $t_1 = (q - v_1) \cdot n_{X_1}$, $\xi_2 = (v_2 - q) \cdot u_{X_2}$, $t_2 = (q - v_2) \cdot n_{X_2}$, and

$$(7.3) \quad \cos \theta \xi_1 + \xi_2 \leq \sin \theta \max(t_1, t_2 / \cos \theta) < \delta \tan \theta.$$

Consequently:

- (i) $0 < \xi_1 < \delta \tan \theta / \cos \theta$ and $0 < \xi_2 < \delta \tan \theta$;
- (ii) $\text{dist}(X_1, X_2) \leq \sin \theta \max(t_1, t_2 / \cos \theta) < \delta \tan \theta$;
- (iii) $q = v_1 + a u_{X_1} + b n_{X_1}$ with $0 < a < \delta \tan \theta / \cos \theta$ and $0 < b < \delta$.

Proof. Write u_i, n_i, c_i as before and $\kappa := \cos \theta$, $\varsigma := \sin \theta$. The four identities follow from $(v_1 - c_1) \cdot u_1 = -\frac{1}{2}$, $(v_1 - c_1) \cdot n_1 = \frac{1}{2}$, $(v_2 - c_2) \cdot u_2 = \frac{1}{2}$, $(v_2 - c_2) \cdot n_2 = \frac{1}{2}$ and $(q - c_i) \cdot n_i = \frac{1}{2} + t_i$. Since $\varphi_2 = \varphi_1 - \theta$,

$$u_2 = \kappa u_1 - \varsigma n_1, \quad n_2 = \varsigma u_1 + \kappa n_1,$$

so $u_1 \cdot u_2 = \kappa$, $n_1 \cdot u_2 = -\varsigma$, $u_1 \cdot n_2 = \varsigma$, $n_1 \cdot n_2 = \kappa$. Moreover

$$X_1 = \{v_1 + a u_1 - b n_1 : a, b \in [0, 1]\}, \quad X_2 = \{v_2 - a u_2 - b n_2 : a, b \in [0, 1]\}.$$

Let $\bar{t} := \max(t_1, t_2 / \kappa)$. Note $\kappa > \cos 2\alpha_{\max} > 0$ and $\bar{t} < \delta / \kappa \leq \frac{1}{2} \cos 2\alpha_{\max} / \cos \theta < \frac{1}{2}$.

A family of points of X_1 . For $\varepsilon \in (0, 1]$ and $\eta > 0$ put

$$\hat{t} := \max(t_1 + \eta, (t_2 + \varsigma \varepsilon) / \kappa), \quad h := \hat{t} - t_1 (\geq \eta), \quad z := v_1 + \varepsilon u_1 - h n_1.$$

(The letters $\bar{t}$, $\hat{t}$, h , κ , ς , a_z , b_z are local to this proof.) As $\varepsilon, \eta \rightarrow 0^+$, $\hat{t} \rightarrow \bar{t} < \frac{1}{2}$; so for small ε, η we have $h \leq 1$ and $z \in X_1$. Using $v_2 - q = \xi_2 u_2 - t_2 n_2$ and $q - z = (\xi_1 - \varepsilon) u_1 + \hat{t} n_1$, the coordinates of z in the frame of X_2 are

$$a_z := (v_2 - z) \cdot u_2 = (\kappa \xi_1 + \xi_2) - \kappa \varepsilon - \varsigma \hat{t}, \quad b_z := (v_2 - z) \cdot n_2 = -t_2 + \varsigma(\xi_1 - \varepsilon) + \kappa \hat{t}.$$

Since $\kappa \hat{t} \geq t_2 + \varsigma \varepsilon$ and $\xi_1 > 0$, we get $b_z \geq \varsigma \xi_1 \geq 0$. Also $b_z \leq \varsigma \xi_1 + \kappa \hat{t}$, which tends to $\varsigma \xi_1 + \kappa \bar{t} < \frac{1}{2} \varsigma^2 + \frac{1}{2} < 1$ by Lemma 7.5 and $\kappa \bar{t} < \delta \leq \frac{1}{2}$; so $b_z \in [0, 1]$ for small ε, η . Finally $a_z \leq \kappa \xi_1 + \xi_2 < \frac{1}{2} \varsigma(\kappa + 1) < 1$ by Lemma 7.5.

Proof of (7.3). Suppose $\kappa \xi_1 + \xi_2 > \varsigma \bar{t}$. As $\varepsilon, \eta \rightarrow 0^+$, $\kappa \varepsilon + \varsigma \hat{t} \rightarrow \varsigma \bar{t}$, so $a_z > 0$ for small ε, η . Then $a_z, b_z \in [0, 1]$, so $z \in X_2$; but $z \in X_1$, contradicting $X_1 \cap X_2 = \emptyset$. Hence $\kappa \xi_1 + \xi_2 \leq \varsigma \bar{t}$. Since $t_1 < \delta \leq \delta / \kappa$ and $t_2 / \kappa < \delta / \kappa$, $\varsigma \bar{t} < \varsigma \delta / \kappa = \delta \tan \theta$.

(i) $\xi_1 = \tau_1 + \frac{1}{2} > 0$ and $\xi_2 = \frac{1}{2} - \tau_2 > 0$ because $\tau_i \in (-\frac{1}{2}, \frac{1}{2})$. Then (7.3) gives $\kappa \xi_1 < \delta \tan \theta$ and $\xi_2 < \delta \tan \theta$.

(ii) Now (7.3) holds. For small ε, η take $z \in X_1$ as above. Since $\hat{t} \geq \bar{t}$, $a_z \leq \varsigma \bar{t} - \kappa \varepsilon - \varsigma \hat{t} \leq -\kappa \varepsilon < 0$, and $b_z \in [0, 1]$. Put $z' := z + a_z u_2$. Then $(v_2 - z') \cdot u_2 = a_z - a_z = 0$ and $(v_2 - z') \cdot n_2 = b_z$, so $z' \in X_2$, and $|z - z'| = |a_z| = \kappa \varepsilon + \varsigma \hat{t} - (\kappa \xi_1 + \xi_2) \leq \kappa \varepsilon + \varsigma \hat{t}$. Letting $\varepsilon, \eta \rightarrow 0^+$ gives $\text{dist}(X_1, X_2) \leq \varsigma \bar{t} < \delta \tan \theta$.

(iii) Take $a = \xi_1$ and $b = t_1$, and use (i) and $0 < t_1 < \delta$. $\square$

Lemma 7.9 (small discs). *Every closed disc of radius $r \leq 10^{-4}$ meets at most 6 squares of $\mathcal{P}$.*

Proof. Let S be a square and $p \in S$, $p = c_S + a u_S + b n_S$ with $|a|, |b| \leq \frac{1}{2}$. Put $\text{sgn } 0 := 1$ and

$$Z_p := \{c_S + a' u_S + b' n_S : a' \text{ between } a \text{ and } a - \frac{1}{2} \text{sgn } a, b' \text{ between } b \text{ and } b - \frac{1}{2} \text{sgn } b\}.$$

If $a \geq 0$, the values a' lie in $[a - \frac{1}{2}, a] \subset [-\frac{1}{2}, \frac{1}{2}]$, and if $a < 0$, in $[a, a + \frac{1}{2}] \subset [-\frac{1}{2}, \frac{1}{2}]$; likewise for b' . So $Z_p \subset S$ is a square of side $\frac{1}{2}$ and area $\frac{1}{4}$ with vertex p , and $Z_p \subset \bar{B}(p, \sqrt{2}/2)$. Let $\bar{B}(o, r)$ be the disc, and for each square S meeting it choose $p_S \in S \cap \bar{B}(o, r)$. The squares Z_{p_S} lie in pairwise disjoint squares, so they are pairwise disjoint, and they lie in $\bar{B}(o, r + \sqrt{2}/2)$. Hence their number is at most $4\pi(r + \sqrt{2}/2)^2 \leq 4\pi \cdot 0.70721^2 < 6.29$. $\square$

7.2. Valley pairs.

Definition 7.10 (valley pairs). A point reached by two paths of F^0 is a *double point*. By Lemma 7.7 its two sources are squares X_1, X_2 ; we label them so that $\varphi_{X_1} > \varphi_{X_2}$ and call $e = \{X_1, X_2\}$ the *valley pair* of the double point. For a valley pair e let Dbl_e be the set of its double points, and put

$$\theta_e := \varphi_{X_1} - \varphi_{X_2}, \quad w_e := \frac{\delta \tan \theta_e}{\cos \theta_e}, \quad \text{Rect}_e := \{v_1 + a u_{X_1} + b n_{X_1} : 0 < a < w_e, 0 < b < \delta\},$$

where v_1 is the upper left vertex of X_1 . Let $\mathcal{Q}_b$ be the set of valley pairs of F^0 , and let $\mathcal{Q}_c$ be the set of valley pairs of the master flow of $\rho\mathcal{P}$, transported to $\mathcal{P}$ by ρ (with Dbl_e and Rect_e transported by ρ ; θ_e and w_e are unchanged).

By Lemma 7.5, $0 < \theta_e < 2\alpha_{\max} \leq 3 \cdot 10^{-6}$, so $\theta_e = \theta(X_1, X_2)$. By Lemma 7.8(iii), $\text{Dbl}_e \subset \text{Rect}_e$; and $|\text{Rect}_e| = \delta w_e$. Both sets $\mathcal{Q}_b, \mathcal{Q}_c$ are finite.

Lemma 7.11 (sums over valley pairs). *Let K be the constant of Lemma 5.5. Then:*

- (a) $\mathcal{Q}_b \cap \mathcal{Q}_c = \emptyset$;
- (b) $\sum_{e \in \mathcal{Q}_b \cup \mathcal{Q}_c} \theta_e \leq \frac{4K}{Q_*(2 - 3 \cdot 10^{-6})} W$;
- (c) $\sum_{e \in \mathcal{Q}_b \cup \mathcal{Q}_c} w_e \leq S_K \delta W$, where $S_9 \leq 56.2501$ and $S_{13} \leq 81.2502$ (rounded up).

Proof. (a) The squares of a pair in $\mathcal{Q}_b$ are sources of points reached by paths of F^0 , hence floor squares; those of a pair in $\mathcal{Q}_c$ are ceiling squares. By Lemma 5.4 no square is both.

(b) Let $e = \{X_1, X_2\} \in \mathcal{Q}_b$. By Lemma 7.8(ii), $\text{dist}(X_1, X_2) < \delta \tan \theta_e < 10^{-5}$, and $\theta_e < 3 \cdot 10^{-6} \leq 10^{-5}$. By [R, Proposition 4.5] with $d = \theta_1 = 10^{-5}$, the waste in $R_e = R(X_1, X_2)$ has area

$$f_e \geq \frac{Q_* \theta_e (2 - \theta_e)}{4} \geq \frac{Q_* (2 - 3 \cdot 10^{-6})}{4} \theta_e.$$

For $e \in \mathcal{Q}_c$ the same holds: apply the argument to $\rho\mathcal{P}$ and use that ρ preserves distances, θ , the rectangles $R(\cdot, \cdot)$ and the waste. By (a), $\mathcal{Q}_b \cup \mathcal{Q}_c$ is a set of pairs of squares at distance $< 10^{-5} \leq 10^{-4}$, so Lemma 5.5 gives $\sum_e f_e \leq KW$, and (b) follows.

(c) For $0 < \theta \leq 3 \cdot 10^{-6}$,

$$\frac{\tan \theta}{\theta \cos \theta} = \frac{\sin \theta}{\theta} \cdot \frac{1}{\cos^2 \theta} \leq \frac{1}{\cos^2(3 \cdot 10^{-6})} \leq 1 + 10^{-11},$$

so $w_e \leq (1 + 10^{-11}) \delta \theta_e$, and (c) follows from (b). Numerically, $4 \cdot 9 / (0.32(2 - 3 \cdot 10^{-6})) = 56.2500843 \dots$ and $4 \cdot 13 / (0.32(2 - 3 \cdot 10^{-6})) = 81.2501218 \dots$; multiplied by $1 + 10^{-11}$ and rounded up they give 56.2501 and 81.2502. $\square$

7.3. Deaths.

Proof of Theorem 7.1(D). Consider first the floor flow.

Cells. By (S2), $\mathcal{D} \setminus \Sigma$ is the union of the pieces I that it contains. Fix such an I . For $x \in I$ the path π_x has the ray steps $j = 0, 1, \dots, J$, where J and the sources X_j do not depend on $x \in I$, with start points $p_j(x)$, cumulative gaps $g_j(x)$ at the start and lengths $t_j^*(x)$, all affine in x . The steps $j < J$ end with contacts that are entry candidates won by π_x and followed by a pass, and step J ends with D. Passes do not change the cumulative gap, so $g_0 = 0$ and $g_{j+1} = g_j + t_j^*$, and $g_J + t_J^* = \delta$ because step J ends at $t_D = \delta - g_J$. For $j = 0, \dots, J$ let

$$C_{I,j} := \{(x, t) : x \in I, g_j(x) < t < g_j(x) + t_j^*(x)\}, \quad \gamma_{I,j}(x, t) := p_j(x) + (t - g_j(x)) n_{X_j}.$$

For fixed $x \in I$ the intervals $(g_j(x), g_j(x) + t_j^*(x))$, $j = 0, \dots, J$, are pairwise disjoint and cover $(0, \delta)$ up to the points $g_1(x), \dots, g_J(x)$. So the sets $C_{I,j}$ are pairwise disjoint, and their union is $I \times (0, \delta)$ up to the graphs of the affine functions $g_1, \dots, g_J$, which are null sets. Summing over I ,

$$(7.4) \quad \sum_{I,j} |C_{I,j}| = \delta |\mathcal{D}|.$$

Jacobian. The map $\gamma_{I,j}$ is affine, with $\partial_x \gamma_{I,j} = p'_j - g'_j n_{X_j} = \lambda_j u_{X_j} - g'_j n_{X_j}$ and $\partial_t \gamma_{I,j} = n_{X_j}$, where $\lambda_j \geq 1$ by (S3). Hence $\det D\gamma_{I,j} = \lambda_j \det[u_{X_j}, n_{X_j}] = \lambda_j$, so $\gamma_{I,j}$ is injective and

$$|\gamma_{I,j}(C_{I,j})| = \lambda_j |C_{I,j}| \geq |C_{I,j}|.$$

Images. For $(x, t) \in C_{I,j}$ the point $\gamma_{I,j}(x, t)$ is reached by π_x from the source X_j with delay $t - g_j(x) \in (0, t_j^*(x))$, that is, in the gap case. By Lemma 5.2(d) it is a waste point of the strip $S_b := (0, k) \times (0, h_{\max})$.

Multiplicity. For a point z let $\text{mult}(z) := \sum_{I,j} \mathbf{1}_{\gamma_{I,j}(C_{I,j})}(z)$; each term equal to 1 corresponds to exactly one preimage $(x, t) \in C_{I,j}$ of z . Two such preimages $(x, t) \in C_{I,j}$ and $(x', t') \in C_{I',j'}$ with $(I, j) \neq (I', j')$ have $x \neq x'$: if $x = x'$, then $I = I'$, and π_x would reach z in the steps j and j' , so $j = j'$ by Lemma 5.2(f). All these x reach z in the gap case, so $\text{mult}(z) \leq M_g(z)$. By Lemma 7.7, $M_g(z) \leq 2$, and if $M_g(z) = 2$, then z is a double point and $z \in \text{Dbl}_e \subset \text{Rect}_e$ for its valley pair $e \in \mathcal{Q}_b$. Thus $\text{mult}(z) \leq 2$; if $\text{mult}(z) \geq 1$, then z is a waste point of S_b by the previous step; and if $\text{mult}(z) = 2$, then $z \in \text{Rect}_e$ for some $e \in \mathcal{Q}_b$. Hence, for every point z ,

$$\text{mult}(z) \leq \mathbf{1}_{\text{waste} \cap S_b}(z) + \sum_{e \in \mathcal{Q}_b} \mathbf{1}_{\text{Rect}_e}(z).$$

Integration. By (7.4) and the two previous steps,

$$\begin{aligned} \delta |\mathcal{D}| &= \sum_{I,j} |C_{I,j}| \leq \sum_{I,j} |\gamma_{I,j}(C_{I,j})| = \int \text{mult}(z) dz \\ &\leq |\text{waste} \cap S_b| + \sum_{e \in \mathcal{Q}_b} |\text{Rect}_e| = |\text{waste} \cap S_b| + \delta \sum_{e \in \mathcal{Q}_b} w_e. \end{aligned}$$

Ceiling flow and conclusion. Applying this to $\rho\mathcal{P}$ and reflecting, $\delta |\tilde{\mathcal{D}}| \leq |\text{waste} \cap S_c| + \delta \sum_{e \in \mathcal{Q}_c} w_e$ with $S_c := \rho(S_b) = (0, k) \times (k - h_{\max}, k)$. Since $h_{\max} < k - h_{\max}$, the strips S_b and S_c are disjoint, so $|\text{waste} \cap S_b| + |\text{waste} \cap S_c| \leq W$. With Lemma 7.11(a),(c),

$$|\mathcal{D}| + |\tilde{\mathcal{D}}| \leq \frac{W}{\delta} + \sum_{e \in \mathcal{Q}_b \cup \mathcal{Q}_c} w_e \leq \frac{W}{\delta} + S_K \delta W = (1 + S_K \delta^2) \frac{W}{\delta}. \quad \square$$

Remark 7.12. Using only $\text{mult} \leq M_g \leq 2$ (Lemma 7.7), the same computation gives $|\mathcal{D}| + |\tilde{\mathcal{D}}| \leq 2W/\delta$ without any result of [R]. The bound $(1 + O(\delta^2))W/\delta$ uses that double points lie in the rectangles Rect_e , together with Lemma 7.11.

7.4. Merges.

Proof of Theorem 7.1(M). Consider the floor flow. For a square Y let $\mathcal{M}_Y$ be the set of $x \in \mathcal{M}$ whose path stops with M at an entry candidate on Y ; then $\mathcal{M}$ is the disjoint union of the sets $\mathcal{M}_Y$. For $x \in \mathcal{M}_Y$ let $q(x) \in \text{relint bot}(Y)$ be that entry candidate.

One loser per entry point. Let $x \in \mathcal{M}_Y$. The paths that reach the entry candidate $(Y, q(x))$ alive are the paths that reach $q(x)$ in the contact case; they include x and the winner of R1 there, which is different from x . By Lemma 7.7 there are exactly two of them, their sources form a valley pair $e \in \mathcal{Q}_b$, and $q(x) \in \text{Dbl}_e$. Since only one of the two paths loses, the map $x \mapsto q(x)$ is injective on $\mathcal{M}_Y$.

Expansion. By (S2) and (S3), $\mathcal{M}_Y \setminus \Sigma$ is a finite union of pieces I , and on each of them q is affine along the line of $\text{bot}(Y)$ with $q' = \mu u_Y$, $\mu \geq 1$. With injectivity,

$$|\mathcal{M}_Y| = \sum_I |I| \leq \sum_I \mathcal{H}^1(q(I)) = \mathcal{H}^1(q(\mathcal{M}_Y \setminus \Sigma)) \leq \sum_e \mathcal{H}^1(\text{bot}(Y) \cap \text{Rect}_e),$$

the last sum over the $e \in \mathcal{Q}_b$ with $\text{Dbl}_e \cap \text{relint bot}(Y) \neq \emptyset$, because $q(x) \in \text{Dbl}_e \subset \text{Rect}_e$.

Length of one intersection. Let $e = \{X_1, X_2\}$. The rectangle Rect_e lies in the strip $\{z : 0 < (z - v_1) \cdot u_{X_1} < w_e\}$. Along the line of $\text{bot}(Y)$, which has the direction u_Y , the function $(z - v_1) \cdot u_{X_1}$ changes at the rate $u_Y \cdot u_{X_1} = \cos(\varphi_Y - \varphi_{X_1})$. Since $|\varphi_{X_1}| < \alpha_{\max}$ and $|\varphi_Y| \leq \pi/4$, we have $|\varphi_Y - \varphi_{X_1}| < \pi/4 + \alpha_{\max}$, so this rate is $> \cos(\pi/4 + \alpha_{\max}) > 0$, and

$$\mathcal{H}^1(\text{bot}(Y) \cap \text{Rect}_e) \leq \frac{w_e}{\cos(\pi/4 + \alpha_{\max})}.$$

At most four squares per pair. Fix $e = \{X_1, X_2\} \in \mathcal{Q}_b$, and let Y be a square with a point $q \in \text{Dbl}_e \cap \text{relint bot}(Y)$. As $q \in \text{Rect}_e$ and $w_e < 3.1 \cdot 10^{-6} \delta$, $|q - v_1| < (w_e^2 + \delta^2)^{1/2} < 1.01 \delta$. By Lemma 7.8(i), $|q - v_2| = (\xi_2^2 + t_2^2)^{1/2} < \delta(\tan^2 \theta_e + 1)^{1/2} < 1.01 \delta$, with v_2 the upper right vertex of X_2 . So the closed disc $\bar{B}(v_1, 3\delta)$ contains $v_1 \in X_1$, $v_2 \in X_2$ and $q \in Y$. Moreover $Y \notin \{X_1, X_2\}$, because $(q - c_{X_i}) \cdot n_{X_i} = \frac{1}{2} + t_i > \frac{1}{2}$ shows $q \notin X_i$. Hence X_1, X_2 and all such squares Y are distinct squares meeting $\bar{B}(v_1, 3\delta)$, a disc of radius $3\delta \leq 3 \cdot 10^{-5}$. By Lemma 7.9 there are at most 6 of them, so at most 4 such squares Y .

Summation. Summing over Y and exchanging the order of summation,

$$\begin{aligned} |\mathcal{M}| &= \sum_Y |\mathcal{M}_Y| \leq \frac{1}{\cos(\pi/4 + \alpha_{\max})} \sum_{e \in \mathcal{Q}_b} w_e \cdot \#\{Y : \text{Dbl}_e \cap \text{relint bot}(Y) \neq \emptyset\} \\ &\leq \frac{4}{\cos(\pi/4 + \alpha_{\max})} \sum_{e \in \mathcal{Q}_b} w_e. \end{aligned}$$

Applying this to $\rho\mathcal{P}$ gives the same bound for $|\tilde{\mathcal{M}}|$ with $\mathcal{Q}_c$. By Lemma 7.11(a),(c),

$$|\mathcal{M}| + |\tilde{\mathcal{M}}| \leq \frac{4}{\cos(\pi/4 + \alpha_{\max})} \sum_{e \in \mathcal{Q}_b \cup \mathcal{Q}_c} w_e \leq \frac{4}{\cos(\pi/4 + \alpha_{\max})} S_K \delta W.$$

Numerically, $4/\cos(\pi/4 + 1.5 \cdot 10^{-6}) = 5.65686273 \dots \leq 5.656863$ (rounded up), and $5.656863 \cdot 56.2501 = 318.1991 \dots \leq 318.20$, $5.656863 \cdot 81.2502 = 459.6212 \dots \leq 459.7$. $\square$

The proof shows more precisely that, for each square Y , the losers at entry candidates on Y whose two sources form the pair e have measure at most $w_e/\cos(\varphi_Y - \varphi_{X_1})$; and the two last gap directions of two merging paths are always different ($\theta_e > 0$, Lemma 7.5).

7.5. Gap overlap.

Proof of Theorem 7.1(G). Fix $y \in (0, h_{\max})$ and consider the floor flow. Only paths alive at y , that is, with termination height $> y$, enter $A_{\text{gap}}(y)$. In particular a path whose cumulative gap reaches δ exactly at a point of height y (a death point at height y) is not counted at y ; this agrees with the

gap case of Definition 5.1, whose delays are $< t^*$, and with the definition of M_g , which counts only points of the open parts of gap segments.

Density. By Lemma 3.29(c), $A_{\text{gap}}(y)$ is the union of finitely many points and finitely many disjoint open intervals J_c ; on each J_c the point $P_y(x)$ lies on the open part of the gap segment of a fixed ray step, X_y is affine with slope $\kappa_c \geq 1$, and

$$\rho_y = \sum_c \kappa_c^{-1} \mathbf{1}_{X_y(J_c)} \leq \sum_c \mathbf{1}_{X_y(J_c)} \quad \text{almost everywhere.}$$

For $z \in \text{Wa}_y$, $\sum_c \mathbf{1}_{X_y(J_c)}(z)$ is the number of $x \in \bigcup_c J_c$ with $P_y(x) = z$, because X_y is injective on each J_c and the intervals J_c are disjoint. For each such x the point z lies on the open part of a gap segment of the trajectory of π_x , that is, π_x reaches z in the gap case; so the sum is at most $M_g(z) \leq 2$ (Lemma 7.7). If it equals 2, then z is a double point and lies in Rect_e for its valley pair $e \in \mathcal{Q}_b$. Hence, almost everywhere on Wa_y ,

$$(\rho_y - 1)_+ \leq \sum_{e \in \mathcal{Q}_b} \mathbf{1}_{\text{Rect}_e}.$$

Length of one intersection. Let $e = \{X_1, X_2\} \in \mathcal{Q}_b$, with v_1 the upper left vertex of X_1 . Along ℓ_y (direction e_1) the function $(z - v_1) \cdot u_{X_1}$ changes at the rate $e_1 \cdot u_{X_1} = \cos \varphi_{X_1} \geq \cos \alpha_{\max}$. So ℓ_y meets the strip $\{0 < (z - v_1) \cdot u_{X_1} < w_e\} \supset \text{Rect}_e$ in a segment of length at most $w_e / \cos \alpha_{\max}$, and

$$\text{Ov}_{\text{gap}}^{F^0}(y) = \int_{\text{Wa}_y} (\rho_y - 1)_+ \leq \sum_{e \in \mathcal{Q}_b} \mathcal{H}^1(\ell_y \cap \text{Rect}_e) \leq \frac{1}{\cos \alpha_{\max}} \sum_{e \in \mathcal{Q}_b} w_e.$$

This holds for every $y \in (0, h_{\max})$, including the heights of horizontal sides of squares and the exceptional heights of Definition 3.32: the waste does not contain the points of the closed squares, and the argument uses no property of y . The bound does not depend on y , so it bounds $\sup_y \text{Ov}_{\text{gap}}^{F^0}(y)$. Applying this to $\rho\mathcal{P}$ gives

$$\sup_y \text{Ov}_{\text{gap}}^{\tilde{F}^0}(y) \leq \frac{1}{\cos \alpha_{\max}} \sum_{e \in \mathcal{Q}_c} w_e,$$

and by Lemma 7.11(a),(c),

$$\sup_y \text{Ov}_{\text{gap}}^{F^0}(y) + \sup_y \text{Ov}_{\text{gap}}^{\tilde{F}^0}(y) \leq \frac{1}{\cos \alpha_{\max}} \sum_{e \in \mathcal{Q}_b \cup \mathcal{Q}_c} w_e \leq \frac{S_K}{\cos \alpha_{\max}} \delta W.$$

Numerically (rounded up),

$$S_9 / \cos(1.5 \cdot 10^{-6}) < 56.2501, \quad S_{13} / \cos(1.5 \cdot 10^{-6}) < 81.2502 \leq 81.26. \quad \square$$

7.6. The bound on Λ_0 and the wall term.

Proof of Corollary 7.2. By definition,

$$\Lambda_0 + \tilde{\Lambda}_0 = (|\mathcal{D}| + |\tilde{\mathcal{D}}|) + (|\mathcal{M}| + |\tilde{\mathcal{M}}|) + (L_E + \tilde{L}_E) + \left(\sup_y \text{Ov}_{\text{gap}}^{F^0}(y) + \sup_y \text{Ov}_{\text{gap}}^{\tilde{F}^0}(y) \right).$$

With $K = 9$, Theorem 7.1 and Proposition 5.10 give

$$\begin{aligned} \Lambda_0 + \tilde{\Lambda}_0 &\leq (1 + 56.2501 \delta^2) \frac{W}{\delta} + (318.20 + 754.8 + 56.2501) \delta W \\ &= (1 + 1185.5002 \delta^2) \frac{W}{\delta} \leq (1 + 1185.6 \delta^2) \frac{W}{\delta}. \end{aligned}$$

With $K = 13$ the same computation gives $81.26 + 459.7 + 1090.3 + 81.26 = 1712.52 \leq 1712.6$. Since $\delta \leq 10^{-5}$, $1185.6 \delta^2 \leq 1.1856 \cdot 10^{-7}$ and $1712.6 \delta^2 \leq 1.7126 \cdot 10^{-7}$, both less than $2 \cdot 10^{-7}$. $\square$

Remark 7.13 (the wall term). The quantity Λ_0 collects the losses of the master flow that are bounded in terms of W . The loss term $\Lambda^{F_s}(y)$ of a scale flow F_s (Section 3) also contains the wall loss of F_s , which is not bounded in terms of W . Proposition 6.10 bounds $\Lambda^{F_s}(y)$, for $s \in [y_0, y_1]$ and $0 < y < s + 2$, by $\Lambda_0 + 2(s + 2)\tan \alpha(s)$; the wall term $2(s + 2)\tan \alpha(s)$ is not part of Λ_0 , and it is treated separately in Section 10.

Remark 7.14 (dependence on [R]). The bounds of Theorem 7.1 with $K = 9$, and hence the constant $C_\Lambda = 1 + 1185.6\delta^2$ of Corollary 7.2, use [R, Proposition 4.5] and [R, Lemma 4.10] (through Lemma 5.5), together with [R, Lemma 4.8(a)] for L_E (Proposition 5.10); [R, Lemma 4.10] is computer-assisted in [R] (Remark 2.2). The bounds with $K = 13$, and $C_\Lambda^{(13)} = 1 + 1712.6\delta^2$, use [R, Lemma 4.9] in place of [R, Lemma 4.10] and do not depend on the computer-assisted lemma.

Remark 7.15 (degenerate cases). The lemmas of Section 7.1 are statements about single points, so contacts at vertices and rays running along a side of a square (which occur for finitely many x , by (S2)) do not affect them. Ties between the event parameters, which may occur for sets of x of positive measure, are decided by the priority rule, and the delay bound of Lemma 5.2(a) holds also at ties. A square resting on the floor gives contacts with delay 0 from the floor, which Lemma 7.4 allows (it only uses delays $< \delta$); a square source always has a positive delay. Contacts at height exactly $h_{\max}$ are processed, and Lemma 5.4 uses only $q_y \leq h_{\max}$. Ties of the cumulative gap at an entry candidate are decided by R1 through the floor abscissa; they do not enter the bounds above, which only use that each entry candidate has exactly one winner. Squares with equal centre heights cause no difficulty (Lemma 3.12). Squares of phase $\pi/4$ are never sources ($\alpha_{\max} < \pi/4$). Two distinct squares with the same phase are never the two sources of one point (Lemma 7.5). No ray step starts on a side wall of the container (Lemma 3.17(b)). The bound (G) holds at every height, including heights of horizontal sides of squares.

8. P6: THE PATH VERSION OF [R, LEMMA 4.14]

This section establishes Theorem 8.3, the analogue of [R, Lemma 4.14] for the master flows of Section 3, and its scale form, Corollary 8.13, which is the form used in Section 10. In [R, Lemma 4.14] a column is a vertical segment with vertical waste less than δ , and two squares that are consecutive on it are consecutive on the whole vertical line. The paths of F^0 follow instead the directions n_X of the squares they pass, so two consecutively entered squares need not be consecutive on vertical lines. The transfer to vertical lines is Lemma 8.7 (the vertical comparison P6.V). It rests on Lemma 8.6: for a pair of consecutively entered squares, the whole region swept by the gap segments over the convex hull of the exit points is waste. The rest of the argument follows the proof of [R, Lemma 4.14]: a triangle inequality along each path, the slit inequality, the waste in the rectangles of [R, Proposition 4.5] and [R, Lemma 4.8(a)], the multiplicity bound of Proposition 2.1, and the Cauchy–Schwarz inequality.

8.1. Setting, notation, and facts used from Section 3. Throughout this section $k \geq 4$ is an integer, $\mathcal{P}$ is a closed packing of $[0, k]^2$ with waste area $W = k^2 - N$, the parameters satisfy (C0) and (C1) of Definition 3.2, so that

$$(8.1) \quad \alpha_{\max} = \alpha(y_0) \leq \bar{\beta} = 1.5 \cdot 10^{-6}, \quad h_{\max} = \frac{k}{2} - 1,$$

$0 < \delta \leq 10^{-5}$, and $F^0 = F(\alpha_{\max}, \delta, h_{\max})$ is the master flow of Definition 3.20. For $x \in (0, k)$, π_x^0 denotes the path of F^0 that starts at $(x, 0)$. The constants A, A', A_{13}, A'_{13} are those of Definition 3.6:

$$A = \frac{4\sqrt{K_w^*/Q_*}}{2 - 3 \cdot 10^{-6}}, \quad A' = \frac{A}{\cos \bar{\beta}} + 10^{-8}, \quad A_{13} = \frac{4\sqrt{K_w/Q_*}}{2 - 3 \cdot 10^{-6}}, \quad A'_{13} = \frac{A_{13}}{\cos \bar{\beta}} + 10^{-8},$$

with $Q_* = 0.32$, $K_w^* = 9$, $K_w = 13$; A is the constant of [R, Lemma 4.13].

Ceiling versions. The ceiling master flow is $\tilde{F}^0 = \rho \circ F^0[\rho\mathcal{P}] \circ \rho$, $\rho(x, y) = (x, k - y)$ (Definition 3.24). By Lemma 3.8, $\rho\mathcal{P}$ is a closed packing with the same waste area W , and ρ preserves inclinations, the function θ below, distances, vertical lines and lengths, and maps the waste of $\rho\mathcal{P}$ onto the waste of $\mathcal{P}$. Every statement below that is established for F^0 and an arbitrary closed packing therefore applies to $\rho\mathcal{P}$, and transported by ρ it gives the corresponding statement for $\tilde{F}^0$, its *ceiling version*. Ceiling objects carry a tilde.

Notation. We use the phases $\varphi_S \in (-\frac{\pi}{4}, \frac{\pi}{4}]$ (with the 45° convention), the frames u_S, n_S , the centres c_S , the sides $\text{top}(S)$, $\text{bot}(S)$ and their relative interiors, and the floor X_0 (with $\varphi_{X_0} = 0$, $u_{X_0} = e_1$, $n_{X_0} = e_2$, “top side” $[0, k] \times \{0\}$) of Definition 3.7, and the identities (3.1). Thus $\text{top}(S) = \{c_S + \frac{1}{2}n_S + \xi u_S : |\xi| \leq \frac{1}{2}\}$, $\text{bot}(S) = \{c_S - \frac{1}{2}n_S + \xi u_S : |\xi| \leq \frac{1}{2}\}$, the relative interiors are given by $|\xi| < \frac{1}{2}$, and $a(S) = |\varphi_S|$. For points $\xi \in \mathbb{R}^2$ we call $\xi \cdot e_1$ the abscissa and $\xi \cdot e_2$ the height of ξ . For squares S_1, S_2 , $\theta(S_1, S_2)$ is the distance of φ_{S_1} and φ_{S_2} in $\mathbb{R}/\frac{\pi}{2}\mathbb{Z}$, and, as in [R], $\theta(X_0, S) := a(S) = \theta(S, S_0)$, which is the same distance with $\varphi_{X_0} = 0$. Hence θ satisfies the triangle inequality on $\mathcal{P} \cup \{X_0\}$, and $\theta(S_1, S_2) \leq |\varphi_{S_2} - \varphi_{S_1}|$.

Let S be a square with inclination $a = a(S)$ and lowest point at height $v_S \geq 0$. Its vertical extent is $[v_S, v_S + \cos a + \sin a] \subset [0, k]$ by [R, Lemma 3.1(a)], and $c_S \cdot e_2 = v_S + \frac{1}{2}(\cos a + \sin a)$ because S is symmetric about its centre. By (3.1), for $|\xi| \leq \frac{1}{2}$,

$$(c_S \pm \frac{1}{2}n_S + \xi u_S) \cdot e_2 = c_S \cdot e_2 \pm \frac{1}{2} \cos \varphi_S + \xi \sin \varphi_S, \quad |\xi \sin \varphi_S| \leq \frac{1}{2} \sin a;$$

hence

$$(8.2) \quad \xi \cdot e_2 \geq v_S + \cos a > 0 \quad (\xi \in \text{top}(S)), \quad \xi \cdot e_2 \leq v_S + \sin a \leq k - \cos a < k \quad (\xi \in \text{bot}(S)).$$

Entries. Following Definition 3.20, the *R1-passed entries* of π_x^0 are the entry candidates (Y, q) of π_x^0 (Definition 3.9: $q \in \text{relint bot}(Y)$ is a contact point of a processed ray step) at which x is the winner (Definition 3.14). We list them as $(Z_1, q_1), \dots, (Z_J, q_J)$, $J = J(x) \geq 0$, call them simply the *entries* of π_x^0 , and say that π_x^0 *enters* Z_j at q_j . For $j < J$ the path passes Z_j (Definition 3.10): it traverses the pass segment $[q_j, q_j + n_{Z_j}]$ and leaves Z_j at the *exit point* $q_j + n_{Z_j} \in \text{relint top}(Z_j)$; Z_J is passed or T-terminates π_x^0 . Contacts with a left or right side, or with a vertex, are end collisions and never entries.

We use the following facts from Section 3.

- (F1) (Lemma 3.15(c).) At most one path of F^0 passes, or is T-terminated at, a given entry point (Y, q) .
- (F2) (Lemma 3.12.) The squares $Z_1, \dots, Z_J$ entered by one path have strictly increasing centre heights; in particular they are pairwise distinct.
- (F3) (Lemmas 3.18(a) and 3.19(a).) There is a finite set $\Sigma \subset [0, k]$ such that on each component (*piece*) of $[0, k] \setminus \Sigma$ the combinatorial type of π_x^0 is constant; in particular its sequence of entries is constant, and so is, for each entry, whether the path passes or is T-terminated. On a piece, the start point $p_j(x)$ of the j -th ray step is an affine function of x with $p'_j(x) = \lambda_j u_{X_j}$, where X_j is the source of that step (X_0 the floor) and $\lambda_j = \prod_{i=1}^j \sec(\varphi_{X_i} - \varphi_{X_{i-1}}) \geq 1$. Each factor is positive, because $|\varphi_{X_i} - \varphi_{X_{i-1}}| \leq \frac{\pi}{4} + |\varphi_{X_{i-1}}| < \frac{\pi}{2}$; so the slope λ_j is positive.

The next lemma recalls what one ray step between two consecutive entries gives.

Lemma 8.1 (one step). *Let $x \in (0, k)$, let $Z_1, \dots, Z_J$ be the squares entered by π_x^0 , in order, and put $Z_0 := X_0$. Fix $1 \leq i \leq J$ and write $X := Z_{i-1}$, $Y := Z_i$. Let p_x be the point where π_x^0 leaves X (its exit point from X if $i \geq 2$, and $p_x = (x, 0)$ if $i = 1$), and let $q_x := q_i$ be its entry point into Y . Then:*

- (a) $q_x = p_x + t_x n_X$ with $0 \leq t_x < \delta$, and $t_x > 0$ if X is a square;
- (b) $q_x \cdot e_2 \leq h_{\max}$; if X is a square, its entry point q_{i-1} also has height $\leq h_{\max}$;
- (c) the open segment $\{p_x + s n_X : 0 < s < t_x\}$ lies in the waste; if $X = X_0$, then the point $p_x = (x, 0)$ lies in no square other than Y ;

(d) if X is a square, then $a(X) < \alpha_{\max} \leq \bar{\beta}$; in all cases $|\varphi_X| < \bar{\beta}$.

Proof. If $i \geq 2$, then π_x^0 has an entry after the one into X , so it passes X , and $a(X) < \alpha_{\max}$ by Definition 3.10; this gives (d), as $\varphi_{X_0} = 0 < \bar{\beta}$. After leaving X at p_x , the path performs a ray step $r(t) = p_x + tn_X$ with source X ; since it has a further entry, this step ends with a processed contact at an entry candidate at which x wins, and this is the entry into Y . So $q_x = r(t_S)$, and we put $t_x := t_S = t^*$. At a processed contact the cumulative gap is $< \delta$ (Lemma 3.17(f)), so $t_x < \delta$; and $t_S > 0$ if the source is a square (Lemma 3.17(b)). This gives (a). A contact is processed only if $t_S = t^* \leq t_H$ (Definition 3.9); as the height increases along the ray at the rate $n_X \cdot e_2 = \cos \varphi_X > 0$, the contact point then has height at most $p_x \cdot e_2 + t_H \cos \varphi_X = h_F = h_{\max}$. The entry into X (if X is a square) is also a processed contact, so (b) follows. The open part of the gap segment $[p_x, q_x]$ lies in the waste by Lemma 3.17(c). If $X = X_0$ and $(x, 0)$ lies in a square Z , then $t_S = 0$, so $q_x = (x, 0)$, and $Z = Y$ because q_x lies in exactly one square (Lemma 3.17(d)). This gives (c). $\square$

Definition 8.2 (consecutive entry pairs). In the notation of Lemma 8.1, the ordered pairs (Z_{i-1}, Z_i) , $1 \leq i \leq J$, are the *consecutive entry pairs* of π_x^0 ; Z_{i-1} is the *first* and Z_i the *second member*. For an ordered pair $e = (X, Y)$ with $X \in \mathcal{P} \cup \{X_0\}$ and $Y \in \mathcal{P}$ put

$$P_e := \{x \in (0, k) : e \text{ is a consecutive entry pair of } \pi_x^0\}, \quad m_e := |P_e|,$$

and let $\mathcal{C} := \{e : P_e \neq \emptyset\}$. For $e = (X, Y)$ put $\theta_e := \theta(X, Y)$ (so $\theta_e = a(Y)$ if $X = X_0$) and $\theta'_e := \min(\theta_e, \bar{\beta})$. The ceiling objects $\tilde{\mathcal{C}}$ and, for $e \in \tilde{\mathcal{C}}$, P_e , m_e , θ_e , θ'_e are defined in the same way from $\tilde{F}^0$; the first member of $e \in \tilde{\mathcal{C}}$ is a square or the ceiling pseudo-square $\tilde{X}_0 = \rho X_0$, whose “side” is $[0, k] \times \{k\}$, and $\theta(\tilde{X}_0, Y) := a(Y)$.

Since $\mathcal{P}$ is finite, $\mathcal{C}$ and $\tilde{\mathcal{C}}$ are finite. By (F2) the consecutive entry pairs of one path are pairwise distinct, as their second members are.

Theorem 8.3 (path version of [R, Lemma 4.14]). *Let $k \geq 4$ be an integer, $\mathcal{P}$ a closed packing of $[0, k]^2$ with waste area W , $0 < \delta \leq 10^{-5}$, and assume (C0) and (C1). Let $\mathcal{X} \subset (0, k)$ be a Lebesgue measurable set and $\zeta : \mathcal{X} \rightarrow [0, \bar{\beta}]$ a measurable function such that for every $x \in \mathcal{X}$ the path π_x^0 enters a square Z_x (that is, one of its $R1$ -passed entries is an entry into Z_x ; Z_x may be passed or T -terminate the path) with $a(Z_x) \geq \zeta(x)$. Let $\mathcal{X}' \subset (0, k)$ and $\zeta' : \mathcal{X}' \rightarrow [0, \bar{\beta}]$ satisfy the same with $\tilde{F}^0$ in place of F^0 .*

$$(a) \quad \int_{\mathcal{X}} \zeta(x) dx + \int_{\mathcal{X}'} \zeta'(x) dx \leq A'W, \text{ where } A' = \frac{A}{\cos \bar{\beta}} + 10^{-8} < 10.6068.$$

(b) More strongly, independently of $\mathcal{X}, \zeta, \mathcal{X}', \zeta'$,

$$\sum_{e \in \mathcal{C} \cup \tilde{\mathcal{C}}} \theta'_e m_e \leq \left(\frac{A}{\cos \bar{\beta}} + 8.5 \cdot 10^{-10} \right) W.$$

(c) Without [R, Lemma 4.10], using [R, Lemma 4.9] in its place, (a) and (b) hold with A replaced by $A_{13} < 12.7476$, A' by $A'_{13} < 12.7476$, and $8.5 \cdot 10^{-10}$ by $1.22 \cdot 10^{-9}$.

All numerical bounds are rounded up; $A < 10.6067$.

Remark 8.4 (dependence on [R, Lemma 4.10]). Parts (a) and (b) use the multiplicity bound $K_w^* = 9$ of Proposition 2.1, which rests on the proof of [R, Lemma 4.10] and [R, Remark 4.11]; the proof of [R, Lemma 4.10] is computer-assisted (Remark 2.2). Part (c) uses instead the bound $K_w = 13$ of Proposition 2.1, which rests on the analytic [R, Lemma 4.9], and does not use [R, Lemma 4.10]. Apart from this, the proofs of (a)–(c) are the same. Through Corollary 8.13, part (a) enters the proof of Theorem A and part (c) the proof of Theorem A₁₃ (Section 11).

The rest of the section contains the density lemma (§8.2), the hull slit (§8.3), the vertical comparison (§8.4), the disjointness of the vertical slits (§8.5), the slit inequality, the pair waste and the multiplicity (§8.6), the proof of Theorem 8.3 (§8.7), the scale form (§8.8), and remarks on the rules and on degenerate cases (§8.9).

8.2. Density. For $e = (X, Y) \in \mathcal{C}$ put $o_X := c_X + \frac{1}{2}n_X$ if X is a square and $o_{X_0} := (0, 0)$. For $x \in P_e$ let p_x be the point where π_x^0 leaves X (Lemma 8.1), and define the *exit coordinate* $w_X(x)$ by $p_x = o_X + w_X(x)u_X$. Thus $|w_X(x)| < \frac{1}{2}$ if X is a square, and $w_{X_0}(x) = x$. Put $I_e := w_X(P_e)$.

Lemma 8.5 (density). *Let $e = (X, Y) \in \mathcal{C}$. Then P_e and I_e are Borel sets, w_X is injective on P_e , and $|I_e| \geq m_e$.*

Proof. Pieces. By (F3) the sequence of entries of π_x^0 is constant on each piece, so each piece is either contained in P_e or disjoint from it. Hence P_e is the union of the pieces $I \subset P_e$ and of a subset of the finite set Σ ; it is a Borel set, and $m_e = \sum_{I \subset P_e} |I|$. If $X = X_0$, then w_X is the identity and $I_e = P_e$, and there is nothing more to show. Let X be a square and $I \subset P_e$ a piece. On I , X is the source X_j of a fixed ray step j , and $p_x = p_j(x)$; by (F3), $\frac{d}{dx}w_X(x) = p'_j(x) \cdot u_X = \lambda_j \geq 1$. So w_X is affine on I with positive slope λ_j , and $w_X(I)$ is an interval of length $\lambda_j|I| \geq |I|$.

Injectivity. Let $x \neq x'$ be points of P_e with $w_X(x) = w_X(x')$. Both paths pass X and leave it at the same exit point; since the exit point is the entry point plus n_X , they entered X at the same point $q := c_X - \frac{1}{2}n_X + w_X(x)u_X$. Two paths would then pass X from the same entry point (X, q) , contradicting (F1). So w_X is injective on P_e .

Measure. By injectivity, the intervals $w_X(I)$, $I \subset P_e$ a piece, are pairwise disjoint, and I_e is their union together with the finitely many points $w_X(P_e \cap \Sigma)$. Hence I_e is a Borel set and $|I_e| = \sum_I \lambda_{j(I)}|I| \geq \sum_I |I| = m_e$. $\square$

8.3. The frame of a pair and the hull slit. Fix $e = (X, Y) \in \mathcal{C}$ and write $u := u_X$, $n := n_X$, $\varphi := \varphi_X$, $o := o_X$. Every point of the plane is $o + wu + sn$ for a unique $(w, s) \in \mathbb{R}^2$, and $(w, s) \mapsto o + wu + sn$ is an isometry. (In this subsection and in Lemma 8.1(c), s denotes a coordinate or a length parameter, not a scale; the lines ℓ_w below are lines in the direction n_X , not horizontal lines ℓ_y .) Put $\ell_w := \{o + wu + sn : s \in \mathbb{R}\}$ and let $L_Y := \{\xi : (\xi - c_Y) \cdot n_Y = -\frac{1}{2}\}$ be the line of $\text{bot}(Y)$. By Lemma 8.1(d), $|\varphi| < \bar{\beta}$, so $|\varphi_Y - \varphi| < \frac{\pi}{4} + \bar{\beta} < \frac{\pi}{2}$ and, by (3.1), $n \cdot n_Y = \cos(\varphi_Y - \varphi) > 0$. Hence ℓ_w meets L_Y in exactly one point $o + wu + t(w)n$, where

$$(8.3) \quad t(w) = t_0 + \kappa_e w, \quad \kappa_e := -\frac{u \cdot n_Y}{n \cdot n_Y} = \tan(\varphi_Y - \varphi), \quad |\kappa_e| < \tan\left(\frac{\pi}{4} + \bar{\beta}\right) < 1.00001,$$

for a constant t_0 (the last bound is rounded up). Along ℓ_w , in the direction n , the function $\xi \mapsto (\xi - c_Y) \cdot n_Y$ increases at the rate $n \cdot n_Y > 0$; since $Y \subset \{\xi : (\xi - c_Y) \cdot n_Y \geq -\frac{1}{2}\}$,

$$(8.4) \quad \text{no point } o + wu + sn \text{ with } s < t(w) \text{ belongs to } Y.$$

For $x \in P_e$, Lemma 8.1(a) gives $q_x = o + w_X(x)u + t_x n \in \text{bot}(Y) \subset L_Y$; hence

$$(8.5) \quad t(w_X(x)) = t_x \in [0, \delta], \quad \text{and} \quad t(w_X(x)) > 0 \text{ if } X \text{ is a square.}$$

Let $w_1 := \inf I_e$ and $w_2 := \sup I_e$. Then $[w_1, w_2] \subset [-\frac{1}{2}, \frac{1}{2}]$ if X is a square, and $[w_1, w_2] \subset [0, k]$ if $X = X_0$.

Lemma 8.6 (the hull slit is waste). *Let $e = (X, Y) \in \mathcal{C}$.*

- (i) *For every $w \in [w_1, w_2]$: $o + wu \in \text{top}(X)$ ($o + wu = (w, 0) \in [0, k] \times \{0\}$ if $X = X_0$), $o + wu + t(w)n \in \text{bot}(Y)$, and $0 \leq t(w) \leq \delta$. For every $w \in (w_1, w_2)$ moreover $o + wu \in \text{relint top}(X)$ ($0 < w < k$ if $X = X_0$), $o + wu + t(w)n \in \text{relint bot}(Y)$, $t(w) < \delta$, and $t(w) > 0$ if X is a square.*

- (ii) *The set*

$$\mathcal{O}_e := \{o + wu + sn : w_1 < w < w_2, 0 < s < t(w)\}$$

is open and contained in the waste.

Proof. (i) Let $w \in (w_1, w_2)$. By the definition of w_1 and w_2 there are $p_1, p_2 \in I_e$ with $p_1 < w < p_2$; write $w = (1 - \mu)p_1 + \mu p_2$ with $0 < \mu < 1$, and $p_j = w_X(x_j)$ with $x_j \in P_e$. By Lemma 8.1(a) and (8.5), $o + p_j u = p_{x_j} \in \text{relint top}(X)$ (respectively $0 < p_j < k$ if $X = X_0$),

$o + p_j u + t(p_j)n = q_{x_j} \in \text{relint bot}(Y)$, and $0 \leq t(p_j) < \delta$, with $t(p_j) > 0$ if X is a square. The maps $w' \mapsto o + w'u$ and $w' \mapsto o + w'u + t(w')n$ are affine, and $\text{relint top}(X)$, $\text{relint bot}(Y)$ and $(0, k)$ are convex. So the images of w lie in $\text{relint top}(X)$ (respectively $0 < w < k$) and in $\text{relint bot}(Y)$, and $t(w) = (1 - \mu)t(p_1) + \mu t(p_2)$ lies in $[0, \delta)$, and in $(0, \delta)$ if X is a square. For $w \in I_e$ the claims of the first sentence hold by Lemma 8.1(a) and (8.5). If $w_1 = w_2$, then $I_e = \{w_1\}$ and we are done. If $w_1 < w_2$, each endpoint w_j is a limit of points of (w_1, w_2) ; since $\text{top}(X)$, $[0, k] \times \{0\}$ and $\text{bot}(Y)$ are closed and t is continuous, we obtain $o + w_j u \in \text{top}(X)$ (respectively $\in [0, k] \times \{0\}$), $o + w_j u + t(w_j)n \in \text{bot}(Y)$ and $0 \leq t(w_j) \leq \delta$.

(ii) The set $\{(w, s) : w_1 < w < w_2, 0 < s < t(w)\}$ is open since t is continuous, and $\mathcal{O}_e$ is its image under an isometry, so $\mathcal{O}_e$ is open. Each point $o + wu + sn$ of $\mathcal{O}_e$ lies on the segment from $o + wu \in [0, k]^2$ to $o + wu + t(w)n \in \text{bot}(Y) \subset [0, k]^2$ (by (i)), so $\mathcal{O}_e \subset [0, k]^2$; being open, $\mathcal{O}_e \subset (0, k)^2$. If X is a square, then $X \subset \{\xi : (\xi - c_X) \cdot n \leq \frac{1}{2}\}$ while $(o + wu + sn - c_X) \cdot n = \frac{1}{2} + s > \frac{1}{2}$, so $\mathcal{O}_e \cap X = \emptyset$. By (8.4), $\mathcal{O}_e \cap Y = \emptyset$.

Suppose now that a square Z , with $Z \neq Y$ and $Z \neq X$, contains a point $v = o + wu + sn \in \mathcal{O}_e$, so $w_1 < w < w_2$ and $0 < s < t(w)$.

Claim 1. For every $p \in [w_1, w_2]$ and every point $z = o + pu + \sigma n$ of Z we have $\sigma < t(p)$; moreover $\sigma > 0$ if X is a square, and $\sigma \geq 0$ if $X = X_0$. Let X be a square. The disjoint compact convex sets X and Z are strictly separated by a line: there are $\nu \in \mathbb{R}^2$ and $c \in \mathbb{R}$ with $\nu \cdot \xi < c$ for $\xi \in X$ and $\nu \cdot \xi > c$ for $\xi \in Z$. By (i), $o + wu \in X$, and $v \in Z$, so $s(\nu \cdot n) = \nu \cdot v - \nu \cdot (o + wu) > 0$, whence $\nu \cdot n > 0$. By (i), $o + pu \in X$ for every $p \in [w_1, w_2]$, including the endpoints (this is where the closed hull is needed), so $\sigma(\nu \cdot n) = \nu \cdot z - \nu \cdot (o + pu) > 0$, and $\sigma > 0$. If $X = X_0$, then $o = (0, 0)$, $n = e_2$, and $\sigma = z \cdot e_2 \geq 0$ because $Z \subset [0, k]^2$. For the upper bound, separate Z and Y in the same way: $\nu' \cdot \xi < c'$ on Z and $\nu' \cdot \xi > c'$ on Y . By (i), $o + wu + t(w)n \in Y$; as $(o + wu + t(w)n) - v = (t(w) - s)n$ with $t(w) - s > 0$, we get $\nu' \cdot n > 0$. By (i), $o + pu + t(p)n \in Y$ for every $p \in [w_1, w_2]$, so $(t(p) - \sigma)(\nu' \cdot n) = \nu' \cdot (o + pu + t(p)n) - \nu' \cdot z > 0$, and $\sigma < t(p)$.

Claim 2. $Z \cap \ell_p = \emptyset$ for every $p \in I_e$; consequently, if $p_1, p_2 \in I_e$ and $p_1 < w < p_2$, then every point of Z has the form $o + w'u + \sigma n$ with $p_1 < w' < p_2$. Let $p = w_X(x)$ with $x \in P_e$, and let $z = o + pu + \sigma n \in Z$. By Claim 1 and (8.5), $\sigma < t(p) = t_x$. If $\sigma > 0$, then $z = p_x + \sigma n$ lies on the open segment of Lemma 8.1(c), which lies in the waste; this is impossible for a point of Z . If $\sigma \leq 0$, then by Claim 1 $X = X_0$ and $\sigma = 0$, so $z = (p, 0) = p_x$ lies in the square $Z \neq Y$, which contradicts Lemma 8.1(c). This proves the first assertion. The function $\xi \mapsto (\xi - o) \cdot u$ is continuous on the connected set Z , so its image is an interval; it contains w (the value at v) and, by the first assertion, neither p_1 nor p_2 . Hence it lies in (p_1, p_2) .

Claim 3: a contradiction. Choose $p_1, p_2 \in I_e$ with $p_1 < w < p_2$ (possible since $w_1 < w < w_2$). By Claim 2, every $z \in Z$ is $o + w'u + \sigma n$ with $w' \in (p_1, p_2) \subset (w_1, w_2)$, and then Claim 1 and (i) give $0 \leq \sigma < t(w') < \delta$. So the set $\{(z - o) \cdot n : z \in Z\}$ is contained in $[0, \delta)$, an interval of length $\delta < 1$. But for every unit vector $\mathbf{m}$ the set $\{\mathbf{m} \cdot z : z \in Z\}$ is an interval of length $|\mathbf{m} \cdot u_Z| + |\mathbf{m} \cdot n_Z| \geq (\mathbf{m} \cdot u_Z)^2 + (\mathbf{m} \cdot n_Z)^2 = 1$: every orthogonal projection of a unit square to a line has length at least 1. With $\mathbf{m} = n$ this is a contradiction. Hence no square meets $\mathcal{O}_e$, and $\mathcal{O}_e$ lies in the waste. $\square$

The lemma concerns the geometry of the hull only: points of $(w_1, w_2) \setminus I_e$ need not be exit coordinates of any path of P_e , and paths leaving X there may have been terminated later. Claim 3 plays the role, for the direction n_X , of the middle-chord argument in the proof of [R, Lemma 4.12(a)].

8.4. The vertical comparison. For $e = (X, Y) \in \mathcal{C}$ let $A_e := X$ if X is a square and $A_e := [0, k] \times \{0\}$ if $X = X_0$, and let $V_{x'} := \{x'\} \times \mathbb{R}$. Let E_e^v be the set of $x' \in [0, k]$ such that $V_{x'}$ meets A_e and Y , every point of $A_e \cap V_{x'}$ has height at most the height of every point of $Y \cap V_{x'}$, and the open segment between the highest point of $A_e \cap V_{x'}$ and the lowest point of $Y \cap V_{x'}$ meets no square; for $x' \in E_e^v$ let $g_e^v(x')$ be the length of this segment. This is the set E_{AB} and the gap function of

the vertical version of [R, Lemma 4.12] with $A = A_e$ below and $B = Y$ above. Put

$$\chi_e := \cos \varphi_X - \kappa_e \sin \varphi_X, \quad \Delta_e := \frac{\delta |\sin \varphi_X|}{\chi_e}.$$

(Δ_e is a length; it is unrelated to the phase difference Δ used in Section 5.)

Lemma 8.7 (vertical comparison P6.V). *Let $e = (X, Y) \in \mathcal{C}$. Then $\chi_e > 0.99999849$ (rounded down), and there is an open interval $J'_e \subset E_e^v \cap (0, k)$, possibly empty, with the following properties.*

- (a) *For $x' \in J'_e$, the gap between A_e and Y on $V_{x'}$ (as in the definition of E_e^v) is the open segment from a point $b(x')$ to the point $c(x') = b(x') + g_e^v(x') e_2$, where $b(x') \in \text{relint top}(X)$ is the highest point of $X \cap V_{x'}$ ($b(x') = (x', 0)$ if $X = X_0$) and $c(x') \in \text{relint bot}(Y)$ is the lowest point of $Y \cap V_{x'}$; this open segment lies in $\mathcal{O}_e$, and $0 \leq g_e^v(x') < \delta/\chi_e < 1.0000016 \delta$ (rounded up).*
- (b) *On J'_e the function g_e^v is affine with slope $\tan \varphi_Y - \tan \varphi_X$. Either $g_e^v > 0$ on J'_e , or $g_e^v \equiv 0$ on J'_e ; the latter happens only if $X = X_0$, $\varphi_Y = 0$ and $\text{bot}(Y) \subset [0, k] \times \{0\}$, and then $\theta_e = 0$.*
- (c) $|J'_e| \geq \cos \varphi_X (m_e - \Delta_e) \geq \cos \bar{\beta} \cdot m_e - 1.0000016 \delta \bar{\beta}$ (rounded up).

The ceiling version holds for $e \in \tilde{\mathcal{C}}$, with “highest” and “lowest” exchanged: there the gap is the open vertical segment between the highest point of the second member (below) and the lowest point of the first member (above), or the point (x', k) if the first member is $\tilde{X}_0$.

Proof. Write $\varphi := \varphi_X$, $\kappa := \kappa_e$, $\chi := \chi_e$, and use the frame of §8.3.

The constants. By Lemma 8.1(d), $|\varphi| < \bar{\beta} = 1.5 \cdot 10^{-6}$, and $|\kappa| < 1.00001$ by (8.3). Hence $\chi \geq \cos \varphi - |\kappa| |\sin \varphi| > \cos(1.5 \cdot 10^{-6}) - 1.00001 \cdot 1.5 \cdot 10^{-6} > 0.99999849$, and $1/\chi < 1/0.99999849 < 1.0000016$.

Construction. For $w \in (w_1, w_2)$ put $b_w := o + wu$, $\lambda_w := t(w)/\chi \geq 0$ and $c_w := b_w + \lambda_w e_2$. By (3.1), $c_w = o + \hat{w}(w)u + \lambda_w \cos \varphi n$ with $\hat{w}(w) := w + \lambda_w \sin \varphi$. Since t is affine, $t(\hat{w}(w)) = t(w) + \kappa \lambda_w \sin \varphi = \lambda_w (\chi + \kappa \sin \varphi) = \lambda_w \cos \varphi$, so $c_w = o + \hat{w}(w)u + t(\hat{w}(w))n \in L_Y$. Moreover

$$\hat{w}(w) = w + \frac{(t_0 + \kappa w) \sin \varphi}{\chi} = \frac{\cos \varphi}{\chi} w + \frac{t_0 \sin \varphi}{\chi}$$

is an increasing affine function of w . Let $G_e := \{w \in (w_1, w_2) : \hat{w}(w) \in (w_1, w_2)\}$, an open interval (possibly empty) as the intersection of two open intervals, and

$$J'_e := \{b_w \cdot e_1 : w \in G_e\} = \{o \cdot e_1 + w \cos \varphi : w \in G_e\},$$

an open interval with $|J'_e| = \cos \varphi |G_e|$. For $x' \in J'_e$ let $w(x') \in G_e$ be the unique w with $b_w \cdot e_1 = x'$, and put $b(x') := b_{w(x')}$, $c(x') := c_{w(x')}$.

(a). Fix $w \in G_e$ and $0 < \mu < \lambda_w$. By (3.1), $b_w + \mu e_2 = o + (w + \mu \sin \varphi)u + \mu \cos \varphi n$. The first coordinate lies between w and $\hat{w}(w)$, both in (w_1, w_2) , hence in (w_1, w_2) . The second satisfies $\mu \cos \varphi > 0$, and $\mu \cos \varphi < t(w + \mu \sin \varphi) = t(w) + \kappa \mu \sin \varphi$, because this is equivalent to $\mu \chi < t(w)$, that is, to $\mu < \lambda_w$. So $b_w + \mu e_2 \in \mathcal{O}_e$, and the open vertical segment from b_w to c_w lies in $\mathcal{O}_e$, hence in the waste (Lemma 8.6(ii)).

Let $x' := b_w \cdot e_1$. If X is a square, then $b_w \in \text{relint top}(X) \subset X$ by Lemma 8.6(i), and for $\mu > 0$ we have $(b_w + \mu e_2 - c_X) \cdot n = \frac{1}{2} + \mu \cos \varphi > \frac{1}{2}$, so $b_w + \mu e_2 \notin X$; thus b_w is the highest point of $X \cap V_{x'}$. If $X = X_0$, then $b_w = (w, 0) = (x', 0)$ is the point of $A_e \cap V_{x'}$. Since $\hat{w}(w) \in (w_1, w_2)$, Lemma 8.6(i) gives $c_w = o + \hat{w}(w)u + t(\hat{w}(w))n \in \text{relint bot}(Y) \subset Y$. For $\mu > 0$, $(c_w - \mu e_2 - c_Y) \cdot n_Y = -\frac{1}{2} - \mu \cos \varphi_Y < -\frac{1}{2}$, because $\cos \varphi_Y \geq \cos \frac{\pi}{4} > 0$ (also when $\varphi_Y = \frac{\pi}{4}$); so $c_w - \mu e_2 \notin Y$, and c_w is the lowest point of $Y \cap V_{x'}$. Consequently every point of $A_e \cap V_{x'}$ has height $\leq b_w \cdot e_2$, every point of $Y \cap V_{x'}$ has height $\geq c_w \cdot e_2 \geq b_w \cdot e_2$, and the open segment between b_w and c_w meets no square. So $x' \in E_e^v$ and $g_e^v(x') = |c_w - b_w| = \lambda_w = t(w)/\chi < \delta/\chi$ by Lemma 8.6(i). Finally $x' \in (0, k)$: if $\lambda_w > 0$, the segment lies in $\mathcal{O}_e \subset (0, k)^2$; if $\lambda_w = 0$, then $t(w) = 0$, so $X = X_0$ by Lemma 8.6(i), and $x' = w \in (w_1, w_2) \subset (0, k)$.

(b). For $x' \in J'_e$, $b(x')$ lies on the line of $\text{top}(X)$ (on $\mathbb{R} \times \{0\}$ if $X = X_0$) and $c(x')$ on L_Y . These lines have directions u_X and u_Y and are not vertical, as $\cos \varphi_X, \cos \varphi_Y > 0$; their heights above the abscissa x' are therefore affine functions of x' with slopes $\tan \varphi_X$ and $\tan \varphi_Y$, and $g_e^v(x') = c(x') \cdot e_2 - b(x') \cdot e_2$ is affine with slope $\tan \varphi_Y - \tan \varphi_X$. Next, t is affine and, by Lemma 8.6(i), nonnegative on $[w_1, w_2]$; if $t(w) = 0$ for some $w \in (w_1, w_2)$, then $t \equiv 0$ on $[w_1, w_2]$. For a square X this is excluded by Lemma 8.6(i). For $X = X_0$ it means $\kappa_e = 0$ and $t_0 = 0$, that is, $\varphi_Y = 0$ and $L_Y = \mathbb{R} \times \{0\}$; then $\theta_e = a(Y) = 0$. Since $g_e^v(x') = t(w(x'))/\chi$, this proves (b).

(c). For $w \in (w_1, w_2)$ we have $t(w) < \delta$, hence $|\hat{w}(w) - w| = \lambda_w |\sin \varphi| \leq \delta |\sin \varphi|/\chi = \Delta_e$. If $\sin \varphi \geq 0$, then $w \leq \hat{w}(w) \leq w + \Delta_e$, so $(w_1, w_2 - \Delta_e) \subset G_e$; if $\sin \varphi < 0$, then $w - \Delta_e \leq \hat{w}(w) \leq w$, so $(w_1 + \Delta_e, w_2) \subset G_e$. In both cases $|G_e| \geq (w_2 - w_1 - \Delta_e)_+$. Since $I_e \subset [w_1, w_2]$, Lemma 8.5 gives $w_2 - w_1 \geq |I_e| \geq m_e$. Therefore

$$|J'_e| = \cos \varphi |G_e| \geq \cos \varphi (w_2 - w_1 - \Delta_e)_+ \geq \cos \varphi (m_e - \Delta_e).$$

Finally $\cos \varphi \geq \cos \bar{\beta}$, and $\cos \varphi \cdot \Delta_e \leq \delta |\sin \varphi|/\chi < 1.0000016 \delta \sin \bar{\beta} \leq 1.0000016 \delta \bar{\beta}$, so $|J'_e| \geq \cos \bar{\beta} \cdot m_e - 1.0000016 \delta \bar{\beta}$. (If $w_1 = w_2$, then $G_e = J'_e = \emptyset$ and $m_e \leq |I_e| = 0$, and (c) holds trivially.)

The ceiling version follows by applying the above to $\rho\mathcal{P}$ and transporting by ρ , which preserves abscissae and vertical lengths and exchanges “highest” and “lowest”. $\square$

Remark 8.8 (the exceptional set). If $w_1 < w_2$, the abscissae $o \cdot e_1 + w \cos \varphi_X$, $w \in (w_1, w_2)$, form an open interval of length $\cos \varphi_X (w_2 - w_1)$. By the proof of Lemma 8.7(c), J'_e contains all of it except one end interval of length at most $\cos \varphi_X \Delta_e < 1.0000016 \delta \bar{\beta}$, at the end towards which the vertical direction drifts in the frame of X ; only in this end interval can the construction of Lemma 8.7 fail. Points of the hull that are not exit coordinates (holes of I_e) cause no loss, since Lemma 8.6 concerns the whole hull.

8.5. Disjointness of the vertical slits. For $e \in \mathcal{C}$ put

$$\mathcal{S}_e := \{(x', y) : x' \in J'_e, b(x') \cdot e_2 < y < c(x') \cdot e_2\}, \quad s_e := \int_{J'_e} g_e^v(x') dx',$$

with b, c as in Lemma 8.7(a); for $e \in \tilde{\mathcal{C}}$ let J'_e and s_e be defined through $\rho\mathcal{P}$, and let $\mathcal{S}_e$ be the image under ρ of the corresponding set for $\rho\mathcal{P}$. Let $\mathcal{K}$ be the set of squares that are members of some $e \in \mathcal{C}$, and $\tilde{\mathcal{K}}$ the set of squares that are members of some $e \in \tilde{\mathcal{C}}$.

Lemma 8.9 (disjoint vertical slits). (a) $\mathcal{K} \cap \tilde{\mathcal{K}} = \emptyset$.

(b) Each $\mathcal{S}_e$, $e \in \mathcal{C} \cup \tilde{\mathcal{C}}$, is an open subset of the waste with $|\mathcal{S}_e| = s_e$.

(c) The sets $\mathcal{S}_e$, $e \in \mathcal{C} \cup \tilde{\mathcal{C}}$, are pairwise disjoint. Consequently $\sum_{e \in \mathcal{C} \cup \tilde{\mathcal{C}}} s_e \leq W$.

Proof. (a) Let $S \in \mathcal{K}$, a member of $e \in \mathcal{C}$, and let $x \in P_e$. If S is the second member, it contains the entry point q_x , of height $\leq h_{\max} = \frac{k}{2} - 1$; if it is the first member, it contains its entry point, also of height $\leq h_{\max}$ (Lemma 8.1(b)). By the ceiling version, every $S \in \tilde{\mathcal{K}}$ contains a point of height $\geq k - h_{\max} = \frac{k}{2} + 1$. A square of $\mathcal{K} \cap \tilde{\mathcal{K}}$ would have vertical extent at least 2, whereas its vertical extent is $\cos a + \sin a \leq \sqrt{2}$ by [R, Lemma 3.1(a)].

(b) Let $e \in \mathcal{C}$. On the open interval J'_e the functions $x' \mapsto b(x') \cdot e_2$ and $x' \mapsto c(x') \cdot e_2$ are affine (proof of Lemma 8.7(b)), so $\mathcal{S}_e$ is open and, by Tonelli's theorem, $|\mathcal{S}_e| = \int_{J'_e} (c(x') - b(x')) \cdot e_2 dx' = s_e$. By Lemma 8.7(a), $\mathcal{S}_e \subset \mathcal{O}_e$, which lies in the waste. For $e \in \tilde{\mathcal{C}}$ the same follows by reflection, since ρ maps the waste of $\rho\mathcal{P}$ onto the waste of $\mathcal{P}$ and preserves areas.

(c) For $x' \in (0, k)$ let $\mathcal{U}_{x'} := \{y \in (0, k) : (x', y) \text{ lies in no square}\}$, an open subset of $\mathbb{R}$. For $e \in \mathcal{C} \cup \tilde{\mathcal{C}}$ and $x' \in J'_e$, the section $\mathcal{S}_e^{x'} := \{y : (x', y) \in \mathcal{S}_e\}$ is an open interval (empty if $g_e^v(x') = 0$). Let $e = (X, Y) \in \mathcal{C}$ and $\mathcal{S}_e^{x'} \neq \emptyset$. Then $\mathcal{S}_e^{x'} \subset \mathcal{U}_{x'}$ by (b); its upper endpoint is $c(x') \cdot e_2$ with $c(x') \in Y$, and its lower endpoint is $b(x') \cdot e_2$ with $b(x') \in X$ if X is a square and $b(x') \cdot e_2 = 0$ if $X = X_0$. So neither endpoint lies in $\mathcal{U}_{x'}$, and $\mathcal{S}_e^{x'}$ is a connected component of $\mathcal{U}_{x'}$. By reflection, for $e = (X, Y) \in \tilde{\mathcal{C}}$ every nonempty section $\mathcal{S}_e^{x'}$ is a component of $\mathcal{U}_{x'}$ whose lower endpoint is the

height of a point of Y and whose upper endpoint is the height of a point of X if X is a square, and equals k if $X = \tilde{X}_0$.

Let $e \neq e'$ in $\mathcal{C} \cup \tilde{\mathcal{C}}$ and suppose that $(x', y) \in \mathcal{S}_e \cap \mathcal{S}_{e'}$. Then $\mathcal{S}_e^{x'}$ and $\mathcal{S}_{e'}^{x'}$ are the component of $\mathcal{U}_{x'}$ that contains y , so they have the same endpoints; the corresponding points on $V_{x'}$ coincide.

Case $e = (X, Y)$, $e' = (X', Y') \in \mathcal{C}$. The common upper endpoint $c(x')$ lies in Y and in Y' , so $Y = Y'$, since distinct squares are disjoint. If X is a square, then $b(x') \in \text{relint top}(X)$ has height > 0 by (8.2); so the common lower endpoint has height 0 if and only if $X = X_0$, and likewise for X' . Hence either $X = X' = X_0$, or $b(x') \in X \cap X'$ and $X = X'$. In both cases $e = e'$, a contradiction.

Case $e, e' \in \tilde{\mathcal{C}}$. This is the previous case for $\rho\mathcal{P}$.

Case $e = (X, Y) \in \mathcal{C}$, $e' = (X', Y') \in \tilde{\mathcal{C}}$. The upper endpoint of $\mathcal{S}_e^{x'}$ is the height of $c(x') \in \text{relint bot}(Y)$, which is $< k$ by (8.2). So the upper endpoint of $\mathcal{S}_{e'}^{x'}$ is not k ; hence X' is a square and contains $c(x')$. Then $X' = Y \in \mathcal{K} \cap \tilde{\mathcal{K}}$, contradicting (a).

So the sets $\mathcal{S}_e$ are pairwise disjoint subsets of the waste, and $\sum_e s_e = |\bigcup_e \mathcal{S}_e| \leq W$; the family is finite. $\square$

8.6. Slit inequality, pair waste and multiplicity.

Lemma 8.10 (slit inequality). *For every $e \in \mathcal{C} \cup \tilde{\mathcal{C}}$,*

$$(8.6) \quad \theta'_e (2 - \bar{\beta}) |J'_e|^2 \leq 4 s_e.$$

Proof. Let $e = (X, Y) \in \mathcal{C}$, and assume that the interval $J := J'_e$ is not empty (otherwise there is nothing to show). By parts (a) and (b) of Lemma 8.7, the function $g := g_e^v$ is affine and nonnegative on J , with

$$|g'| = |\tan \varphi_Y - \tan \varphi_X| \geq |\varphi_Y - \varphi_X| \geq \theta_e,$$

by the mean value theorem ($\tan' = \sec^2 \geq 1$) and since $\theta_e \leq |\varphi_Y - \varphi_X|$; for $X = X_0$ this reads $|\tan \varphi_Y| \geq |\varphi_Y| = a(Y) = \theta_e$. Let x_* be an endpoint of $\bar{J}$ at which the continuous extension of g to $\bar{J}$ is smallest; it is ≥ 0 . Then $g(x') \geq |g'| |x' - x_*| \geq \theta_e |x' - x_*|$ on J , so $s_e = \int_J g \geq \theta_e |J|^2 / 2$. As $\theta'_e \leq \theta_e$ and $2 - \bar{\beta} \leq 2$, we get $\theta'_e (2 - \bar{\beta}) |J|^2 \leq 2\theta_e |J|^2 \leq 4s_e$. (This is the monotone case of [R, Lemma 4.1].) For $e \in \tilde{\mathcal{C}}$ the claim follows by reflection. $\square$

For two squares S_1, S_2 , and for a square S_1 and $S_2 = S_0 = \partial[0, k]^2$, let $R(S_1, S_2)$ be the rectangle defined before [R, Proposition 4.3]; for $R(S_1, S_0)$ the point of S_0 nearest to the centre of S_1 used there is fixed once and for all for each square S_1 . For $e = (X, Y) \in \mathcal{C} \cup \tilde{\mathcal{C}}$ put

$$R_e := R(X, Y) \text{ if } X \text{ is a square,} \quad R_e := R(Y, S_0) \text{ if } X \in \{X_0, \tilde{X}_0\}, \quad f_e := |R_e \cap \text{waste}|.$$

Lemma 8.11 (pair waste). *Let $e = (X, Y) \in \mathcal{C} \cup \tilde{\mathcal{C}}$. If X is a square, then $\text{dist}(X, Y) < \delta$; if $X = X_0$ (resp. $X = \tilde{X}_0$), then Y is at distance less than δ from the bottom side $[0, k] \times \{0\}$ (resp. the top side $[0, k] \times \{k\}$), and hence from S_0 . Moreover*

$$(8.7) \quad f_e \geq \frac{Q_* \theta'_e (2 - \bar{\beta})}{4}.$$

Proof. Let $e \in \mathcal{C}$ and $x \in P_e$. By Lemma 8.1(a), $|q_x - p_x| = t_x < \delta$, where $q_x \in Y$, and $p_x \in X$ if X is a square, while $p_x = (x, 0)$ lies on the bottom side if $X = X_0$. This gives the distance bounds; for $e \in \tilde{\mathcal{C}}$ they follow by reflection, which preserves distances and maps the bottom side onto the top side. Note that $\theta(Y, S_0) = a(Y) = \theta_e$ if $X \in \{X_0, \tilde{X}_0\}$.

If $\theta_e \leq 10^{-5}$, [R, Proposition 4.5], applied with $S_1 = X$, $S_2 = Y$ (respectively $S_1 = Y$, $S_2 = S_0$), $d = \delta \leq 10^{-5}$ and $\theta_1 = 10^{-5}$, gives $f_e \geq Q_* \theta_e (2 - \theta_e) / 4 \geq Q_* \theta'_e (2 - \bar{\beta}) / 4$; the last step holds because $\theta'_e \leq \theta_e$, $\theta'_e \leq \bar{\beta}$ and $x \mapsto x(2 - x)$ is increasing on $[0, 1]$, so that $\theta'_e (2 - \bar{\beta}) \leq \theta'_e (2 - \theta'_e) \leq \theta_e (2 - \theta_e)$. If $\theta_e > 10^{-5}$, [R, Lemma 4.8(a)], with $d = \delta \leq 10^{-4}$, gives $f_e \geq \theta_e (2 - \theta_e) / 40 \geq 10^{-5} (2 - 10^{-5}) / 40 > 4.99 \cdot 10^{-7}$ (as $\theta_e \leq \frac{\pi}{4} < 1$), while $Q_* \theta'_e (2 - \bar{\beta}) / 4 \leq 0.32 \cdot 1.5 \cdot 10^{-6} \cdot 2 / 4 = 2.4 \cdot 10^{-7}$.

If $X \in \{X_0, \tilde{X}_0\}$ and Y is near a corner of $[0, k]^2$, the rectangle $R(Y, S_0)$ may stand on the left or the right side of $[0, k]^2$ rather than on the bottom or the top side. This does not matter here: [R, Proposition 4.5] and [R, Lemma 4.8(a)] concern $R(S_1, S_0)$ for any square S_1 at distance less than d from S_0 , with $\theta(S_1, S_0) = a(S_1)$ (as noted in the proof of [R, Lemma 4.13]), and Proposition 2.1 allows this situation (see the remark following it). $\square$

For $e = (X, Y) \in \mathcal{C} \cup \tilde{\mathcal{C}}$ let $\bar{e} := \{\hat{X}, Y\}$ be the unordered pair with $\hat{X} := X$ if X is a square, $\hat{X} := [0, k] \times \{0\}$ (the bottom side) if $X = X_0$, and $\hat{X} := [0, k] \times \{k\}$ (the top side) if $X = \tilde{X}_0$.

Lemma 8.12 (multiplicity). *The unordered pairs $\bar{e}$, $e \in \mathcal{C} \cup \tilde{\mathcal{C}}$, are pairwise distinct, and*

$$(8.8) \quad \sum_{e \in \mathcal{C} \cup \tilde{\mathcal{C}}} f_e \leq K_w^* W = 9W.$$

Without [R, Lemma 4.10], $\sum_{e \in \mathcal{C} \cup \tilde{\mathcal{C}}} f_e \leq K_w W = 13W$.

Proof. Distinctness. Let $e \neq e'$ with $\bar{e} = \bar{e}'$. If $e = (X, Y)$ and e' are both in $\mathcal{C}$, then, as X_0 is never a second member, $e' = (Y, X)$ with X and Y squares; by (F2), $c_Y \cdot e_2 > c_X \cdot e_2$ (from e) and $c_X \cdot e_2 > c_Y \cdot e_2$ (from e'), which is impossible. Both in $\tilde{\mathcal{C}}$: by reflection. If $e \in \mathcal{C}$ and $e' \in \tilde{\mathcal{C}}$, the second member of e is a square of $\bar{e} = \bar{e}'$, so it lies in $\mathcal{K} \cap \tilde{\mathcal{K}}$, contradicting Lemma 8.9(a).

The bound. Let $\mathfrak{F} := \{\bar{e} : e \in \mathcal{C} \cup \tilde{\mathcal{C}}\}$, a family of distinct unordered pairs. Each pair consists of two squares, or of a square and one of the two opposite sides $[0, k] \times \{0\}$, $[0, k] \times \{k\}$, and its members are at distance less than $\delta \leq 10^{-4}$ (Lemma 8.11); the rectangle that Proposition 2.1 attaches to $\bar{e}$ is R_e . Proposition 2.1, with $d := \delta$ and σ, σ' the bottom and the top side, gives $\sum_e f_e \leq K_w^* W = 9W$, and $\sum_e f_e \leq K_w W = 13W$ by the analytic argument. That proposition is the form of the multiplicity bound given by the proof of [R, Lemma 4.10], as stated in [R, Remark 4.11] (“The proof bounds the number of *all* pairs of squares (or of a square and one of two opposite sides) at distance less than d whose rectangles contain p , without using the graph G ”), and of [R, Lemma 4.9] by the paragraph following its proof; the formal statements of [R, Lemmas 4.9 and 4.10] concern edges of the visibility graphs, and the pairs $\bar{e}$ need not be such edges. $\square$

8.7. Proof of Theorem 8.3. *Step 1: the chain.* Let $x \in \mathcal{X}$, let $Z_1, \dots, Z_J$ be the squares entered by π_x^0 , in order, and $Z_0 := X_0$. By (F2), $Z_x = Z_j$ for exactly one $j \in \{1, \dots, J\}$. By the triangle inequality for θ ,

$$\zeta(x) \leq a(Z_j) = \theta(Z_0, Z_j) \leq \sum_{i=1}^j \theta(Z_{i-1}, Z_i).$$

With $\theta' := \min(\theta, \bar{\beta})$ also $\sum_{i=1}^j \theta'(Z_{i-1}, Z_i) \geq \zeta(x)$: if some term has $\theta(Z_{i-1}, Z_i) \geq \bar{\beta}$, its capped value is $\bar{\beta} \geq \zeta(x)$; otherwise no term is capped. The pairs (Z_{i-1}, Z_i) , $1 \leq i \leq J$, are pairwise distinct and are exactly the $e \in \mathcal{C}$ with $x \in P_e$; the terms with $i > j$ are nonnegative. Hence

$$\zeta(x) \leq \vartheta(x) := \sum_{e \in \mathcal{C}} \theta'_e \mathbf{1}_{P_e}(x).$$

ϑ is a nonnegative Borel function on $(0, k)$ (Lemma 8.5; the sum is finite), so

$$(8.9) \quad \int_{\mathcal{X}} \zeta \leq \int_0^k \vartheta = \sum_{e \in \mathcal{C}} \theta'_e m_e, \quad \text{and likewise} \quad \int_{\mathcal{X}'} \zeta' \leq \sum_{e \in \tilde{\mathcal{C}}} \theta'_e m_e$$

by the ceiling version.

Step 2: vertical comparison. Let $e = (X, Y) \in \mathcal{C} \cup \tilde{\mathcal{C}}$. By Lemma 8.7(c), $m_e \leq |J'_e| / \cos \varphi_X + \Delta_e$, where $\cos \varphi_X \geq \cos \bar{\beta}$ and $\Delta_e = \delta |\sin \varphi_X| / \chi_e < 1.0000016 \delta \bar{\beta}$. Hence

$$(8.10) \quad \sum_{e \in \mathcal{C} \cup \tilde{\mathcal{C}}} \theta'_e m_e \leq \frac{1}{\cos \bar{\beta}} \sum_{e \in \mathcal{C} \cup \tilde{\mathcal{C}}} \theta'_e |J'_e| + 1.0000016 \delta \bar{\beta} \sum_{e \in \mathcal{C} \cup \tilde{\mathcal{C}}} \theta'_e.$$

Step 3: Cauchy-Schwarz. For $e \in \mathcal{C} \cup \tilde{\mathcal{C}}$, multiplying (8.6) and (8.7) gives $Q_* \theta_e'^2 (2 - \bar{\beta})^2 |J_e'|^2 \leq 16 s_e f_e$, that is, $\theta_e' |J_e'| \leq \frac{4}{(2 - \bar{\beta}) \sqrt{Q_*}} \sqrt{s_e f_e}$. By the Cauchy-Schwarz inequality, Lemma 8.9(c) and (8.8),

$$\begin{aligned} \sum_e \theta_e' |J_e'| &\leq \frac{4}{(2 - \bar{\beta}) \sqrt{Q_*}} \left(\sum_e s_e \right)^{1/2} \left(\sum_e f_e \right)^{1/2} \\ &\leq \frac{4}{(2 - \bar{\beta}) \sqrt{Q_*}} \sqrt{W \cdot 9W} = \frac{4\sqrt{9/Q_*}}{2 - \bar{\beta}} W \leq A W, \end{aligned}$$

the sums being over $\mathcal{C} \cup \tilde{\mathcal{C}}$ and the last step using $2 - \bar{\beta} \geq 2 - 3 \cdot 10^{-6}$.

Step 4: the error term. By (8.7) and (8.8),

$$\sum_{e \in \mathcal{C} \cup \tilde{\mathcal{C}}} \theta_e' \leq \frac{4}{Q_*(2 - \bar{\beta})} \sum_e f_e \leq \frac{36}{0.32(2 - 1.5 \cdot 10^{-6})} W < 56.26 W,$$

so the last term of (8.10) is at most $1.0000016 \cdot 10^{-5} \cdot 1.5 \cdot 10^{-6} \cdot 56.26 W < 8.5 \cdot 10^{-10} W$ (rounded up).

Step 5: conclusion. Steps 2–4 give (b). Adding the two inequalities of (8.9) and using (b) gives $\int_{\mathcal{X}} \zeta + \int_{\mathcal{X}'} \zeta' \leq (A/\cos \bar{\beta} + 8.5 \cdot 10^{-10})W \leq A'W$, which is (a). For the numerical bounds: $A = 4\sqrt{9/0.32}/(2 - 3 \cdot 10^{-6}) < 10.6067$ and $1/\cos \bar{\beta} < 1 + 1.2 \cdot 10^{-12}$, hence $A' < 10.6067(1 + 1.2 \cdot 10^{-12}) + 10^{-8} < 10.60671 < 10.6068$ (all rounded up).

Step 6: part (c). Replace (8.8) by $\sum_e f_e \leq 13W$ (Lemma 8.12). Step 3 gives $\sum_e \theta_e' |J_e'| \leq 4\sqrt{13/Q_*} W/(2 - \bar{\beta}) \leq A_{13}W$, and Step 4 gives $\sum_e \theta_e' \leq 52W/(0.32(2 - 1.5 \cdot 10^{-6})) < 81.26W$ and an error term at most $1.0000016 \cdot 10^{-5} \cdot 1.5 \cdot 10^{-6} \cdot 81.26W < 1.22 \cdot 10^{-9}W$. Hence $\sum_e \theta_e' m_e \leq (A_{13}/\cos \bar{\beta} + 1.22 \cdot 10^{-9})W$ and $\int_{\mathcal{X}} \zeta + \int_{\mathcal{X}'} \zeta' \leq A'_{13}W$. Numerically $A_{13} = 4\sqrt{13/0.32}/(2 - 3 \cdot 10^{-6}) < 12.74757$ and $A'_{13} < 12.74757(1 + 1.2 \cdot 10^{-12}) + 10^{-8} < 12.7476$ (rounded up). $\square$

8.8. The scale form. Recall from Definition 3.22 the tilt time T^* : with (Z_j, q_j) , $1 \leq j \leq J(x)$, the R1-passed entries of π_x^0 and $\eta_j := q_j \cdot e_2$,

$$s_j := \max(y_0, \eta_j - 2, (c_0/a(Z_j))^2) \text{ if } a(Z_j) > 0, \quad s_j := +\infty \text{ if } a(Z_j) = 0,$$

and $T^*(x) := \min_j s_j$ if this minimum is at most $y_1 = \frac{k}{2} - 3$; otherwise (in particular if $J(x) = 0$) $x \notin \text{dom } T^*$. Here $\alpha(s) = c_0 s^{-1/2}$ and $\alpha_{\max} = \alpha(y_0)$ (Definition 3.1). Only R1-passed entries enter; deaths, merges and end collisions do not. The ceiling tilt time $\tilde{T}^* = T^*[\rho\mathcal{P}]$ is defined from $\tilde{F}^0$ (Definition 3.24).

Corollary 8.13 (scale form). *Let $k \geq 4$ be an integer, $\mathcal{P}$ a closed packing of $[0, k]^2$ with waste area W , $0 < \delta \leq 10^{-5}$, and assume (C0) and (C1). Then $\text{dom } T^*$ and $\text{dom } \tilde{T}^*$ are Borel subsets of $(0, k)$, T^* and $\tilde{T}^*$ are Borel functions on them with values in $[y_0, y_1]$, and*

$$\int_{\text{dom } T^*} \alpha(T^*(x)) dx + \int_{\text{dom } \tilde{T}^*} \alpha(\tilde{T}^*(x)) dx \leq A'W, \quad A' < 10.6068.$$

Without [R, Lemma 4.10], using [R, Lemma 4.9] in its place, the same holds with $A'_{13} < 12.7476$ in place of A' .

Proof. Borel measurability. By Lemma 3.31(a),(b), $\text{dom } T^*$ is a finite union of intervals and points, and T^* is a Borel function on it (on each piece of F^0 the entries Z_j are fixed and the η_j are affine, so T^* is continuous on the intersection of $\text{dom } T^*$ with each piece). Its values lie in $[y_0, y_1]$ because $s_j \geq y_0$. The ceiling tilt time is treated by reflection.

The hypotheses of Theorem 8.3. Let $x \in \text{dom } T^*$ and choose j with $T^*(x) = s_j$. As $s_j \leq y_1 < \infty$, we have $a(Z_j) > 0$. Since α is decreasing and $s_j \geq \max(y_0, (c_0/a(Z_j))^2)$,

$$\begin{aligned} 0 < \alpha(T^*(x)) = c_0 s_j^{-1/2} &\leq \min\left(c_0 y_0^{-1/2}, c_0((c_0/a(Z_j))^2)^{-1/2}\right) \\ &= \min(\alpha_{\max}, a(Z_j)) \leq \min(\bar{\beta}, a(Z_j)), \end{aligned}$$

by (C1). So $\zeta := \alpha \circ T^*$ is a Borel function from $\mathcal{X} := \text{dom } T^*$ to $[0, \bar{\beta}]$ (α is continuous on $(0, \infty)$), and π_x^0 enters $Z_x := Z_j$ (an R1-passed entry at $q_j \in \text{relint bot}(Z_j)$) with $a(Z_x) \geq \zeta(x)$. The same holds for $\mathcal{X}' := \text{dom } \tilde{T}^*$ and $\zeta' := \alpha \circ \tilde{T}^*$ with $\tilde{F}^0$. Theorem 8.3(a), respectively (c), gives the claim. $\square$

All scale flows F_s are truncations of the single master flow F^0 , in which R1 is decided once (Definition 3.21 and Remark 3.25); accordingly Corollary 8.13 involves the paths of F^0 only, and the injectivity of Lemma 8.5 is that of this one family of paths.

8.9. Remarks.

Remark 8.14 (where the rules R1 and R2 enter). (1) R1 is used only in Lemma 8.5, through (F1) applied at the first member X of a pair: two floor points with the same exit point from X entered X at the same point, and at most one path passes X from there. A path that passes X is the R1 winner at its entry point into X ; as $a(X) < \alpha_{\max}$, this does not depend on the order in which M and T are decided at an entry candidate (compare Remark 3.34). Nothing about the entry into the second member is used in Lemma 8.5.

(2) R2 enters in two ways. First, the first member X of every pair with $X \neq X_0$ is a passed square, so $|\varphi_X| < \alpha_{\max} \leq \bar{\beta}$ (Lemma 8.1(d)); this makes Δ_e of order $\delta\bar{\beta}$ in Lemma 8.7(c). Secondly, the second member Y of every pair, in particular Z_x in the last pair of the chain in Step 1 of §8.7, is entered through $\text{relint bot}(Y)$, since side and vertex contacts are never entries. Hence, over the hull, the boundary of Y facing X is the single line L_Y (Lemma 8.6), and $c(x')$ is the lowest point of Y on $V_{x'}$ (Lemma 8.7(a)).

Remark 8.15 (degenerate cases). The general conventions are those of Remark 3.35. *Vertices and grazing contacts*: a first contact at a vertex terminates the path (with E, or with D or W at a tie) and is never an entry, so it produces no pairs; only contacts with $\text{relint bot}(Y)$ are used. *Ties* between events are resolved by the priority $D > W > \text{contact} > H$; the proofs use only the consequences $g + t_S < \delta$, $t_S < t_W$ and $t_S \leq t_H$ at processed contacts (Definition 3.9 and Lemma 3.17(f)); they enter through Lemma 8.1(a),(b)), and a contact at height exactly $h_{\max}$ is processed and covered by Lemma 8.1(b). *Holes* of I_e , and exit coordinates of paths that are later D-, M- or E-terminated, do not matter, because Lemma 8.6 concerns the whole hull. *A square resting with a side on the floor* gives the case $g_e^v \equiv 0$ of Lemma 8.7(b), with $\theta_e = 0$ and $\mathcal{S}_e = \emptyset$; empty sections are excluded in the proof of Lemma 8.9(c). *Inclination* $\frac{\pi}{4}$: for the second member only $\cos \varphi_Y > 0$, $|\varphi_Y - \varphi_X| < \frac{\pi}{2}$, $|\kappa_e| < 1.00001$ and $\theta_e \leq |\varphi_Y - \varphi_X|$ are used, all valid under the 45° convention; the first member always has $|\varphi_X| < \bar{\beta}$. *Walls*: a wall event terminates the path (W), and by the priority $W > \text{contact}$ of Definition 3.9 a contact with $t_S = t_W$ is not processed; so only contacts reached before a wall event form pairs, and $\mathcal{O}_e \subset (0, k)^2$ by Lemma 8.6(ii).

9. P7: THE PER-SCALE COUNT

In this section we fix a scale $s \in [y_0, y_1]$. We compare the measure $N(s)$ of the set of floor points whose path is T_s -terminated in the scale flow F_s with the number of squares of a family $\mathcal{Z}(s)$ (Theorem 9.1), and we bound $|\mathcal{Z}(s)|$ from below by an integral over lines of kind (iii) (Theorem 9.2). Both theorems have ceiling versions, and both are used in Section 10. Besides the definitions and lemmas of Section 3, the proofs use Theorem 4.11, Theorem 6.9, Proposition 6.10 and [R, Lemmas 3.1, 3.4 and 3.5].

9.1. Setting, notation and statements. *Standing assumptions.* Throughout this section $k \geq 4$ is an integer, $\mathcal{P}$ is a closed packing of $[0, k]^2$, $\delta = 10^{-5}$ (Definition 3.1), and the parameters $c_0, c_1, y_0, \varepsilon, \omega_0$ satisfy the constraints (C0)–(C7) of Definition 3.2. We write $y_1 = k/2 - 3$, $h_{\max} = k/2 - 1$, $\alpha(t) = c_0 t^{-1/2}$ and $\hat{\beta}(t) = c_1 t^{-3/4}$ for $t > 0$, and $\alpha_{\max} = \alpha(y_0)$. The proofs of the results of this section use directly only the constraints

- (C0) $c_0, c_1 > 0$, y_0 is a positive integer, $0 < \varepsilon < 1$ and $\omega_0 > 0$;
- (C1) $\alpha_{\max} = c_0 y_0^{-1/2} \leq 1.5 \cdot 10^{-6}$;
- (C2) $\hat{\beta}(y_0) = c_1 y_0^{-3/4} \leq 1.5 \cdot 10^{-6}$,

together with the hypotheses of the results quoted from Sections 3, 4 and 6, which are among the standing assumptions. Constraint (C3) is not used (Remark 9.14).

Lines. We use the following notation of Section 3 (Definitions 3.1 and 3.5), repeated for convenience. For $y \in \mathbb{R}$ put $\ell_y := \mathbb{R} \times \{y\}$; a square S *meets the line y* if $S \cap \ell_y \neq \emptyset$ (since $S \subset [0, k]^2$, this is the same as meeting $[0, k] \times \{y\}$). For a square S , $c_S(y)$ is the length of the closed chord $S \cap \ell_y$, $\Phi(S) = \{y : 0 < c_S(y) < 1\}$, and $\mathcal{R}(S)$ is the ramp set of [R, Lemma 3.1]; $\omega(y)$ is the line waste and

$$E(y) = \sum_{S: c_S(y) \geq 1} (c_S(y) - 1).$$

Since $c_S(y)$ is the length of the section of S at height y , $\int_{\mathbb{R}} c_S(y) dy = \text{area}(S) = 1$ (Cavalieri's principle). Further, $d(y) = \min(y, k - y)$ and

$$\beta(y) := \hat{\beta}(d(y)) = c_1 d(y)^{-3/4} \quad (0 < y < k),$$

so that $\beta(t) = \hat{\beta}(t)$ for $0 < t \leq k/2$. Finally $\text{dist}(y, \mathbb{Z}) = \min(\{y\}, 1 - \{y\})$, so that for every $w > 0$

$$\{y\} \in [w, 1 - w] \iff \text{dist}(y, \mathbb{Z}) \geq w.$$

Put

$$w_0 := 0.50001 c_0^2 \left(1 + \frac{1.0001}{y_0}\right) + \delta + 1.0001 c_1 ((1 - \varepsilon) y_0)^{-3/4} + 10^{-12},$$

$$H := \{y \in (0, k) : y_0 \leq d(y) \leq (1 - \varepsilon) y_1, \{y\} \in [w_0, 1 - w_0]\}.$$

Every $y \in H$ is of exactly one of the following kinds:

- (i) $\omega(y) \geq \omega_0$;
- (ii) $\omega(y) < \omega_0$, and the line y meets a square S with $a(S) \geq \beta(y)$;
- (iii) $\omega(y) < \omega_0$, and every square S meeting the line y has $a(S) < \beta(y)$.

The corresponding subsets of H are Y_i , Y_{ii} and Y_{iii} , and $Y_b := Y_{iii} \cap (0, k/2)$, $Y_t := Y_{iii} \cap (k/2, k)$. For $s \in [y_0, y_1]$ put

$$\mathcal{W}_s := [(1 - \varepsilon)s, s], \quad V_s := [(1 - \varepsilon)s - 1.0001, s + 1.0001], \quad |V_s| = \varepsilon s + b_W, \quad b_W := 2.0002,$$

$$\mathcal{Z}(s) := \{Z \in \mathcal{P} : \Phi(Z) \cap Y_b \cap \mathcal{W}_s \neq \emptyset\}, \quad \tilde{\mathcal{Z}}(s) := \{Z \in \mathcal{P} : \Phi(Z) \cap Y_t \cap (k - \mathcal{W}_s) \neq \emptyset\},$$

where $k - Y := \{k - y : y \in Y\}$ for $Y \subset \mathbb{R}$. For every $Z \in \mathcal{Z}(s)$ we fix a *witness* $y_Z \in \Phi(Z) \cap Y_b \cap \mathcal{W}_s$.

Flows. The master flow is $F^0 = F(\alpha_{\max}, \delta, h_{\max})$ (Definition 3.20); R1 is decided in F^0 only. The *R1-passed entries* of π_x^0 are the entry candidates of π_x^0 at which x is the winner of R1 (Definition 3.20). For $s \in [y_0, y_1]$ the path π_x of the scale flow F_s (written π_x^s in Definition 3.21) is the path π_x^0 of F^0 , truncated at the first, in path order, of the following events:

- (a) T_s : an R1-passed entry (Y, q) of π_x^0 with $a(Y) \geq \alpha(s)$; the path ends at q ;
- (b) H_s : a gap point at height $s + 2$, reached in a ray step with $t_H^{(s)} := (s + 2 - p_y)/d_y < \min(t_S, t_W, t_D)$, or an exit point of height $\geq s + 2$;
- (c) a termination of π_x^0 in F^0 , which is then inherited.

Here p, d, t_S, t_W, t_D are the start point, the direction and the parameters of the ray step (Definition 3.9). A T-termination of π_x^0 in F^0 happens at an R1-passed entry (Y, q) with $a(Y) \geq \alpha_{\max} \geq \alpha(s)$; by Definition 3.21, if it is reached in F_s it is a T_s event, so that “T-terminated in F_s ” and “ T_s -terminated in F_s ” mean the same. We put

$$\mathcal{T}_s := \{x \in (0, k) : \pi_x \text{ is } T_s\text{-terminated in } F_s\}, \quad N(s) := |\mathcal{T}_s|.$$

For $F \in \{F^0, F_s\}$ the *trajectory* Γ_x of π_x is the union of the segments $\{p + td : 0 \leq t \leq t^*\}$ of its ray steps, where t^* is the length of the step in F , and of its pass segments $[q, q + n_Y]$; its *termination point* $e(x)$ is its last point in path order, and $\tau_F(x)$ is the height of $e(x)$, except that $\tau_F(x) := h_F$ if π_x is H-completed; here $h_{F^0} = h_{\max}$ and $h_{F_s} = s + 2$ (Definition 3.16). For $y \in (0, h_F)$ and $\tau_F(x) \geq y$, $\Gamma_x \cap \ell_y$ is a single point $P_y(x)$ (Definition 3.28). A path of F *makes an alive contact with* $\text{relint bot}(Y)$ if one of its ray steps ends in the contact branch of the priority rule at a point $q \in \text{relint bot}(Y)$, that is, if (Y, q) is an entry candidate reached by the path; this refers to the moment before R1 is applied.

As in Definitions 3.26, 3.28 and 3.30: for $\star \in \{T, D, E, M, W\}$ and $y \in \mathbb{R}$ let $L_\star^F(y) := |\{x : \pi_x \text{ is } \star\text{-terminated in } F, \tau_F(x) < y\}|$, and for $y \in (0, h_F)$

$$\mu_Z^F(y) := |\{x : \tau_F(x) > y, P_y(x) \in \text{int } Z\}|, \quad \text{Sh}^F(y) := \sum_{S \in \mathcal{P}} (c_S(y) - \mu_S^F(y))_+,$$

and $\Lambda^F(y) := [L_D^F + L_E^F + L_M^F + L_W^F](y) + \text{Ov}_{\text{gap}}^F(y)$. With $\mathcal{D}, \mathcal{M}, \mathcal{E}$ the sets of floor points that are D-, M- and E-terminated in F^0 and $L_E := |\mathcal{E}|$,

$$\Lambda_0 := |\mathcal{D}| + |\mathcal{M}| + L_E + \sup_{0 < y < h_{\max}} \text{Ov}_{\text{gap}}^{F^0}(y), \quad \Lambda^*(s) := \Lambda_0 + 2(s + 2) \tan \alpha(s).$$

As noted in Section 3.8, $\Lambda_0 \leq 2k < \infty$.

Ceiling. With $\rho(x, y) = (x, k - y)$, the ceiling flows are $\tilde{F}^0 = \rho \circ F^0[\rho\mathcal{P}] \circ \rho$ and $\tilde{F}_s = \rho \circ F_s[\rho\mathcal{P}] \circ \rho$, and $\tilde{N}(s), \tilde{\Lambda}_0, \tilde{T}^*$ are the quantities $N(s), \Lambda_0, T^*$ computed for the packing $\rho\mathcal{P}$ (Definition 3.24); $\tilde{\Lambda}^*(s) := \tilde{\Lambda}_0 + 2(s + 2) \tan \alpha(s)$.

Theorem 9.1 (per-scale count). *Assume the standing assumptions of Section 9.1, and let $s \in [y_0, y_1]$ and $b_W = 2.0002$.*

$$\begin{aligned} (a) \quad N(s) &\geq \frac{|\mathcal{Z}(s)|}{\varepsilon s + b_W} - \Lambda^*(s), \text{ where } \Lambda^*(s) = \Lambda_0 + 2(s + 2) \tan \alpha(s). \\ (b) \quad \tilde{N}(s) &\geq \frac{|\tilde{\mathcal{Z}}(s)|}{\varepsilon s + b_W} - \tilde{\Lambda}^*(s), \text{ where } \tilde{\Lambda}^*(s) = \tilde{\Lambda}_0 + 2(s + 2) \tan \alpha(s). \end{aligned}$$

Theorem 9.2 (lower bound for $|\mathcal{Z}(s)|$). *Assume the standing assumptions of Section 9.1, and let $s \in [y_0, y_1]$.*

(a) *We have*

$$\int_{Y_b \cap \mathcal{W}_s} (1 - \omega_0 - E(y))_+ dy \leq \sum_{Z \in \mathcal{Z}(s)} \int_{\Phi(Z) \cap Y_b \cap \mathcal{W}_s} c_Z(y) dy \leq |\mathcal{Z}(s)| \hat{\beta}((1 - \varepsilon)s),$$

and consequently

$$|\mathcal{Z}(s)| \geq \frac{((1 - \varepsilon)s)^{3/4}}{c_1} \int_{Y_b \cap \mathcal{W}_s} (1 - \omega_0 - E(y))_+ dy.$$

$$(b) \quad |\tilde{\mathcal{Z}}(s)| \geq \frac{((1 - \varepsilon)s)^{3/4}}{c_1} \int_{Y_t \cap (k - \mathcal{W}_s)} (1 - \omega_0 - E(y))_+ dy.$$

All sets of heights occurring here are finite unions of intervals and all integrands are Borel functions (Lemma 9.12); $\Lambda_0, \tilde{\Lambda}_0 \leq 2k$. The proofs of the floor parts (a) are given in Sections 9.6 and 9.7, and the ceiling parts (b) are deduced by reflection in Section 9.8.

9.2. Parameter facts.

Lemma 9.3. *Let $s \in [y_0, y_1]$.*

- (a) *If $y \in H$, then $d(y) \geq y_0$ and $0 < \beta(y) \leq \hat{\beta}(y_0) \leq 1.5 \cdot 10^{-6}$.*
- (b) *If $y \in H$, then $d(y) \leq (1 - \varepsilon)y_1 < k/2 - 1$. Moreover $k/2 \notin H$, and $d(y) = y$ for every $y \in H \cap (0, k/2)$.*
- (c) *If $Z \in \mathcal{Z}(s)$, then*

$$0 < a(Z) < \beta(y_Z) = \hat{\beta}(y_Z) \leq \min \{ \hat{\beta}((1 - \varepsilon)s), \hat{\beta}(y_0) \} \leq 1.5 \cdot 10^{-6}.$$

- (d) *If $Z \in \mathcal{Z}(s)$, then $\cos a(Z) + \sin a(Z) \leq 1 + a(Z) < 1.0001$.*
- (e) *If $Z \in \mathcal{Z}(s)$ and $0 < q_y \leq s + 1.0001$, then*

$$w(s, Z) := 0.50001 \alpha(s)^2 q_y + \delta + \sin a(Z) + (1 - \cos a(Z)) < w_0 - 10^{-12} < w_0.$$

- (f) $\alpha(s) \leq \alpha_{\max}$, $s + 2 \leq h_{\max}$ and $s < k/2$.

The quantity $w(s, Z)$ in (e) is the width $w(s, Z; q)$ of Theorem 4.11 at a point q of height q_y .

Proof. (a) By the definition of H , $d(y) \geq y_0 > 0$, so $\beta(y) = c_1 d(y)^{-3/4} > 0$. As $\hat{\beta}$ is decreasing, $\beta(y) = \hat{\beta}(d(y)) \leq \hat{\beta}(y_0)$, and $\hat{\beta}(y_0) \leq 1.5 \cdot 10^{-6}$ is (C2).

(b) Since $y_1 \geq s \geq y_0 > 0$ and $\varepsilon > 0$, we have $(1 - \varepsilon)y_1 < y_1 = k/2 - 3 < k/2 - 1$. As $d(k/2) = k/2 > (1 - \varepsilon)y_1$, the point $k/2$ is not in H . For $0 < y < k/2$ we have $d(y) = y$.

(c) Since $y_Z \in \Phi(Z)$, $c_Z(y_Z) > 0$; thus $\Phi(Z) \neq \emptyset$ and Z meets the line y_Z . By [R, Lemma 3.4], $\Phi(Z) = \emptyset$ if $a(Z) = 0$; hence $a(Z) > 0$. As $y_Z \in Y_{iii}$ and Z meets the line y_Z , the definition of kind (iii) gives $a(Z) < \beta(y_Z)$. As $y_Z \in Y_b \subset H \cap (0, k/2)$, part (b) gives $d(y_Z) = y_Z$; hence $\beta(y_Z) = \hat{\beta}(y_Z)$, and $y_Z = d(y_Z) \geq y_0$ by the definition of H . Moreover $y_Z \geq (1 - \varepsilon)s > 0$ because $y_Z \in \mathcal{W}_s$. Since $\hat{\beta}$ is decreasing, $\hat{\beta}(y_Z) \leq \min \{ \hat{\beta}((1 - \varepsilon)s), \hat{\beta}(y_0) \}$, and $\hat{\beta}(y_0) \leq 1.5 \cdot 10^{-6}$ by (C2).

(d) For $a \geq 0$ we have $\cos a \leq 1$ and $\sin a \leq a$; and $a(Z) < 1.5 \cdot 10^{-6}$ by (c).

(e) Since $\alpha(s)^2 = c_0^2/s$, $q_y \leq s + 1.0001$ and $s \geq y_0$,

$$0.50001 \alpha(s)^2 q_y \leq 0.50001 c_0^2 \frac{s + 1.0001}{s} \leq 0.50001 c_0^2 \left(1 + \frac{1.0001}{y_0} \right).$$

Put $a := a(Z)$. By (c), $0 < a < 1.5 \cdot 10^{-6} < 2 \cdot 10^{-4}$. Using $\sin a \leq a$ and $1 - \cos a \leq a^2/2 \leq 10^{-4}a$, then (c) and $s \geq y_0$,

$$\sin a + (1 - \cos a) \leq 1.0001 a < 1.0001 \hat{\beta}((1 - \varepsilon)s) \leq 1.0001 \hat{\beta}((1 - \varepsilon)y_0) = 1.0001 c_1 ((1 - \varepsilon)y_0)^{-3/4}.$$

Adding δ and these two bounds gives $w(s, Z) < w_0 - 10^{-12}$.

(f) α is decreasing and $s \geq y_0$; and $s \leq y_1 = k/2 - 3$. □

Remark 9.4. The inequality $\sin a + (1 - \cos a) \leq 1.0001 a$ used in Lemma 9.3(e) holds for $0 \leq a \leq 2 \cdot 10^{-4}$, by $\sin a \leq a$ and $1 - \cos a \leq a^2/2$, but not for all inclinations: for $a = \pi/4$ the left side equals 1, while $1.0001 \cdot \pi/4 = 0.7854767 \dots$. There it is applied only with $0 < a < 1.5 \cdot 10^{-6}$.

Remark 9.5. For later use we note that Lemma 9.3(a),(b) give the hypotheses of two lemmas of [R] on the lines of H ; these lemmas are not used in this section. Let $y \in Y_{ii}$. Then $0 \leq \beta(y) \leq 1.5 \cdot 10^{-6}$, $\omega(y) < \omega_0 < \frac{1}{2}$ by (C4), and the line y meets a square of inclination $\geq \beta(y)$; these are the hypotheses of [R, Lemma 4.13] with $\bar{\beta} = 1.5 \cdot 10^{-6}$, the set Y_{ii} and the function β . Let $y \in Y_{iii}$ and $\alpha := \beta(y)$. Then $0 < \alpha \leq 1.5 \cdot 10^{-6}$, $y \in (0, k)$, $d(y) \leq k/2 - 1$, and every square meeting the line y has inclination $< \alpha$; as $\delta = 10^{-5}$, these are the hypotheses of [R, Lemma 4.15], which gives $E(y) \leq 0.50001 \beta(y)(A + \beta(y)/\delta)W$ with the constant A of [R, Lemma 4.13].

9.3. Tilt terminations in the scale flows. Fix $x \in (0, k)$. At an R1-passed entry (Y, q) , π_x^0 passes Y if $a(Y) < \alpha_{\max}$, and is T-terminated at q otherwise (rule R2, Definition 3.10); R1 is applied before R2 (Definitions 3.14 and 3.10). As in Definition 3.20, let

$$(Z_1, q_1), \dots, (Z_J, q_J), \quad J = J(x) \geq 0,$$

be the R1-passed entries of π_x^0 in path order, and $\eta_j := (q_j)_y$; only the last one can be a T-termination. These are the entries in the definition of T^* (Definition 3.22): $s_j := \max\{y_0, \eta_j - 2, (c_0/a(Z_j))^2\}$ if $a(Z_j) > 0$, $s_j := +\infty$ if $a(Z_j) = 0$, and $T^*(x) := \min_j s_j$ if this minimum is at most y_1 ; otherwise $T^*(x)$ is undefined. We write $\{T^* \leq s\}$ for the set of x for which $T^*(x)$ is defined and $T^*(x) \leq s$.

We use two simple facts about trajectories. First, every segment of a trajectory of F^0 or F_s is parallel to n_X , where X is the floor or a square passed by the path; a passed square has $a(X) < \alpha_{\max}$ by R2, so $n_X \cdot e_2 = \cos \varphi_X \geq \cos \alpha_{\max} > 0$. Hence the height is nondecreasing along a trajectory and strictly increasing along every segment of positive length (Lemma 3.17(a)). Second, a ray step ends with D, W, H, E, or at an entry candidate, and the path continues after the step only if the step ends at an R1-passed entry at which the path passes the square and is not completed at its exit point (Definitions 3.9–3.10). Hence, before the first R1-passed entry and between two consecutive ones, π_x^0 performs exactly one ray step: the step from $(x, 0)$ ends at q_1 , and for $j \geq 2$ the step ending at q_j starts at the exit point $q_{j-1} + n_{Z_{j-1}}$ of Z_{j-1} .

Lemmas 9.6 and 9.7 below restate, in the form used in this section, Lemma 3.23 (with Lemma 3.19 for the scale flows) and Corollary 3.27, under the same conventions; we include the proofs so that this section can be read on its own. Precisely, Lemma 9.6(a), (b) are Lemma 3.23(a), (b); Lemma 9.6(c) contains Lemma 3.23(d), (e); Lemma 9.6(d) restates Lemma 3.23(f), the finite set Σ_s constructed there possibly differing by finitely many points from the one of Lemma 3.23(f) (only its finiteness and the stated properties are used); and Lemma 9.7 is Corollary 3.27.

Lemma 9.6. *Let $s \in [y_0, y_1]$.*

- (a) $\eta_1 < \eta_2 < \dots < \eta_J$. For $j \geq 2$, every point of the trajectory of π_x^0 that precedes q_j has height $< \eta_j$; for $j = 1$ such points have height $\leq \eta_1$.
- (b) π_x is T-terminated in F_s if and only if there is $j \leq J(x)$ with $a(Z_j) \geq \alpha(s)$ and $\eta_j \leq s + 2$. In this case the termination point is q_{j^*} , where j^* is the least such index. In particular, every T-termination in F_s , including an inherited T-termination of F^0 , is the T_s -truncation at q_{j^*} .
- (c) $\mathcal{T}_s = \{x : \exists j \leq J(x) : a(Z_j) \geq \alpha(s), \eta_j \leq s + 2\}$, and this set is a finite union of intervals and points. If $y_0 \leq s \leq s' \leq y_1$, then $\mathcal{T}_s \subset \mathcal{T}_{s'}$. Moreover $\{T^* \leq s\} = \mathcal{T}_s$. Consequently $s \mapsto N(s)$ is nondecreasing on $[y_0, y_1]$ and $N(s) = |\{T^* \leq s\}|$.
- (d) There is a finite set $\Sigma_s \subset [0, k]$, containing the finite set $\Sigma_0 := \Sigma_{F^0}$ of Lemma 3.18 for F^0 , such that on each component I of $[0, k] \setminus \Sigma_s$ the following holds: the number of ray steps and pass segments of π_x in F_s and its termination kind are constant; for each ray step the source X is constant, and its start point $p(x)$ is an affine function of x with $p'(x) = \lambda u_X$, $|\lambda| \geq 1$; and the termination point $e(x)$ and the termination height $\tau_{F_s}(x)$ are affine functions of x . In particular, for $\star \in \{T, D, E, M, W\}$ and $y \in \mathbb{R}$, the set $\{x : \pi_x \text{ is } \star\text{-terminated in } F_s, \tau_{F_s}(x) < y\}$ is a finite union of intervals and points.

Proof. (a) Let $j \geq 2$. The ray step ending at q_j starts at the exit point $p = q_{j-1} + n_{Z_{j-1}}$ of Z_{j-1} . The pass segment $[q_{j-1}, p]$ has length 1, so $p_y > \eta_{j-1}$. The source of the ray step is the square Z_{j-1} , which contains p and is disjoint from $Z_j \ni q_j$; hence the segment $[p, q_j]$ has positive length and $\eta_j > p_y$. Points preceding p have height $\leq p_y$, and the points of the segment $[p, q_j)$ have height $< \eta_j$. Hence every point preceding q_j has height $< \eta_j$; in particular $\eta_{j-1} < \eta_j$. For $j = 1$, the points preceding q_1 lie on the vertical segment from $(x, 0)$ to q_1 and have height $\leq \eta_1$.

(b) For a ray step with start point p and direction d we use $t_H^{(s)} = (s + 2 - p_y)/d_y$. If such a step of π_x^0 ends in F^0 at a processed contact q , then $t_S = t^*$ while neither D nor W applies, so $t_S < t_D$ and $t_S < t_W$; thus $\min(t_S, t_W, t_D) = t_S$, and

$$(9.1) \quad \text{the } H_s \text{ condition } t_H^{(s)} < \min(t_S, t_W, t_D) \text{ holds in this step} \iff q_y > s + 2.$$

($\Leftarrow$) Let j be the least index with $a(Z_j) \geq \alpha(s)$ and $\eta_j \leq s + 2$. We show that none of the three truncation events of F_s occurs before q_j , and that a T_s event occurs at q_j .

- π_x^0 is not terminated in F^0 before q_j , because it reaches the entry candidate (Z_j, q_j) and wins R1 there.
- No T_s event occurs before q_j : the R1-passed entries before q_j are the (Z_i, q_i) with $i < j$; by (a), $\eta_i < \eta_j \leq s + 2$, so the minimality of j gives $a(Z_i) < \alpha(s)$.
- No H_s event occurs before q_j . By (a), the exit points preceding q_j (there are none if $j = 1$) have height $< \eta_j \leq s + 2$. The ray steps up to q_j are the steps ending at $q_1, \dots, q_j$; each ends at a processed contact of height $\eta_i \leq \eta_j \leq s + 2$, so by (9.1) the H_s condition fails in each of them. (If $\eta_i = s + 2$, then $t_S = t_H^{(s)}$ and the contact is processed, since contact precedes H.)

Hence π_x coincides in F_s with π_x^0 up to q_j , with the same cumulative gap, and reaches the R1-passed entry (Z_j, q_j) ; the decision of R1 is the one taken in F^0 (Definition 3.21). As $a(Z_j) \geq \alpha(s)$, π_x is T_s -terminated at q_j .

($\Rightarrow$) Let π_x be T-terminated in F_s . By the definition of F_s , its termination point is either a T_s -truncation point or a T-termination point of F^0 ; in both cases it is an R1-passed entry $(Y, q) = (Z_i, q_i)$ of π_x^0 with $a(Y) \geq \alpha(s)$ (in the second case $a(Y) \geq \alpha_{\max} \geq \alpha(s)$ by Lemma 9.3(f)). The path is not truncated by H_s before reaching q_i ; in particular the H_s condition fails in the ray step ending at q_i , and (9.1) gives $\eta_i \leq s + 2$. So the index i has the property in (b). By ($\Leftarrow$), π_x is T_s -terminated at q_{j^*} , j^* the least index with this property. Since the termination point is unique and the q_j are distinct by (a), $i = j^*$. This proves (b), including the last sentence.

(c) The description of $\mathcal{T}_s$ is (b). On each component I of $[0, k] \setminus \Sigma_0$, the combinatorial type of π_x^0 is constant, including the outcome of R1 at each entry candidate, and the contact points are affine in x (Lemma 3.18). Hence J and the squares Z_j are constant on I , and each η_j is an affine function on I . Therefore

$$\mathcal{T}_s \cap I = \bigcup_{j: a(Z_j) \geq \alpha(s)} \{x \in I : \eta_j(x) \leq s + 2\}$$

is a finite union of intervals; together with the finite set Σ_0 this shows that $\mathcal{T}_s$ is a finite union of intervals and points. If $s \leq s'$, then $\alpha(s') \leq \alpha(s)$ and $s + 2 \leq s' + 2$, so the condition in (b) for s implies the condition for s' ; thus $\mathcal{T}_s \subset \mathcal{T}_{s'}$. Finally let $s \in [y_0, y_1]$. If $\min_j s_j \leq s$, then $\min_j s_j \leq y_1$, so $T^*(x)$ is defined and equals $\min_j s_j$; conversely $T^*(x) \leq s$ means $\min_j s_j \leq s$. Hence $x \in \{T^* \leq s\}$ if and only if $s_j \leq s$ for some j . Since $y_0 \leq s$, and since for $a(Z_j) > 0$

$$\left(\frac{c_0}{a(Z_j)}\right)^2 \leq s \iff a(Z_j) \geq c_0 s^{-1/2} = \alpha(s),$$

the condition $s_j \leq s$ holds if and only if $a(Z_j) > 0$, $\eta_j \leq s + 2$ and $a(Z_j) \geq \alpha(s)$. Indices with $a(Z_j) = 0$ satisfy neither $s_j \leq s$ nor $a(Z_j) \geq \alpha(s)$, because $s_j = +\infty$ and $\alpha(s) > 0$. So $\{T^* \leq s\} = \mathcal{T}_s$ by (b). The last sentence of (c) follows.

(d) Fix a component I of $[0, k] \setminus \Sigma_0$. On I , the combinatorial type of π_x^0 is constant; the start points, the cumulative gaps, the step lengths $t_i^*(x)$, the contact points and the termination point of π_x^0 are affine in x ; and a ray step with source X has $p'(x) = \lambda u_X$ with $|\lambda| \geq 1$ (Lemmas 3.18 and 3.19). By definition, π_x is π_x^0 truncated at the first, in path order, of the following events:

- a T_s event at an R1-passed entry (Z_j, q_j) with $a(Z_j) \geq \alpha(s)$; the set of such indices j is constant on I ;

- an H_s event in the i -th ray step. If the event of this step in F^0 is D, W or a contact, then $t_i^* = \min(t_S, t_W, t_D)$ (in F^0 the rule H applies only if $t_H < \min(t_S, t_W, t_D)$, where $t_H = (h_{\max} - p_y)/d_y$), and the H_s condition is the affine strict inequality $t_H^{(s)}(x) < t_i^*(x)$. If the event of this step in F^0 is H, then $t_H^{(s)} \leq t_H < \min(t_S, t_W, t_D)$ because $s + 2 \leq h_{\max}$ (Lemma 9.3(f)), and the H_s condition holds;
- an H_s event at the exit point of the i -th pass, whose height $q_{i,y}(x) + \cos \varphi_{Y_i}$ is affine; the condition is that this height is $\geq s + 2$.

If none of these events occurs, π_x inherits the termination of π_x^0 , which is then of kind D, W, E or M: an H-termination of π_x^0 (at a gap point of height $h_{\max}$, or at an exit point of height $\geq h_{\max} \geq s + 2$) satisfies the condition of the second or the third item, and a T-termination of π_x^0 occurs at an R1-passed entry (Z_j, q_j) with $a(Z_j) \geq \alpha_{\max} \geq \alpha(s)$, which is an event of the first item. Let Σ_s be the union of Σ_0 and of the zeros, in the components of $[0, k] \setminus \Sigma_0$, of the finitely many affine functions $t_H^{(s)} - t_i^*$ and $q_{i,y} + \cos \varphi_{Y_i} - (s + 2)$ that do not vanish identically. On each component of $[0, k] \setminus \Sigma_s$ all the conditions above have constant truth values, so the position (the step) and the kind of the first truncation, or its absence, are constant. The ray steps of π_x in F_s are ray steps of π_x^0 , the last one possibly shortened by H_s at a gap point, so their sources are constant and $p' = \lambda u_X$ with $|\lambda| \geq 1$. The termination point is affine: it is $q_j(x)$ for T_s , $p_i(x) + t_H^{(s)}(x) d_i$ for H_s at a gap point, the exit point $q_i(x) + n_{Y_i}$ for H_s at an exit, and the termination point of π_x^0 otherwise. The termination height is the height of $e(x)$, or the constant $s + 2$ for H_s ; it is affine. Therefore $\{x : \pi_x \text{ is } \star\text{-terminated in } F_s, \tau_{F_s}(x) < y\}$ meets each component of $[0, k] \setminus \Sigma_s$ in an interval or in the empty set, and the last assertion follows since Σ_s is finite. $\square$

Lemma 9.7. *Let $s \in [y_0, y_1]$. Then $L_T^{F_s}(y) \leq N(s)$ for every $y \in \mathbb{R}$.*

Proof. By Lemma 9.6(b), the set of x for which π_x is T-terminated in F_s is $\mathcal{T}_s$. Hence $L_T^{F_s}(y) = |\{x \in \mathcal{T}_s : \tau_{F_s}(x) < y\}| \leq |\mathcal{T}_s| = N(s)$. $\square$

Remark 9.8. Lemma 9.7 uses that an inherited T-termination of F^0 is a T_s event of F_s (Definition 3.21) and is therefore counted in $N(s)$. Lemma 9.6 uses that R1 is applied before R2, so that T-terminations occur only at R1-passed entries. Lemma 9.6(c) gives the equality $\{T^* \leq s\} = \mathcal{T}_s$, and not only an inclusion, together with the monotonicity of $N(s)$; both rest on the fact that R1 is decided once, in F^0 , for all scales.

9.4. Squares without alive bottom contacts.

Lemma 9.9. *Let $s \in [y_0, y_1]$, and let $Y \in \mathcal{P}$ be a square such that no path of F_s makes an alive contact with $\text{relint bot}(Y)$.*

- For every $x \in (0, k)$, the trajectory Γ_x of π_x in F_s meets Y in at most one point. If it does, this point is the termination point $e(x)$; it lies in ∂Y , it is the end point $p + t_{SD}$ of the last ray step of π_x and the first point of Y on that ray, and π_x is D-, W- or E-terminated there.*
- If $a(Y) > 0$, then for every $y \in \mathbb{R}$ the set $\{x : e(x) \in \partial Y \cap \ell_y\}$ is finite.*
- $\mu_Y^{F_s}(y) = 0$ for every $y \in (0, s + 2)$; more precisely, there is no x with $\tau_{F_s}(x) > y$ and $P_y(x) \in Y$. If $a(Y) > 0$, then also $|\{x : \tau_{F_s}(x) \geq y, P_y(x) \in Y\}| = 0$ for every $y \in (0, s + 2)$.*

Proof. (a) Consider a ray step of π_x in F_s , with start point p , direction $d = n_X$, where X is the floor or the square last passed, parameters t_S, t_W, t_D , and $t_H := t_H^{(s)}$. Its length in F_s is $t^* = \min(t_S, t_W, t_D, t_H)$: if the H_s condition holds, the step is cut at $t_H < \min(t_S, t_W, t_D)$; otherwise $t_H \geq \min(t_S, t_W, t_D)$, the event of the step in F^0 is not H (as in the proof of Lemma 9.6(d)), and the step ends at $\min(t_S, t_W, t_D)$. In particular $t^* \leq t_S$. Write $r(t) = p + td$.

If X is a square, then $t^* > 0$ (this is also part of Lemma 3.17(b)). Indeed $p \in \text{relint top}(X) \subset X$, and: $t_S > 0$, because X is disjoint from each of the finitely many other squares, which are closed, so $r(t)$ lies in no other square for small $t \geq 0$; $t_D = \delta - g > 0$, because the contact at which the path

entered X was processed, so its cumulative gap g is $< \delta$ (D precedes contact), and a pass does not change g ; $t_W > 0$, because the abscissa of p lies in $(0, k)$: along $\text{top}(X) = \{c_X + \frac{1}{2}n_X + \xi u_X : |\xi| \leq \frac{1}{2}\}$ the abscissa $c_{X,x} - \frac{1}{2} \sin \varphi_X + \xi \cos \varphi_X$ is strictly increasing in ξ and lies in $[0, k]$ for $\xi = \pm \frac{1}{2}$, since $X \subset [0, k]^2$, so it lies in $(0, k)$ for $|\xi| < \frac{1}{2}$; $t_H > 0$, because a ray step starts at an exit point p only if the path was not H-completed there, that is, only if $p_y < s + 2$.

We now go through the points of Γ_x .

- (i) *Points $r(t)$ with $0 < t < t^*$.* Since $t < t_S$, $r(t)$ lies in no square other than X . If X is a square, then $(r(t) - c_X) \cdot n_X = \frac{1}{2} + t > \frac{1}{2}$, so $r(t) \notin X$. Thus $r(t)$ lies in no square.
- (ii) *The start point p .* If X is a square, then $p \in X$ and $X \neq Y$: the path passed X , which requires an alive contact with $\text{relint bot}(X)$, and there is none with $\text{relint bot}(Y)$. As $X \cap Y = \emptyset$, $p \notin Y$. If X is the floor, then $p = (x, 0)$, and if $p \in Y$, then $t_S = 0 = t^*$, so that p is also the end point of the step and is treated in (iii).
- (iii) *The end point $r(t^*)$.* If $t^* < t_S$, then $r(t^*)$ lies in no square: by the argument of (i), using $t^* > 0$ when X is a square. Let $t^* = t_S$ and $r(t_S) \in Y$. Then Y is the only square containing $r(t_S)$, and $r(t) \notin Y$ for $0 \leq t < t_S$. Hence $r(t_S) \in \partial Y$: for $t_S > 0$ it is a limit of points $r(t) \notin Y$, and for $t_S = 0$ it is the point $(x, 0)$ of height 0, which lies on ∂Y because $Y \subset [0, k]^2$. At the parameter $t^* = t_S$ the rules are applied in the order D, W, contact (the rule H and the H_s condition do not apply, since $t^* = t_S$). If the contact branch is reached, then $r(t_S) \notin \text{relint bot}(Y)$ by hypothesis, so the event is E. In every case π_x terminates at $r(t_S)$, by D, W or E.
- (iv) *Pass segments $[q, q + n_{X'}]$.* They lie in the passed square X' , and $X' \neq Y$ as in (ii); so they do not meet Y .

The trajectory Γ_x is the union of the points considered in (i)–(iv); the truncation points of F_s are among them (T_s points are entry points into squares $X' \neq Y$, and H_s points are gap points as in (iii) with $t^* < t_S$, or exit points, which lie on pass segments as in (iv)). Hence the only point of Γ_x that can lie in Y is a point $r(t_S)$ as in the second case of (iii), and π_x terminates there. This proves (a).

(b) Let $a(Y) > 0$ and $y \in \mathbb{R}$. Let I be a component of $[0, k] \setminus \Sigma_s$ (Lemma 9.6(d)) and $I_Y := \{x \in I : e(x) \in Y\}$; assume $I_Y \neq \emptyset$. By (a), for $x \in I_Y$ the path π_x is D-, W- or E-terminated at the end point of its last ray step. As the termination kind and the index of the last ray step are constant on I , for every $x \in I$ the termination point is the end point $e(x) = p(x) + t^*(x)d$ of the last ray step, whose source X and direction $d = n_X$ are constant on I ; p is affine with $p' = \lambda u_X$, $|\lambda| \geq 1$, and e is affine. Since $d \perp u_X$, $e(x) \cdot u_X = p(x) \cdot u_X$ has derivative $\lambda \neq 0$; hence e is injective on I . For $x \in I_Y$, by (a), $e(x) = p(x) + t_S(x)d$ is the first point of Y on the ray $\{p(x) + td : t \geq 0\}$.

- (1) *Vertices.* As e is injective on I , at most four $x \in I$ have $e(x)$ at a vertex of Y . (This suffices here. In fact no $x \in I$ has $e(x)$ at a vertex: I lies in a piece of F^0 , the ray steps of π_x in F_s are initial parts of those of π_x^0 , and by Lemma 3.18(c) for F^0 the lines carrying them contain no vertex of a square.) Every other $x \in I_Y$ has $e(x) \in \partial Y$ (by (a)) not a vertex, so $e(x)$ lies in the relative interior of one of the four sides of Y . For a side σ of Y let $I_\sigma := \{x \in I_Y : e(x) \in \text{relint } \sigma\}$.
- (2) *If $I_\sigma \neq \emptyset$, then d is not parallel to σ .* Let $x \in I_\sigma$. First, $p(x) \notin Y$: if X is a square, then $p(x) \in X$ and $X \cap Y = \emptyset$ (as in (a)(ii)); if X is the floor and $p(x) = (x, 0) \in Y$, then $t_S = 0$ and $e(x) = (x, 0)$ is a point of Y at height 0, i.e. a lowest point of Y ; since $a(Y) > 0$, the lowest point of Y is a vertex, contradicting $e(x) \in \text{relint } \sigma$. If d were parallel to σ , the line of the ray would contain σ , because it meets σ at $e(x)$; that line meets Y exactly in σ , and a ray starting at $p(x) \notin \sigma$ first meets σ at one of its end points, which is a vertex. This contradicts $e(x) \in \text{relint } \sigma$.
- (3) *An affine parametrization.* Let $I_\sigma \neq \emptyset$, and let L_σ be the line containing σ . For $x \in I$ let $h(x)$ be the intersection point of the line $\{p(x) + td : t \in \mathbb{R}\}$ with L_σ ; it exists and is unique

by (2). If ν is a unit normal of L_σ and $L_\sigma = \{z : \nu \cdot z = c\}$, then $h(x) = p(x) + \frac{c - \nu \cdot p(x)}{\nu \cdot d} d$, an affine function of x . Since $d = n_X \perp u_X$, we have $h(x) \cdot u_X = p(x) \cdot u_X$, whose derivative is $\lambda \neq 0$; hence $h' \neq 0$, and h' is parallel to σ because $h(x) \in L_\sigma$. For $x \in I_\sigma$, $e(x)$ lies on the ray and on L_σ , so $e(x) = h(x)$.

- (4) *Monotone height.* The side σ is parallel to u_Y or to n_Y , and the second components of these vectors are $\sin \varphi_Y$ and $\cos \varphi_Y$. Since $0 < a(Y) = |\varphi_Y| \leq \pi/4$, both are nonzero. Hence the height of $h(x)$ is a strictly monotone affine function on I , and at most one $x \in I_\sigma$ has $e(x) \in \ell_y$.

Thus each component of $[0, k] \setminus \Sigma_s$ contains at most eight points x with $e(x) \in \partial Y \cap \ell_y$: at most four with $e(x)$ at a vertex, and at most one in each I_σ . There are finitely many components, and Σ_s is finite; this proves (b).

(c) Let $y \in (0, s + 2)$, $\tau_{F_s}(x) > y$ and $P_y(x) \in Y$. By (a), $P_y(x) \in \Gamma_x \cap Y$ is the termination point $e(x)$, and π_x is D-, W- or E-terminated there, so $\tau_{F_s}(x)$ is the height of $e(x)$, namely y ; this contradicts $\tau_{F_s}(x) > y$. Hence no such x exists, and in particular $\mu_Y^{F_s}(y) = 0$. If $a(Y) > 0$, $\tau_{F_s}(x) \geq y$ and $P_y(x) \in Y$, then in the same way $e(x) = P_y(x) \in \partial Y \cap \ell_y$, and by (b) there are only finitely many such x . $\square$

Remark 9.10. Part (b) can also be obtained without Lemma 3.18, with the explicit bound $2(|\mathcal{P}| + 1)$. Since no side of Y is horizontal when $a(Y) > 0$, the line ℓ_y meets ∂Y in at most two points. By (a), each x in the set of (b) ends a ray step at a point $z \in \partial Y \cap \ell_y$, so $z = p + t n_X$ with p the start point and X the source of that step. For fixed z and X there is at most one such x . If X is the floor, then $x = z_x$. If X is a square, then p is the unique point of the line of $\text{top}(X)$ with $z - p$ parallel to n_X ; so all such paths leave X at the same exit point p , hence enter X at the same point $p - n_X \in \text{relint bot}(X)$, where only one path is the winner of R1. As there are at most $|\mathcal{P}| + 1$ sources, the set in (b) has at most $2(|\mathcal{P}| + 1)$ elements.

Remark 9.11. The hypothesis $a(Y) > 0$ in (b) and in the last assertion of (c) cannot be omitted. Under the standing assumptions (so that $y_0 \leq y_1$), let $\mathcal{P}$ consist of the two squares $X = [1, 2] \times [0, 1]$ and $Y = [1, 2] \times [1 + \delta, 2 + \delta]$, and let $s \in [y_0, y_1]$. For $x \in (1, 2)$, the first ray step of π_x has $t_S = 0$ at the point $(x, 0) \in \text{relint bot}(X)$; this contact is processed ($t_D = \delta > 0$ and $t_W = +\infty$), x is the only path reaching it, and since $a(X) = 0 < \alpha(s) \leq \alpha_{\max}$ the path passes X and leaves it at $(x, 1)$ with cumulative gap 0. In the next ray step $t_S = t_D = \delta$, $t_W = +\infty$ and $t_H^{(s)} = s + 1 > \delta$, so D applies and π_x terminates at $(x, 1 + \delta) \in \text{bot}(Y)$. For $x \notin [1, 2]$ the vertical ray from $(x, 0)$ meets neither square, and for $x \in \{1, 2\}$ the path is E-terminated at the vertex $(x, 0)$ of X . So no path of F_s makes an alive contact with $\text{relint bot}(Y)$, but every $x \in (1, 2)$ has $e(x) \in \partial Y \cap \ell_{1+\delta}$, and $|\{x : \tau_{F_s}(x) \geq 1 + \delta, P_{1+\delta}(x) \in Y\}| = 1$. In this example $\mu_Y^{F_s} \equiv 0$, in accordance with the first assertion of (c), which needs no assumption on $a(Y)$. The squares of $\mathcal{Z}(s)$, to which Lemma 9.9 is applied below, satisfy $a(Z) > 0$ (Lemma 9.3(c)).

9.5. Measurability.

Lemma 9.12. *Let $s \in [y_0, y_1]$.*

- (a) *The set $\mathcal{T}_s$ and the sets $\{x : \pi_x \text{ is } \star\text{-terminated in } F_s, \tau_{F_s}(x) < y\}$, for $\star \in \{T, D, E, M, W\}$ and $y \in \mathbb{R}$, are finite unions of intervals and points; in particular $N(s) \in [0, k]$ and $L_\star^{F_s}(y)$ are well defined. Moreover $\Lambda_0 < \infty$.*
- (b) *For every square S , c_S is piecewise linear with finitely many pieces. The functions ω (on $[0, k]$) and E are piecewise linear with finitely many pieces, and the sets $H, Y_i, Y_{ii}, Y_{iii}, Y_b, Y_l, \mathcal{W}_s, V_s$ and $\Phi(S)$ are finite unions of intervals. In particular $y \mapsto (1 - \omega_0 - E(y))_+$ and $y \mapsto c_S(y) \mathbf{1}_{\Phi(S)}(y)$ are Borel functions.*
- (c) *On $[y_0, y_1]$ the function $s \mapsto N(s)$ is nondecreasing, hence Borel; the function $s \mapsto |\mathcal{Z}(s)|$ is a finite sum of indicator functions of finite unions of intervals, hence Borel; and $(y, s) \mapsto \mathbf{1}_{[y \in Y_b \cap \mathcal{W}_s]}$ is a Borel function on $\mathbb{R} \times [y_0, y_1]$.*

The same statements hold for the ceiling flows and for $\tilde{Z}(s)$, Y_t and $k - \mathcal{W}_s$.

Proof. (a) The first assertion is Lemma 9.6(c),(d). The sets $\mathcal{D}$, $\mathcal{M}$, $\mathcal{E}$ are finite unions of intervals and points by Lemma 3.18 (see also Lemma 3.31(a)), and $\Lambda_0 \leq 2k < \infty$ (Section 3.8).

(b) If $a(S) = 0$, then c_S is the indicator function of an interval of length 1. If $a(S) > 0$, then c_S vanishes outside the vertical extent of S and is affine between consecutive heights of vertices of S , because each end point of the chord moves along a fixed side between two such heights. Since $\mathcal{P}$ is finite and the chords $S \cap \ell_y$, $S \in \mathcal{P}$, are pairwise disjoint subsets of $[0, k] \times \{y\}$, the functions $\omega = k - \sum_S c_S$ (on $[0, k]$) and $E = \sum_S (c_S - 1)_+$ are piecewise linear with finitely many pieces; hence $\{\omega < \omega_0\}$, $\{\omega \geq \omega_0\}$ and $\Phi(S) = \{0 < c_S < 1\}$ are finite unions of intervals. For a square S let

$$A_S := \{y \in (0, k) : S \text{ does not meet the line } y, \text{ or } a(S) < \beta(y)\}.$$

If $a(S) = 0$, then $A_S = (0, k)$, since $\beta > 0$. If $a(S) > 0$, then $a(S) < \beta(y)$ if and only if $d(y) < (c_1/a(S))^{4/3}$, so

$$A_S = ((0, k) \setminus [v_S, v_S + \cos a(S) + \sin a(S)]) \cup \{y \in (0, k) : d(y) < (c_1/a(S))^{4/3}\},$$

where v_S is the height of the lowest point of S ; this is a finite union of intervals. The set $\{y \in (0, k) : y_0 \leq d(y) \leq (1 - \varepsilon)y_1\}$ consists of at most two intervals, and $\{y \in (0, k) : \text{dist}(y, \mathbb{Z}) \geq w_0\}$ is a finite union of intervals (one condition in each of the finitely many intervals $[m, m + 1]$, $m \in \mathbb{Z}$); so H is a finite union of intervals. Then $Y_{iii} = H \cap \{\omega < \omega_0\} \cap \bigcap_{S \in \mathcal{P}} A_S$, $Y_{ii} = (H \cap \{\omega < \omega_0\}) \setminus Y_{iii}$, $Y_i = H \cap \{\omega \geq \omega_0\}$, Y_b and Y_t are finite unions of intervals. The Borel property of the two functions follows.

(c) The monotonicity of N is Lemma 9.6(c). For $Z \in \mathcal{P}$ the set $G_Z := \Phi(Z) \cap Y_b$ is a finite union of intervals by (b). For an interval J , the set of s with $J \cap \mathcal{W}_s \neq \emptyset$ is the intersection of $\{s : (1 - \varepsilon)s \leq \sup J\}$ and $\{s : s \geq \inf J\}$, with strict inequality where the corresponding end point of J does not belong to J ; as both end points of $\mathcal{W}_s$ increase with s , this is an interval. Hence $\{s : Z \in \mathcal{Z}(s)\}$ is a finite union of intervals, and $|\mathcal{Z}(s)| = \sum_{Z \in \mathcal{P}} \mathbf{1}[Z \in \mathcal{Z}(s)]$ is Borel. Finally $\mathbf{1}[y \in Y_b \cap \mathcal{W}_s] = \mathbf{1}_{Y_b}(y) \mathbf{1}_C(y, s)$ with the closed set $C = \{(y, s) : (1 - \varepsilon)s \leq y \leq s\}$.

The ceiling statements follow in the same way, or by applying (a)–(c) to the packing $\rho\mathcal{P}$ (Section 9.8). $\square$

9.6. The per-scale count.

Proof of Theorem 9.1(a). Fix $s \in [y_0, y_1]$. If $\mathcal{Z}(s) = \emptyset$, the claim reads $N(s) \geq -\Lambda^*(s)$, which holds because $N(s) \geq 0$ and $\Lambda^*(s) \geq 0$. Let $\mathcal{Z}(s) \neq \emptyset$.

Step 1: $Z \subset \mathbb{R} \times V_s$ for every $Z \in \mathcal{Z}(s)$. Let $a := a(Z)$, and let v be the height of the lowest point of Z . By [R, Lemma 3.1(a)], the vertical extent of Z is $[v, v + \cos a + \sin a]$; it contains y_Z because $c_Z(y_Z) > 0$. By Lemma 9.3(d), its length $\cos a + \sin a$ is less than 1.0001. Hence, as $y_Z \in \mathcal{W}_s$,

$$v > y_Z - 1.0001 \geq (1 - \varepsilon)s - 1.0001, \quad v + \cos a + \sin a < y_Z + 1.0001 \leq s + 1.0001.$$

Thus $Z \subset \mathbb{R} \times V_s$, and $c_Z(y) = 0$ for $y \notin V_s$.

Step 2: for $Z \in \mathcal{Z}(s)$, no path of F_s makes an alive contact with $\text{relint bot}(Z)$, and $\mu_Z^{F_s}(y) = 0$ for all $y \in (0, s + 2)$. These assertions are also contained in Theorem 4.13(b),(d), and Step 3 below is Theorem 4.13(e); we derive them here from Theorem 4.11, using only (C1) and (C2). Suppose that a path of F_s makes an alive contact with $\text{relint bot}(Z)$ at a point q .

- Since $q \in Z$, Step 1 gives $q_y \leq s + 1.0001 \leq s + 2$.
- $q_y > 0$. Indeed $a(Z) > 0$ by Lemma 9.3(c), so $\sin \varphi_Z \neq 0$, and along $\text{bot}(Z) = \{c_Z - \frac{1}{2}n_Z + \xi u_Z : |\xi| \leq \frac{1}{2}\}$ the height $c_{Z,y} - \frac{1}{2} \cos \varphi_Z + \xi \sin \varphi_Z$ is a strictly monotone function of ξ . Hence a point of $\text{relint bot}(Z)$ lies strictly above one of the two end points of $\text{bot}(Z)$, whose height is ≥ 0 because $Z \subset [0, k]^2$.

- By Theorem 4.11(a), which applies because (Q1) follows from (C1), $s \in [y_0, y_1]$, and $q \in \text{bot}(Z)$ lies on the trajectory of a path of F_s , every $y \in \mathcal{R}(Z)$ satisfies $\text{dist}(y, \mathbb{Z}) < w(s, Z)$, with $w(s, Z)$ as in Lemma 9.3(e). By the two previous items and Lemma 9.3(e), $w(s, Z) < w_0$.
- On the other hand, $a(Z) > 0$, so $y_Z \in \Phi(Z) \subset \mathcal{R}(Z)$ by [R, Lemma 3.4]; and $y_Z \in H$, so $\text{dist}(y_Z, \mathbb{Z}) \geq w_0$. This contradicts the previous item.

Hence no such contact exists; in particular no path of F_s enters Z . Lemma 9.9, applied with $Y = Z$, gives $\mu_Z^{F_s}(y) = 0$ for every $y \in (0, s+2)$; as $a(Z) > 0$, the same holds if in the definition of μ the interior of Z is replaced by Z or the condition $\tau_{F_s}(x) > y$ by $\tau_{F_s}(x) \geq y$.

Step 3: a lower bound for the shadow. For every $y \in (0, s+2)$, by Step 2 and since all terms are nonnegative,

$$\text{Sh}^{F_s}(y) = \sum_{S \in \mathcal{P}} (c_S(y) - \mu_S^{F_s}(y))_+ \geq \sum_{Z \in \mathcal{Z}(s)} (c_Z(y) - 0)_+ = \sum_{Z \in \mathcal{Z}(s)} c_Z(y).$$

Step 4: the shadow inequality. Let $\Omega_s := \text{Exc} \cap (0, s+2)$, where $\text{Exc} = \text{Exc}_{F_0}$ is the finite set of exceptional heights (Definition 3.32 and Lemma 3.33) off which Theorem 6.9(f) holds. For every $y \in (0, s+2) \setminus \Omega_s$,

$$\sum_{Z \in \mathcal{Z}(s)} c_Z(y) \leq \text{Sh}^{F_s}(y) \leq L_T^{F_s}(y) + \Lambda^{F_s}(y) \leq N(s) + \Lambda_0 + 2(s+2) \tan \alpha(s) = N(s) + \Lambda^*(s).$$

The first inequality is Step 3; the second is Theorem 6.9(f) for $F = F_s$, whose height bound is $h_{F_s} = s+2$; the third follows from Lemma 9.7 and Proposition 6.10, which applies since $y < s+2$.

Step 5: integration. Let $c_{\mathcal{Z}(s)}(y) := \sum_{Z \in \mathcal{Z}(s)} c_Z(y)$ (the total length of the chords of the squares of $\mathcal{Z}(s)$), a finite sum of bounded nonnegative Borel functions. By Step 1 and since $Z \subset [0, k]^2$, $c_{\mathcal{Z}(s)}$ vanishes outside $V_s \cap [0, k]$. As $s+1.0001 < k$, we have $V_s \cap [0, k] \subset [0, s+1.0001] \subset [0, s+2)$. Hence, by Step 4, $c_{\mathcal{Z}(s)}(y) \leq N(s) + \Lambda^*(s)$ for every $y \in V_s \cap [0, k]$ outside the finite set $\Omega_s \cup \{0\}$; for $y \in V_s$ with $y < 0$ this holds as well, because $c_{\mathcal{Z}(s)}(y) = 0$. Since $\int_{\mathbb{R}} c_Z = 1$ for every square,

$$|\mathcal{Z}(s)| = \sum_{Z \in \mathcal{Z}(s)} \int_{\mathbb{R}} c_Z(y) dy = \int_{V_s} c_{\mathcal{Z}(s)}(y) dy \leq |V_s| (N(s) + \Lambda^*(s)) = (\varepsilon s + b_W)(N(s) + \Lambda^*(s)).$$

Dividing by $\varepsilon s + b_W > 0$ gives Theorem 9.1(a). $\square$

Remark 9.13. (1) The proof does not use any measurability of $y \mapsto \text{Sh}^{F_s}(y)$: it uses the pointwise bound of Step 4 outside a finite set and integrates only the piecewise linear function $c_{\mathcal{Z}(s)}$.

(2) In Step 1 the number 1.0001 may be replaced by $\max_{Z \in \mathcal{Z}(s)} (\cos a(Z) + \sin a(Z)) < 1 + 1.5 \cdot 10^{-6}$. Hence Theorem 9.1(a) also holds with $b_W = 2 + 3 \cdot 10^{-6}$; we keep $b_W = 2.0002$.

(3) Of the definition of $\mathcal{Z}(s)$, the proof of Theorem 9.1(a) uses only the following properties of each $Z \in \mathcal{Z}(s)$: there is a point $y_Z \in \Phi(Z) \cap H \cap \mathcal{W}_s$ with $y_Z < k/2$, and $a(Z) < \beta(y_Z)$. The condition $y_Z < k/2$ gives $d(y_Z) = y_Z$, hence $\beta(y_Z) = \hat{\beta}(y_Z)$ and $y_Z \geq y_0$ (Lemma 9.3(c)); these properties enter the height bound of Step 1 through Lemma 9.3(d) and the inequality $w(s, Z) < w_0$ of Step 2 through Lemma 9.3(e). The condition $\omega(y_Z) < \omega_0$ of kind (iii) is not used here; it is used in Theorem 9.2.

Remark 9.14. (1) Constraint (C3), which says that $\hat{\beta}((1-\varepsilon)s) < \alpha(s)$ for all $s \geq y_0$, is not used in this section. Theorem 4.11 applies to every square Z and does not require $a(Z) < \alpha(s)$, and Step 2 shows that a square of $\mathcal{Z}(s)$ receives no alive contact from F_s at all; so it does not matter whether R2 would stop a path at such a square.

(2) Constraint (Q2) is not used either: Step 2 uses (C1), through (Q1) in Theorem 4.11, and (C2), through Lemma 9.3(c)–(e), for every $\varepsilon \in (0, 1)$. Only the statement about alive contacts with $\text{relint bot}(Z)$ is needed in this paper.

9.7. The lower bound for $|\mathcal{Z}(s)|$.

Proof of Theorem 9.2(a). Fix $s \in [y_0, y_1]$.

Step 1: one line. Let $y \in Y_b \cap \mathcal{W}_s$. Then $y \in H \subset (0, k)$, so [R, Lemma 3.5] applies ($k \geq 2$). Since $\omega(y) < \omega_0$ for $y \in Y_{iii}$,

$$\sum_{S: 0 < c_S(y) < 1} c_S(y) \geq 1 - \omega(y) - E(y) > 1 - \omega_0 - E(y).$$

The left side is nonnegative, so it is $\geq (1 - \omega_0 - E(y))_+$.

Step 2: the squares in the sum. Every square S in the sum of Step 1 satisfies $y \in \Phi(S) \cap Y_b \cap \mathcal{W}_s$, so $S \in \mathcal{Z}(s)$ by definition. Hence, for every $y \in Y_b \cap \mathcal{W}_s$,

$$(1 - \omega_0 - E(y))_+ \leq \sum_{Z \in \mathcal{Z}(s)} c_Z(y) \mathbf{1}_{\Phi(Z)}(y).$$

Step 3: integration. The set $Y_b \cap \mathcal{W}_s$ is a finite union of intervals and both sides are Borel functions (Lemma 9.12). Integrating over $Y_b \cap \mathcal{W}_s$ and interchanging the integral with the finite sum,

$$\int_{Y_b \cap \mathcal{W}_s} (1 - \omega_0 - E(y))_+ dy \leq \sum_{Z \in \mathcal{Z}(s)} \int_{\Phi(Z) \cap Y_b \cap \mathcal{W}_s} c_Z(y) dy \leq \sum_{Z \in \mathcal{Z}(s)} \int_{\Phi(Z)} c_Z(y) dy.$$

Each square is counted once here. A square may have short chords on many lines of $Y_b \cap \mathcal{W}_s$; its contribution is the integral of these chords over all such lines, which is at most $\int_{\Phi(Z)} c_Z$, while $|\mathcal{Z}(s)|$ counts the square once.

Step 4: the area of the short chords. Let $Z \in \mathcal{Z}(s)$. By Lemma 9.3(c), $0 < a(Z) < \hat{\beta}((1 - \varepsilon)s)$, and by [R, Lemma 3.4], $\int_{\Phi(Z)} c_Z = \sin a(Z) \cos a(Z) \leq a(Z)$. Hence the right side of Step 3 is at most $|\mathcal{Z}(s)| \hat{\beta}((1 - \varepsilon)s)$; if $\mathcal{Z}(s) = \emptyset$, all three quantities in Theorem 9.2(a) are 0. This proves the chain of inequalities. Since $\hat{\beta}((1 - \varepsilon)s) = c_1((1 - \varepsilon)s)^{-3/4} > 0$, dividing by it gives the lower bound for $|\mathcal{Z}(s)|$. $\square$

Remark 9.15. The term $E(y)$ is kept inside the positive part. In Section 10 it is bounded by means of [R, Lemma 4.15], whose hypotheses on the lines of kind (iii) are recorded in Remark 9.5.

9.8. The ceiling versions.

Proof of Theorems 9.1(b) and 9.2(b). Let $\rho(x, y) = (x, k - y)$ and $\mathcal{P}' := \rho\mathcal{P}$. Quantities computed for $\mathcal{P}'$ are marked with a prime.

- (1) ρ is an isometry of the plane that maps $[0, k]^2$ onto itself. Hence $\mathcal{P}'$ is a closed packing of $[0, k]^2$ with the same number of squares and the same waste area.
- (2) For every square S , ρS is a square with $a(\rho S) = a(S)$, since ρ maps horizontal and vertical directions to horizontal and vertical directions. Since $\rho S \cap \ell_y = \rho(S \cap \ell_{k-y})$, we have $c'_{\rho S}(y) = c_S(k - y)$, and therefore $\Phi(\rho S) = k - \Phi(S)$.
- (3) The square ρS meets the line y if and only if S meets the line $k - y$. Moreover $\omega'(y) = \omega(k - y)$, $E'(y) = E(k - y)$, $d(k - y) = d(y)$ and $\beta(k - y) = \beta(y)$.
- (4) H is invariant under $y \mapsto k - y$: d is invariant, and since k is an integer, $\mathbb{Z}$ is invariant under $t \mapsto k - t$, so $\text{dist}(k - y, \mathbb{Z}) = \text{dist}(y, \mathbb{Z})$. This is where the integrality of k is used.
- (5) By (2)–(4), the kind of $y \in (0, k)$ with respect to $\mathcal{P}'$ is the kind of $k - y$ with respect to $\mathcal{P}$. Hence $Y'_{iii} = k - Y_{iii}$, and $Y'_b = Y'_{iii} \cap (0, k/2) = k - Y_t$.
- (6) For $Z \in \mathcal{P}$: $\rho Z \in \mathcal{Z}'(s)$ if and only if $(k - \Phi(Z)) \cap (k - Y_t) \cap \mathcal{W}_s \neq \emptyset$, that is, if and only if $\Phi(Z) \cap Y_t \cap (k - \mathcal{W}_s) \neq \emptyset$. Hence $\mathcal{Z}'(s) = \rho \tilde{\mathcal{Z}}(s)$ and $|\mathcal{Z}'(s)| = |\tilde{\mathcal{Z}}(s)|$.
- (7) By the definition of the ceiling flows, $\tilde{N}(s) = N'(s)$ and $\tilde{\Lambda}_0 = \Lambda'_0$, so $\tilde{\Lambda}^*(s) = \Lambda'^*(s)$; the floor coordinate x is the same, as $\rho(x, 0) = (x, k)$.

- (8) All results used in Sections 9.2–9.7 (the lemmas of Section 3, Theorems 4.11 and 6.9, Proposition 6.10 and the results of [R]) hold for every closed packing of $[0, k]^2$, and the constraints on the parameters do not depend on the packing. Hence Theorems 9.1(a) and 9.2(a) hold for $\mathcal{P}'$. Theorem 9.1(b) follows from (6) and (7). For Theorem 9.2(b), the substitution $y \mapsto k - y$, together with (3) and (5), gives

$$\int_{Y'_b \cap \mathcal{W}_s} (1 - \omega_0 - E'(y))_+ dy = \int_{Y_t \cap (k - \mathcal{W}_s)} (1 - \omega_0 - E(y))_+ dy,$$

and (6) gives $|\mathcal{Z}'(s)| = |\tilde{\mathcal{Z}}(s)|$. $\square$

Corollary 9.16. *Lemmas 9.6, 9.7, 9.9 and 9.12 hold for the ceiling flows. In particular, for $s \in [y_0, y_1]$, the set of x for which $\tilde{T}^*(x)$ is defined and $\tilde{T}^*(x) \leq s$ is the set of x whose ceiling path is T_s -terminated in $\tilde{F}_s$, the function $s \mapsto \tilde{N}(s)$ is nondecreasing on $[y_0, y_1]$, and $\tilde{N}(s)$ is the measure of $\{x : \tilde{T}^*(x) \leq s\}$.*

Proof. By the definition of the ceiling flows, these statements are the statements of Lemmas 9.6–9.12 for the closed packing $\rho\mathcal{P}$, to which the lemmas apply as in step (8) of the preceding proof. $\square$

10. THE INTEGRATION INEQUALITY

In this section we prove the inequality (★) of Theorem 10.3. It bounds the waste W from below by an explicit expression in k and the parameters. Horizontal lines are weighted with the line measure ℓ of Definition 3.5; we write

$$\ell(Y) = \int_Y d(y)^{-3/4} dy, \quad \int_Y g d\ell := \int_Y g(y) d(y)^{-3/4} dy,$$

where $Y \subset H$ is a Borel set and $g \geq 0$ is a Borel function. The heights $y \in H$ are of three kinds. Lines of the first kind carry line waste. Lines of the second kind meet a square of large inclination; they are paid for by [R, Lemma 4.13]. On lines of the third kind the excess of the long chords is paid for by [R, Lemma 4.15], and the short chords are counted by the per-scale bounds of Section 9; these bounds are integrated over the scales s against $|\alpha'(s)| ds$ and compared with the scale form of Section 8.

10.1. Standing assumptions and notation. Throughout Sections 10–12, $k \geq 4$ is an integer, $\mathcal{P}$ is a closed packing of $[0, k]^2$, $N = |\mathcal{P}|$ and $W = k^2 - N$; $\delta = 10^{-5}$, $\bar{\beta} = 1.5 \cdot 10^{-6}$ and $b_W = 2.0002$ (Definition 3.1); and the parameters $\pi = (c_0, c_1, y_0, \varepsilon, \omega_0)$ satisfy the constraints (C0)–(C7) of Definition 3.2 for the given k . We use the notation of Section 3 and recall the parts that are needed.

- (N1) (Definition 3.1.) $\alpha(s) = c_0 s^{-1/2}$ for $s > 0$ and $\alpha_{\max} = \alpha(y_0)$; $\hat{\beta}(t) = c_1 t^{-3/4}$ for $t > 0$; $d(y) = \min(y, k - y)$ and $\beta(y) = \hat{\beta}(d(y))$ for $0 < y < k$; $b^* = \hat{\beta}((1 - \varepsilon)y_0)$; $y_1 = k/2 - 3$, $h_{\max} = k/2 - 1$;

$$w_0 = 0.50001 c_0^2 \left(1 + \frac{1.0001}{y_0}\right) + \delta + 1.0001 b^* + 10^{-12}, \quad h_0 = 1 - 2w_0.$$

- (N2) (Definition 3.5.) With $\{y\}$ the fractional part of y ,

$$H = \{y \in (0, k) : y_0 \leq d(y) \leq (1 - \varepsilon)y_1, \{y\} \in [w_0, 1 - w_0]\},$$

$H_b = H \cap (0, \frac{k}{2})$ and $H_t = H \cap (\frac{k}{2}, k)$. A height $y \in H$ is of kind (i) if $\omega(y) \geq \omega_0$; of kind (ii) if $\omega(y) < \omega_0$ and the line y meets a square S with $a(S) \geq \beta(y)$; of kind (iii) if $\omega(y) < \omega_0$ and every square meeting the line y has $a(S) < \beta(y)$. The sets of heights of the three kinds are Y_i, Y_{ii}, Y_{iii} , and $Y_b = Y_{iii} \cap (0, \frac{k}{2})$, $Y_t = Y_{iii} \cap (\frac{k}{2}, k)$. For $s \in [y_0, y_1]$, $\mathcal{W}_s = [(1 - \varepsilon)s, s]$,

$$\mathcal{Z}(s) = \{Z \in \mathcal{P} : \Phi(Z) \cap Y_b \cap \mathcal{W}_s \neq \emptyset\}, \quad \tilde{\mathcal{Z}}(s) = \{Z \in \mathcal{P} : \Phi(Z) \cap Y_t \cap (k - \mathcal{W}_s) \neq \emptyset\},$$

where $k - Y = \{k - y : y \in Y\}$ for $Y \subset \mathbb{R}$. Here $\omega(y)$ is the line waste, $c_S(y)$ the length of the chord $S \cap (\mathbb{R} \times \{y\})$, $\Phi(S) = \{y : 0 < c_S(y) < 1\}$ and $E(y) = \sum_{S \in \mathcal{P} : c_S(y) \geq 1} (c_S(y) - 1)$ ([R, Lemmas 3.4 and 3.5]).

- (N3) (Definitions 3.20, 3.21 and 3.24.) F^0 is the master flow, with paths π_x^0 and R1-passed entries (Z_j, q_j) , $j \leq J(x)$, of heights $\eta_j = (q_j)_y$. For $s \in [y_0, y_1]$, F_s is the scale flow, with paths π_x^s ; $\mathcal{T}_s$ is the set of x for which π_x^s is T-terminated (equivalently, T_s -terminated), and $N(s) = |\mathcal{T}_s|$. The ceiling versions carry a tilde.
- (N4) (Definition 3.22.) $s_j = \max(y_0, \eta_j - 2, (c_0/a(Z_j))^2)$ if $a(Z_j) > 0$ and $s_j = +\infty$ if $a(Z_j) = 0$; $T^*(x) = \min_j s_j$ on $\text{dom } T^* = \{x : \min_j s_j \leq y_1\}$; $\tilde{T}^*$ is its ceiling version.
- (N5) (Definition 3.30.) $\Lambda_0 = |\mathcal{D}| + |\mathcal{M}| + L_E + \sup_{0 < y < h_{\max}} \text{Ov}_{\text{gap}}^{F^0}(y)$ and $\Lambda^*(s) = \Lambda_0 + 2(s + 2) \tan \alpha(s)$, with ceiling versions $\tilde{\Lambda}_0$, $\tilde{\Lambda}^*(s) = \tilde{\Lambda}_0 + 2(s + 2) \tan \alpha(s)$.

Lemma 10.1. *Under the standing assumptions:*

- (a) $0 < \alpha(s) \leq \alpha(y_0) \leq 1.5 \cdot 10^{-6}$ for every $s \geq y_0$; (Q1) and (C6) hold;
- (b) $0 < w_0 < \frac{1}{2}$ and $0 < h_0 < 1$;
- (c) $y_0 + 1 \leq (1 - \varepsilon)y_1 < y_1 < \frac{k}{2} - 1$, so $[y_0, y_1]$ is a nondegenerate interval;
- (d) for $y \in H$: $y_0 \leq d(y) \leq (1 - \varepsilon)y_1 < \frac{k}{2} - 1$ and $0 < \beta(y) \leq \hat{\beta}(y_0) \leq 1.5 \cdot 10^{-6}$; moreover $\frac{k}{2} \notin H$, $d(y) = y$ on H_b and $d(y) = k - y$ on H_t ;
- (e) $H = Y_i \sqcup Y_{ii} \sqcup Y_{iii}$ and $Y_{iii} = Y_b \sqcup Y_t$ (disjoint unions).

Proof. (a) α is decreasing and positive, and $\alpha(y_0) \leq 1.5 \cdot 10^{-6}$ is (C1); (Q1) and (C6) follow from (C1) (Lemma 3.3(a)). (b) $w_0 > 0$ is clear and $w_0 < \frac{1}{2}$ is (C5). (c) The first inequality is (C7); then $y_1 > 0$ and $(1 - \varepsilon)y_1 < y_1 = \frac{k}{2} - 3$. (d) The bounds on $d(y)$ are the definition of H and (c); $\beta(y) = c_1 d(y)^{-3/4} \leq c_1 y_0^{-3/4}$, which is at most $1.5 \cdot 10^{-6}$ by (C2). As $d(k/2) = k/2 > (1 - \varepsilon)y_1$, $\frac{k}{2} \notin H$; the formulas for d are its definition. (e) The three kinds are defined by mutually exclusive conditions which together cover H , and $Y_{iii} \subset H \subset (0, k) \setminus \{\frac{k}{2}\}$. $\square$

The constraints enter this section as follows: (C0) everywhere (the integrality of y_0 in Step 2 below); (C1) in Lemma 10.1(a) and Step 11; (C2) in Steps 4 and 5 (the hypotheses of [R, Lemmas 4.13 and 4.15]); (C4) in Step 4; (C5) in Step 2; (C7) in Lemma 10.1(c) and Step 2. Constraint (C3) is not used in Sections 10–12, and neither is (Q2). The results of Sections 3–9 quoted below are stated under hypotheses that are part of the standing assumptions.

10.2. The constants. We use the constants of Definition 3.6: with $Q_* = 0.32$ and $K_w^* = 9$,

$$A = \frac{4\sqrt{K_w^*/Q_*}}{2 - 3 \cdot 10^{-6}} \quad (\text{the constant of [R, Lemma 4.13]}),$$

$$A' = \frac{A}{\cos \beta} + 10^{-8} \quad (\text{the constant of Corollary 8.13}),$$

$$C_\Lambda = 1 + 1185.6 \delta^2 \quad (\text{the constant of Corollary 7.2}),$$

and, for the given π and k ,

$$\text{inv } F = \frac{4(1 - \varepsilon)^{3/4}(1 - (1 - \varepsilon)^{3/4})}{3\varepsilon(1 + b_W/(\varepsilon y_0))}, \quad \Gamma = 1.00002 \left(2c_1 A y_0^{-1/2} + \frac{4}{5} \frac{c_1^2}{\delta} y_0^{-5/4} \right),$$

$$\Omega_W = \int_{y_0}^{y_1} 2(s + 2) \tan \alpha(s) \frac{c_0}{2} s^{-3/2} ds, \quad K_W = \frac{4c_1}{c_0 \text{inv } F} \cdot 1.000001 c_0^2,$$

$$B = \frac{1}{\omega_0 y_0^{3/4}} + \frac{A}{2c_1} + \frac{2c_1}{c_0 \text{inv } F} \left(A' + \frac{\alpha(y_0) C_\Lambda}{\delta} \right) + \Gamma.$$

In addition we put

$$C_{\text{wall}} := \frac{4c_1}{c_0 \operatorname{inv} F} \quad (\text{so that } K_W = 1.000001 c_0^2 C_{\text{wall}}),$$

$$\gamma(y) := 0.50001 \beta(y) \left(A + \frac{\beta(y)}{\delta} \right) \quad (y \in H), \quad \Gamma_H := \int_H \gamma \, d\ell.$$

Thus Γ_H is an integral, which depends on k through H , while Γ is the closed-form expression above, which depends only on π ; Lemma 10.5 shows $\Gamma_H \leq \Gamma$, and it is Γ that enters B . (These are the numbers Γ and Γ_H of Section 3.2; they are not trajectories, which always carry a floor point as an index.) Ω_W depends on k through y_1 ; $\operatorname{inv} F$, Γ , B , C_{wall} and K_W depend only on π and the fixed constants.

Lemma 10.2. $0 < \operatorname{inv} F < (1 + b_W/(\varepsilon y_0))^{-1} < 1$.

Proof. Put $v := (1 - \varepsilon)^{1/4} \in (0, 1)$, so that $\varepsilon = 1 - v^4$ and $\operatorname{inv} F \cdot (1 + b_W/(\varepsilon y_0)) = 4v^3(1 - v^3)/(3(1 - v^4)) > 0$. The polynomial identity

$$3(1 - v^4) - 4v^3(1 - v^3) = 4v^6 - 3v^4 - 4v^3 + 3 = (v - 1)^2(4v^4 + 8v^3 + 9v^2 + 6v + 3)$$

(expand the right-hand side) shows that $3(1 - v^4) - 4v^3(1 - v^3) > 0$ for $0 < v < 1$. Hence $\operatorname{inv} F \cdot (1 + b_W/(\varepsilon y_0)) < 1$, and $1 + b_W/(\varepsilon y_0) > 1$. $\square$

10.3. Results used. Besides the definitions recalled above we use the following results; under the standing assumptions their hypotheses are satisfied (Lemma 10.1).

- (U1) [R, Lemma 4.13]: if $0 < \bar{\beta}_1 \leq 1.5 \cdot 10^{-6}$, $Y \subset [0, k]$, $\beta_1 : Y \rightarrow [0, \bar{\beta}_1]$, and for every $y \in Y$ we have $\omega(y) < \frac{1}{2}$ and the line y meets a square of inclination $\geq \beta_1(y)$, then $\int_Y 2\beta_1(y) \, dy \leq AW$.
- (U2) [R, Lemma 4.15]: if $0 < \delta \leq 10^{-5}$, $0 < \alpha_1 \leq 1.5 \cdot 10^{-6}$, $y \in (0, k)$, $d(y) \leq \frac{k}{2} - 1$, and every square meeting the line y has inclination $< \alpha_1$, then $E(y) \leq 0.50001 \alpha_1 (A + \alpha_1/\delta)W$. (Its proof in [R] uses [R, Lemma 4.14] with $\bar{\beta} := \alpha_1$ and $\bar{y}_x := d(y)$, and through it the pair estimates [R, Proposition 4.5 and Lemma 4.10] with $d = \delta$; see Section 2.)
- (U3) Corollary 7.2: $\Lambda_0 + \tilde{\Lambda}_0 \leq C_\Lambda W/\delta$.
- (U4) Corollary 8.13: $\operatorname{dom} T^*$ and $\operatorname{dom} \tilde{T}^*$ are Borel sets, T^* and $\tilde{T}^*$ are Borel functions on them with values in $[y_0, y_1]$, and

$$\int_{\operatorname{dom} T^*} \alpha(T^*(x)) \, dx + \int_{\operatorname{dom} \tilde{T}^*} \alpha(\tilde{T}^*(x)) \, dx \leq A'W.$$

- (U5) Theorem 9.1: for every $s \in [y_0, y_1]$,

$$N(s) \geq \frac{|\mathcal{Z}(s)|}{\varepsilon s + b_W} - \Lambda^*(s), \quad \tilde{N}(s) \geq \frac{|\tilde{\mathcal{Z}}(s)|}{\varepsilon s + b_W} - \tilde{\Lambda}^*(s).$$

- (U6) Theorem 9.2: for every $s \in [y_0, y_1]$, with $f(y) := (1 - \omega_0 - E(y))_+$,

$$|\mathcal{Z}(s)| \geq G_b(s) := \frac{((1 - \varepsilon)s)^{3/4}}{c_1} \int_{Y_b \cap \mathcal{W}_s} f(y) \, dy,$$

$$|\tilde{\mathcal{Z}}(s)| \geq G_t(s) := \frac{((1 - \varepsilon)s)^{3/4}}{c_1} \int_{Y_t \cap (k - \mathcal{W}_s)} f(y) \, dy.$$

- (U7) Lemma 3.23(c),(e): every first contact point on the trajectory of π_x^s has height $\leq s + 2$, and $\{x \in \operatorname{dom} T^* : T^*(x) \leq s\} = \mathcal{T}_s$ for $s \in [y_0, y_1]$; and Lemma 3.31(b),(c): the set $\{(x, s) : x \in \operatorname{dom} T^*, T^*(x) \leq s\}$ is a Borel subset of $\mathbb{R}^2$, and $s \mapsto N(s)$ is non-decreasing. The same holds for the ceiling versions.

Here $|\mathcal{Z}(s)|$ is the number of squares in $\mathcal{Z}(s)$. The function f should not be confused with the waste areas f_e of [R] or with the auxiliary function $f(a)$ of Lemma 4.6. In (U1) and (U2) the threshold functions of [R] are renamed β_1 , α_1 , since β and α denote the thresholds of (N1).

10.4. The inequality.

Theorem 10.3 (integration inequality). *Let $k \geq 4$ be an integer, $\mathcal{P}$ a closed packing of $[0, k]^2$ with waste $W = k^2 - |\mathcal{P}|$, and let $\pi = (c_0, c_1, y_0, \varepsilon, \omega_0)$ satisfy (C0)–(C7). Then*

$$(\star) \quad (1 - \omega_0) \ell(H) \leq BW + C_{\text{wall}} \Omega_W.$$

Moreover

$$(10.1) \quad \ell(H) \geq 8h_0 \left(((1 - \varepsilon)y_1)^{1/4} - (y_0 + 1)^{1/4} \right),$$

$$(10.2) \quad \Omega_W \leq 1.000001 c_0^2 \left(\log \frac{y_1}{y_0} + \frac{2}{y_0} \right),$$

and consequently, with $y_1 = \frac{k}{2} - 3$,

$$(10.3) \quad W \geq \Psi_\pi(k) := \frac{1}{B} \left[8h_0(1 - \omega_0) \left(((1 - \varepsilon)y_1)^{1/4} - (y_0 + 1)^{1/4} \right) - K_W \left(\log \frac{y_1}{y_0} + \frac{2}{y_0} \right) \right].$$

Remark 10.4. B is increasing in each of A, A', C_Λ (also through Γ), while $C_{\text{wall}}, K_W, h_0$ and Ω_W do not depend on them. Since $W \geq 0$, $(\star)$ remains true when B is computed with any numbers $\bar{A} \geq A, \bar{A}' \geq A', \bar{C}_\Lambda \geq C_\Lambda$ in place of A, A', C_Λ ; dividing by the corresponding $\bar{B} > 0$ shows that (10.3) also remains true with $\bar{B}$ in place of B .

10.5. Proof of Theorem 10.3. The proof has twelve steps. The standing assumptions are in force.

Step 1 (measurability). For every square S , c_S is a piecewise linear function of y with finitely many pieces (it is continuous if $a(S) > 0$, and the indicator function of a closed interval of length 1 if $a(S) = 0$). The chords of the squares on a line $y \in [0, k]$ are pairwise disjoint subsets of $[0, k] \times \{y\}$, so $\omega(y) = k - \sum_{S \in \mathcal{P}} c_S(y)$ there, and $E(y) = \sum_{S \in \mathcal{P}} (c_S(y) - 1)_+$. Hence ω, E and $f = (1 - \omega_0 - E)_+$ are Borel functions, and $0 \leq f \leq 1 - \omega_0 < 1$. The set $\{y : y_0 \leq d(y) \leq (1 - \varepsilon)y_1\}$ is a union of two closed intervals and $\{y \in (0, k) : \{y\} \in [w_0, 1 - w_0]\}$ is a finite union of intervals, so H is a finite union of intervals. For a square S , the set of y such that the line y meets S is a closed interval (the vertical extent of S), and the set of $y \in (0, k)$ with $a(S) \geq \beta(y)$ is empty if $a(S) = 0$ and equals $\{y : d(y) \geq (c_1/a(S))^{4/3}\}$ if $a(S) > 0$. Therefore $Y_i, Y_{ii}, Y_{iii}, Y_b, Y_t$ are Borel sets. Since $d \geq y_0 > 0$ on H , $\ell(Y) \leq y_0^{-3/4} k < \infty$ for every Borel $Y \subset H$. By (U7), the set $\{(x, s) : x \in \text{dom } T^*, T^*(x) \leq s\}$ is a Borel subset of $\mathbb{R}^2$, and N is a non-decreasing, hence Borel, function on $[y_0, y_1]$ with values in $[0, k]$; the same holds for the ceiling versions.

Step 2 (lower bound for $\ell(H)$; proof of (10.1)). *Reflection.* The map $y \mapsto k - y$ maps $(0, \frac{k}{2})$ onto $(\frac{k}{2}, k)$ and preserves d . If $y \notin \mathbb{Z}$, then $\{k - y\} = 1 - \{y\}$ because k is an integer, so $\{y\} \in [w_0, 1 - w_0]$ if and only if $\{k - y\} \in [w_0, 1 - w_0]$; if $y \in \mathbb{Z}$, then $\{y\} = \{k - y\} = 0 \notin [w_0, 1 - w_0]$ because $w_0 > 0$. Hence $H_t = k - H_b$, and since d is preserved, $\ell(H_t) = \ell(H_b)$. As $\frac{k}{2} \notin H$ (Lemma 10.1(d)), $\ell(H) = 2\ell(H_b)$.

Unit intervals. Put $j_0 := \lceil y_0 \rceil = y_0$ (an integer by (C0)) and $j_1 := \lfloor (1 - \varepsilon)y_1 \rfloor$. Let m be an integer with $j_0 \leq m \leq j_1 - 1$. Then $[m + w_0, m + 1 - w_0] \subset H_b$: indeed $m + w_0 > j_0 = y_0$; $m + 1 - w_0 \leq j_1 - w_0 < j_1 \leq (1 - \varepsilon)y_1 < \frac{k}{2}$, so $d(y) = y \in [y_0, (1 - \varepsilon)y_1]$ on this interval; and since $0 < w_0 < \frac{1}{2}$ (Lemma 10.1(b)), every y in it satisfies $m < y < m + 1$ and $\{y\} = y - m \in [w_0, 1 - w_0]$.

Comparison. Since $y^{-3/4}$ is decreasing and the interval has length $1 - 2w_0 = h_0 > 0$,

$$\int_{m+w_0}^{m+1-w_0} y^{-3/4} dy \geq h_0 (m + 1 - w_0)^{-3/4} \geq h_0 (m + 1)^{-3/4} \geq h_0 \int_{m+1}^{m+2} y^{-3/4} dy.$$

Summation. The intervals $[m + w_0, m + 1 - w_0]$, $j_0 \leq m \leq j_1 - 1$, are pairwise disjoint, so

$$\ell(H_b) \geq h_0 \int_{j_0+1}^{j_1+1} y^{-3/4} dy = 4h_0 \left((j_1 + 1)^{1/4} - (j_0 + 1)^{1/4} \right).$$

If $j_1 \leq j_0$ the sum is empty, but then the right-hand side is ≤ 0 and the inequality still holds. Finally $j_1 + 1 > (1 - \varepsilon)y_1$, $j_0 + 1 = y_0 + 1$ and $h_0 > 0$, so $\ell(H) = 2\ell(H_b) \geq 8h_0(((1 - \varepsilon)y_1)^{1/4} - (y_0 + 1)^{1/4})$, which is (10.1).

Step 3 (kind (i)). On H we have $d(y) \geq y_0$, hence $d(y)^{-3/4} \leq y_0^{-3/4}$. Therefore

$$W = \int_0^k \omega(y) dy \geq \int_{Y_i} \omega(y) dy \geq \omega_0 |Y_i| \geq \omega_0 y_0^{3/4} \ell(Y_i), \quad \text{that is} \quad \ell(Y_i) \leq \frac{W}{\omega_0 y_0^{3/4}}.$$

Step 4 (kind (ii)). Apply (U1) with $\bar{\beta}_1 := 1.5 \cdot 10^{-6}$, $Y := Y_{ii}$ and $\beta_1 := \beta|_{Y_{ii}}$. By Lemma 10.1(d), β_1 takes values in $(0, 1.5 \cdot 10^{-6}]$. For $y \in Y_{ii}$ we have $\omega(y) < \omega_0 < \frac{1}{2}$ by (C4), and the line y meets a square of inclination $\geq \beta(y)$ by definition of kind (ii). Since $\beta(y) = c_1 d(y)^{-3/4}$,

$$2c_1 \ell(Y_{ii}) = \int_{Y_{ii}} 2\beta(y) dy \leq AW, \quad \text{that is} \quad \ell(Y_{ii}) \leq \frac{A}{2c_1} W.$$

Step 5 (kind (iii): the excess of the long chords).

Lemma 10.5. $\Gamma_H \leq \Gamma$, and $E(y) \leq \gamma(y)W$ for every $y \in Y_{iii}$.

Proof. Let $y \in Y_{iii}$. Apply (U2) with $\alpha_1 := \beta(y)$: we have $\delta = 10^{-5}$, $0 < \beta(y) \leq 1.5 \cdot 10^{-6}$ and $d(y) \leq (1 - \varepsilon)y_1 < \frac{k}{2} - 1$ by Lemma 10.1(d), and every square meeting the line y has inclination $< \beta(y)$ by definition of kind (iii). This gives $E(y) \leq \gamma(y)W$.

On H_b we have $d(y) = y \geq y_0$, so $\gamma(y) d(y)^{-3/4} = 0.50001 (Ac_1 y^{-3/2} + c_1^2 \delta^{-1} y^{-9/4})$ and

$$\int_{H_b} \gamma d\ell \leq 0.50001 \int_{y_0}^{\infty} (Ac_1 y^{-3/2} + c_1^2 \delta^{-1} y^{-9/4}) dy = 0.50001 \left(2Ac_1 y_0^{-1/2} + \frac{4}{5} \frac{c_1^2}{\delta} y_0^{-5/4} \right).$$

The substitution $y \mapsto k - y$ maps H_t onto H_b (Step 2) and preserves d , hence γ and ℓ ; so $\int_{H_t} \gamma d\ell = \int_{H_b} \gamma d\ell$. Since $\frac{k}{2} \notin H$ and $2 \cdot 0.50001 = 1.00002$, $\Gamma_H = \int_{H_b} \gamma d\ell + \int_{H_t} \gamma d\ell \leq \Gamma$. $\square$

For $y \in Y_{iii}$ we have $1 - \omega_0 - E(y) \leq f(y)$, so by Lemma 10.5, $1 - \omega_0 \leq f(y) + \gamma(y)W$. Integrating over Y_{iii} with respect to ℓ and using $\gamma \geq 0$, $Y_{iii} \subset H$ and Lemma 10.5,

$$(10.4) \quad (1 - \omega_0) \ell(Y_{iii}) \leq \int_{Y_{iii}} f d\ell + \Gamma_H W \leq \int_{Y_{iii}} f d\ell + \Gamma W.$$

This is the integral of a pointwise inequality in which the same W appears for every y ; no disjointness is used.

Step 6 (the lower bounds for $|\mathcal{Z}(s)|$ and $|\tilde{\mathcal{Z}}(s)|$). By (U6), $|\mathcal{Z}(s)| \geq G_b(s)$ and $|\tilde{\mathcal{Z}}(s)| \geq G_t(s)$ for every $s \in [y_0, y_1]$ (these bounds rest on [R, Lemmas 3.4 and 3.5]). The functions G_b and G_t are continuous on $[y_0, y_1]$. Indeed, since $0 \leq f \leq 1$, for $s, s' \in [y_0, y_1]$

$$\left| \int_{Y_b \cap \mathcal{W}_s} f - \int_{Y_b \cap \mathcal{W}_{s'}} f \right| \leq |\mathcal{W}_s \triangle \mathcal{W}_{s'}| \leq (1 - \varepsilon)|s - s'| + |s - s'| \leq 2|s - s'|,$$

and the same holds for the windows $k - \mathcal{W}_s$; the factor $((1 - \varepsilon)s)^{3/4}/c_1$ is continuous. In particular G_b and G_t are bounded Borel functions on $[y_0, y_1]$.

Step 7 (the inclusion $\mathcal{T}_s \subset \{T^* \leq s\}$). By (U7), $\{x \in \text{dom } T^* : T^*(x) \leq s\} = \mathcal{T}_s$ for $s \in [y_0, y_1]$. Only the inclusion $\mathcal{T}_s \subset \{T^* \leq s\}$ is used below; we recall why it holds. Let $x \in \mathcal{T}_s$. The path π_x^s coincides with π_x^0 up to its termination, and it is T-terminated at an R1-passed entry (Z_j, q_j) of π_x^0 (by the definition of T_s , and, for a T-termination inherited from F^0 , because R1 is applied before R2, Definitions 3.14 and 3.10); moreover $a(Z_j) \geq \alpha(s) = c_0 s^{-1/2} > 0$, that is $(c_0/a(Z_j))^2 \leq s$. The point q_j is a first contact point on the trajectory of π_x^s , so $\eta_j \leq s + 2$ by (U7). Since also $y_0 \leq s$, we get $s_j = \max(y_0, \eta_j - 2, (c_0/a(Z_j))^2) \leq s \leq y_1$; hence $x \in \text{dom } T^*$ and $T^*(x) \leq s$. Consequently

$$|\{x \in \text{dom } T^* : T^*(x) \leq s\}| \geq N(s), \quad |\{x \in \text{dom } \tilde{T}^* : \tilde{T}^*(x) \leq s\}| \geq \tilde{N}(s)$$

(the second by the same argument for the reflected packing).

Step 8 (first use of Tonelli's theorem). For $s > 0$ we have $|\alpha'(s)| = \frac{c_0}{2} s^{-3/2}$. If $\tau \in [y_0, y_1]$, then

$$\int_{y_0}^{y_1} \mathbf{1}[\tau \leq s] \frac{c_0}{2} s^{-3/2} ds = \int_{\tau}^{y_1} \frac{c_0}{2} s^{-3/2} ds = c_0(\tau^{-1/2} - y_1^{-1/2}) \leq c_0 \tau^{-1/2} = \alpha(\tau).$$

Since T^* takes values in $[y_0, y_1]$, applying this with $\tau = T^*(x)$, integrating over $x \in \text{dom } T^*$ and using Tonelli's theorem for the nonnegative Borel function $(x, s) \mapsto \mathbf{1}[T^*(x) \leq s] \frac{c_0}{2} s^{-3/2}$ on $\text{dom } T^* \times [y_0, y_1]$ (Step 1), and then Step 7, we get

$$\int_{\text{dom } T^*} \alpha(T^*(x)) dx \geq \int_{y_0}^{y_1} |\{T^* \leq s\}| \frac{c_0}{2} s^{-3/2} ds \geq \int_{y_0}^{y_1} N(s) \frac{c_0}{2} s^{-3/2} ds.$$

With the same argument for $\tilde{T}^*$ and (U4),

$$(10.5) \quad A'W \geq \int_{y_0}^{y_1} (N(s) + \tilde{N}(s)) \frac{c_0}{2} s^{-3/2} ds.$$

Step 9 (the per-scale bound; the loss and wall terms). Let $s \in [y_0, y_1]$. By (U5) and Step 6,

$$N(s) + \tilde{N}(s) \geq \Theta(s) := \frac{G_b(s) + G_t(s)}{\varepsilon s + b_W} - (\Lambda_0 + \tilde{\Lambda}_0) - 4(s+2) \tan \alpha(s).$$

The right-hand side may be negative; we do not take positive parts but integrate the lower bound as it stands. By (U3), $\Lambda_0 + \tilde{\Lambda}_0$ is finite, so Θ is continuous and bounded on $[y_0, y_1]$ (Step 6), and we may multiply by $\frac{c_0}{2} s^{-3/2}$, integrate over $[y_0, y_1]$ and split the integral into three terms. The loss term satisfies

$$(\Lambda_0 + \tilde{\Lambda}_0) \int_{y_0}^{y_1} \frac{c_0}{2} s^{-3/2} ds = (\Lambda_0 + \tilde{\Lambda}_0) c_0 (y_0^{-1/2} - y_1^{-1/2}) \leq \alpha(y_0) (\Lambda_0 + \tilde{\Lambda}_0) \leq \alpha(y_0) C_\Lambda \frac{W}{\delta}$$

by (U3), and the wall term is $\int_{y_0}^{y_1} 4(s+2) \tan \alpha(s) \frac{c_0}{2} s^{-3/2} ds = 2\Omega_W$ (one Ω_W for each family). With (10.5) and

$$\mathcal{J} := \int_{y_0}^{y_1} \frac{G_b(s) + G_t(s)}{\varepsilon s + b_W} \frac{c_0}{2} s^{-3/2} ds$$

we obtain

$$(10.6) \quad \mathcal{J} \leq \left(A' + \frac{\alpha(y_0) C_\Lambda}{\delta} \right) W + 2\Omega_W.$$

Step 10 (second use of Tonelli's theorem; the constant $\text{inv } F$). By Lemma 10.1(d), every $y \in Y_b$ satisfies $y_0 \leq y = d(y) \leq (1 - \varepsilon)y_1$, so

$$\{s \in [y_0, y_1] : (1 - \varepsilon)s \leq y \leq s\} = [y, y/(1 - \varepsilon)].$$

The function $(y, s) \mapsto \mathbf{1}_{Y_b}(y) \mathbf{1}[(1 - \varepsilon)s \leq y \leq s] f(y) s^{3/4} s^{-3/2} / (\varepsilon s + b_W)$ is a nonnegative Borel function on $\mathbb{R} \times [y_0, y_1]$ (the set $\{(y, s) : (1 - \varepsilon)s \leq y \leq s\}$ is closed), so by Tonelli's theorem

$$\int_{y_0}^{y_1} \frac{G_b(s)}{\varepsilon s + b_W} \frac{c_0}{2} s^{-3/2} ds = \frac{c_0(1 - \varepsilon)^{3/4}}{2c_1} \int_{Y_b} f(y) \left[\int_y^{y/(1 - \varepsilon)} \frac{s^{-3/4}}{\varepsilon s + b_W} ds \right] dy.$$

For $s \geq y \geq y_0$ we have $\varepsilon s + b_W = \varepsilon s (1 + b_W/(\varepsilon s)) \leq \varepsilon s (1 + b_W/(\varepsilon y_0))$, hence

$$\int_y^{y/(1 - \varepsilon)} \frac{s^{-3/4}}{\varepsilon s + b_W} ds \geq \frac{1}{\varepsilon (1 + b_W/(\varepsilon y_0))} \int_y^{y/(1 - \varepsilon)} s^{-7/4} ds = \frac{4(1 - (1 - \varepsilon)^{3/4})}{3\varepsilon (1 + b_W/(\varepsilon y_0))} y^{-3/4}.$$

Multiplying by $(1 - \varepsilon)^{3/4}$ gives $\text{inv } F \cdot y^{-3/4} = \text{inv } F \cdot d(y)^{-3/4}$. Hence the bottom part of $\mathcal{J}$ is at least $\frac{c_0 \text{inv } F}{2c_1} \int_{Y_b} f d\ell$. For the top part, write $u := k - y = d(y)$ for $y \in Y_t$; then $y \in k - \mathcal{W}_s$ if and

only if $(1-\varepsilon)s \leq u \leq s$, and $y_0 \leq u \leq (1-\varepsilon)y_1$; the same computation gives at least $\frac{c_0 \operatorname{inv} F}{2c_1} \int_{Y_t} f \, d\ell$. Since $Y_{iii} = Y_b \sqcup Y_t$ (Lemma 10.1(e)),

$$(10.7) \quad \mathcal{J} \geq \frac{c_0 \operatorname{inv} F}{2c_1} \int_{Y_{iii}} f \, d\ell.$$

Step 11 (upper bound for Ω_W ; proof of (10.2)). The function $\tan x/x$ is increasing on $(0, \pi/2)$ (the Taylor coefficients of $\tan$ are positive), and $\tan(1.5 \cdot 10^{-6}) \leq 1.000001 \cdot 1.5 \cdot 10^{-6}$; hence $\tan x \leq 1.000001 x$ for $0 \leq x \leq 1.5 \cdot 10^{-6}$. For $s \geq y_0$ we have $0 < \alpha(s) \leq 1.5 \cdot 10^{-6}$ (Lemma 10.1(a)), so

$$2(s+2) \tan \alpha(s) \frac{c_0}{2} s^{-3/2} \leq 1.000001 c_0^2 (s+2) s^{-2} = 1.000001 c_0^2 \left(\frac{1}{s} + \frac{2}{s^2} \right).$$

Integrating over $[y_0, y_1]$ gives

$$\Omega_W \leq 1.000001 c_0^2 \left(\log \frac{y_1}{y_0} + 2(y_0^{-1} - y_1^{-1}) \right) \leq 1.000001 c_0^2 \left(\log \frac{y_1}{y_0} + \frac{2}{y_0} \right).$$

Step 12 (summation). From (10.6) and (10.7), and since $\operatorname{inv} F > 0$ (Lemma 10.2),

$$(10.8) \quad \int_{Y_{iii}} f \, d\ell \leq \frac{2c_1}{c_0 \operatorname{inv} F} \left(A' + \frac{\alpha(y_0) C_\Lambda}{\delta} \right) W + C_{\text{wall}} \Omega_W.$$

By Lemma 10.1(e), $\ell(H) = \ell(Y_i) + \ell(Y_{ii}) + \ell(Y_{iii})$; as $0 < 1 - \omega_0 \leq 1$, Steps 3 and 4, (10.4) and (10.8) give

$$\begin{aligned} (1 - \omega_0) \ell(H) &\leq \ell(Y_i) + \ell(Y_{ii}) + (1 - \omega_0) \ell(Y_{iii}) \\ &\leq \left[\frac{1}{\omega_0 y_0^{3/4}} + \frac{A}{2c_1} + \frac{2c_1}{c_0 \operatorname{inv} F} \left(A' + \frac{\alpha(y_0) C_\Lambda}{\delta} \right) + \Gamma \right] W + C_{\text{wall}} \Omega_W. \end{aligned}$$

The bracket is B , so $(\star)$ holds. Inserting (10.1) (the coefficient $(1 - \omega_0)$ of $\ell(H)$ is positive) and (10.2) (the coefficient C_{wall} of Ω_W is positive) into $(\star)$ and dividing by $B > 0$ gives (10.3), because $C_{\text{wall}} \cdot 1.000001 c_0^2 = K_W$. This completes the proof of Theorem 10.3. $\square$

10.6. Remarks on the proof.

Remark 10.6 (directions of the inequalities). Step 2 gives a lower bound for $\ell(H)$, which enters $(\star)$ with the positive coefficient $1 - \omega_0$. Steps 3–5 give upper bounds for $\ell(Y_i)$, $\ell(Y_{ii})$ and $\int_{Y_{iii}} E \, d\ell$. Step 6 gives lower bounds for $|\mathcal{Z}(s)|$, $|\tilde{\mathcal{Z}}(s)|$; Step 7 the inequality $|\{T^* \leq s\}| \geq N(s)$; Step 8 bounds the left-hand side of (U4) from below; in Step 9 the subtracted loss and wall terms are bounded from above; Step 10 bounds $\mathcal{J}$ from below, using $1/(\varepsilon s + b_W) \geq 1/(\varepsilon s(1 + b_W/(\varepsilon y_0)))$ for $s \geq y_0$; Step 11 bounds Ω_W from above (using $\tan x \leq 1.000001x$ and dropping $-2/y_1$). The only positive part taken is $f = (1 - \omega_0 - E)_+$, in (U6); in Step 5 it is used through $1 - \omega_0 - E \leq f$.

Remark 10.7 (how the waste is used). The upper bounds in terms of W that enter B are: the line waste in Step 3 ($\int_{Y_i} \omega \leq \int_0^k \omega = W$), [R, Lemma 4.13] in Step 4, [R, Lemma 4.15] applied separately to each $y \in Y_{iii}$ in Step 5, and Corollaries 8.13 and 7.2 in Steps 8–9 (these two together give the third term of B). Each holds with the waste W of the whole packing, and $(\star)$ is obtained by adding them. No disjointness between the parts of the waste used by different bounds is claimed or needed; the price is that their contributions to B are added. The floor and ceiling families are combined within Corollaries 7.2 and 8.13, each of which bounds a sum over both families by one multiple of W , and each of these two corollaries is used once; Theorems 9.1 and 9.2 are applied to each family separately. The wall term $C_{\text{wall}} \Omega_W$ does not involve W .

Remark 10.8 (integrality of k). In this section the integrality of k is used in Step 2 (the reflection $y \mapsto k - y$ preserves the condition $\{y\} \in [w_0, 1 - w_0]$) and, for the ceiling family, through the results quoted in (U5) and (U6).

11. PROOF OF THEOREMS A AND A_{13} AND OF COROLLARY A

By Theorem 10.3, every closed packing of $[0, k]^2$ satisfies $W \geq \Psi_\pi(k)$ whenever π satisfies (C0)–(C7) for k . In this section we choose explicit parameters, show that the bound stays above $c_* k^{1/4}$ for all k beyond a threshold (Proposition 11.2), and deduce Theorems A and A_{13} and Corollary A. The numerical values used are listed in Section 12, together with the direction in which each displayed value is rounded; they are recomputed by the script described in Section 12.7. The notation is that of Sections 3 and 10. As in Section 1, $M(k)$ is the largest number of squares in a closed packing of $[0, k]^2$ (it exists, since $N \leq k^2$), and $W_{\min}(k) = k^2 - M(k)$ is the least waste of a closed packing of $[0, k]^2$.

11.1. Upper bounds for the constants. Put

$$\bar{A} := 10.6067, \quad \bar{A}' := 10.6068, \quad \bar{C}_\Lambda := 1.0000002.$$

By Table 1, $A < \bar{A}$, $A' < \bar{A}'$ and $C_\Lambda < \bar{C}_\Lambda$. Let $\bar{B}$ denote the number B of Definition 3.6 computed with $\bar{A}, \bar{A}', \bar{C}_\Lambda$ in place of A, A', C_Λ (also in Γ), and let $\bar{\Psi}_\pi(k)$ be $\Psi_\pi(k)$ of (10.3) with $\bar{B}$ in place of B .

Lemma 11.1. *If π satisfies (C0)–(C7) for the integer $k \geq 4$, then every closed packing of $[0, k]^2$ satisfies $W \geq \bar{\Psi}_\pi(k)$.*

Proof. This is Theorem 10.3 together with Remark 10.4. $\square$

All explicit bounds below are computed with $\bar{B}$. In the asymptotic statements (Section 11.4) the exact constants A, A', C_Λ are used, which is allowed by Theorem 10.3 itself.

11.2. Monotonicity in k . For real k with $y_0 + 1 \leq (1 - \varepsilon)(\frac{k}{2} - 3)$ we use the right-hand side of (10.3) as the definition of $\Psi_\pi(k)$ (with $y_1 = y_1(k) := \frac{k}{2} - 3$), and likewise for $\bar{\Psi}_\pi(k)$.

Proposition 11.2. *Let π satisfy (C0)–(C6), let $c_* > 0$, and let $k_0 > 0$ be a real number with $y_0 + 1 \leq (1 - \varepsilon)(\frac{k_0}{2} - 3)$. Let $\hat{B}$ be either B or $\bar{B}$, and $\hat{\Psi}_\pi$ the corresponding function (10.3). Put*

$$D_1 := \frac{h_0(1 - \omega_0)(1 - \varepsilon)^{1/4}}{\hat{B}} - \frac{c_*}{4 \cdot 2^{3/4}}.$$

Suppose that

- (i) $\hat{\Psi}_\pi(k_0) \geq c_* k_0^{1/4}$;
- (ii) $D_1 > 0$;
- (iii) $y_1(k_0)^{1/4} D_1 \geq K_W / (2\hat{B})$.

Then $\hat{\Psi}_\pi(k) \geq c_ k^{1/4}$ for every real $k \geq k_0$.*

Proof. For $k \geq k_0$ we have $y_1(k) \geq y_1(k_0) > y_0 > 0$, so $\hat{\Psi}_\pi$ is differentiable on $[k_0, \infty)$. Let $\phi(k) := \hat{\Psi}_\pi(k) - c_* k^{1/4}$ and write $y_1 = y_1(k)$, so that $dy_1/dk = \frac{1}{2}$ and $k = 2y_1 + 6$. Then

$$\phi'(k) = \frac{h_0(1 - \omega_0)(1 - \varepsilon)^{1/4}}{\hat{B}} y_1^{-3/4} - \frac{K_W}{2\hat{B} y_1} - \frac{c_*}{4} (2y_1 + 6)^{-3/4}.$$

Since $(2y_1 + 6)^{-3/4} < (2y_1)^{-3/4} = 2^{-3/4} y_1^{-3/4}$,

$$\phi'(k) > y_1^{-3/4} D_1 - \frac{K_W}{2\hat{B} y_1} = y_1^{-1} \left(y_1^{1/4} D_1 - \frac{K_W}{2\hat{B}} \right) \geq y_1^{-1} \left(y_1(k_0)^{1/4} D_1 - \frac{K_W}{2\hat{B}} \right) \geq 0,$$

where we used $D_1 > 0$ and $y_1 \geq y_1(k_0)$ in the second inequality and (iii) in the last. As $\phi(k_0) \geq 0$ by (i), the mean value theorem gives $\phi(k) \geq 0$ for all $k \geq k_0$. $\square$

Remark 11.3. If (C7) holds for k_0 , it holds for every $k \geq k_0$ (Lemma 3.3(e)). Hence, if π satisfies (C0)–(C6), (C7) holds for k_0 and (i)–(iii) hold with $\hat{B} = \bar{B}$, then by Lemma 11.1 and Proposition 11.2 every closed packing of $[0, k]^2$, for every integer $k \geq \max(k_0, 4)$, satisfies $W \geq \bar{\Psi}_\pi(k) \geq c_* k^{1/4}$. Note that the proof of Proposition 11.2 uses (i) only at the end: if (ii) holds and (iii) holds at some k_1 in place of k_0 , then ϕ is increasing on $[k_1, \infty)$.

11.3. The explicit part of Theorem A. We prove: *for every integer $k \geq k_0 := 4.62 \cdot 10^{12}$, every closed packing of $[0, k]^2$ satisfies $W \geq 0.1 k^{1/4}$; consequently $W \geq \lceil 0.1 k^{1/4} \rceil \geq 147$.*

Let

$$\pi^* = (c_0, c_1, y_0, \varepsilon, \omega_0) = (0.3078, 0.2754, 42\,108\,000\,000, 5.8 \cdot 10^{-6}, 1.7 \cdot 10^{-5})$$

as in Section 3. By Lemma 3.4(a) and Table 2 (column “Theorem A”), π^* satisfies (C0)–(C6), it satisfies (C7) for $k_0 = 4.62 \cdot 10^{12}$, and conditions (i)–(iii) of Proposition 11.2 hold with $\hat{B} = \bar{B}$ and $c_* = 0.1$: indeed $\bar{\Psi}_{\pi^*}(k_0) \geq 146.62429 > 146.60896 \geq 0.1 k_0^{1/4}$, $D_1 \geq 0.0086429 > 0$ and $y_1(k_0)^{1/4} D_1 \geq 10.655 > 0.0044029 \geq K_W/(2\bar{B})$. By Remark 11.3, $W \geq 0.1 k^{1/4}$ for every integer $k \geq k_0$ and every closed packing of $[0, k]^2$. Since W is an integer, $W \geq \lceil 0.1 k^{1/4} \rceil$, and $\lceil 0.1 k^{1/4} \rceil \geq \lceil 0.1 k_0^{1/4} \rceil = 147$ because $146 < 0.1 k_0^{1/4} \leq 146.60896$. $\square$

11.4. The asymptotic constant. Let $\lambda := 2\delta + 2 \cdot 10^{-12}$ (as in Theorem A), and put

$$h_0^\infty(c) := 1 - \lambda - 1.00002 c^2, \quad c_0^{\text{opt}} := \sqrt{\frac{1 - \lambda}{5.0001}},$$

$$(11.1) \quad c_\infty := \frac{4 \cdot 2^{-1/4} h_0^\infty(c_0^{\text{opt}}) \sqrt{c_0^{\text{opt}}}}{\sqrt{AA'}} = \frac{16 \cdot 2^{-1/4}}{5\sqrt{AA'}} (1 - \lambda)^{5/4} 5.0001^{-1/4}.$$

The right-hand side is the constant c_∞ of Theorem A. Here A, A' are the exact constants of Definition 3.6; numerically $c_\infty = 0.1696524112 \dots$ (Table 1). For π satisfying (C0)–(C6) let

$$\Psi_\infty(\pi) := \lim_{k \rightarrow \infty} \frac{\Psi_\pi(k)}{k^{1/4}} = \frac{8 \cdot 2^{-1/4} h_0(1 - \omega_0)(1 - \varepsilon)^{1/4}}{B};$$

the limit exists and has this value because $((1 - \varepsilon)(\frac{k}{2} - 3))^{1/4}/k^{1/4} \rightarrow 2^{-1/4}(1 - \varepsilon)^{1/4}$, while $(y_0 + 1)^{1/4}/k^{1/4} \rightarrow 0$ and $\log(y_1/y_0)/k^{1/4} \rightarrow 0$.

Lemma 11.4. *On $(0, \infty)$ the function $c \mapsto h_0^\infty(c)\sqrt{c}$ attains its maximum exactly at $c = c_0^{\text{opt}}$, and $h_0^\infty(c_0^{\text{opt}}) = \frac{4}{5}(1 - \lambda)$. In particular $4 \cdot 2^{-1/4} h_0^\infty(c)\sqrt{c}/\sqrt{AA'} \leq c_\infty$ for every $c > 0$.*

Proof. The derivative of $(1 - \lambda - 1.00002 c^2)\sqrt{c}$ is $(1 - \lambda - 5.0001 c^2)/(2\sqrt{c})$, which is positive for $c < c_0^{\text{opt}}$ and negative for $c > c_0^{\text{opt}}$. At $c = c_0^{\text{opt}}$ we have $1.00002 c^2 = (1 - \lambda)/5$, so $h_0^\infty = \frac{4}{5}(1 - \lambda)$, and the two expressions in (11.1) agree. $\square$

Proposition 11.5 (the asymptotic part of Theorem A). $\liminf_{k \rightarrow \infty} \frac{W_{\min}(k)}{k^{1/4}} \geq c_\infty$.

Proof. Let π satisfy (C0)–(C6). Then (C7) holds for all sufficiently large k , since $(1 - \varepsilon)(\frac{k}{2} - 3) \rightarrow \infty$ as $k \rightarrow \infty$ (and once it holds, it holds for all larger k , Remark 11.3), and Theorem 10.3 gives $W_{\min}(k) \geq \Psi_\pi(k)$ for those k . Hence $\liminf_k W_{\min}(k)/k^{1/4} \geq \Psi_\infty(\pi)$, and it suffices to find parameters with $\Psi_\infty(\pi)$ arbitrarily close to c_∞ .

For an integer $n \geq 2$ let $\pi_n := (c_0^{\text{opt}}, c_1^{\text{opt}}, n, n^{-1/2}, n^{-1/4})$ with $c_1^{\text{opt}} := \sqrt{A c_0^{\text{opt}}/(4A')}$. As $n \rightarrow \infty$: $\alpha(n) = c_0^{\text{opt}} n^{-1/2} \rightarrow 0$; $\hat{\beta}(n) = c_1^{\text{opt}} n^{-3/4} \rightarrow 0$ and $b^* = c_1^{\text{opt}}((1 - n^{-1/2})n)^{-3/4} \rightarrow 0$; the left-hand side of (C3) tends to c_1^{opt} while its right-hand side $c_0^{\text{opt}} n^{1/4}$ tends to ∞ ; $n^{-1/4} < \frac{1}{2}$ for

$n > 16$; and $w_0 \rightarrow 0.50001(c_0^{\text{opt}})^2 + \delta + 10^{-12} < 0.10002 < \frac{1}{2}$, because $(c_0^{\text{opt}})^2 = (1 - \lambda)/5.0001 < 0.2$. Hence π_n satisfies (C0)–(C6) for all large n ((C6) follows from (C1), Lemma 3.3(a)).

Along this family $h_0 \rightarrow 1 - 1.00002(c_0^{\text{opt}})^2 - 2\delta - 2 \cdot 10^{-12} = h_0^\infty(c_0^{\text{opt}})$ and $(1 - \omega_0)(1 - \varepsilon)^{1/4} \rightarrow 1$. Since $\varepsilon = n^{-1/2} \rightarrow 0$ and $(1 - \varepsilon)^{3/4}(1 - (1 - \varepsilon)^{3/4})/\varepsilon \rightarrow \frac{3}{4}$, while $b_W/(\varepsilon n) = b_W n^{-1/2} \rightarrow 0$, we have $\text{inv } F \rightarrow 1$. Further $1/(\omega_0 n^{3/4}) = n^{-1/2} \rightarrow 0$, $\alpha(n)C_\Lambda/\delta \rightarrow 0$ and $\Gamma \rightarrow 0$. Therefore

$$B \rightarrow \frac{A}{2c_1^{\text{opt}}} + \frac{2c_1^{\text{opt}}A'}{c_0^{\text{opt}}} = 2\sqrt{\frac{AA'}{c_0^{\text{opt}}}}, \quad \Psi_\infty(\pi_n) \rightarrow \frac{8 \cdot 2^{-1/4}h_0^\infty(c_0^{\text{opt}})}{2\sqrt{AA'/c_0^{\text{opt}}}} = c_\infty.$$

This proves the proposition. $\square$

The value c_∞ is the best constant that the bounds (10.3) can give, if the fixed constants δ , $\bar{\beta}$, b_W , C_Λ and the numerical constants in w_0 , Γ and (10.2) are kept as they are; it is a supremum and is not attained.

Proposition 11.6. *Let Ψ_π , Ψ_∞ and c_∞ be computed with the same constants A , A' .*

- (a) *If π satisfies (C0)–(C7) for k , then $\Psi_\pi(k) < c_\infty k^{1/4}$.*
- (b) *If π satisfies (C0)–(C6), then $\Psi_\infty(\pi) < c_\infty$; and $\sup_\pi \Psi_\infty(\pi) = c_\infty$, the supremum being over all π satisfying (C0)–(C6).*

The same holds with $\bar{A}$, $\bar{A}'$ (and $\bar{B}$) in place of A , A' (and B), c_∞ being replaced by the value c_∞^+ of (11.1) computed with $\bar{A}$, $\bar{A}'$.

Proof. First, $h_0 < h_0^\infty(c_0)$, because $w_0 > 0.50001c_0^2 + \delta + 10^{-12}$ (the terms $0.50001c_0^2 \cdot 1.0001/y_0$ and $1.0001b^*$ are positive); and $h_0 > 0$ by (C5), so $h_0^\infty(c_0) > 0$. Second, all four terms of B are positive and $0 < \text{inv } F < 1$ (Lemma 10.2), so by the inequality between arithmetic and geometric means

$$B > \frac{A}{2c_1} + \frac{2c_1A'}{c_0 \text{inv } F} \geq \frac{A}{2c_1} + \frac{2c_1A'}{c_0} \geq 2\sqrt{\frac{AA'}{c_0}}.$$

(a) By (C7), $y_1 > y_0$, so $\log(y_1/y_0) + 2/y_0 > 0$ and the subtracted term in (10.3) is positive; also $(y_0 + 1)^{1/4} > 0$, $0 < 1 - \omega_0 < 1$, $(1 - \varepsilon)^{1/4} < 1$ and $y_1 < k/2$. Hence, as $h_0 > 0$,

$$\Psi_\pi(k) < \frac{8h_0(k/2)^{1/4}}{B} < \frac{8 \cdot 2^{-1/4}h_0^\infty(c_0)k^{1/4}}{2\sqrt{AA'/c_0}} = \frac{4 \cdot 2^{-1/4}h_0^\infty(c_0)\sqrt{c_0}}{\sqrt{AA'}} k^{1/4} \leq c_\infty k^{1/4}$$

by Lemma 11.4. (b) In the same way

$$\Psi_\infty(\pi) = \frac{8 \cdot 2^{-1/4}h_0(1 - \omega_0)(1 - \varepsilon)^{1/4}}{B} < \frac{4 \cdot 2^{-1/4}h_0^\infty(c_0)\sqrt{c_0}}{\sqrt{AA'}} \leq c_\infty.$$

The family π_n in the proof of Proposition 11.5 gives $\Psi_\infty(\pi_n) \rightarrow c_\infty$, so the supremum is c_∞ , and by the strict inequality it is not attained. The last sentence follows by the same argument, since only $A, A' > 0$ and the form of B were used. $\square$

Theorem A consists of the explicit statement of Section 11.3 and Proposition 11.5; this completes its proof.

11.5. Theorem A₁₃: the variant without the computer-assisted lemma. By Remark 2.2, the computer-assisted [R, Lemma 4.10] enters the proof of Theorem 10.3 only through the multiplicity bound $K_w^* = 9$, namely through the constant A of [R, Lemmas 4.13–4.15] and, through Proposition 2.1, through the constants C_Λ of Corollary 7.2 (Sections 5 and 7) and A' of Corollary 8.13. With the analytic [R, Lemma 4.9] ($K_w = 13$) in all these places, [R, Lemmas 4.13–4.15] hold with A replaced by A_{13} (Remark 2.2, [R, Remark 7.5]), Corollary 7.2 holds with $C_\Lambda^{(13)} = 1 + 1712.6\delta^2$, and Corollary 8.13 holds with A'_{13} (Definition 3.6 for $A_{13}, A'_{13}, C_\Lambda^{(13)}$). Consequently Theorem 10.3,

Remark 10.4 and Sections 11.1–11.4 hold verbatim with $A_{13}, A'_{13}, C_{\Lambda}^{(13)}$ in place of A, A', C_{Λ} , and with the upper bounds

$$\bar{A}_{13} := 12.7476, \quad \bar{A}'_{13} := 12.7476, \quad \bar{C}_{\Lambda}^{(13)} := 1.0000002$$

(Table 1) in place of $\bar{A}, \bar{A}', \bar{C}_{\Lambda}$. Nothing else in the proof depends on [R, Lemma 4.10].

Proof of Theorem A₁₃. Explicit part. Let $k_0 := 2.17 \cdot 10^{13}$ and

$$\pi_{13}^* = (c_0, c_1, y_0, \varepsilon, \omega_0) = (0.3674, 0.3012, 60\,000\,000\,000, 5 \cdot 10^{-6}, 1.4 \cdot 10^{-5})$$

as in Section 3. By Lemma 3.4(b) and Table 2 (column “Theorem A₁₃”), π_{13}^* satisfies (C0)–(C6), (C7) for k_0 , and (i)–(iii) of Proposition 11.2 with $c_* = 0.1$ and $\hat{B} = \bar{B}$ computed from $\bar{A}_{13}, \bar{A}'_{13}, \bar{C}_{\Lambda}^{(13)}$: $\bar{\Psi}_{\pi_{13}^*}(k_0) \geq 215.83530 > 215.83156 \geq 0.1 k_0^{1/4}$, $D_1 \geq 0.0055790 > 0$ and $y_1(k_0)^{1/4} D_1 \geq 10.125 > 0.0052311 \geq K_W/(2\bar{B})$. As in Section 11.3, every closed packing of $[0, k]^2$ with an integer $k \geq k_0$ satisfies $W \geq 0.1 k^{1/4}$, hence $W \geq \lceil 0.1 k^{1/4} \rceil \geq \lceil 0.1 k_0^{1/4} \rceil = 216$, because $215 < 0.1 k_0^{1/4} \leq 215.83156$.

Asymptotic part. Propositions 11.5 and 11.6 with A_{13}, A'_{13} give $\liminf_k W_{\min}(k)/k^{1/4} \geq c_{\infty}^{(13)}$, where $c_{\infty}^{(13)}$ is (11.1) with A_{13}, A'_{13} , as in Theorem A₁₃; numerically $c_{\infty}^{(13)} = 0.1411593387 \dots$ (Table 1). $\square$

11.6. Corollary A. We prove: *for all integers $c \geq 0$ and $k \geq \max(4.62 \cdot 10^{12}, 10^4 c^4 + 1)$ one has $s(k^2 - c) = k$. Moreover $c^*(k) = W_{\min}(k) - 1 \geq \lceil 0.1 k^{1/4} \rceil - 1$ for every integer $k \geq 4.62 \cdot 10^{12}$; in particular $c^*(k) \geq 146$ there, and for $0 \leq c \leq 146$ the condition on k is just $k \geq 4.62 \cdot 10^{12}$. Without [R, Lemma 4.10], the same holds with $2.17 \cdot 10^{13}$ in place of $4.62 \cdot 10^{12}$, with 215 in place of 146. Here $s(n)$ and $c^*(k)$ are as in Section 1.*

Proof. By [R, Lemma 2.1], for integers $k \geq 2$ and $n \geq 1$ we have $s(n) < k$ if and only if $n \leq M(k)$, and $c^*(k) = k^2 - M(k) - 1 = W_{\min}(k) - 1$. Moreover $s(n) \leq k$ whenever $n \leq k^2$: the squares $[i, i+1] \times [j, j+1]$ ($0 \leq i, j < k$) have pairwise disjoint interiors, and any $n \leq k^2$ of them lie in $[0, k]^2$.

Let $k \geq 4.62 \cdot 10^{12}$ and $c \geq 0$ be integers with $k \geq 10^4 c^4 + 1$. Then $k > 10^4 c^4 = (c/0.1)^4$, so $c < 0.1 k^{1/4}$. By Theorem A, $W_{\min}(k) \geq 0.1 k^{1/4} > c$. Hence $n := k^2 - c$ satisfies $n > k^2 - W_{\min}(k) = M(k)$, and $n \geq 1$ because $c < 0.1 k^{1/4} < k^2 - 1$. By [R, Lemma 2.1], $s(n) \geq k$; and $s(n) \leq k$ since $n \leq k^2$. So $s(k^2 - c) = k$.

Since $W_{\min}(k)$ is an integer and $W_{\min}(k) \geq 0.1 k^{1/4}$, we get $W_{\min}(k) \geq \lceil 0.1 k^{1/4} \rceil$, so $c^*(k) \geq \lceil 0.1 k^{1/4} \rceil - 1 \geq \lceil 0.1 (4.62 \cdot 10^{12})^{1/4} \rceil - 1 = 146$ (Section 11.3). Finally, $10^4 c^4 + 1 \leq 10^4 \cdot 146^4 + 1 = 4\,543\,718\,560\,001 < 4.62 \cdot 10^{12}$ for $0 \leq c \leq 146$.

Without [R, Lemma 4.10] the same argument applies with Theorem A₁₃ in place of Theorem A: $\lceil 0.1 (2.17 \cdot 10^{13})^{1/4} \rceil - 1 = 215$, and $10^4 \cdot 215^4 + 1 = 21\,367\,506\,250\,001 < 2.17 \cdot 10^{13}$. $\square$

Remark 11.7. The condition $k \geq 10^4 c^4 + 1$ is equivalent to $c < 0.1 k^{1/4}$ for integers $c \geq 0$ and $k \geq 1$, because $10^4 c^4$ is an integer. For $c = 147$ it reads $k \geq 10^4 \cdot 147^4 + 1 = 4\,669\,488\,810\,001$, which exceeds $4.62 \cdot 10^{12}$; so 146 is the largest c for which the threshold $4.62 \cdot 10^{12}$ alone suffices (and 215 in the variant, since $10^4 \cdot 216^4 + 1 = 21\,767\,823\,360\,001 > 2.17 \cdot 10^{13}$). For large c the further thresholds of Section 12.5 give better conditions.

12. CONSTANTS AND THRESHOLDS

This section lists the numerical values used in Sections 10 and 11 and in the statements of Section 1, with the direction in which every displayed number is rounded. All of them are recomputed by the script described in Section 12.7. The parameter values are fixed numbers; the proofs use only the inequalities displayed below and do not depend on how the values were chosen. We do not claim that the thresholds $4.62 \cdot 10^{12}$ and $2.17 \cdot 10^{13}$ are the least ones that (10.3) can give (for fixed

parameters, see Section 12.6), and we do not prove bounds of the form $W \geq c_* k^{1/4}$ for all k below them; Table 4 gives bounds at single values of k .

12.1. Conventions.

- An equality sign between a quantity and a decimal number ($\varepsilon y_0 = 244\,226.4$) means exact equality.
- A decimal number followed by dots, $x = d \dots$, is the truncation of x : $d \leq x < d + u$, where u is one unit of the last displayed digit of d (all such quantities are positive).
- An upper bound $x \leq v$ is the value of x rounded up at the last displayed digit ($v - u < x \leq v$); a lower bound $x \geq v$ is the value rounded down ($v \leq x < v + u$).
- $x < v$ (as in $A < 10.6067$) denotes a chosen bound which is not necessarily obtained by rounding at the last digit.

In every inequality of a proof the displayed values are used in the safe direction: quantities that are subtracted or appear in a denominator are bounded from above, and the others from below. For example, the lower bound $\bar{\Psi}_{\pi^*}(k_0) \geq 146.62429$ and the upper bound $0.1 k_0^{1/4} \leq 146.60896$ together give condition (i) of Proposition 11.2.

12.2. Fixed constants. The fixed constants are

$$\delta = 10^{-5}, \quad \bar{\beta} = 1.5 \cdot 10^{-6}, \quad b_W = 2.0002, \quad Q_* = 0.32,$$

$$\lambda = 2\delta + 2 \cdot 10^{-12} = 2.0000002 \cdot 10^{-5}, \quad c_0^{\text{opt}} = \sqrt{\frac{1-\lambda}{5.0001}} = 0.4472046513 \dots,$$

and the constants depending on the multiplicity bound K ($K = K_w^* = 9$ with [R, Lemma 4.10], $K = K_w = 13$ with [R, Lemma 4.9] only) are given in Table 1, where

$$A = \frac{4\sqrt{K/Q_*}}{2 - 3 \cdot 10^{-6}}, \quad A' = \frac{A}{\cos \bar{\beta}} + 10^{-8}, \quad C_\Lambda = \begin{cases} 1 + 1185.6 \delta^2 & (K = 9), \\ 1 + 1712.6 \delta^2 & (K = 13). \end{cases}$$

The bounds $\bar{A}, \bar{A}', \bar{C}_\Lambda$ (and their analogues for $K = 13$) are the numbers used in all explicit computations (Sections 11.1 and 11.5).

TABLE 1. Constants depending on the multiplicity bound K ; c_∞^+ denotes c_∞ computed with the upper bounds $\bar{A}, \bar{A}'$ (resp. $\bar{A}_{13}, \bar{A}'_{13}$).

	$K = 9$ (with [R, Lemma 4.10])	$K = 13$ ([R, Lemma 4.9] only)
A	$10.6066176277 \dots < 10.6067 = \bar{A}$	$12.7475679053 \dots < 12.7476 = \bar{A}_{13}$
A'	$10.6066176377 \dots < 10.6068 = \bar{A}'$	$12.7475679153 \dots < 12.7476 = \bar{A}'_{13}$
C_Λ	$= 1.00000011856$	$= 1.00000017126$
$\bar{C}_\Lambda$, resp. $\bar{C}_\Lambda^{(13)}$	1.0000002	1.0000002
c_∞ (exact $\bar{A}, \bar{A}'$)	$0.1696524112 \dots$	$0.1411593387 \dots$
c_∞^+	$0.1696502940 \dots$	$0.1411589833 \dots$

The value $C_\Lambda = 1 + 1185.6 \delta^2$ comes from the bound $56.2501 + 318.20 + 56.2501 + 754.8 \leq 1185.6$ for the coefficients of δ^2 in Sections 5 and 7; for $K = 13$ the corresponding coefficients give $81.26 + 459.7 + 81.26 + 1090.3 \leq 1712.6$. The constant $\bar{A}'_{13}$ also bounds the expression $4\sqrt{13/Q_*}/((2 - \bar{\beta}) \cos \bar{\beta}) + 10^{-8}$, because $2 - \bar{\beta} \geq 2 - 3 \cdot 10^{-6}$.

12.3. The parameters of Theorems A and A_{13} . Table 2 lists the parameters π^* (Theorem A) and π_{13}^* (Theorem A_{13}), the values that establish the constraints (C1)–(C7), the constituents of $\bar{B}$, and the quantities in conditions (i)–(iii) of Proposition 11.2 with $c_* = 0.1$. The column for Theorem A uses $\bar{A}, \bar{A}', \bar{C}_\Lambda$, the column for Theorem A_{13} uses $\bar{A}_{13}, \bar{A}'_{13}, \bar{C}_\Lambda^{(13)}$. Constraint (C4) holds since $\omega_0 < \frac{1}{2}$; (C6) and (Q1) follow from (C1) (Lemma 3.3(a)); (C7) holds because $y_0 + 1$ is smaller than the displayed lower bound for $(1 - \varepsilon)(k_0/2 - 3)$. The constraints (C3) and (Q2), which are not used, hold as well (see also Lemma 3.4).

TABLE 2. The parameters and the values at k_0 for Theorems A and A_{13} .

	Theorem A	Theorem A_{13}
c_0	0.3078	0.3674
c_1	0.2754	0.3012
y_0	42 108 000 000	60 000 000 000
ε	$5.8 \cdot 10^{-6}$	$5 \cdot 10^{-6}$
ω_0	$1.7 \cdot 10^{-5}$	$1.4 \cdot 10^{-5}$
k_0	$4.62 \cdot 10^{12}$	$2.17 \cdot 10^{13}$
(C1): $\alpha(y_0)$	$\leq 1.4999830 \cdot 10^{-6}$	$\leq 1.4999043 \cdot 10^{-6}$
(C2): $\hat{\beta}(y_0)$	$\leq 2.9628 \cdot 10^{-9}$	$\leq 2.4846 \cdot 10^{-9}$
(C3): $c_1(1 - \varepsilon)^{-3/4}$	≤ 0.27541	≤ 0.30121
(C3): $c_0 y_0^{1/4}$	≥ 139.43	≥ 181.83
(C5): w_0	≤ 0.04738138	≤ 0.06750274
(C7): $(1 - \varepsilon)(k_0/2 - 3)$	$\geq 2.3099866 \cdot 10^{12}$	$\geq 1.0849945 \cdot 10^{13}$
b^*	$\leq 2.9628 \cdot 10^{-9}$	$\leq 2.4846 \cdot 10^{-9}$
h_0	≥ 0.9052372592	≥ 0.8649945353
εy_0	$= 244\,226.4$	$= 300\,000$
$\text{inv } F$	≥ 0.99998818	≥ 0.99999020
$1/(\omega_0 y_0^{3/4})$	≤ 0.00063282	≤ 0.00058920
$\bar{A}/(2c_1)$	≤ 19.256900	≤ 21.161355
$\frac{2c_1}{c_0 \text{ inv } F} (\bar{A}' + \alpha(y_0)\bar{C}_\Lambda/\delta)$	≤ 19.249235	≤ 21.147482
Γ	$\leq 2.8472 \cdot 10^{-5}$	$\leq 3.1351 \cdot 10^{-5}$
$\bar{B}$	≤ 38.506796	≤ 42.309457
$C_{\text{wall}} = 4c_1/(c_0 \text{ inv } F)$	≤ 3.5789897	≤ 3.2792918
$K_W = 1.000001 c_0^2 C_{\text{wall}}$	≤ 0.3390769	≤ 0.4426483
bound (10.2) for Ω_W at k_0	≤ 0.37941519	≤ 0.70158383
C_{wall} times this bound	≤ 1.3579231	≤ 2.3006981
bound (10.1) for $\ell(H)$ at k_0	≥ 5647.4855	≥ 9134.3029
$\bar{\Psi}_\pi(k_0)$	≥ 146.62429	≥ 215.83530
$0.1 k_0^{1/4}$	≤ 146.60896	≤ 215.83156
D_1	≥ 0.0086429	≥ 0.0055790
$y_1(k_0)^{1/4} D_1$	≥ 10.655	≥ 10.125
$K_W/(2\bar{B})$	≤ 0.0044029	≤ 0.0052311
$\lceil 0.1 k_0^{1/4} \rceil$	147	216
$\Psi_\infty(\pi)$ (computed with $\bar{B}$)	0.1581428...	0.1375313...

The table shows directly that conditions (i)–(iii) of Proposition 11.2 hold: $146.62429 > 146.60896$, $0.0086429 > 0$, $10.655 > 0.0044029$ for Theorem A, and $215.83530 > 215.83156$, $0.0055790 > 0$, $10.125 > 0.0052311$ for Theorem A_{13} . For the exact values, the four terms of $\bar{B}$ add up to $\bar{B}$, and $\bar{\Psi}_\pi(k_0)$ equals $(1 - \omega_0)$ times the bound (10.1), minus C_{wall} times the bound (10.2), divided by $\bar{B}$. The displayed values are rounded separately; in particular the four displayed terms of $\bar{B}$, each rounded up, add up to slightly more than the displayed upper bound for $\bar{B}$.

12.4. The asymptotic constant; the variant without [R, Lemma 4.10]. By Propositions 11.5 and 11.6,

$$c_\infty = \frac{4 \cdot 2^{-1/4} h_0^\infty(c_0^{\text{opt}}) \sqrt{c_0^{\text{opt}}}}{\sqrt{AA'}} = \frac{16 \cdot 2^{-1/4}}{5\sqrt{AA'}} (1 - \lambda)^{5/4} 5.0001^{-1/4}$$

is the supremum, over all parameters satisfying (C0)–(C6), of the limits $\Psi_\infty(\pi)$ of $\Psi_\pi(k)/k^{1/4}$; it is not attained. It depends on the fixed constants only through A , A' and λ (the numbers 5.0001 and 1.00002 come from the factor 0.50001 in w_0). The constants C_Λ and b_W , and the terms $1/(\omega_0 y_0^{3/4})$ and Γ , do not affect c_∞ , since along the family π_n of Proposition 11.5 they enter only through quantities tending to 0; they affect the thresholds. The values of c_∞ are in Table 1; with the exact constants, $c_\infty = 0.1696524112\dots$ for $K = 9$ (Theorem A) and $c_\infty^{(13)} = 0.1411593387\dots$ for $K = 13$ (Theorem A₁₃). With the rounded constants $\bar{A}$, $\bar{A}'$ one gets the slightly smaller c_∞^+ . The parameters of Table 2 are tuned to the thresholds k_0 , not to the asymptotics; their limit ratios $\Psi_\infty(\pi)$ (last row) are smaller than c_∞ , as they must be by Proposition 11.6.

Theorem A (both parts) and Corollary A with the threshold $4.62 \cdot 10^{12}$ depend on the computer-assisted [R, Lemma 4.10] through the constants A , A' , C_Λ (Remark 2.2 and Section 11.5). Theorem A₁₃ and the corresponding form of Corollary A (threshold $2.17 \cdot 10^{13}$) use [R, Lemma 4.9] instead and do not depend on [R, Lemma 4.10].

12.5. Further thresholds and bounds at given k . The proof of Section 11.3 applies verbatim to the pairs (c_*, k_0) and parameters of Table 3: for each of them π satisfies (C0)–(C6) and (C7) for k_0 , and conditions (i)–(iii) of Proposition 11.2 hold (the row $\bar{\Psi}_\pi(k_0)/k_0^{1/4}$ shows (i) in the form $\bar{\Psi}_\pi(k_0)/k_0^{1/4} \geq c_*$). Hence every closed packing of $[0, k]^2$ with an integer $k \geq k_0$ satisfies $W \geq c_* k^{1/4}$; the last column uses $\bar{A}_{13}$, $\bar{A}'_{13}$, $\bar{C}_\Lambda^{(13)}$ and does not depend on [R, Lemma 4.10].

TABLE 3. Further thresholds: $W \geq c_* k^{1/4}$ for all integers $k \geq k_0$.

	$K = 9$				$K = 13$
c_*	0.12	0.14	0.15	0.16	0.12
k_0	$2.17 \cdot 10^{13}$	$2.01 \cdot 10^{14}$	$1.18 \cdot 10^{15}$	$2.77 \cdot 10^{16}$	$3.78 \cdot 10^{14}$
c_0	0.3674	0.4086	0.4243	0.4376	0.4153
c_1	0.3009	0.3173	0.3234	0.3284	0.3203
y_0	$6 \cdot 10^{10}$	$7.422 \cdot 10^{10}$	$8.003 \cdot 10^{10}$	$8.5109 \cdot 10^{10}$	$7.667 \cdot 10^{10}$
ε	$5 \cdot 10^{-6}$	$4.7 \cdot 10^{-6}$	$4.6 \cdot 10^{-6}$	$4.5 \cdot 10^{-6}$	$4.6 \cdot 10^{-6}$
ω_0	$1.5 \cdot 10^{-5}$	$1.5 \cdot 10^{-5}$	$1.4 \cdot 10^{-5}$	$1.4 \cdot 10^{-5}$	$1.3 \cdot 10^{-5}$
$\bar{\Psi}_\pi(k_0)/k_0^{1/4}$	≥ 0.1200448	≥ 0.1400019	≥ 0.1500058	≥ 0.1600075	≥ 0.1200086
D_1	≥ 0.0067036	≥ 0.0041135	≥ 0.0027026	≥ 0.0012500	≥ 0.0029558
$y_1(k_0)^{1/4} D_1$	≥ 12.166	≥ 13.024	≥ 13.320	≥ 13.560	≥ 10.959
$K_W/(2\bar{B})$	≤ 0.0062733	≤ 0.0077586	≤ 0.0083679	≤ 0.0088999	≤ 0.0066855

For example, with $(c_*, k_0) = (0.16, 2.77 \cdot 10^{16})$ the argument of Section 11.6 gives $s(k^2 - c) = k$ for all integers $c \geq 0$ and $k \geq 2.77 \cdot 10^{16}$ with $k > (c/0.16)^4$; for $c \geq 2065$ the second condition implies the first, because $(2065/0.16)^4 > 2.77 \cdot 10^{16}$ (while $(2064/0.16)^4 < 2.77 \cdot 10^{16}$).

At a single value of k no monotonicity is needed: if π satisfies (C0)–(C7) for k , then Lemma 11.1 gives $W \geq \bar{\Psi}_\pi(k)$, and since W is an integer, $W \geq \lceil \bar{\Psi}_\pi(k) \rceil$. Table 4 lists such bounds; each parameter set satisfies (C0)–(C7) for the k of its row.

12.6. Sensitivity of the threshold. Condition (i) of Proposition 11.2 for Theorem A holds with a small margin. Keep π^* and $k_0 = 4.62 \cdot 10^{12}$, and keep the numerator of $\bar{\Psi}_{\pi^*}(k_0)$ fixed. Then (i)

TABLE 4. Lower bounds for W at given k .

K	k	c_0	c_1	y_0	ε	ω_0	$\bar{B} \leq$	$\bar{\Psi}_\pi(k) \geq$	$W \geq$
9	10^{13}	0.3421	0.2903	$5.2015 \cdot 10^{10}$	$5.3 \cdot 10^{-6}$	$1.6 \cdot 10^{-5}$	36.52537	196.77	197
9	10^{14}	0.3993	0.3136	$7.0863 \cdot 10^{10}$	$4.8 \cdot 10^{-6}$	$1.5 \cdot 10^{-5}$	33.80809	426.16	427
9	10^{16}	0.4345	0.3273	$8.393 \cdot 10^{10}$	$4.5 \cdot 10^{-6}$	$1.4 \cdot 10^{-5}$	32.40967	1575.76	1576
9	10^{20}	0.4457	0.3324	$2.458 \cdot 10^{11}$	$2.7 \cdot 10^{-6}$	$9.5 \cdot 10^{-6}$	31.91017	16751.28	16752
9	10^{30}	0.4472	0.3343	$5.473 \cdot 10^{14}$	$5.7 \cdot 10^{-8}$	$5.3 \cdot 10^{-7}$	31.72495	5363347.52	5363348
13	10^{13}	0.3421	0.2907	$5.2015 \cdot 10^{10}$	$5.3 \cdot 10^{-6}$	$1.4 \cdot 10^{-5}$	43.84613	163.91	164
13	10^{14}	0.3993	0.314	$7.0863 \cdot 10^{10}$	$4.8 \cdot 10^{-6}$	$1.3 \cdot 10^{-5}$	40.58423	355.01	356
13	10^{16}	0.4345	0.3276	$8.393 \cdot 10^{10}$	$4.5 \cdot 10^{-6}$	$1.3 \cdot 10^{-5}$	38.90551	1312.66	1313
13	10^{20}	0.4458	0.3325	$1.928 \cdot 10^{11}$	$3 \cdot 10^{-6}$	$9.5 \cdot 10^{-6}$	38.33686	13948.51	13949
13	10^{30}	0.4472	0.3343	$4.287 \cdot 10^{14}$	$6.4 \cdot 10^{-8}$	$5.3 \cdot 10^{-7}$	38.12805	4462691.45	4462692

holds exactly when the bracket does not exceed

$$B_{\text{crit}} := \frac{(1 - \omega_0) \cdot (\text{bound (10.1) at } k_0) - C_{\text{wall}} \cdot (\text{bound (10.2) at } k_0)}{0.1 k_0^{1/4}} \geq 38.510824,$$

and the relative margin is $(B_{\text{crit}} - \bar{B})/\bar{B} = 1.0463 \dots \cdot 10^{-4}$, less than $1.05 \cdot 10^{-4}$.

Consequently Theorem A, with the same π^* and k_0 , remains valid if one of the constants entering (★) is replaced by a larger value as follows (all other constants as in Table 1):

- (a) A' by any value up to 10.60905 (with $A \leq 10.6067$);
- (b) A by any value up to 10.60891 (with $A' \leq 10.6068$);
- (c) A and A' both, by values with $A \leq 10.60781$ and $A' \leq A + 10^{-4}$; in particular $A = A' \leq 10.60781$ suffices;
- (d) C_Λ by any value up to 1.0150;
- (e) b_W in Theorem 9.1 by any value up to 53.09;
- (f) the wall term $C_{\text{wall}}\Omega_W$ by up to 1.434 times the bound obtained from (10.2).

An additional loss of the form $C'W/\delta$ in Theorem 9.1 acts as an increase of C_Λ by C' . In each case it suffices to evaluate conditions (i)–(iii) at the extreme value: increasing any of these constants does not decrease $\bar{B}$, K_W or the wall term, and increases at least one of them; hence it decreases $\bar{\Psi}_{\pi^*}(k_0)$ (whose numerator stays positive), does not increase D_1 , and decreases

$$y_1(k_0)^{1/4} D_1 - \frac{K_W}{2\bar{B}} = \frac{y_1(k_0)^{1/4} h_0 (1 - \omega_0) (1 - \varepsilon)^{1/4} - K_W/2}{\bar{B}} - \frac{y_1(k_0)^{1/4} c_*}{4 \cdot 2^{3/4}}$$

as long as the numerator of the first fraction is positive, which holds at the extreme values. The listed values are the largest possible ones for this π^* , rounded down: the largest values for which (i) still holds are 10.6090514... in (a), 10.6089191... in (b), 10.6078176... for A in (c) with $A' = A + 10^{-4}$, 1.0150101... in (d), 53.0942... in (e) and 1.43499... in (f).

If, on the other hand, C_Λ were 2, then for the same π^* conditions (ii) and (iii) would still hold at $k = 4.62 \cdot 10^{12}$, so by the proof of Proposition 11.2 (Remark 11.3) the function $\bar{\Psi}_{\pi^*}(k) - 0.1 k^{1/4}$ would be increasing on $[4.62 \cdot 10^{12}, \infty)$; it would be negative at $4.62 \cdot 10^{12}$ and vanish at a single point $4.8449 \dots \cdot 10^{12}$. Thus condition (i) would fail at $4.62 \cdot 10^{12}$, and for this π^* the threshold would move to $4.845 \cdot 10^{12}$ (rounded up), where conditions (i)–(iii) and (C7) hold. We do not discuss other choices of π for this case. The asymptotic constant c_∞ depends only on A and A' (Section 12.4).

In terms of k the margin is as follows. Keep π^* and the bounds $\bar{A}, \bar{A}', \bar{C}_\Lambda$. Conditions (ii) and (iii) of Proposition 11.2 hold already at the least integer k for which (C7) holds, namely $k = 84\,216\,488\,464$, so by the proof of Proposition 11.2 (Remark 11.3) the function $\bar{\Psi}_{\pi^*}(k) - 0.1 k^{1/4}$ is increasing wherever (C7) holds. Hence condition (i) holds exactly for $k \geq k_*$, where

$$k_* = 4.616673 \dots \cdot 10^{12}, \quad 4\,616\,673\,963\,026 < k_* \leq 4\,616\,673\,963\,027,$$

and with this π^* the argument gives $W \geq 0.1 k^{1/4}$ for every integer $k \geq 4616673963027$, but not for any smaller k (this says nothing about W itself for smaller k). The threshold $k_0 = 4.62 \cdot 10^{12}$ of Theorem A is k_* rounded up to three significant digits: $k_0/k_* - 1 = 7.20 \dots \cdot 10^{-4}$.

For Theorem A₁₃ the corresponding relative margin is $(B_{\text{crit}} - \bar{B})/\bar{B} = 1.7385 \dots \cdot 10^{-5}$ (with π_{13}^* , $k_0 = 2.17 \cdot 10^{13}$ and the constants for $K = 13$). In terms of k : with π_{13}^* , conditions (ii) and (iii) hold already at the least integer k for which (C7) holds, $k = 120\,000\,600\,012$, and condition (i) holds exactly for $k \geq k_*^{(13)}$, where

$$k_*^{(13)} = 2.169597 \dots \cdot 10^{13}, \quad 21\,695\,977\,343\,082 < k_*^{(13)} \leq 21\,695\,977\,343\,083;$$

the threshold $2.17 \cdot 10^{13}$ is $k_*^{(13)}$ rounded up to three significant digits, and $2.17 \cdot 10^{13}/k_*^{(13)} - 1 = 1.85 \dots \cdot 10^{-4}$.

12.7. The script. The script `k14_constants_check.py`, which accompanies this manuscript, recomputes every numerical constant displayed in Sections 1–12 (in particular the constants in Theorems A, A₁₃ and Corollary A, the numbers in the comparison with [R] in Section 1, the numerical constants of Sections 2–9, and all numbers of Sections 10–12) and checks each displayed value in its stated rounding direction; it consists of 555 such checks. It requires Python 3 and the `mpmath` library and is run as

`python k14_constants_check.py`

Every quantity is computed with 50 significant digits and, in addition, enclosed in interval arithmetic with outward rounding (`mpmath.iv`, 50 digits). The endpoints of the enclosures are converted exactly to rational numbers and compared with the displayed decimals according to the conventions of Section 12.1. The constraints (C0)–(C4), (C7), (Q1), (Q2) are evaluated in exact rational arithmetic (after raising both sides to a suitable power), displayed values that are rational functions of displayed decimals (such as S_9 in Section 7) are compared in exact rational arithmetic, and conditions (i)–(iii) of Proposition 11.2 and all other inequalities in interval arithmetic. It prints one line for each check and a summary line at the end.

VERIFICATION STATUS

This paper was developed with AI assistance (see the section on the use of AI). Each part of the proof (Sections 4–10) was first written as a separate draft, and each draft received one AI review and one AI numerical test, made independently in separate sessions with their own code; the assembled argument received a review and an end-to-end numerical test. The reviews judged the drafts correct, with minor and wording items only, and no test found a counterexample; the resulting changes concern statements and proof details (for instance, a hypothesis that was used but not stated is now constraint (C6)) and are included here. The present text was written from the drafts, and the script of Section 12.7 recomputes every displayed constant. A blank-slate AI review of the present text, divided into twelve parts reviewed separately and accompanied by numerical searches for counterexamples, found no critical or major issue and eighteen minor ones (wording, the use of the proofs in [R], notation, and two constants displayed with a last digit larger than necessary); all of them have been addressed in this version. That review also compared the statements quoted from [R] with [R] (version 1.2), recomputed the constants independently, and recomputed the finite check in the proof of [R, Lemma 4.9] with a new program in interval arithmetic. All reviewers are AI systems of the same family and may share blind spots, and the geometric tests use small analogues, not the actual parameter regime. The proofs use [R] (version 1.2), itself not peer reviewed. Theorem A also uses the computer-assisted [R, Lemma 4.10], whose box computations were replayed by the Squares Project [LevR] with the same programs but have not been reproduced by an independent implementation; Theorem A₁₃ does not use it. The programs, the numerical tests and a summary of the reviews are available at <https://github.com/squarepacker/k2-minus-c-power>. No human has checked the argument. This paper has not been peer reviewed.

Acknowledgements. I thank J. Levy and the Squares Project for an AI review of version 1.1 of [R] and for replaying the computation of [R, Lemma 4.10], on which Theorem A depends [LevR].

Use of AI. Claude (Anthropic) was used extensively in developing the arguments, writing the text and the programs (including the numerical tests described above), and in checking them. It is not an author; the author takes full responsibility for the contents. The paper has not been refereed.

Licence. This text is distributed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0); the accompanying programs are distributed under the MIT licence.

REFERENCES

- [AB] M. Z. Arslanov, H. D. Bui, Note on “Efficient packings of unit squares in a large square”, *Discrete Comput. Geom.* **76** (2026), no. 2, 805–812 (published online 3 August 2025), doi:10.1007/s00454-025-00767-w.
- [Bui] H. D. Bui, Square packing with $O(x^{0.6})$ wasted area, arXiv:2508.04603v2 (2025, revised 2026).
- [CG] F. Chung, R. Graham, Efficient packings of unit squares in a large square, *Discrete Comput. Geom.* **64** (2020), no. 3, 690–699, doi:10.1007/s00454-019-00088-9.
- [CG09] F. Chung, R. Graham, Packing equal squares into a large square, *J. Combin. Theory Ser. A* **116** (2009), no. 6, 1167–1175, doi:10.1016/j.jcta.2009.02.005.
- [Dan] E. Daniel, *square-packing*, GitHub repository, <https://github.com/evand/square-packing>, file `s12/search/FRIEDMAN.md`, commit `7ff3b21` (accessed 5 October 2026).
- [EG] P. Erdős, R. L. Graham, On packing squares with equal squares, *J. Combin. Theory Ser. A* **19** (1975), no. 1, 119–123, doi:10.1016/0097-3165(75)90099-0.
- [Fri] E. Friedman, Packing unit squares in squares: a survey and new results, *Electron. J. Combin.*, Dynamic Survey DS7 (1998, revised 2009), doi:10.37236/28.
- [LevR] J. Levy, *The Squares Project*, GitHub repository, <https://github.com/jlevy/squares>: AI review of version 1.1 of [R], file `docs/project/reviews/review-2026-10-05-squarepacker-k2-minus-c.md`, commit `6953107`, and issue `#368` (comments of 5 and 6 October 2026, UTC; accessed 7 October 2026).
- [McC] R. McClenagan, Optimally packing a large square by unit squares, arXiv:2602.01484 (2026).
- [R] S. Ryu, Packing $k^2 - c$ unit squares: $s(k^2 - c) = k$ for all large k , preprint, version 1.2, 2026, doi:10.5281/zenodo.23194104; programs and data: doi:10.5281/zenodo.23194031 and <https://github.com/squarepacker/k2-minus-c> (release v1.2).
- [RV] K. F. Roth, R. C. Vaughan, Inefficiency in packing squares with unit squares, *J. Combin. Theory Ser. A* **24** (1978), no. 2, 170–186, doi:10.1016/0097-3165(78)90005-5.